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Camila de Castro Pinheiro

**POTENCIAL DE ANTIVENENO DE MOLÉCULAS NATURAIS
NA NEUTRALIZAÇÃO DE ATIVIDADES TÓXICO-
FARMACOLÓGICAS CAUSADAS POR VENENOS DE
SERPENTES**

**Niterói/RJ
2026**

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Tese de Doutorado apresentada ao Programa de Pós-Graduação em Ciências Biomédicas (Fisiologia e Farmacologia) da Universidade Federal Fluminense, como parte dos requisitos necessários à obtenção do título de Doutor em Ciências Biomédicas. Área de concentração: Farmacologia.

Orientador: Prof. Dr. André Lopes Fuly
Coorientador: Prof. Dr. Luiz Carlos Simas Pereira Junior

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Ao meu pai Elias que sempre me ensinou a voar. A minha mãe Dilza que sempre foi meu lar pra retornar.

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*As pessoas podem tirar tudo de você, menos
o seu conhecimento.*

Immanuel Kant

RESUMO

Castro-Pinheiro, Camila. **Potencial antiveneno de moléculas naturais na neutralização de atividades tóxico-farmacológicas causadas por venenos de serpentes**. 2026. Tese de Doutorado (Doutorado em Ciências Biomédicas – Fisiologia e Farmacologia) – Instituto Biomédico, Universidade Federal Fluminense, Niterói, 2026.

O envenenamento por serpentes é um problema de saúde pública em países tropicais. No Brasil, as serpentes *Bothrops* são responsáveis por 90% dos acidentes ofídicos. Seu veneno é rico em proteases responsáveis por alterações locais e sistêmicas graves, como distúrbios da coagulação, hemorragia e necrose tecidual. A soroterapia é o tratamento preconizado pelo Ministério da Saúde, porém sua eficácia é limitada na neutralização dos efeitos locais, prevalentes em casos de envenenamento botrópico. Neste cenário, o presente estudo busca por produtos naturais com potencial antiveneno que possam servir como terapias coadjuvantes ou alternativas a soroterapia. Extratos de plantas *Aesculus hippocastanum* e *Melilotus officinalis*, assim como polissacarídeos isolados das algas marinhas *Fucus vesiculosus*, *Undaria pinnatifida*, *Chlorella sorokiniana*, *Scenedesmus acuminatus*, *Scenedesmus obliqs* e *Nannocholis* sp foram testados *in vitro* contra as atividades coagulante, proteolítica e fosfolipásica dos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*. Os resultados sugerem que diferentes espécies de plantas e polissacarídeos isolados de algas e microalgas foram capazes de inibir significativamente as atividades coagulante, proteolítica e fosfolipásica dos venenos em diferentes níveis a depender do tipo da espécie de serpente, origem e natureza química dos produtos testados. Dessa maneira, o estudo demonstra que os produtos naturais testados constituem fontes promissoras de moléculas com potencial biotecnológico, podendo ser utilizados como agentes auxiliares à soroterapia no tratamento de envenenamentos por serpentes do gênero *Bothrops*.

Palavras-chave: Veneno de serpentes, Bothrops, Plantas, Polissacarídeos, Bioprospecção, Produtos naturais

ABSTRACT

Castro-Pinheiro, Camila. **Antivenom potential of natural molecules in the neutralization of toxic-pharmacological activities caused by snake venoms.** 2026. Doctorate Thesis (Doctorate in Biomedical Sciences – Physiology and Pharmacology) – Biomedical Institute, Fluminense Federal University, Niterói, 2026.

Snakebite envenoming (SBE) is a public health problem in tropical countries. In Brazil, Bothrops snakes are responsible for over 90% SBE. Their venom is rich in proteases responsible for severe local and systemic alterations, such as coagulation disorders, hemorrhage, and tissue necrosis. Antivenom serotherapy is the treatment recommended by the Brazilian Ministry of Health; however, its efficacy is limited in neutralizing local effects, which are prevalent in cases of Bothrops SBE. Thus, the present study looking for natural products with antivenom potential that may serve as adjuvant or alternative therapies to serotherapy. Extracts from *Aesculus hippocastanum*, and *Melilotus officinalis*, as well as isolated products from marine algae *Fucus vesiculosus* and *Undaria pinnatifida*, and microalgae *Chlorella sorokiniana*, *Scenedesmus acuminatus*, *Scenedesmus obliquus*, and *Nannochloris* sp., were tested in vitro against the coagulant, proteolytic, and phospholipase activities of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms. The results demonstrate that different plant species and polysaccharides isolated from algae and microalgae were able to significantly inhibit venom proteolysis, coagulation, and phospholipase A₂ activities at different levels depending on the snake species, as well as on the origin and chemical nature of the tested products. Therefore, the study demonstrates that the tested natural products represent promising sources of molecules with biotechnological potential and may be used as auxiliary agents to serotherapy in the treatment of envenomings caused by Bothrops.

Keywords: Snake Venom, Bothrops, Plants, Polysaccharide, Bioprospecting, Natural products.

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LISTA DE ABREVIATURAS E SIGLAS

1509	<i>Nannochloris</i> sp <i>Naumann</i>
3FTx	Toxina de três dedos
AH	<i>Aesculus hippocastanum</i>
Bthtx II	Botrotoxina II
BuOH	Butanol
C1	<i>Chlorella sorokiniana</i>
CLP	Leticina do tipo C
COX-2	Ciclooxigenase-2
CRISP	Proteína secretora rica em cisteína
CTL	Proteína semelhante a lectina do tipo C
DCM	Diclorometano
DTx	Dendrotoxina
E	Etanólico
EROS	Espécies reativas de oxigênio
EtOAc	Acetato de etila
FDA	Administração de alimentos e medicamentos
FVF	<i>Fucus vesiculosus</i>
GRAS	Geralmente reconhecido como seguro
HEX	Hexano
IL 1 β	Interleucina 1 β
IL 6	Interleucina 6
LAO	L-amino-oxidase

mAbs	Anticorpos monoclonais
MeOH	Metanol
MO	<i>Melilotus officinalis</i>
MS	Ministério da Saúde
Myo	Miotoxina
NF-κB	Fator nuclear kappa B
OMS	Organização Mundial de Saúde
PLAs ₂	Fosfolipase A ₂
S302	<i>Scenedesmus obliquus</i>
S309	<i>Scenedesmus acuminatus</i>
SC	<i>Siparuna cymosa</i>
SD	<i>Siparuna decipiens</i>
SF	<i>Siparuna ficoides</i>
SR	<i>Siparuna reginae</i>
SVMPs	Metaloproteinases de serpentes
SVSPs	Serinoproteases de serpentes
TNFα	Fator de necrose tumoral alfa
UPF	<i>Undaria Pinnatifida</i>

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INTRODUÇÃO GERAL - O ENVENENAMENTO

1 Envenenamento por serpentes como problema de saúde pública global

O envenenamento por serpentes é reconhecido pela Organização Mundial da Saúde (OMS) como uma Doença Tropical Negligenciada, afetando populações majoritariamente rurais com vulnerabilidade socioeconômica e com dificuldade ao acesso a serviços de saúde. Estima-se que globalmente ocorram entre 1,8 e 2,7 milhões de acidentes ofídicos por ano, resultando em até 138 mil mortes e aproximadamente 400 mil casos de sequelas físicas permanentes. Ainda assim, este número tende a ser subestimado, pois existe um alto número de subnotificações associados ao envenenamento por serpentes em regiões tropicais e subtropicais (Gutiérrez *et al.*, 2017).

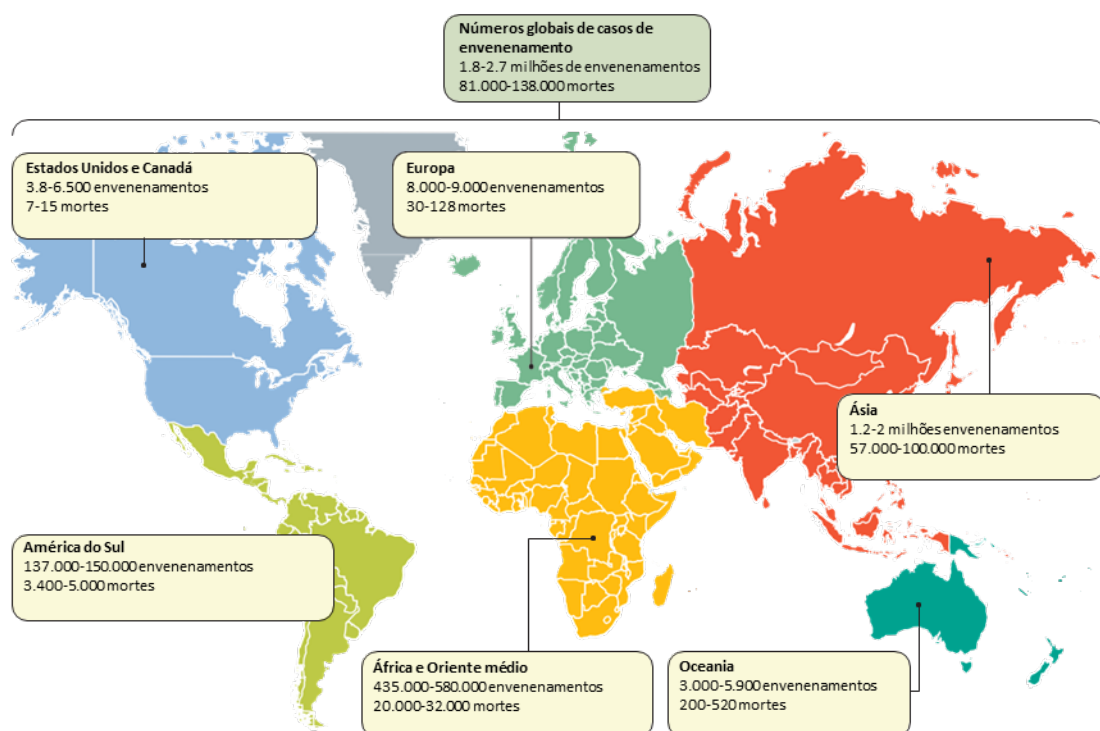


Figura 1. Distribuição geográfica do número estimado de casos de envenenamento por serpentes e mortes associadas. Adaptado de Gutiérrez 2017.

A maioria dos casos de envenenamento nas Américas ocorre em áreas rurais, entre trabalhadores agrícolas, populações ribeirinhas e comunidades indígenas. No Brasil e em outros países sul-americanos como Colômbia, Peru e Venezuela, a ampla distribuição das serpentes, assim como a vulnerabilidade social e a distância do local onde ocorreu o envenenamento dos centros de saúde contribuem para índices persistentes de morbidade (Chippaux, 2017). A região amazônica representa o epicentro dessa problemática seja pela sua extensão territorial, diversidade ecológica ou dificuldade de acesso aos serviços de saúde, e estes são fatores que constituem barreiras importantes para que ocorra a assistência médica adequada aos casos; sendo a demora no atendimento a grande responsável pelo aumento do risco de complicações sistêmicas graves. Sendo assim, o envenenamento por serpentes permanece como um grande desafio de saúde pública (Rocha e Gomides, 2024).

2 Principais gêneros de serpentes causadores de acidentes no Brasil

O Brasil possui uma das maiores diversidades de serpentes do mundo, com mais de 400 espécies, sendo cerca de 70 consideradas de importância médica. Entre elas, podemos destacar principalmente as pertencentes à família *Viperidae*, especialmente os gêneros *Crotalus*, *Bothrops* e *Lachesis*, responsáveis pela grande maioria dos acidentes ofídicos. Cerca de aproximadamente 90% dos acidentes são causados por serpentes do gênero *Bothrops*. As espécies *B. jararaca*, *B. jararacussu* e *B. neuwiedi* popularmente conhecidas por jararaca, jararacuçu e jararaca pintada, respectivamente, estão entre as espécies que mais causam acidentes ofídicos. Estas serpentes são altamente adaptáveis a diferentes ambientes, incluindo matas, áreas agrícolas e regiões periurbanas, o que contribui significativamente para alta incidência de casos de envenenamento em seres humanos (Romanazzi *et al.*, 2025).

O gênero *Lachesis* apresenta uma menor frequência de acidentes, porém são acidentes de maior letalidade. As espécies desse gênero, em especial a *Lachesis muta*, popularmente conhecida como surucucu-pico-de-jaca, habitam tipicamente florestas densas e de difícil acesso, como a Amazônia (Diniz-Sousa *et al.*, 2020). Embora representem cerca de 1 a 3% dos acidentes no país, os casos envolvendo

Lachesis costumam evoluir com maior intensidade clínica, devido ao volume de veneno inoculado e a composição de seu veneno. O envenenamento laquéutico caracteriza-se por manifestações sistêmicas como hipotensão, vômitos, diarreia, sudorese intensa e alterações hemostáticas severas (Cremones *et al.*, 2016; Sachett *et al.*, 2022).

3 Etiologia dos acidentes ofídicos no Brasil

A origem dos acidentes ofídicos no Brasil está intimamente relacionada ao comportamento biológico das serpentes, à ecologia das espécies, aos fatores ambientais e às atividades humanas que favorecem o contato entre humanos e animais. A família *Viperidae* é composta pelos gêneros *Bothrops*, *Lachesis* e *Crotalus*. Essas espécies apresentam hábitos predominantemente terrestres, com comportamento defensivo quando ameaçadas (Alcântara *et al.*, 2018; Matos e Ignotti, 2020).

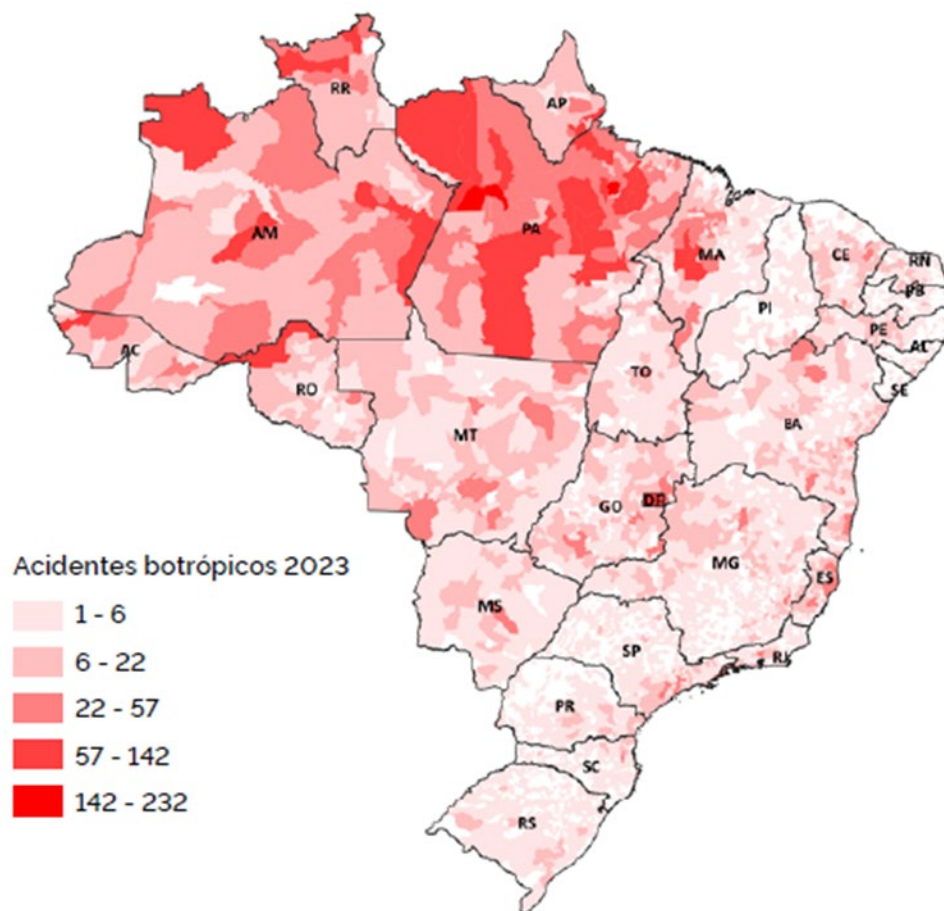


Figura 2. Distribuição geográfica dos acidentes ofídicos causados por serpentes *Bothrops* no Brasil no ano de 2023. Adaptado de Boletim epidemiológico- Ministério da Saúde 2023.

O gênero *Bothrops* possui ampla distribuição geográfica e elevada capacidade de adaptação ecológica. Essas serpentes ocupam ambientes variados, desde florestas úmidas até áreas mais habitadas pelo homem. A maior parte dos acidentes ocorre em períodos de maior atividade desses animais, como durante a estação chuvosa, quando há aumento da oferta de alimento e deslocamento das serpentes para áreas abertas. Além disso, as práticas agrícolas aumentam a exposição humana (Alcântara *et al.*, 2018; Duque *et al.*, 2023; Matos e Ignotti, 2020).

4 Composição do veneno

Os venenos de serpentes constituem misturas complexas de biomoléculas biologicamente ativas, cuja composição pode variar significativamente entre espécies, populações, idade dos animais e fatores ambientais como a presa disponível para alimentação. Os venenos da família *Viperidae* apresentam um conteúdo predominantemente proteico de cerca 90-99% do peso seco total, que são responsáveis por interferir nos sistemas hemostático, cardiovascular, muscular, imune, renal e respiratório (Tasoulis e Isbister, 2023).

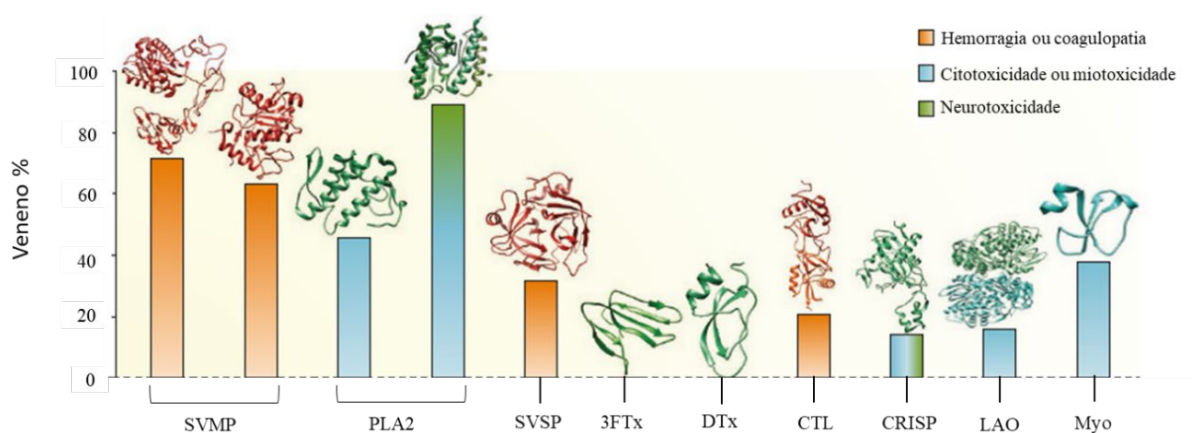


Figura 3. Níveis de toxinas no veneno de serpentes da família *Viperidae* expressos como % do proteoma total do veneno. As cores representam os principais efeitos biológicos dose dependente de cada toxina. SVMPs, metaloproteinase de veneno de serpente; PLAs₂, fosfolipase A₂; SVSPs, serinoproteinase de veneno de serpente; 3FTx, toxina de três dedos; DTx, dendrotoxina; CTL, proteína semelhante à lectina do tipo C; CRISP, proteína secretora rica em cisteína; LAO, l-amino oxidase; Myo, miotoxina; Adaptado de Gutiérrez 2017.

A composição proteica dos venenos viperídeos é majoritariamente composta por quatro grandes classes: metaloproteinases de veneno de serpentes (SVMPs), serino proteases de veneno de serpentes (SVSPs), fosfolipases A₂ (PLAs₂) e lectinas tipo C (CLP). Além dessas, os venenos também têm proteínas não enzimáticas com alto poder tóxico, como peptídeos vasoativos, fatores pró-inflamatórios e proteínas ligantes de carboidratos. As lectinas tipo C são proteínas não enzimáticas que se ligam a carboidratos específicos e participam de processos de regulação na hemostasia, como na agregação e adesão plaquetária. Elas podem tanto estimular quanto

bloquear a agregação das plaquetas, dependendo da estrutura tridimensional e do tipo de interação molecular estabelecida (Damm *et al.*, 2021; Offor *et al.*, 2022; Tasoulis e Isbister, 2023).

As SVMs constituem uma das classes mais abundantes nos venenos botrópicos, sendo responsáveis por uma variedade de efeitos locais, como hemorragia, necrose tecidual e degradação da matriz extracelular. Sua ação proteolítica é voltada para componentes estruturais dos vasos sanguíneos, o que leva ao extravasamento de sangue e à progressão das lesões (Offor *et al.*, 2022). Em contrapartida, as SVSPs são essenciais para os distúrbios hemostáticos sistêmicos observados no envenenamento da família *Viperidae*. Elas mimetizam enzimas endógenas da cascata de coagulação, atuando como ativadoras de trombina (trombina-símile), do plasminogênio ou de fatores que interferem diretamente na formação e estabilidade do fibrinogênio. Em venenos de *Bothrops*, as SVSPs alteram drasticamente o equilíbrio hemostático, resultando em consumo de fibrinogênio, incoagulabilidade sanguínea e sangramentos espontâneos (Rao *et al.*, 2024; Romanazzi *et al.*, 2025).

As PLAs₂ desempenham papel central em processos inflamatórios, na mionecrose e degradação de membranas celulares. Elas são responsáveis por liberar ácidos graxos e lisofosfolípidios, iniciando cascatas inflamatórias e danos musculares extensos. O conjunto de toxinas presentes no veneno de serpentes, portanto, reflete uma estratégia evolutiva altamente eficaz que combina múltiplos efeitos fisiopatológicos, garantindo a captura de presas e a defesa do animal (Lomonte *et al.*, 2020; Offor *et al.*, 2022; Tasoulis e Isbister, 2023).

5 Efeitos toxicológicos do envenenamento por Bothrops

O envenenamento provocado por serpentes do gênero *Bothrops* apresenta quadro clínico característico, marcado por efeitos locais e sistêmicos. No local da picada, os sintomas mais comuns incluem dor intensa, edema progressivo, equimose e hemorragia. Esses sinais decorrem principalmente da ação das SVMPs e PLAs₂, que desencadeiam degradação vascular, lise de células musculares e ativação inflamatória exacerbada. A evolução do quadro pode envolver formação de edema bolhoso, necrose extensa e até necessidade de procedimentos cirúrgicos para descompressão e desbridamento (Gutiérrez *et al.*, 2017).

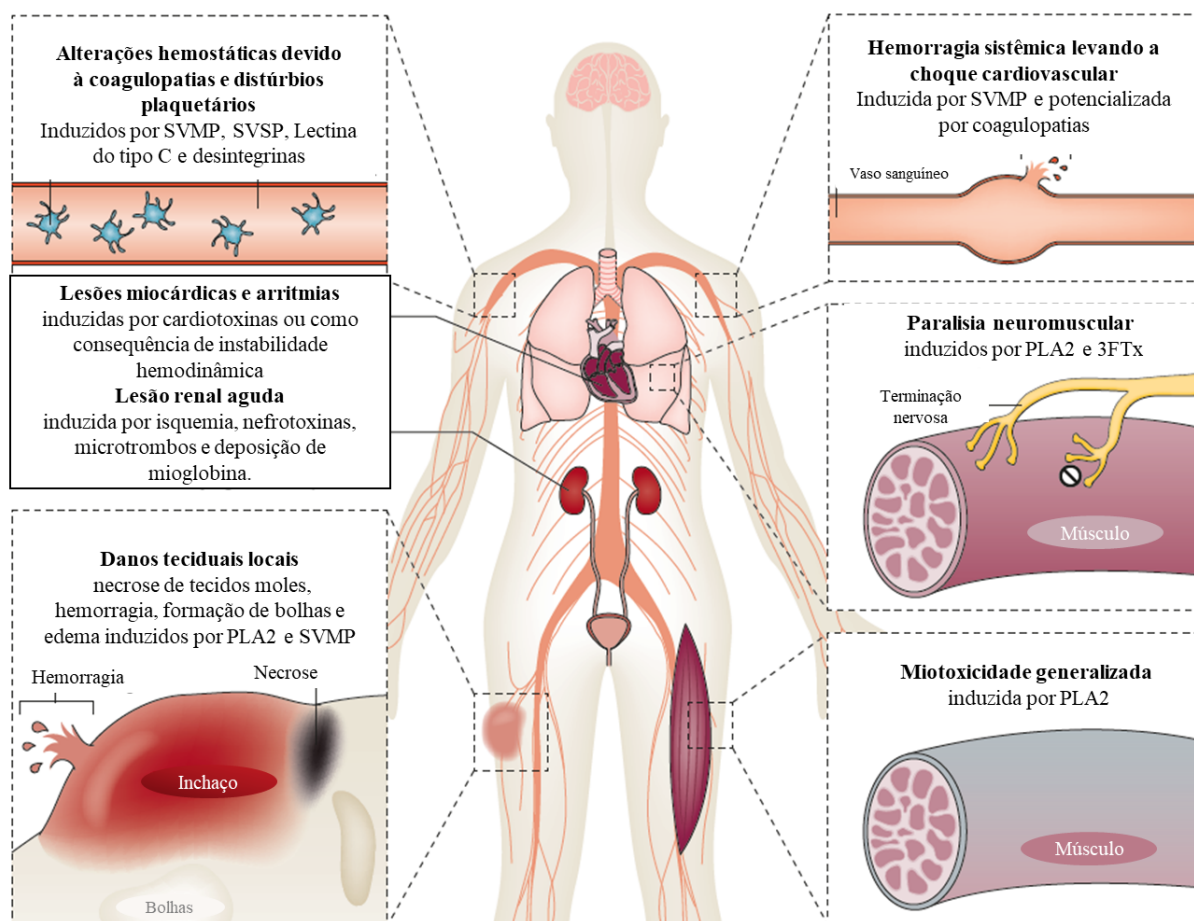


Figura 4. Efeitos locais e sistêmicos decorrentes de envenenamento por serpentes. Adaptado de Gutiérrez 2017

No contexto sistêmico, o envenenamento é conhecido por gerar distúrbios hemostáticos, incluindo consumo acelerado de fibrinogênio, incoagulabilidade sanguínea e sangramentos espontâneos. Esses efeitos são mediados por serinoproteases com atividade trombina-símile, que convertem fibrinogênio em fibrina de forma desregulada, além de proteínas que interferem na integridade e função das plaquetas (Cavalcante *et al.*, 2023; Larréché *et al.*, 2021). Outros sintomas sistêmicos incluem hipotensão, insuficiência renal aguda, alterações na função hepática e manifestações cardiovasculares variadas. A gravidade do envenenamento depende de fatores como quantidade de veneno inoculado, local da picada e o tempo decorrido até a administração do antiveneno (Cavalcante *et al.*, 2023; Gutiérrez *et al.*, 2009).

Um aspecto relevante nos envenenamentos por *Bothrops* é a presença de efeitos locais que não são completamente neutralizados pela soroterapia heteróloga. Lesões teciduais extensas, necrose, perda funcional e sequelas permanentes são decorrentes da ação direta das toxinas nos tecidos profundos, onde os anticorpos do soro não conseguem reverter. Essa limitação terapêutica motiva o estudo de agentes naturais com potencial de neutralizar toxinas específicas antes que causem danos irreversíveis (Barbosa *et al.*, 2024; Gutiérrez *et al.*, 2009).

6 Soroterapia

O tratamento preconizado pelo Ministério da Saúde (MS) para acidentes ofídicos baseia-se na administração de antivenenos heterólogos, produzidos principalmente em equinos. Esses soros são formulados a partir da imunização dos animais com venenos específicos, resultando na geração de anticorpos capazes de neutralizar as principais toxinas circulantes no organismo humano. A soroterapia monovalente ou polivalente é reconhecida como o único tratamento efetivo para reduzir a mortalidade e minimizar complicações sistêmicas dos envenenamentos (Hamza *et al.*, 2021).

No Brasil, há produção de soro antibotrópico, anticrotálico, antilaquético e antielapídico, que podem ser produzidos isolados ou combinados a depender do tipo de soro. Em casos de acidentes envolvendo serpentes do gênero *Bothrops*, o soro

antibotrópico é o indicado. Em situações de dúvida sobre a espécie agressora para a melhor escolha do soro antiveneno, recomenda-se consultar a região geográfica do local da picada bem outras características do envenenamento, como os sintomas apresentados pela vítima. A administração deve ocorrer o mais rápido possível, para maior eficácia do tratamento (Resiere *et al.*, 2023). No “site” do Ministério da Saúde é possível acessar a lista de hospitais de referência para atendimento de acidentes por animais peçonhentos bem como procedimentos mais específicos para o correto tratamento (Brasil, 2024). Ressalta-se que no caso dos acidentes botrópicos, principalmente não se deve fazer torniquetes, cortar o local acometido pela picada e nem chupar o veneno (Brasil, 2024)

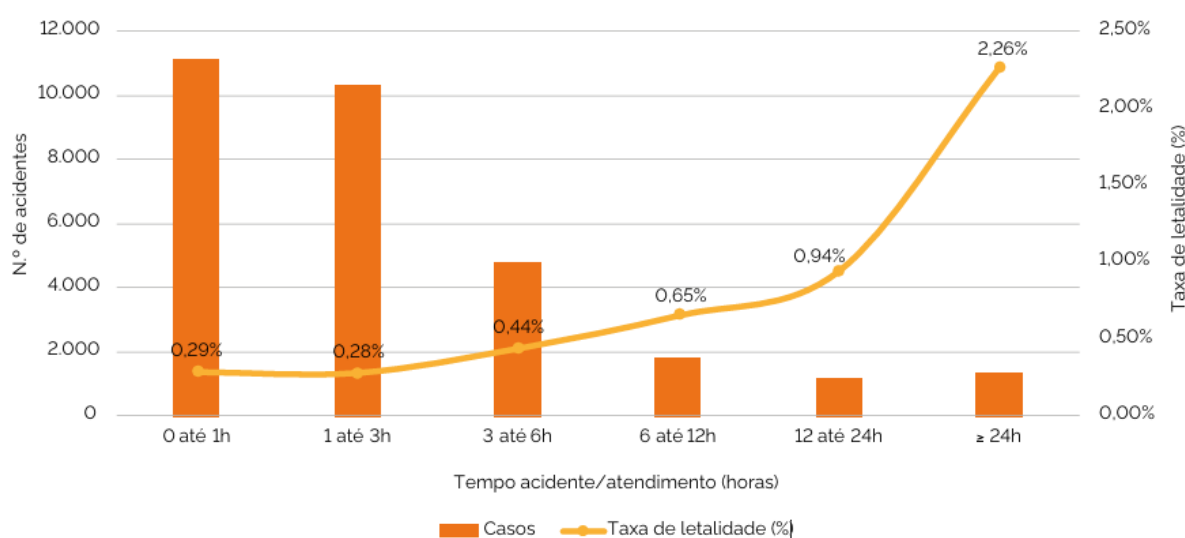


Figura 5. Gráfico com o número total de acidentes ofídicos mostrando a relação entre o tempo do acidente e o tempo que leva para o paciente receber atendimento médico e a taxa de letalidade. Boletim epidemiológico, Ministério da Saúde 2023

Apesar da eficácia da soroterapia na neutralização de toxinas circulantes e na prevenção de complicações sistêmicas, o antiveneno possui pouca ou nenhuma capacidade de reverter danos locais decorrentes do envenenamento, como a necrose tecidual e lesões inflamatórias. Além disso, a soroterapia pode gerar um quadro de hipersensibilidade imediata, levando o paciente a ter uma morte decorrente de reações adversas, em especial o choque anafilático. Outro grande desafio da

soroterapia é o alto custo de logística de produção, distribuição e armazenamento dos soros (necessita de temperatura entre 2 °C e 8 °C), especialmente em regiões remotas e de difícil acesso ao sistema de saúde (Alangode *et al.*, 2020; de Silva *et al.*, 2016). Para a administração dos antivenenos nos pacientes é necessário exclusivamente de um ambiente hospitalar com suporte de emergência com recursos e profissionais da área da saúde qualificados.

7 Terapias alternativas para o tratamento do envenenamento

Devido às limitações da soroterapia convencional, há um interesse por terapias complementares e alternativas capazes de reverter ou diminuir os efeitos locais do envenenamento. Produtos naturais de origem vegetal e marinha têm se destacado como fontes promissoras de moléculas bioativas com propriedades antiveneno (Haque *et al.* 2022). Os extratos de plantas, sobretudo os ricos em flavonoides, taninos, alcaloides e terpenoides demonstram potencial para inibir atividades proteolíticas, fosfolipásicas e hemorrágicas de diversos venenos (Adrião *et al.*, 2022; Páez *et al.*, 2025).

As algas marinhas têm emergido como importantes fontes de polissacarídeos extracelulares (EPS), também conhecidos como exopolissacarídeos, bem como polissacarídeos sulfatados, como os fucoidans, carragenanas e ulvanas capazes de modular processos inflamatórios e interferir na atividade de enzimas-chave dos venenos. Esses polissacarídeos apresentam mecanismos de ação variados, incluindo interação direta com toxinas, quelação de íons essenciais à atividade enzimática e modulação de vias celulares envolvidas na resposta inflamatória (El Asri *et al.*, 2025). A complexidade dos acidentes ofídicos causados pelo gênero *Bothrops* evidencia a necessidade urgente de expandir o conhecimento científico sobre a fisiopatologia dos venenos e sobre novas estratégias terapêuticas. Embora a soroterapia represente a intervenção mais eficaz na redução da mortalidade, suas limitações quanto à reversão de efeitos locais e sua incapacidade de neutralizar toxinas específicas dos venenos abrem espaço para o desenvolvimento de alternativas complementares de tratamento. Nesse cenário, produtos naturais oriundos de plantas e algas surgem como fontes ricas e pouco exploradas de compostos bioativos com ação antiveneno, pois apesar do crescente interesse científico poucos estudos exploram sistematicamente sua capacidade antiveneno (Alangode *et al.*, 2020).

CAPÍTULO I- EFEITO DE PRODUTOS NATURAIS MARINHOS CONTRA AÇÕES TÓXICO-FARMACOLÓGICAS DE VENENOS DE SERPENTES DA FAMÍLIA *Bothrops*

1. Introdução

1.1 Produtos naturais marinhos como fontes emergentes de biotecnologia

Os oceanos constituem o maior e mais diverso bioma do planeta, abrigando uma ampla variedade de organismos capazes de sintetizar moléculas bioativas exclusivas. A complexidade ecológica dos ambientes marinhos, marcada por variações intensas de luz, temperatura, salinidade, pressão, disponibilidade de nutrientes e competição biológica impulsiona a evolução de mecanismos metabólicos altamente especializados, refletidos na diversidade estrutural e funcional de seus metabólitos secundários. Nas últimas décadas, os produtos naturais marinhos têm se destacado como fontes promissoras de compostos com propriedades farmacológicas relevantes, incluindo atividades anti-inflamatória, antioxidante, antiviral, anticoagulante e imunomoduladora (Ghosh *et al.*, 2022; Pradhan *et al.*, 2020).

Dentre os organismos marinhos, as algas assumem papel central devido à sua versatilidade metabólica e à elevada capacidade de biossíntese de moléculas estruturais e bioativas. Polissacarídeos sulfatados, carotenoides, ficobiliproteínas, terpenoides, alcaloides, flavonoides marinhos e peptídeos bioativos estão entre os principais metabólitos de interesse biotecnológico. Esses compostos apresentam mecanismos de ação variados, desde modulação de vias inflamatórias até interação direta com proteínas plasmáticas e celulares, o que os torna candidatos para aplicações médicas emergentes (Ghallab *et al.*, 2024).

Além da composição química, as algas oferecem também vantagem quanto a seu cultivo pela indústria. Sua alta taxa de crescimento e a possibilidade de cultivo controlado em grande escala contribuem para o seu uso sustentável. Em regiões tropicais, subtropicais e temperadas, o cultivo de algas já representa uma das atividades biotecnológicas de maior expansão, com impactos diretos em setores farmacêuticos, nutracêuticos e cosméticos (Chauton *et al.*, 2021; Hosny *et al.*, 2025). No contexto da toxinologia, o interesse por compostos marinhos capazes de

neutralizar mecanismos fisiopatológicos desencadeados por toxinas de venenos vem crescendo. Estudos recentes mostram que polissacarídeos sulfatados extraídos de algas *Chondrus crispus* podem inibir enzimas das famílias de toxinas de venenos de serpentes, incluindo SVMPs, SVSPs e PLAs₂ (Silva *et al.*, 2018), que são responsáveis por danos locais e sistêmicos observados em envenenamentos por serpentes da família Viperidae (Gutiérrez *et al.*, 2021). Esses achados evidenciam o potencial das algas como fontes alternativas ou adjuvantes terapêuticos, especialmente diante das limitações do antiveneno para reverter os efeitos locais decorrentes do envenenamento (Silva *et al.*, 2018).

1.2 Diferenças entre algas de clima temperado, tropical e subtropical

As características metabólicas e estruturais das algas são ligadas às condições ambientais em que elas se desenvolvem. Algas de clima temperado, predominantes em regiões frias ou moderadas, estão expostas a variações significativas de iluminação e temperatura ao longo do ano, pressões ambientais que favorecem a síntese de polissacarídeos complexos como fucoidans e alginatos, amplamente estudados por suas atividades anti-inflamatórias, antioxidantes e anticoagulantes. Nesse grupo, as algas pardas são especialmente relevantes, concentrando moléculas sulfatadas de grande interesse farmacológico (Figuerola *et al.*, 2025; Manikandan *et al.*, 2020; Obluchinskaya *et al.*, 2022).

As algas de clima tropical, por outro lado, convivem com intensa radiação solar, elevada temperatura da água, bem como maior competição biológica. Esses fatores estimulam a produção de compostos fotoprotetores e antioxidantes, incluindo carotenoides de alta eficiência e fenólicos bioativos capazes de neutralizar espécies reativas de oxigênio. Além disso, muitas espécies tropicais sintetizam polissacarídeos menos rígidos e metabolicamente mais dinâmicos, adaptados a ambientes altamente variáveis (Cichoński e Chrzanowski, 2022; Coulombier *et al.*, 2020).

Em regiões subtropicais, observa-se uma combinação intermediária de pressões ambientais, resultando em algas com padrões metabólicos híbridos. Essa diversidade metabólica é especialmente interessante para aplicações biotecnológicas

porque amplia a variedade estrutural dos compostos disponíveis, incluindo polissacarídeos com diferentes graus de sulfatação, pigmentos acessórios e lipídios bioativos com potenciais propriedades anti-inflamatórias e moduladoras de vias de coagulação (Inwongwan *et al.*, 2025; Stengel *et al.*, 2011).

Essas diferenças ecológicas resultam em perfis químicos distintos entre algas temperadas, tropicais e subtropicais. Essa variabilidade é particularmente importante quando se investigam compostos com atividade antiveneno, uma vez que a eficiência de neutralização de toxinas depende diretamente da estrutura molecular dos metabólitos envolvidos. Assim, compreender essas diferenças é essencial para direcionar a seleção de espécies com maior potencial terapêutico (Ojuka *et al.*, 2025; Vasilakis *et al.*, 2025).

1.3 Principais grupos de algas

A diversidade de algas pode ser organizada em três grandes grupos taxonômicos: algas verdes (*Chlorophyta*), algas pardas (*Phaeophyceae*) e algas vermelhas (*Rhodophyta*); cada grupo apresenta perfis fisiológicos, estruturais e bioquímicos particulares, responsáveis por seu potencial na produção de moléculas bioativas (Ghaliaoui *et al.*, 2024).

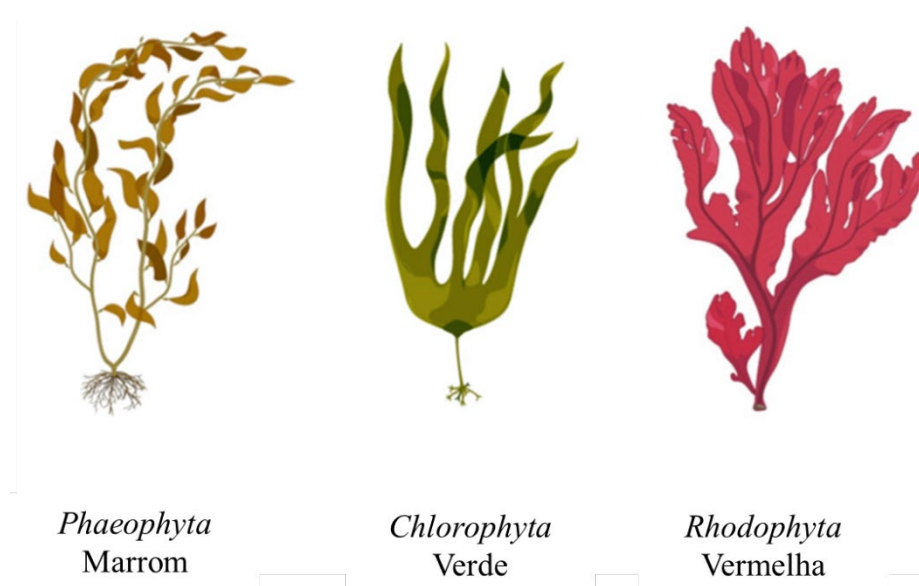


Figura 6. Principais grupos de algas (Perera, C.O., Al-Zahrani, M. 2022).

As algas verdes são ricas em clorofila a e b, carotenoides e lipídios estruturais, além de produzirem polissacarídeos extracelulares com propriedades antioxidantes e imunomoduladoras. Espécies como *Chlorella sorokiniana* e *Scenedesmus obliquus* destacam-se pela facilidade de cultivo, alta produtividade e diversidade metabólica adaptativa, tornando-se excelentes plataformas para exploração de moléculas de interesse farmacológico (Gupta *et al.*, 2016; Piasecka e Baier, 2022).

As algas pardas representam uma das fontes mais abundantes de polissacarídeos sulfatados de alta complexidade, incluindo fucoidans, alginatos e laminarinas, amplamente investigados por suas propriedades anti-inflamatórias, anticoagulantes e moduladoras de coagulação. Espécies como *Fucus vesiculosus* e *Undaria pinnatifida* são clássicos modelos de estudo devido à sua elevada concentração de fucoidans com diferentes graus de sulfatação e atividades biológicas complementares (Cumashi *et al.*, 2007; Obluchinskaya *et al.*, 2022).

As algas vermelhas são conhecidas pela produção de carragenanas e agaranas, polissacarídeos com propriedades antivirais, antioxidantes e imunomoduladoras. A alta densidade de grupos sulfato nesses compostos é particularmente relevante, pois favorece interações com proteínas catiônicas, incluindo toxinas de venenos dependentes de íons metálicos, como muitas metaloproteinases (Maurya *et al.*, 2024; Usov *et al.*, 2022).

1.4 Metabólitos de algas e seu potencial biotecnológico

A grande diversidade química encontrada em algas é fruto da pressão seletiva exercida pelo ambiente marinho, caracterizado por competição intensa, oscilações físicas extremas e abundância de agentes patogênicos. Esses fatores impulsionam a evolução de vias metabólicas complexas responsáveis pela síntese de moléculas com alta reatividade e especificidade biológica. Entre essas moléculas, os polissacarídeos sulfatados destacam-se como os componentes mais estudados devido à sua ampla gama de atividades, incluindo ações anticoagulante, anti-inflamatória, antiviral e antimicrobiana. A presença de grupos sulfato confere elevada densidade negativa à molécula, favorecendo interações eletrostáticas com proteínas catiônicas, receptores

celulares e íons metálicos essenciais para funções enzimáticas (Apostolova *et al.*, 2020; Silva *et al.*, 2018).

Além dos polissacarídeos, as algas produzem um conjunto diversificado de compostos lipofílicos, como carotenoides, ácidos graxos poli-insaturados, terpenos halogenados, fenóis marinhos, ficobiliproteínas e peptídeos bioativos, que exibem propriedades antioxidantes, anti-inflamatórias, citoprotetoras e moduladoras de vias metabólicas humanas. Particularmente, a fucoxantina, abundante em algas pardas, é reconhecida por seus efeitos anti-inflamatórios e moduladores da resposta imunológica, incluindo inibição da produção de citocinas pró-inflamatórias e modulação de espécies reativas de oxigênio (Matin *et al.*, 2024; Su *et al.*, 2019).

Esses metabólitos apresentam uma grande capacidade de interferir em processos biológicos essenciais, como coagulação, inflamação, degradação tecidual e oxidação lipídica, mecanismos altamente relevantes no contexto do envenenamento por serpentes. Toxinas como metaloproteinases (SVMPs), serinoproteases (SVSPs) e fosfolipases A₂ (PLAs₂) dependem de interações estruturais específicas com lipídios ou íons metálicos, atividades que podem ser moduladas por polissacarídeos sulfatados ou compostos antioxidantes derivados de algas (Moreira *et al.*, 2021; Zhao *et al.*, 2025).

A compreensão dessas interações é especialmente importante porque, ao contrário do antiveneno, que atua predominantemente na neutralização sistêmica, muitos metabólitos de algas são capazes de interagir diretamente com toxinas individuais ou com seus cofatores. Alguns polissacarídeos sulfatados, por exemplo, são capazes de se ligar a proteínas catiônicas, modulando a atividade catalítica das SVSPs e inibindo mecanismos de coagulação exacerbada induzida por serinoproteases trombina-símile (Cao *et al.*, 2019; Peipei *et al.*, 2024; Qin *et al.*, 2020). Outros compostos, como fenóis e carotenoides possuem ação antioxidante intensa capaz de atenuar danos teciduais resultantes de inflamação, necrose e estresse oxidativo desencadeados por PLAs₂ (Begum *et al.*, 2021; Goya e Mateos, 2025).

Desse modo, o perfil metabólico das algas confere a essas espécies um potencial biotecnológico excepcional, abrangendo desde a aplicação farmacológica

até novas abordagens terapêuticas, como agentes antitóxicos e adjuvantes terapêuticos.

1.5 Famílias de algas com potencial antiveneno descrito na literatura

Embora a pesquisa sobre compostos marinhos como agentes antiveneno ainda esteja em estágio inicial, estudos destacam um conjunto de famílias de algas com capacidade comprovada de inibir toxinas de venenos ofídicos. A maioria desses estudos concentra-se em algas pardas e verdes, cujos polissacarídeos sulfatados exibem afinidade eletrostática com toxinas catiônicas, modulando sua atividade catalítica ou impedindo sua interação com substratos fisiológicos (Angulo e Lomonte, 2003; Monteiro-Machado *et al.*, 2015).

Algas pardas da ordem Fucales e Laminariales, como *Fucus vesiculosus* (FVF) e *Undaria pinnatifida*, (UPF) são relativamente bem descritas por seus polissacarídeos sulfatados que apresentam inibição significativa de SVMPs, SVSPs e PLAs₂ por métodos *in vitro*. Essas toxinas representam os principais componentes responsáveis pelos efeitos hemorrágico, coagulante e mionecrótico que são característicos dos envenenamentos por serpentes da família *Viperidae* (Angulo e Lomonte, 2003; Monteiro-Machado *et al.*, 2015).

Nas algas verdes, compostos derivados de carboidratos extracelulares têm sido estudados pela capacidade de interagir com proteínas carregadas positivamente, incluindo toxinas enzimáticas. Espécies como *Chlorella sorokiniana*, *Scenedesmus obliquus* e *Scenedesmus acuminatus* são reconhecidas pela presença de polissacarídeos e glicoproteínas extracelulares com propriedades anti-inflamatórias, antioxidantes e quelantes de íons metálicos. Esses mecanismos são particularmente relevantes porque muitas SVMPs dependem de íons zinco para exercer sua atividade catalítica; assim, compostos capazes de se ligar a esse metal podem reduzir significativamente o efeito proteolítico das proteases presentes em abundância em venenos de serpentes (Zhao *et al.*, 2024).

A literatura também destaca algas vermelhas, especialmente do gênero *Gracilaria* por sua produção de carragenanas altamente sulfatadas com efeitos

neutralizantes sobre toxinas da família das PLAs₂. Embora menos estudadas no contexto antiveneno, essas moléculas apresentam elevada afinidade por proteínas catiônicas, o que sugere aplicabilidade promissora (Alencar *et al.*, 2019).

No geral, o conjunto de estudos demonstra que algas possuem mecanismos de neutralização distintos dos observados do antiveneno comercial, atuando não apenas por ligação direta às toxinas, mas também pela modulação de vias inflamatórias e oxidativas ou interação com metais (como Ca⁺² e Zn⁺²), sugerindo possível uso futuro como adjuvantes terapêuticos ou como moléculas-modelo para síntese de novos fármacos.

1.5.1 *Fucus vesiculosus*

A alga parda *Fucus vesiculosus* (Fucales, Phaeophyceae), conhecida popularmente como “bladderwrack”, é nativa de zonas temperadas do hemisfério norte, distribuindo-se amplamente ao longo das costas do Atlântico Norte, Mar Báltico, Groenlândia, Islândia e regiões frias da América do Norte e Europa. Sua preferência ecológica envolve áreas de substrato rochoso expostas às marés, o que contribui para sua adaptação fisiológica a ciclos intensos de hidratação e desidratação, além de variações de luminosidade e salinidade características desses ambientes (Kautsky *et al.*, 2019; Takolander, 2023).

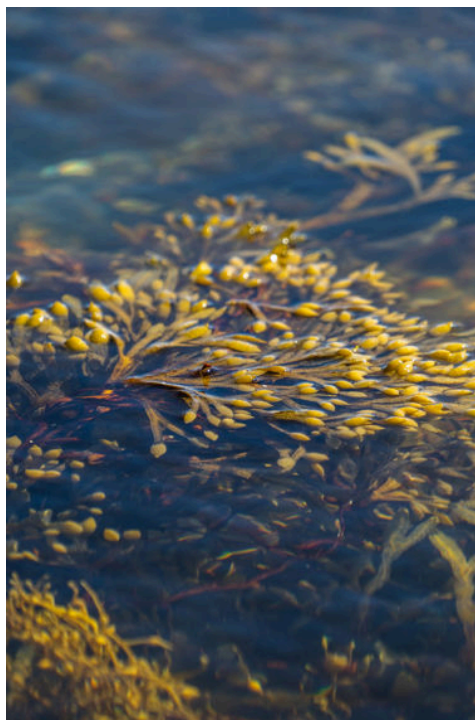


Figura 7. *Fucus vesiculosus*. www.marinova.com.au Acessado em 10/12/25

Cultural e historicamente, *F. vesiculosus* tem sido utilizada na medicina tradicional europeia desde o século XIX, principalmente para o tratamento de bócio devido ao seu conteúdo relativamente elevado de iodo. Com o tempo, seu uso se expandiu para aplicações nutracêuticas, cosméticas e farmacêuticas, incluindo produtos destinados ao tratamento de inflamações, distúrbios metabólicos e alterações hormonais. A espécie também foi incorporada à fitoterapia moderna por suas propriedades antioxidantes, anti-inflamatórias e potencial modulador da coagulação, o que despertou o interesse da comunidade científica para investigação dos mecanismos moleculares envolvidos nesses efeitos (Wang *et al.*, 2019).

A espécie *Fucus vesiculosus* é uma das espécies mais amplamente estudadas no contexto de biotecnologia marinha por conta de sua composição química particularmente rica em polissacarídeos sulfatados, considerados um dos grupos de maior relevância biotecnológica entre as algas pardas. Os fucoidans apresentam estruturas complexas compostas majoritariamente por fucose e grupos sulfato, além de ramificações com manose, xilose e galactose, conferindo uma arquitetura altamente variável e responsiva a influências ambientais e metodológicas de extração.

A presença de grupos sulfato é responsável por propriedades biológicas fundamentais, como interação com proteínas catiônicas, modulação de enzimas dependentes de íons metálicos e inibição de vias de coagulação. Além dos fucoidans, *F. vesiculosus* contém alginatos, laminarinas, compostos fenólicos, carotenoides e minerais essenciais que complementam seu potencial bioativo (Brovko *et al.*, 2025; Obluchinskaya *et al.*, 2022; Lingzhi Wang *et al.*, 2023).

1.5.2 *Undaria pinnatifida*

Undaria pinnatifida, conhecida popularmente como wakame, é uma alga parda pertencente à ordem Laminariales, amplamente reconhecida por sua relevância ecológica, nutricional e biotecnológica. Originária de ambientes temperados do noroeste do oceano Pacífico, principalmente Japão, Coreia e China, essa espécie expandiu sua distribuição geográfica ao longo das últimas décadas, sendo atualmente considerada uma espécie invasora em regiões da Oceania, Europa e América do Sul. Sua capacidade de colonização decorre de características fisiológicas altamente adaptativas, incluindo elevada taxa de crescimento, tolerância a amplitudes térmicas moderadas e adaptação a zonas costeiras de luz variável, tornando-a um organismo extremamente competitivo em ecossistemas marinhos (Dean e Hurd, 2007; South *et al.*, 2017).



Figura 8. *Undaria pinnatifida*. www.marinova.com.au Acessado em 10/12/25

A espécie *U. pinnatifida* tem longa história de uso na culinária e medicina tradicional asiática. No Japão e na Coreia, por exemplo, é consumida há séculos como ingrediente alimentar rico em minerais, fibras dietéticas e compostos bioativos, sendo tradicionalmente utilizada no preparo de sopas, saladas e extratos com fins

terapêuticos. Seu uso medicinal inclui aplicações no manejo de inflamações, distúrbios metabólicos e condições relacionadas ao estresse oxidativo, refletindo o reconhecimento empírico do seu potencial farmacológico. Na indústria, o wakame é amplamente utilizado na formulação de nutracêuticos, cosméticos antioxidantes e produtos voltados à modulação imunológica (Machado *et al.*, 2025; Park *et al.*, 2024; Zeng *et al.*, 2022).

Do ponto de vista químico, *U. pinnatifida* destaca-se pela alta concentração de fucoidans, alginatos, laminarinas e fucoxantina, pigmento carotenóide responsável pela coloração marrom-dourada característica da espécie. Os fucoidans de *U. pinnatifida* apresentam estruturas complexas compostas predominantemente por L-fucose e grupos sulfato, além de pequenas quantidades de galactose, manose e xilose, variando conforme método de extração, idade da planta e condições ambientais. Esses polissacarídeos sulfatados são particularmente valorizados por suas propriedades anti-inflamatórias, antivirais, anticoagulantes, antioxidantes e antitumorais, amplamente documentadas na literatura contemporânea. A fucoxantina, por sua vez, exerce efeitos antioxidantes potentes, atuando na redução de espécies reativas de oxigênio e na modulação de vias pró-inflamatórias, propriedades especialmente relevantes em contextos de dano tecidual associado a toxinas (Koh *et al.*, 2019; Lourenço-Lopes *et al.*, 2022; Skriptsova *et al.*, 2010).

A combinação de ampla disponibilidade, facilidade de cultivo, diversidade de metabólitos bioativos e avanços em métodos de extração e purificação de seus polissacarídeos posiciona *U. pinnatifida* como biorrecurso estratégico na bioprospecção marinha, com forte potencial para gerar moléculas-modelo e terapias adjuvantes, inclusive na neutralização de toxinas ofídicas (Cheong *et al.*, 2025; Zeng *et al.*, 2022).

1.5.3 *Chlorella sorokiniana*

Chlorella sorokiniana é uma microalga verde unicelular pertencente ao filo Chlorophyta, amplamente reconhecida por sua elevada produtividade, robustez fisiológica e capacidade de adaptação a diferentes condições ambientais. Essa espécie é encontrada em ambientes de água doce, embora também ocorra em habitats ligeiramente salobros, distribuindo-se globalmente em regiões tropicais, subtropicais e temperadas. Sua versatilidade ecológica relaciona-se à capacidade de tolerar variações de luz, temperatura e disponibilidade de nutrientes, tornando-a uma das microalgas mais utilizadas em cultivos laboratoriais e industriais (Lizzul *et al.*, 2018).

Culturalmente e industrialmente, a espécie *Chlorella sorokiniana* possui destaque na produção de biomassa com fins nutricionais, farmacêuticos e biotecnológicos. Espécies do gênero *Chlorella* são utilizadas há décadas como suplementos alimentares ricos em proteínas, vitaminas, minerais, carotenoides e antioxidantes. Essa ampla aceitação acontece em parte da segurança toxicológica reconhecida, o que torna *C. sorokiniana* um organismo modelo amplamente empregado em pesquisas de bioprocessos, produção de biofármacos, biorremediação e síntese de compostos bioativos de interesse terapêutico (Dahiya *et al.*, 2021; Panicucci *et al.*, 2025).

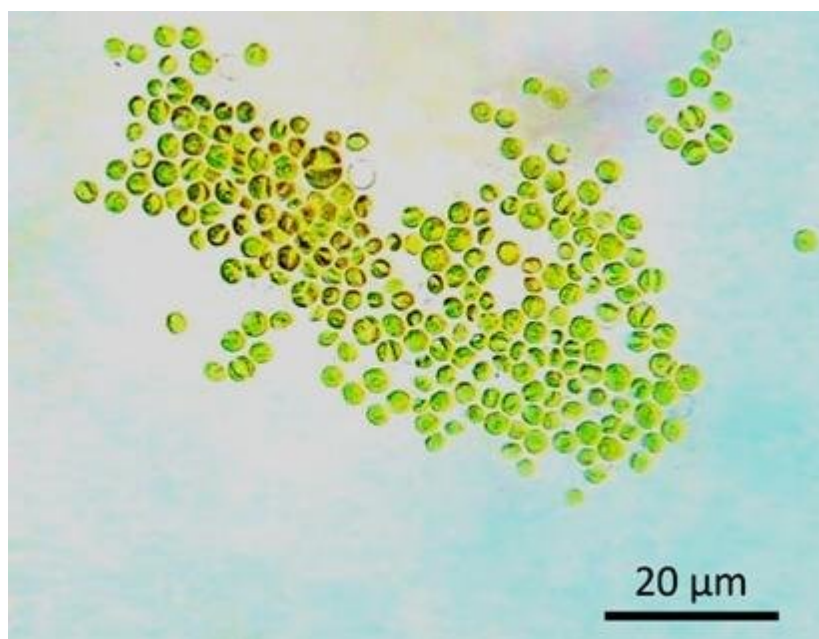


Figura 9. *Chlorella sorokiniana*. (Zawar PA 2016)

A composição química de *Chlorella sorokiniana* é marcada por grande diversidade metabólica. Essa microalga contém carotenoides, clorofilas, polissacarídeos extracelulares, peptídeos bioativos, ácidos graxos poli-insaturados, além de compostos fenólicos que contribuem para sua reconhecida atividade antioxidante e anti-inflamatória. Os polissacarídeos extracelulares (EPS), em particular, apresentam composição variada envolvendo glicose, galactose, arabinose, xilose e ramnose, podendo conter grupos fosfato ou sulfato em algumas condições de cultivo. Esses componentes desempenham funções biológicas importantes, incluindo estimulação de resposta imune, modulação inflamatória e neutralização de espécies reativas de oxigênio (Banskota *et al.*, 2024; Domínguez-Gámez *et al.*, 2024; El-Naggar *et al.*, 2020; Li *et al.*, 2024).

Embora *Chlorella sorokiniana* ainda não tenha sido amplamente investigada especificamente como agente antiveneno, suas características metabólicas, especialmente seu perfil antioxidante, capacidade de modular inflamação e potencial de interação com proteínas catiônicas, a posicionam como candidata importante para exploração biotecnológica em busca de compostos capazes de atuar contra toxinas de venenos de serpentes.

1.5.4 Scenedesmus spp.

Scenedesmus é um gênero de microalgas verdes pertencente ao filo Chlorophyta e à família *Scenedesmaceae*, destacando-se pela elevada eficiência fotossintética, robustez fisiológica e alta versatilidade metabólica. A espécie ocorre predominantemente em ambientes de água doce, sendo comum em lagos, reservatórios, represas e sistemas eutrofizados de regiões tropicais e subtropicais. Sua capacidade de sobreviver sob intensas variações de luminosidade, salinidade moderada e flutuações de temperatura faz de *Scenedesmus* sp. um organismo amplamente utilizado em estudos de bioprocessos, produção de biomassa e síntese de metabólitos bioativos. Além disso, sua elevada taxa de crescimento e tolerância a diferentes condições de cultivo tornam essa espécie um modelo promissor para aplicações industriais em grande escala (Phinyo *et al.*, 2017).

Enquanto algumas espécies de microalgas são historicamente associadas à alimentação humana ou uso medicinal tradicional, o gênero *Scenedesmus* possui aplicação predominantemente tecnológica e biotecnológica. É utilizada em formulações de nutracêuticos, fotobiorreatores voltados à produção de biomassa rica em proteínas, carotenoides e lipídios, além de ser valorizada em processos de biorremediação, devido à sua capacidade de remover nutrientes e metais pesados de meios aquáticos (Arora *et al.*, 2019). O interesse crescente na espécie se deve à sua plasticidade metabólica, que permite modular sua composição química mediante alterações controladas no ambiente de cultivo, como intensidade luminosa, disponibilidade de nitrogênio e presença de agentes estressantes (Ye *et al.*, 2020).

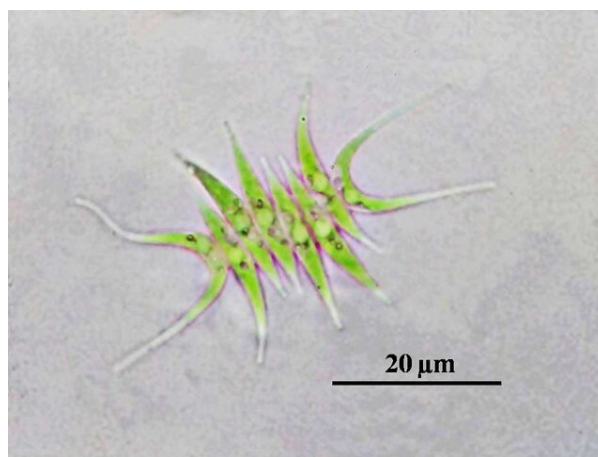


Figura 10. *Scenedesmus acuminatus*. (Unpaprom PA 2015)

A composição bioquímica de *Scenedesmus* spp. é rica e diversificada, incluindo altos teores de carotenoides, clorofilas, polissacarídeos extracelulares, ácidos graxos poli-insaturados, esteróis, proteínas bioativas, peptídeos antioxidantes e compostos fenólicos. Seus polissacarídeos extracelulares apresentam estrutura composta por glicose, galactose, ramnose, arabinose e, em alguns casos, grupos sulfato e fosfato, componentes que conferem marcada atividade biológica aos polissacarídeos extracelulares, incluindo propriedades imunomoduladoras e antioxidantes. Essa diversidade metabólica é amplamente reconhecida na literatura por conferir às espécies do gênero *Scenedesmus* um papel importante como fonte emergente de moléculas bioativas com aplicação farmacêutica e biotecnológica (Patil e Kaliwal, 2019; Xiao e Zheng, 2016).

Os polissacarídeos extracelulares desse gênero merecem destaque por sua capacidade de interagir com proteínas carregadas positivamente. Essa interação química, possivelmente mediada por grupos sulfato ou fosfato presentes na estrutura dos EPS, pode interferir na ação catalítica das toxinas, reduzindo sua capacidade de hidrolisar fosfolipídios ou degradar componentes da matriz extracelular. Além disso, estudos indicam que EPS de microalgas podem atuar como quelantes de metais, especialmente zinco, cofator essencial para a atividade proteolítica de proteases. Dessa forma, a presença de EPS complexos em *Scenedesmus* spp. sugere possível atividade indireta de neutralização de toxinas (Li *et al.*, 2023; Patwal e Baranwal, 2021; Xiao e Zheng, 2016).

Outro ponto relevante é a capacidade dessa microalga de produzir compostos com atividade anti-inflamatória que modulam vias como NF- κ B, COX-2 e citocinas pró-inflamatórias, como IL-1 β , IL-6 e TNF- α . A modulação dessas vias não apenas reduz a extensão dos danos teciduais, mas também atenua processos de necrose e edema, podendo atuar como adjuvante terapêutico no tratamento de lesões locais induzidas por venenos ofídicos (El-Baz *et al.*, 2022; Ghany *et al.*, 2023; Yan *et al.*, 2023)

1.5.5 *Nannochloris* sp.

Nannochloris sp. é uma microalga verde unicelular pertencente ao filo Chlorophyta, conhecida por seu tamanho diminuto, crescimento rápido e elevada versatilidade metabólica. Diferentemente de outras microalgas amplamente estudadas, *Nannochloris* permanece relativamente pouco explorada, o que a coloca como um gênero particularmente interessante para fins de bioprospecção. As espécies do gênero são encontradas tanto em ambientes de água doce quanto marinhos, distribuindo-se globalmente em regiões tropicais, subtropicais e temperadas. Sua habilidade de tolerar variações intensas de salinidade, luminosidade e disponibilidade de nutrientes faz de *Nannochloris* sp. um organismo robusto e adequado para cultivos em grande escala, especialmente em sistemas industriais de produção de biomassa e metabolitos bioativos (Markou *et al.*, 2024; Sanders *et al.*, 2022).

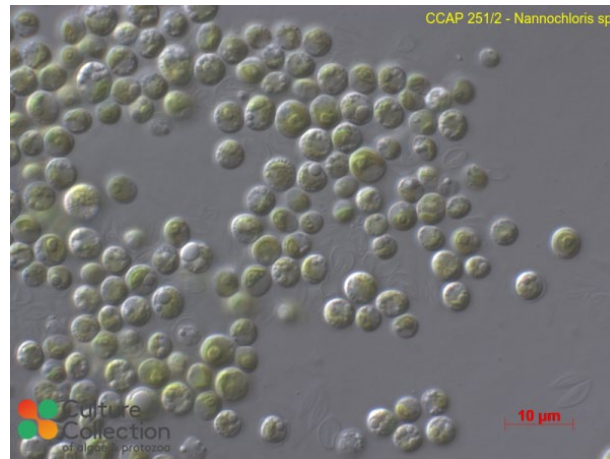


Figura 11. *Nannochloris* sp (https://www.ccap.ac.uk/catalogue/strain-251-2 acessado em 05/02/26)

Do ponto de vista industrial, *Nannochloris* sp. é amplamente reconhecida pelo alto teor de proteínas, lipídios e pigmentos carotenoides, além de sua capacidade de acumular substâncias bioativas em resposta a condições ambientais controladas, como estresse luminoso ou variações de nutrientes. Essa plasticidade metabólica favorece o acúmulo de compostos com interesse farmacêutico, nutracêutico e biotecnológico (Babich *et al.*, 2023; Hulatt *et al.*, 2017).

Em termos de composição química, *Nannochloris* sp. produz uma variedade de metabólitos primários e secundários, incluindo carotenoides (principalmente β -caroteno e luteína), clorofilas, polissacarídeos extracelulares (EPS), aminoácidos essenciais, peptídeos bioativos, ácidos graxos poli-insaturados e compostos fenólicos. Sua produção de EPS merece atenção especial, pois esses polissacarídeos apresentam estrutura heterogênea composta por glicose, ramnose, galactose e traços de xilose, podendo incluir grupos sulfato, característica que se assemelha aos polissacarídeos sulfatados de macroalgas, conhecidos pela forte atividade biológica, fazendo dessa microalga uma fonte altamente atrativa para bioprospecção de novos compostos com possível atividade neutralizante de toxinas ofídicas (Pandeirada *et al.*, 2019; Solovchenko *et al.*, 2014).

2 OBJETIVOS

2.1 Objetivo geral

Avaliar o efeito neutralizante de extratos/bioprodutos de algas marinhas contra atividades tóxicas *in vitro* (fosfolipásica, coagulante e proteolítica) causadas pelo veneno das serpentes *Bothrops jararaca*, *B. jararacussu* e *B. neuwiedi*.

2.2 Objetivos específicos:

- A. Avaliar o efeito de fucoidans isolados das espécies *Fucus vesiculosus* e *Undaria pinnatifida* contra as atividades coagulante, proteolítica e fosfolipásica dos venenos de *Bothrops jararaca*, *B. jararacussu* e *B. neuwiedi*;
- B. Avaliar o efeito de exopolissacarídeos isolados das espécies *C. sorokiniana*, *S. obliquus*, *Nannochloris* sp. *Naumann* e *S. acuminatus* contra as atividades coagulante, proteolítica e fosfolipásica dos venenos de *Bothrops jararaca*, *B. jararacussu* e *B. neuwiedi*.

3 ARTIGOS REFERENTES AO CAPÍTULO I

Os resultados referentes ao capítulo I estão apresentados na forma de dois artigos publicados em revistas de âmbito internacional indexadas.

O primeiro artigo publicado no periódico *Toxins* (Effect of Seaweed-Derived Fucoidans from *Undaria pinnatifida* and *Fucus vesiculosus* on Coagulant, Proteolytic, and Phospholipase A2 Activities of Snake *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* Venom. Castro-Pinheiro C, Junior LCSP, Sanchez EF, da Silva ACR, Dwan CA, Karpiniec SS, Critchley AT, Fuly AL. *Toxins* (Basel). 2024 Apr 12;16(4):188. doi: 10.3390/toxins16040188.) descreve o efeito dos polissacarídeos das espécies *Undaria pinnatifida* (UPF) e *Fucus vesiculosus* (FVF) na inibição das atividades tóxicas dos venenos de *Bothrops jararaca*, *B. jararacussu* e *B. neuwiedi*.

O segundo artigo publicado no periódico *Scientific Reports* (Exploiting exopolysaccharides from microalgae to block the toxic in vitro effects of *Bothrops* sp. venom. Castro-Pinheiro C, Sukhikh S, Babich O, Ivanova S, Sanchez EF, Fuly AL. *Sci Rep*. 2025 Jul 2;15(1):22754. doi: 10.1038/s41598-025-08825-2) descreve o efeito de exopolissacarídeos (EPS) de microalgas na atividade proteolítica, coagulante e fosfolipásica causada pelos venenos das serpentes *B. jararaca*, *B. jararacussu* e *B. neuwiedi*.

Article

Effect of Seaweed-Derived Fucoidans from *Undaria pinnatifida* and *Fucus vesiculosus* on Coagulant, Proteolytic, and Phospholipase A₂ Activities of Snake *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* Venom

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Abstract: Background: Snakebite envenomation (SBE) causes diverse toxic effects in humans, including disability and death. Current antivenom therapies effectively prevent death but fail to block local tissue damage, leading to an increase in the severity of envenomation; thus, seeking alternative treatments is crucial. Methods: This study analyzed the potential of two fucoidan sulfated polysaccharides extracted from brown seaweeds *Fucus vesiculosus* (FVF) and *Undaria pinnatifida* (UPF) against the fibrinogen or plasma coagulation, proteolytic, and phospholipase A₂ (PLA₂) activities of *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. The toxicity of FVF and UPF was assessed by the hemocompatibility test. Results: FVF and UPF did not lyse human red blood cells. FVF and UPF inhibited the proteolytic activity of *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venom by approximately 25%, 50%, and 75%, respectively, while all venoms led to a 20% inhibition of PLA₂ activity. UPF and FVF delayed plasma coagulation caused by the venoms of *B. jararaca* and *B. neuwiedi* but did not affect the activity of *B. jararacussu* venom. FVF and UPF blocked the coagulation of fibrinogen induced by all these Bothropic venoms. Conclusion: FVF and UPF may be of importance as adjuvants for SBE caused by species of *Bothrops*, which are the most medically relevant snakebite incidents in South America, especially Brazil.

Keywords: brown seaweed; snake venom; *Bothrops*; fucoidan; *Undaria pinnatifida*; *Fucus vesiculosus*; antivenom

Key Contribution: Use of fucoidan from seaweed to block toxicity effects of the venom of three species of *Bothrops*.

1. Introduction

Snake venom is a complex mixture of organic and inorganic compounds, of which 90–95% are active proteins and peptides that produce diverse toxic effects in humans. Snakebite envenomation (SBE) is a neglected tropical disease (NTD) recognized by the World Health Organization (WHO) since 2017, mainly to allow funds for the development of strategies to reduce mortality or morbidity by 50% by 2030 [1,2]. SBE occurs worldwide

with 2.7 million incidents, 138,000 deaths, and 400,000 amputations or deformities annually. It is a serious medical problem mainly affecting impoverished, tropical, and subtropical developing regions and a common occupational injury for farmers and agricultural workers, as well as children [3–5]. Epidemiological studies of SBE are scarce, which makes it difficult for governments to develop strategies to improve treatments to prevent deaths or morbidities [1,3]. In Brazil, pit vipers of the genus *Bothrops* are responsible for 87% of the registered incidents, followed by the genus *Crotalus* (rattlesnakes, 7.8%) and *Lachesis* (Bushmasters, 2%). The genus *Bothrops* is the largest among vipers, with around 47 species, and among them, *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* are of medical interest since they cause the highest numbers of severe envenomation incidents [6]. The symptoms of SBE inflicted by these pit vipers share some clinical effects, including pain, inflammation, edema, muscle damage around the affected limb, and drastic systemic disturbances in hemostasis (hemorrhage and coagulopathies), as well as the failure of the kidneys and respiratory and circulatory systems, hypotension, and death [7]. *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* are endemic to South America (mainly Brazil, Paraguay, Bolivia, and Argentina) and are nocturnal and terrestrial [8]. The species of *B. jararaca* and *B. neuwiedi* reach around 120 and 60 cm in length, respectively, while *B. jararacussu* is the largest (160–220 cm). In general, the venoms of *Bothrops* are a mixture of protein/peptide families, such as the snake venom metalloproteases (SVMPs), snake venom serine proteinases (SVSPs), phospholipase A₂ (PLA₂) enzymes, L-amino acid oxidases (LAOs), hyaluronidases, and bradykinin potentiators, which are responsible for the venom toxicity [7,9].

The WHO and Brazilian Ministry of Health, as well as other countries, recommend antivenoms as the only available treatment for SBE [10–13]. In Brazil, the mono- or polyclonal antivenoms are produced at the Institute Butantan (São Paulo, São Paulo, Brazil), Institute Vital Brazil (Niterói, Rio de Janeiro, Brazil), Immunological Production and Research Center (Piraquara, Paraná, Brazil), and Ezequiel Dias Foundation (Belo Horizonte, Minas Gerais, Brazil) through the hyperimmunization of equines, and the antivenoms are intravenously administered to patients after envenomation. Although antivenoms prevent the death of victims, they have some disadvantages, such as inducing allergic reactions (anaphylaxis) and fever and high costs of production, and they are ineffective in preventing tissue necrosis, thus leading to complications, such as amputation or deformity of the affected limb [13,14]. Furthermore, the delay in administering antivenoms may increase the mortality and morbidity of SBE [14]. Given the complexity of SBE and the limitations of antivenoms, seeking alternative or complementary treatments is of great importance. Plants have been used as herbal medicines to treat SBE since ancient times. Currently, the literature reports many species of plants used in whole or in part with antivenom effects [15–18]. On the other hand, research testing antivenom molecules derived from marine environments is rare. Seaweed belongs to the Kingdom Protista and is classified into three groups: Chlorophyceae (green algae), Rhodophyceae (red algae), and Phaeophyceae (brown algae). There are approximately 1500 species of brown algae; however, some species of the class Phaeophyceae, such as *Ecklonia*, *Laminaria*, *Undaria*, *Fucus*, *Ascophyllum*, and *Himanthalia*, are perhaps the most studied due to the high content of bioactive compounds, including alkaloids, terpenoids, phytosterols, carotenoids, polyphenols, sterols, proteins, and sulfated polysaccharides [19,20]. These compounds have a wide range of pharmacological properties, including anticoagulant, antioxidant, antiviral, anti-inflammatory, anticancer, antifungal, antibacterial [21–23], and antivenom [24–27] activities, as well as ecological functions [28]. Indeed, seaweed has been used by the pharmaceutical and nutraceutical industries to promote human health or treat diseases [20,29]. Thus, the successful applicability of seaweeds makes them good candidates for drug development to treat a wide range of diseases, including SBE. The literature has described the antagonist effects of polysaccharides from red algae *Laurencia dendroidea* [24], *Palisada flagellifera* [25], and *Chondrus crispus* [26] and green alga *Gayralia oxysperma* [27] against the toxic activities of the pit vipers *Lachesis muta*, *B. jararaca*, and *B. jararacussu*. However, less is known about the inhibitory properties of fucoidans from brown algae. Although data are limited, two studies

found that fucoidan from *F. vesiculosus* neutralized necrosis of tissue muscle caused by crude venom and purified myotoxic PLA₂ from *Bothrops asper* and other crotaline venoms, including *Cerrophidion godmani*, *Atropoides mummifer*, and *Bothriechis schlegelii* [30,31].

Fucoidans are sulfated polysaccharides and have structural backbones primarily composed of sulfated fucose groups. The polysaccharides may also contain galactose, xylose, and arabinose in significant proportions, while other compounds may also be present in extracts, including uronic acid and the monosaccharides glucose, mannose, and rhamnose, in different ratios, depending on the species [20]. The pharmacological properties are reported to be dependent on the molecular weight, chemical structure, sugar composition, and the position of the sulfate group [20].

Building on the current knowledge of fucoidans in SBE, this study assessed high molecular weight fucoidan extracts derived from two different species, *F. vesiculosus* (FVF) and *U. pinnatifida* (UPF), against proteolytic, plasma, or fibrinogen coagulation, and PLA₂ activities induced by *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom.

2. Results

2.1. Toxicity of FVF and UPF

The in vitro toxicity of FVF and UPF was assayed using the hemocompatibility test. Treating red blood cells (RBCs) with water led to 100% lysis, while treatment with saline produced no lysis. The incubation of FVF and UPF (1500 µg/mL) lysed around 2% of RBC. According to the literature [32], values of hemolysis below 10% mean that the compound is non-toxic, and thus FVF and UPF can be classified as non-hemotoxic molecules.

2.2. Inhibitory Effect of FVF and UPF against the Proteolytic Activity of the Venom of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*

B. jararaca, *B. jararacussu*, and *B. neuwiedi* venoms (10–50 µg/mL) hydrolyzed the substrate azocasein in a concentration-dependent manner. One effective concentration (EC) of each venom (30 µg/mL) was incubated for 5 min at 37 °C with saline (positive group) or with 300 µg/mL FVF or UPF. As seen in Figure 1, the proteolytic activity of *B. jararaca* (Figure 1A) and *B. jararacussu* (Figure 1B) venom was diminished by approximately 25% and 50% by FVF and UPF, respectively. FVF or UPF inhibited the proteolytic activity of *B. neuwiedi* venom by 75% (Figure 1C). FVF or UPF alone, in the absence of venom, did not hydrolyze azocasein.

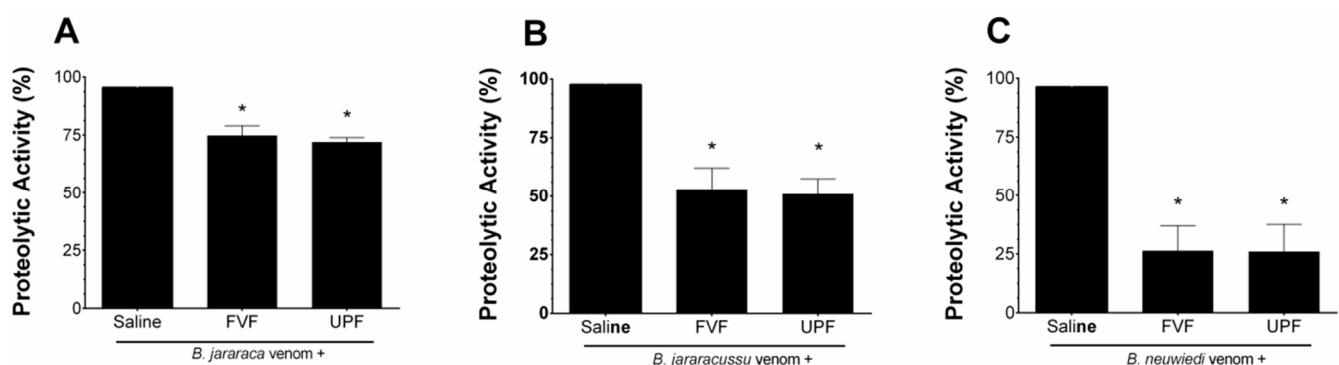


Figure 1. Inhibitory effect of FVF or UPF on the proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. A total of 30 µg/mL of *B. jararaca* (A), *B. jararacussu* (B), and *B. neuwiedi* (C) venoms were incubated with saline or 300 µg/mL FVF or UPF for 5 min at 37 °C. An aliquot of each mixture was then added to the reaction, and the proteolytic activity was determined. A total of 100% proteolytic activity was obtained with venom plus saline. Results are expressed as means ± SEM ($n = 6$). * $p < 0.05$ when compared to each venom plus saline.

2.3. Inhibitory Effect of FVF and UPF on the Coagulant Activity of the Venom of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*

B. jararaca, *B. jararacussu*, and *B. neuwiedi* venom induces plasma and fibrinogen coagulation in a concentration-dependent manner, and the amount of venom able to clot plasma or fibrinogen at 60 s was determined as the minimum coagulating concentration (MCC). One unit of the MCC of *B. jararaca* (30 µg/mL), *B. jararacussu* (60 µg/mL), and *B. neuwiedi* (30 µg/mL) venom was incubated with FVF or UPF for 5 min at 37 °C. An aliquot of each mixture was then added to the plasma. As seen in Figure 2A,C (white columns), *B. jararaca* and *B. neuwiedi* venom (30 µg/mL) mixed with saline (positive groups) clotted plasma at around 60–70 s. However, in the presence of 300 µg/mL FVF or UPF, the coagulation of plasma occurred at 95 s (Figure 2A,C—white columns). In experiments using fibrinogen (Figure 2A–C, black columns), the coagulation caused by 10 µg/mL *B. jararaca* or *B. neuwiedi* venom in the presence of FVF and UPF (100 µg/mL) did not occur until 600 s, which was the maximum period of observation. The fibrinogen coagulation caused by *B. jararacussu* venom (20 µg/mL) was inhibited by 200 µg/mL FVF or UPF (Figure 2B, black columns). The coagulation of plasma caused by *B. jararacussu* venom (60 µg/mL) was not inhibited by FVF or UPF (600 µg/mL). FVF or UPF alone did not clot plasma or fibrinogen.

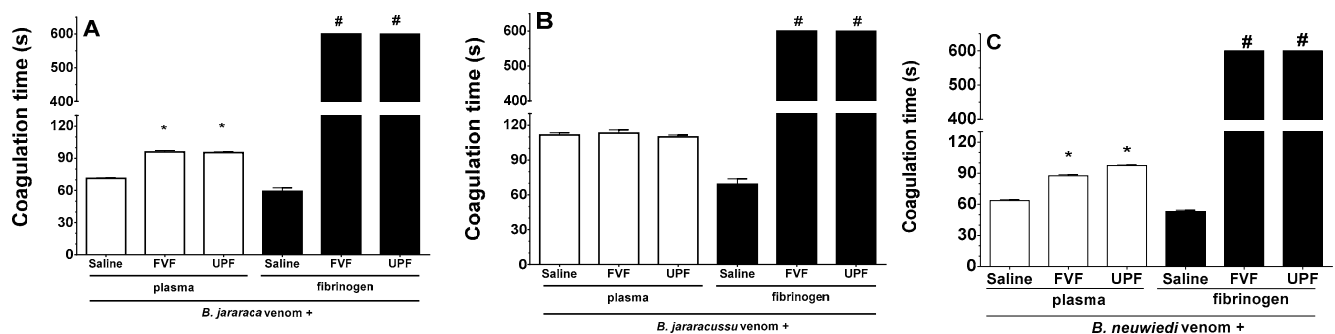


Figure 2. Inhibitory effect of FVF and UPF on the coagulant activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *B. jararaca* (A), *B. jararacussu* (B), and *B. neuwiedi* (C) venom was incubated for 5 min at 37 °C with saline or with FVF or UPF at a 1:10 ratio of venom to polysaccharide. After incubation, an aliquot was added to plasma (white columns) or fibrinogen (black columns), and the coagulation time was monitored in seconds (s). Results are expressed as means $\pm$ SEM ($n = 8$). # means that plasma or fibrinogen did not clot by 600 s of monitoring. * $p < 0.05$ when compared to each venom plus saline.

2.4. Inhibitory Effect of FVF and UPF on the PLA₂ Activity of the Venom of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*

B. jararaca, *B. jararacussu*, and *B. neuwiedi* venom was incubated with saline, and reads of 0.7–0.8 at the absorbance of 740 nm were considered 100% PLA₂ activity. Each venom (50 µg/mL) was incubated with 25 µg/mL FVF or UPF. The inhibition of PLA₂ activity after 30 min was around 20%, regardless of the venom tested (Figure 3). There were no statistically significant differences among the groups containing fucoidans. PLA₂ activity of venoms treated with 20 mM of EDTA vanished, and FVF or UPF in the absence of venom did not have PLA₂ activity.

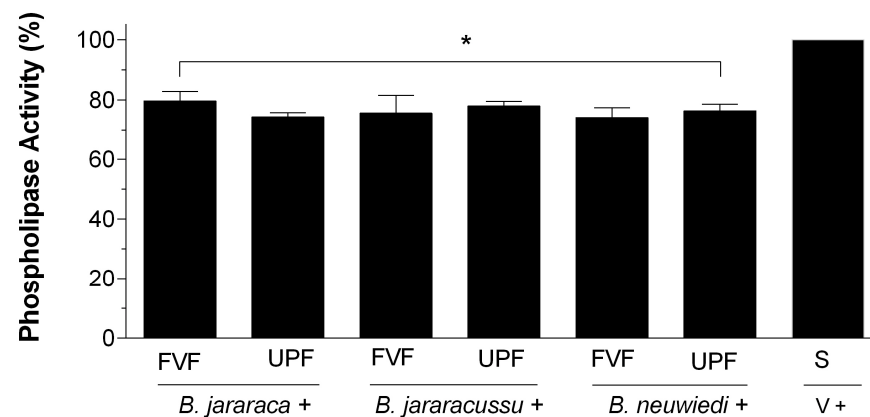


Figure 3. PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom in the presence of FVF or UPF. A total of 50 µg/mL *B. jararaca*, *B. jararacussu*, or *B. neuwiedi* venom was incubated with 25 µg/mL FVF or UPF for 5 min at 37 °C. An aliquot of each mixture was then added to the reaction, and PLA₂ activity was measured. A total of 100% PLA₂ activity was achieved by incubating each venom (V) with saline (S) in the absence of FVF or UPF. Results are expressed as mean ± SEM of $n = 6$, considering * $p < 0.05$ when compared to the positive group (V + S).

3. Discussion

This study provides evidence that two fucoidan extracts derived from *F. vesiculosus* and *U. pinnatifida*, namely FVF and UPF, respectively, inhibited proteolytic, plasma, or fibrinogen coagulant and the PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom but with different profiles. These activities are directly involved in local tissue damage and systemic hemotoxicity in SBE. The proteolytic, hemorrhagic, and coagulating activities are caused by the SVMP and SVSP families of enzymes [7,33,34], while edema, inflammation, and muscle damage effects are mainly due to PLA₂ enzymes [35,36]. Nevertheless, SVMP, SVSP, and PLA₂ are considered the most important and the major group of active proteins of viper venoms and are thus responsible for producing the most serious symptoms in SBE [34,35]. Therefore, inhibition of the toxic effects due to such a family of enzymes by FVF and UPF is of great importance. In the literature, there are differences in the toxicity of venoms between species of *Bothrops*. The venom of *B. jararaca* is highly hemorrhagic due to the presence of SVMPs, e.g., jararhagin [37] and bothropasin [38]. On the other hand, *B. jararacussu* venom is highly myotoxic due to the activity of two molecules with a PLA₂ structure: the enzymatically inactive bothropstoxin I (Bthtx-I) and the enzymatically active bothropstoxin II (Bthtx-II) [39,40]. *B. neuwiedi* venom has high coagulant, hemorrhagic, and myotoxic activities [41].

Antivenoms do not neutralize such toxic activities caused by *Bothropic* venoms [41] or other vipers [42] and may even induce hypersensitive reactions in patients [14]. Thus, molecules able to neutralize the toxic activities of snake venoms are required to improve the recovery of victims of SBE or to prevent sequelae. The SVMP and SVSP families include many enzymes with chemically unique features and intriguing mechanisms of action [33,34]. Therefore, blocking these venom enzymes, which have multiple and complex actions, using a single molecule is a challenge. The sulfated polysaccharide fucoidans derived from seaweed may be a good choice due to their safety and efficacy in humans and attractive physical and chemical properties [43]. Moreover, the ease of development of pharmaceutical forms using fucoidans means that they are good candidates for developing medicines [43,44]. Because conventional antivenoms have limitations [14], Fuly et al. [26] developed a gel containing a galactan polysaccharide from the red alga *Chondrus crispus* that protected mice from hemorrhaging caused by the venom of *B. jararaca* and *B. jararacussu* when the gel was applied topically before or after the injection of venom. The literature also describes the use of an ointment or cream containing fucoidan from the brown algae *F. vesiculosus* and *U. pinnatifida* to treat inflammatory skin diseases, thrombosis, wounds,

and burns [45]. These authors also studied the pharmacokinetics of fucoidan after topical application of such formulations and showed that fucoidan penetrated the skin efficiently and had a long half-life in plasma, striated muscle, and skin, with no accumulation in plasma even after the repeated application of fucoidan for five days [45]. SBE is an inflammatory illness, producing serious local toxic effects in humans, and is thus a transdermal formulation using fucoidan is a good strategy to aid commercial antivenoms. Our group has developed a gel using carrageenan from *C. crispus* [26], and the development of this type of formulation using FVF or UPF is a promising pathway for treating the local toxic effects of SBE. Fucoidans of different molecular weights have been investigated to discover how they are metabolized in the processes of absorption, distribution, metabolism, and excretion from marine sponge-, fungal-, and algal-derived compounds [46]. Moreover, fucoidans have high chemical and physical stability, biocompatibility, and biodegradability and show no side effects in humans [47]. On the other hand, a disadvantage of fucoidans is the lack of gelation ability, but this limitation can be overcome by combining fucoidans with polymers, such as chitosan, which may provide them with a positive charge.

The effect of FVF and UPF against the proteolytic, coagulant, and PLA₂ activities of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom may be due to the presence and quantity of sulfate groups within their structure. They have a negatively charged structure, allowing them to bind to the positively charged active regions of the SVMPs, SVSPs, and PLA₂ enzymes of *Bothrops* venom. The toxic activities of PLA₂ enzymes of snake venoms depend on the cofactor Ca²⁺, as well as some specific basic amino acid residues of their active site, such as histidine at position 48 and others located at the C-terminal region of PLA₂ enzymes [35,36]. On the other hand, SVMPs are the dominant components of these venoms and contain Zn²⁺ in the active site. SVSPs affect the coagulation cascade due to the action of thrombin-like enzymes (TLEs) that have fibrinogenolytic activity, leading to fibrin formation through the cleavage of fibrinogen [34,48]. However, other approaches to inhibit SVSPs have been investigated, such as murine monoclonal antibodies (mAbs) and synthetic peptides based on the sequence of their substrates or natural inhibitors [41]. However, mAbs failed to neutralize the toxins of *B. alternatus* and *B. neuwiedi* venoms [41]. Da Silva et al. [49] designed two synthetic peptides that inhibited SVSPs of *B. jararaca* crude venom and the purified serine protease Batroxobin, but not human trypsin, which is a serine protease. Undoubtedly, such peptides are a promising therapeutic tool for improving the treatment of SBE caused by *Bothrops* species.

The sulfate groups attached to the structure of FVF and UPF are crucial for fucoidans to inhibit the plasma coagulation caused by the venoms of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*. The negatively charged structure of fucoidans may allow them to bind to the metals Ca²⁺ and Zn²⁺ within the active site of enzymes. Some commercial compounds able to chelate divalent metals, such as EDTA and O-phenanthroline, inhibit the toxic effects of SVMPs. Other reagents, such as p-bromophenacyl bromide (p-BPB) and EDTA, react irreversibly with PLA₂ enzymes, thus inhibiting the toxic effects dependent on the catalytic activity of PLA₂, i.e., myotoxicity, hemolysis, and ADP- or collagen-induced platelet aggregation [50,51]. Nonetheless, such commercial inhibitors are considered toxic to humans or, in some cases, aquatic environments. In this manuscript, the PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms was inhibited by EDTA. Therefore, FVF and UPF may share the inhibitory mechanism with such commercial reagents, without producing toxicity to humans or animals.

The molecular weight of FVF and UPF may be an important feature in the blocking of snake venom enzymes. It is reasonable to postulate that larger chains of polysaccharides would have a higher quantity of sulfate groups and, thus, the ability of FVF and UPF to bind venom enzymes would be higher. Overall, the mechanism of inhibition of compounds FVF and UPF against SVMPs, SVSPs, and PLA₂ enzymes appears to occur through the binding to divalent metals, and, thus, this information should be considered when developing compounds with antivenom effects. A synthetic inhibitor of PLA₂ enzymes, Varespladib, was considered in 2019 by the U.S. Food and Drug Administration (FDA) as a drug for

SBE [52]. Varespladib is able to bind the hydrophobic channel of enzymatically or non-enzymatically active PLA₂ enzymes, leading to the inhibition of myotoxicity, hemorrhage, and edema caused by such enzymes [53,54]. Thus, a combination of some compounds with FVF and UPF may be a good strategy to enhance the efficacy of neutralization of symptoms of SBE.

Thus, the chelating action of FVF and UPF fucoidans and the size of their chains are relevant to their antivenom activity. However, the exact mechanism of inhibition by FVF and UPF was not addressed in this work because the experiments were performed using crude venom instead of purified enzymes. Future experiments will explore the mechanism of action of FVF and UPF using purified SVMP, SVSP, and PLA₂ enzymes.

In this work, 1500 µg/mL FVF or UPF lysed around 2% of RBCs, which is an acceptable value for a candidate molecule [32]. The biotechnology company Marinova provides fucoidans to markets, such as food, medical, and pharmaceutical research, and thus a meticulous quality assurance protocol is performed to ensure the purity and safety for human consumption. FVF and UPF are produced in Marinova's International Organization for Standardization 9001 (ISO9001) accredited analytical laboratory and chemically characterized using the carbohydrate content and linkage analysis methods that are described by the FDA.

FVF and UPF have been approved by the FDA as being in the "Generally Recognized As Safe" (GRAS) category of food ingredients [55]. Moreover, in some European countries, such as Austria, Belgium, France, Poland, Spain, and the United Kingdom, preparations containing *F. vesiculosus* are also allowed [56]. A fucoidan from the alga *Cladosiphon okamuranus* was given orally to humans and showed no toxicity at doses of 4 g per day for two weeks. Moreover, other administration routes of fucoidan, such as subcutaneous injection, were also assayed and did not have any negative impact on homeostasis [56].

4. Conclusions

This manuscript evidenced the inhibitory effect of FVF and UPF against proteolytic, plasma, or fibrinogen clotting and PLA₂ activities caused by *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms. Despite inhibiting such toxic activities with similar efficacy, FVF and UPF inhibited toxic activities caused by *B. neuwiedi* venom more efficiently. Research using fucoidan has demonstrated potential antivenom activities; however, existing data were limited, warranting further research and, hence, the present study. The utilization of FVF and UPF as an antivenom involves many challenges, such as differences in the growth conditions of algae and pollution of the marine environment, as well as a wide range of species that contribute to the heterogeneity of their properties. Nevertheless, FVF and UPF may contribute to the manufacture and development of new active topical compounds against the toxic effects of venoms and aid commercial antivenoms in treating SBE caused by *Bothrops* species.

5. Materials and Methods

5.1. Reagents and Venoms

The lyophilized venoms of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* were provided by the serpentarium of the Ezequiel Dias Foundation (FUNED), Belo Horizonte, Minas Gerais, Brazil. The venom was diluted in saline and stored at −20 °C until it was assayed. Snake venom collection was conducted under the authorization of the Brazilian National System for Genetic Heritage Management and Associated Traditional Knowledge (SISGEN), process number A39CTRI 04E. Human fibrinogen, ethylenediaminetetraacetic acid (EDTA), trichloroacetic acid, taurocholic acid sodium salt, and azocasein were purchased from Sigma Chemical Co. (St. Louis, MO, USA). All the other reagents or solvents were research grade.

5.2. Toxicity of FVP and UPF

The toxicity of FVP and UPF was assessed by the in vitro hemocompatibility test, as described previously [32]. The fucoidan (1500 µg/mL) and saline solutions were incu-

bated with a 13% (*v/v*) suspension of RBC for 3 h at 37 °C. Samples were centrifuged for 3 min at 1800× *g*, and the lysis of RBC was measured by the release of hemoglobin at an absorbance (*A*) of 578 nm using a microplate reader (VersaMax, Molecular Devices, CA, USA). Hemolysis rates of 100% and 0% were obtained by adding distilled water or saline to RBC, respectively.

5.3. Fucoïdan Material

Fucus vesiculosus fucoidan (FVF, batch No. DPFVF2021001) and *Undaria pinnatifida* fucoidan (UPF; batch No. DPGFS2019537) were provided by Marinova (Cambridge, TAS, Australia). The materials were extracted using a proprietary aqueous extraction process. The fucoidan purity of both samples was >95% (dry weight). The fucoidan component and compositional profile for each material can be found in Table 1. The calculation of fucoidan purity requires several inputs that are determined using a range of spectrophotometric assays. The total carbohydrate content of a hydrolyzed sample was assessed using the phenol–sulfuric technique developed by [57], while the uronic acid concentration was determined in the presence of 3-phenylphenol, based on the method by [58]. Sulfate content was analyzed using a BaSO₄ precipitation method [59]. UPF and FVF were dissolved in saline solution to perform the assays.

Table 1. Absolute mass percentages of the components of FVF and UPF.

Components	FVF	UPF
Fucoidan	96.1	97
Total carbohydrates	71.7	60.5
Fucose	49.5	28.2
Galactose	2.7	25.6
Uronic acid	3.8	0.8
Polyphenol	<2	<2
Sulfate	30.7	30.4
Cations	~5	6.8

5.4. Effect of FVF and UPF on the Proteolytic Activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* Venom

The proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was performed according to [60]. Different concentrations of each venom (10–50 µg/mL) were incubated with 400 µL azocasein (0.2% *w/v* dissolved in 20 mM Tris-HCl, 8 mM CaCl₂, pH 8.8) for 90 min at 37 °C in a final volume of 800 µL. After 90 min, 400 µL of trichloroacetic acid (10%) was added to the medium to stop the reaction. The tubes were centrifuged at 12,000 rpm for 3 min. A total of 1.0 mL of the supernatant was removed and transferred to tubes containing 0.5 mL NaOH 2N. The tubes were read in a spectrophotometer at 420 nm. The amount of venom (µg/mL) able to produce reads of 0.2 (i.e., 70–80% of the maximal activity) was determined as an arbitrary unit, i.e., the effective concentration (EC). One EC of each venom was incubated with FVF or UPF or saline (positive groups) for 5 min at 37 °C at a 1:10 ratio (*w/w*) of venom to polysaccharide. After incubation, an aliquot was removed and added to the reaction, and the proteolytic activity was measured, as described above. Negative controls were performed by adding saline or fucoidan to the medium in the absence of venom.

5.5. Effect of FVF and UPF on the Coagulant Activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* Venom

The coagulant activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was determined using a digital Amelung coagulometer, model KC4A (Labcon, Heppenheim, Germany). A total of 200 µL of a pool of citrated human plasma from the blood bank of

the Hospital Universitário Antônio Pedro of the Federal Fluminense University (HUAP-UFF), under the authorization of the Committee for Ethical in Experimentation (CEP-UFF, CAAE: 28941314.0.0000.5243), or commercial fibrinogen (2 mg/mL) were kept for 1 min at 37 °C. Then, different concentrations of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom (2–100 µg/mL) were added to plasma or fibrinogen, and the coagulation time was monitored in seconds in the coagulometer. The amount of venom (µg/mL) able to clot plasma or fibrinogen at 60 s was determined as an arbitrary unit, i.e., the minimum coagulant concentration (MCC). Then, one MCC of each venom was incubated for 5 min at 37 °C with saline (positive group) or UPF or FVF at a 1:10 ratio of venom to polysaccharide (*w/w*). An aliquot of each mixture was added to plasma or fibrinogen, and coagulation was monitored as described above. The negative control groups contained solely saline or FVF or UPF in the reaction in the absence of any venom.

5.6. Effect of FVF and UPF on the Phospholipase A₂ Activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* Venom

The PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was measured as in [61]. Fresh egg yolk obtained from a local supermarket was filtered and diluted in saline to a final volume of 100 mL. One volume of this solution was diluted in nine parts saline. The medium reaction contained 580 µL saline, 25 µL sodium taurocholate 0.4%, 25 µL Tris-HCl (200 mM), pH 7.5, 20 µL CaCl₂ (0.5 M), and 50 µL of each venom in the presence or absence of FVF or UPF. The enzymatic reaction was initiated by adding 300 µL of the egg yolk substrate solution, and the tubes were read at 740 nm after 30 min. A total of 100% PLA₂ activity of venoms was achieved by incubating venom with saline, in the absence of FVF or UPF, while 0% PLA₂ activity was determined in a solution containing solely saline or FVF or UPF in the absence of venom. A total of 20 mM of EDTA (final concentration) was added to the reaction in the presence of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom and in the absence of FVF or UPF, followed by the PLA₂ activity.

5.7. Statistical Analyses

Results are expressed as means ± standard error of the mean (SEM) obtained with the number of experiments indicated in each result using the GraphPad Prism[®] program. The statistical significance of differences among experimental groups was evaluated using Student's *t*-test or the Mann–Whitney test. *p*-values < 0.05 were considered statistically significant.

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Institutional Review Board Statement: Plasma was obtained from the blood bank of the Hospital Universitário Antônio Pedro of the Federal Fluminense University (HUAP-UFF) under the authorization of the Committee for Ethical in Experimentation (CEP-UFF), CAAE: 28941314.0.0000.5243). The collection and extraction of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms was conducted under the authorization of the Brazilian National System for Genetic Heritage Management and Associated Traditional Knowledge (SISGEN) (process number A39CTRI 04E).

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Data Availability Statement: Data are contained within the article.

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References

1. World Health Organization. *Target Product Profiles for Animal Plasma-Derived Antivenoms: Antivenoms for Treatment of Snakebite Envenoming in Sub-Saharan Africa*; WHO: Geneva, Switzerland, 2023; 25p.
2. World Health Organization. *WHO Guidelines for the Production, Control and Regulation of Snake Antivenom Immunoglobulins*; WHO: Geneva, Switzerland, 2017; 388p.
3. Williams, D.J.; Faiz, M.A.; Abela-Ridder, B.; Ainsworth, S.; Bulfone, T.C.; Nickerson, A.D.; Habib, A.G.; Junghanss, T.; Fan, H.W.; Turner, M.; et al. Strategy for a globally coordinated response to a priority neglected tropical disease: Snakebite envenoming. *PLoS Negl. Trop. Dis.* **2019**, *13*, e0007059. [CrossRef] [PubMed]
4. Gutiérrez, J.M.; Calvete, J.J.; Habib, A.G.; Harrison, R.A.; Williams, D.J.; Warrell, D.A. Snakebite envenoming. *Nat. Rev. Dis. Primers* **2017**, *3*, 17063. [CrossRef]
5. Chippaux, J.P. Snakebite envenomation turns again into a neglected tropical disease! *J. Venom. Anim. Toxins Incl. Trop. Dis.* **2017**, *23*, 38–39. [CrossRef] [PubMed]
6. Queiroz, G.P.; Pessoa, L.A.; Portaro, F.C.V.; Furtado, M.d.F.D.; Tambourgi, D.V. Interspecific variation in venom composition and toxicity of Brazilian snakes from *Bothrops* genus. *Toxicon* **2008**, *52*, 842–851. [CrossRef] [PubMed]
7. Cavecci-Mendonça, B.; Luciano, K.M.; Vaccas, T.; de Oliveira, L.A.; Clemente, E.F.; Rossini, B.C.; Vieira, J.C.S.; de Barros, L.C.; Biondi, I.; de Magalhães, P.P.; et al. Preliminary insights of Brazilian snake venom metalloproteomics. *Toxins* **2023**, *15*, 648. [CrossRef] [PubMed]
8. Chippaux, J.P. Incidence and mortality due to snakebite in the Americas. *PLoS Negl. Trop. Dis.* **2017**, *11*, e0005662. [CrossRef] [PubMed]
9. de Oliveira, F.N.; Sachetto, A.T.A.; Santoro, M.L. Two-dimensional blue native/SDS polyacrylamide gel electrophoresis for analysis of Brazilian *Bothrops* snake venoms. *Toxins* **2022**, *14*, 661. [CrossRef] [PubMed]
10. Camey, K.U.; Velarde, D.T.; Sanchez, E.F. Pharmacological characterization and neutralization of the venoms used in the production of Bothropic antivenom in Brazil. *Toxicon* **2002**, *40*, 501–509. [CrossRef] [PubMed]
11. de Lima, M.E.; Fortes-Dias, C.L.; Carlini, C.R.; Guimarães, J.A. Toxinology in Brazil: A big challenge for a rich biodiversity. *Toxicon* **2010**, *56*, 1084–1091. [CrossRef]
12. WHO. Snakebite Envenoming. 2021. Available online: <https://www.who.int/news-room/fact-sheets/detail/snakebite-envenoming> (accessed on 17 February 2021).
13. Waidyanatha, S.; Silva, A.; Siribaddana, S.; Isbister, G.K. Long-term effects of snake envenoming. *Toxins* **2019**, *11*, 193. [CrossRef]
14. Solano, G.; Ainsworth, S.; Sánchez, A.; Villalta, M.; Sánchez, P.; Durán, G.; Gutiérrez, J.A.; León, G. Analysis of commercially available snake antivenoms reveals high contents of endotoxins in some products. *Toxicon X* **2024**, *211*, 100187–100189. [CrossRef] [PubMed]
15. Gutiérrez, J.M.; Albulescu, L.O.; Clare, R.H.; Casewell, N.R.; Abd El-Aziz, T.M.; Escalante, T.; Rucavado, A. The search for natural and synthetic inhibitors that would complement antivenoms as therapeutics for snakebite envenoming. *Toxins* **2021**, *13*, 451. [CrossRef]
16. Marshall, W.R.; Soffa, A.N. Considerations for the development of a field-based medical device for the administration of adjunctive therapies for snakebite envenoming. *Toxicon X* **2023**, *20*, 100169–100190.
17. Liaqat, A.; Mallhi, T.H.; Khan, Y.H.; Khokhar, A.; Chaman, S.; Ali, M. Anti-Snake venom properties of medicinal plants: A comprehensive systematic review of literature. *Braz. J. Pharm. Sci.* **2022**, *58*, e191124. [CrossRef]
18. Adrião, A.A.X.; Dos Santos, A.O.; de Lima, E.J.S.P.; Maciel, J.B.; Paz, W.H.P.; da Silva, F.M.A.; Pucca, M.B.; Moura-da-Silva, A.M.; Monteiro, W.M.; Sartim, M.A.; et al. Plant-derived toxin inhibitors as potential candidates to complement antivenom treatment in snakebite envenomations. *Front. Immunol.* **2022**, *9*, 842576–842605. [CrossRef] [PubMed]
19. Mekinić, I.G.; Skroza, D.; Šimat, V.; Hamed, I.; Čagalj, I.M.; Perković, Z.P. Phenolic content of brown algae (Pheophyceae) species: Extraction, identification, and quantification. *Biomolecules* **2019**, *22*, 244. [CrossRef]

20. Lomartire, S.; Gonçalves, A.M.M. An overview of potential seaweed-derived bioactive compounds for pharmaceutical applications. *Mar. Drugs* **2022**, *20*, 141. [\[CrossRef\]](#)
21. Costa, B.B.; Gianelli, J.L.D.; Moreira, T.A.; Soares, A.R.; Glauser, B.F.; Mourão, P.A.S.; Neto, C.; Barros, C.M.; Cinelli, L.P. Partial characterization and anticoagulant activity of sulfated galactan from the green seaweed *Halimeda opuntia*. *An. Acad. Bras. Cienc.* **2023**, *95*, e20211002. [\[CrossRef\]](#) [\[PubMed\]](#)
22. Yinyi, F.; Jiao, H.; Sun, J.; Okoye, C.O.; Zhang, H.; Li, Y.; Lu, X.; Wang, Q.; Liu, J. Structure-activity relationships of bioactive polysaccharides extracted from macroalgae towards biomedical application: A review. *Carbohydr. Polym.* **2024**, *324*, 121533–121558.
23. El-Sheekh, M.M.; Ward, F.; Deyab, M.A.; Al-Zahrani, M.; Touliabah, H.E. Chemical composition, antioxidant, and antitumor activity of fucoidan from the brown alga *Dictyota dichotoma*. *Molecules* **2023**, *19*, 7175. [\[CrossRef\]](#)
24. Ferreira, L.G.; da Silva, A.C.R.; Nosedá, M.D.; Fuly, A.L.; de Carvalho, M.M.; Fujii, M.T.; Sanchez, E.F.; Carneiro, J.; Duarte, M.E.R. Chemical structure and snake antivenom properties of sulfated agarans obtained from *Laurencia dendroidea* (Ceramiales, Rhodophyta). *Carbohydr. Polym.* **2019**, *15*, 136–144. [\[CrossRef\]](#) [\[PubMed\]](#)
25. da Silva, A.C.R.; Ferreira, L.G.; Duarte, M.E.R.; Nosedá, M.D.; Sanchez, E.F.; Fuly, A.L. Sulfated Galactan from *Palisada flagellifera* Inhibits Toxic Effects of *Lachesis muta* Snake Venom. *Mar. Drugs* **2015**, *13*, 3761–3775. [\[CrossRef\]](#) [\[PubMed\]](#)
26. da Silva, A.C.R.; Pereira, K.K.G.; Critchley, A.T.; Sanchez, E.F.; Fuly, A.L. Potential utilization of a lambda carrageenan polysaccharide, derived from a cultivated, clonal strain of the red seaweed *Chondrus crispus* (Irish moss) against toxic actions of venom of *Bothrops jararaca* and *B. jararacussu* snakes. *J. Appl. Phycol.* **2020**, *32*, 4309–4320. [\[CrossRef\]](#)
27. da Silva, A.C.R.; Duarte, M.E.R.; Nosedá, M.D.; Ferreira, L.G.; Cassolato, J.E.F.; Sanchez, E.F.; Fuly, A.L. Potential utilization of a polysaccharide from the marine algae *Gayralia oxysperma*, as an antivenom for Viperidae snakebites. *Mar. Drugs* **2018**, *16*, 412. [\[CrossRef\]](#) [\[PubMed\]](#)
28. Obluchinskaya, E.D.; Pozharitskaya, O.N.; Shevyrin, V.A.; Kovaleva, E.G.; Flisyuk, E.V.; Shikov, A.N. Optimization of Extraction of phlorotannins from the Arctic *Fucus vesiculosus* using natural deep eutectic solvents and their HPLC profiling with Tandem High-Resolution Mass Spectrometry. *Mar. Drugs* **2023**, *21*, 263. [\[CrossRef\]](#) [\[PubMed\]](#)
29. Tagliapietra, B.L.; Clerici, M.T.P.S. Brown algae and their multiple applications as functional ingredient in food production. *Food Res. Int.* **2023**, *167*, 112655–112665. [\[CrossRef\]](#) [\[PubMed\]](#)
30. Azofeifa, K.; Angulo, Y.; Lomonte, B. Ability of fucoidan to prevent muscle necrosis induced by snake venom myotoxins: Comparison of high- and low-molecular weight fractions. *Toxicon* **2008**, *51*, 373–380. [\[CrossRef\]](#) [\[PubMed\]](#)
31. Angulo, Y.; Lomonte, B. Inhibitory effect of fucoidan on the activities of crotaline snake venom myotoxic phospholipases A₂. *Biochem. Pharmacol.* **2003**, *66*, 1993–2000. [\[CrossRef\]](#) [\[PubMed\]](#)
32. Bauer, M.; Lautenschlaeger, C.; Kempe, K.; Tauhardt, L.; Schubert, U.S.; Fischer, D. Poly(2-ethyl-2-oxazoline) as alternative for the stealth polymer poly(ethylene glycol): Comparison of in vitro cytotoxicity and hemocompatibility. *Macromol. Biosci.* **2012**, *12*, 986–998. [\[CrossRef\]](#)
33. Olaoba, O.T.; dos Santos, P.K.; Selistre-de-Arujo, H.S.; de Souza, D.H.F. Snake Venom Metalloproteinases (SVMPs): A structure-function update. *Toxicon X* **2020**, *21*, 100052–100067. [\[CrossRef\]](#)
34. Serrano, S.M. The long road of research on snake venom serine proteinases. *Toxicon* **2013**, *62*, 19–26. [\[CrossRef\]](#) [\[PubMed\]](#)
35. de Paula, R.C.; Castro, H.C.; Rodrigues, C.R.; Melo, P.A.; Fuly, A.L. Structural and pharmacological features of phospholipases A₂ from snake venoms. *Protein Pept. Lett.* **2009**, *16*, 899–907. [\[CrossRef\]](#) [\[PubMed\]](#)
36. Aloulou, A.; Rahier, R.; Arhab, Y.; Noiriél, A.; Abousalham, A. Phospholipases: An Overview. *Methods Mol. Biol.* **2018**, *1835*, 69–105. [\[PubMed\]](#)
37. Laing, G.D.; Moura-da-Silva, A.M. Jararhagin and its multiple effects on hemostasis. *Toxicon* **2005**, *45*, 987–996. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Molina, D.A.; Guerra-Duarte, C.; Naves de Souza, D.L.; Costal-Oliveira, F.; Ávila, G.R.; Soccol, V.T.; Machado-de-Ávila, R.A.; Chávez-Olórtegui, C. Identification of a linear B-cell epitope in the catalytic domain of bothropasin, a metalloproteinase from *Bothrops jararaca* snake venom. *Mol. Immunol.* **2018**, *104*, 20–26. [\[CrossRef\]](#) [\[PubMed\]](#)
39. Cardoso, F.F.; Salvador, G.H.M.; Cavalcante, W.L.G.; Dal-Pai, M.; Fontes, M.R.M. BthTX-I, a phospholipase A₂-like toxin, is inhibited by the plant cinnamic acid derivative: Chlorogenic acid. *Biochim. Biophys. Acta Proteins Proteom.* **2024**, *1872*, 140988–141001. [\[CrossRef\]](#) [\[PubMed\]](#)
40. Borges, R.J.; Salvador, G.H.M.; Campanelli, H.B.; Pimenta, D.C.; de Oliveira, N.M.; Usón, I.; Fontes, M.R.M. BthTX-II from *Bothrops jararacussu* venom has variants with different oligomeric assemblies: An example of snake venom phospholipases A₂ versatility. *Int. J. Biol. Macromol.* **2021**, *191*, 255–266. [\[CrossRef\]](#) [\[PubMed\]](#)
41. Belo, A.A.; de Souza, D.L.N.; de Melo-Braga, M.N.; de Souza, L.L.; Molina, M.D.A.; de Melo, P.D.V.; Larsen, M.R.; Guerra-Duarte, C.; Chávez-Olórtegui, C. Production of a murine mAb against *Bothrops alternatus* and *B. neuwiedi* snake venoms and its use to isolate a thrombin-like serine protease fraction. *Int. J. Biol. Macromol.* **2022**, *214*, 530–541. [\[CrossRef\]](#) [\[PubMed\]](#)
42. Thumtecho, S.; Burlet, N.J.; Ljungars, A.; Laustsen, A.H. Towards better antivenoms: Navigating the road to new types of snakebite envenoming therapies. *J. Venom. Anim. Toxins Incl. Trop. Dis.* **2023**, *29*, e20230057. [\[CrossRef\]](#)
43. Haggag, Y.A.; Elrahman, A.A.A.; Ulber, R.; Zayed, A. Fucoidan in Pharmaceutical Formulations: A Comprehensive Review for Smart Drug Delivery Systems. *Mar. Drugs* **2023**, *21*, 112. [\[CrossRef\]](#)
44. Trujillo, E.; Monreal-Escalante, E.; Ramos-Vega, A.; Angulo, C. Macroalgae: Marine players in vaccinology. *Algal Res.* **2024**, *78*, 103392–103403. [\[CrossRef\]](#)

45. Pozharitskaya, O.N.; Shikov, A.N.; Obluchinskaya, E.D.; Vuorela, H. The pharmacokinetics of fucoidan after topical application to rats. *Mar. Drugs* **2019**, *17*, 687. [CrossRef]
46. Shikov, A.N.; Flisyuk, E.V.; Obluchinskaya, E.D.; Pozharitskaya, O.N. Pharmacokinetics of marine-derived drugs. *Mar. Drugs* **2020**, *18*, 557. [CrossRef] [PubMed]
47. George, A.; Shrivastav, P.S. Fucoidan, a brown seaweed polysaccharide in nanodrug delivery. *Drug Deliv. Transl. Res.* **2023**, *13*, 2427–2446. [CrossRef] [PubMed]
48. Ullah, A.; Masood, R.; Ali, I.; Ullah, K.; Ali, H.; Akbar, H.; Betzel, C. Thrombin-like enzymes from snake venom: Structural characterization and mechanism of action. *Int. J. Biol. Macromol.* **2018**, *15*, 788–811. [CrossRef]
49. da Silva, G.M.; de Souza, D.H.B.; Waitman, K.B.; Ebram, M.C.; Fessel, M.R.; Zainescu, I.C.; Portaro, F.C.; Heras, M.; de Andrade, S.A. Design, synthesis, and evaluation of *Bothrops* venom serine protease peptidic inhibitors. *J. Venom. Anim. Toxins Incl. Trop. Dis.* **2021**, *27*, e20200066. [CrossRef] [PubMed]
50. Fuly, A.L.; Calil-Elias, S.; Zingali, R.B.; Guimarães, J.A.; Melo, P.A. Myotoxic activity of an acidic phospholipase A₂ isolated from *Lachesis muta* (Bushmaster) snake venom. *Toxicon* **2000**, *38*, 961–972. [CrossRef]
51. Fuly, A.L.; de Miranda, A.L.; Zingali, R.B.; Guimarães, J.A. Purification and characterization of a phospholipase A₂ isoenzyme isolated from *Lachesis muta* snake venom. *Biochem. Pharmacol.* **2002**, *63*, 1589–1597. [CrossRef]
52. Lewin, M.R.; Carter, R.W.; Matteo, I.A.; Samuel, S.P.; Rao, S.; Fry, B.G.; Bickler, P.E. Varespladib in the Treatment of Snakebite Envenoming: Development History and Preclinical Evidence Supporting Advancement to Clinical Trials in Patients Bitten by Venomous Snakes. *Toxins* **2022**, *14*, 783. [CrossRef]
53. Salvador, G.H.M.; Pinto, Ê.K.R.; Ortolani, P.L.; Fortes-Dias, C.L.; Cavalcante, W.L.G.; Soares, A.M.; Lomonte, B.; Lewin, M.R.; Fontes, M.R.M. Structural basis of the myotoxic inhibition of the *Bothrops pirajai* PrTX-I by the synthetic Varespladib. *Biochimie* **2023**, *207*, 1–10. [CrossRef]
54. Lewin, M.; Samuel, S.; Merkel, J.; Bickler, P. Varespladib (LY315920) Appears to Be a Potent, Broad-Spectrum, Inhibitor of Snake Venom Phospholipase A₂ and a Possible Pre-Referral Treatment for Envenomation. *Toxins* **2016**, *25*, 248. [CrossRef] [PubMed]
55. Re: GRAS Notice, No. GRN 000661. Available online: <https://www.fda.gov/downloads/Food/IngredientsPackagingLabeling/GRAS/NoticeInventory/ucm549588.pdf> (accessed on 10 February 2024).
56. Citkowska, A.; Szekealska, M.; Winnicka, K. Possibilities of Fucoidan Utilization in the Development of Pharmaceutical Dosage Forms. *Mar. Drugs* **2019**, *17*, 458. [CrossRef] [PubMed]
57. DuBois, M.K.; Gilles, K.J.; Hamilton, J.K.; Rebers, P.A.; Smith, F.S. Colorimetric method for determination of sugars and related substances. *Anal. Chem.* **1956**, *28*, 350–356. [CrossRef]
58. Filisetti-Cozzi, T.M.; Carpita, N.C. Measurement of uronic acids without interference from neutral sugars. *Anal. Biochem.* **1991**, *97*, 157–162. [CrossRef] [PubMed]
59. Dodgson, K.S.; Price, R.G. A note on the determination of the ester sulphate content of sulphated polysaccharides. *Biochem. J.* **1962**, *84*, 106–110. [CrossRef] [PubMed]
60. Garcia, E.S.; Guimarães, J.A.; Prado, J.L. Purification and characterization of a sulfhydryl-dependent protease from *Rhodnius prolixus* midgut. *Arch. Biochem. Biophys.* **1978**, *188*, 315–322. [CrossRef]
61. Marinetti, G.V. The action of phospholipase A₂ on lipoproteins. *Biochim. Biophys. Acta Lipids Lipid Metab.* **1965**, *98*, 554–565. [CrossRef]

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OPEN Exploiting exopolysaccharides from microalgae to block the toxic in vitro effects of *Bothrops* sp. venom

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Snakebite envenomation causes various toxic effects in humans, including death. Antivenoms are effectively prevent death but cannot completely block local muscle tissue damage, resulting in amputation and disability; thus, complementary therapies are needed. In this study, the ability of exopolysaccharides extracted from the microalgae *Chlorella sorokiniana*, *Scenedesmus obliquus*, *Nannochloris* sp. Naumann, and *Scenedesmus acuminatus* to inhibit the proteolytic, plasma coagulant, and phospholipase A₂ (PLA₂) activities of *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms was assessed. Exopolysaccharides from *C. sorokiniana*, *S. obliquus*, *Nannochloris* sp. Naumann, and *S. acuminatus* inhibited the proteolytic activity of *B. jararaca* venom by 25%, 99%, 97%, and 13%, respectively; *B. jararacussu* venom by 14%, 50%, 46%, and 12%, respectively; and *B. neuwiedi* venom by 1%, 78%, 62%, and 1%, respectively. Exopolysaccharides from *S. obliquus* and *Nannochloris* sp. Naumann prevented the coagulation induced by snake venom and decreased the PLA₂ activity by 15% and 30%, respectively. None of the four exopolysaccharides lysed red blood cells and, thus, can be considered as nonhemotoxic compounds. Therefore, microalgae exopolysaccharides may be a candidate as complementary therapies for envenomation by snakes in regions where such incidents are common.

Keywords Microalgae, Snake venoms, Antidote, Neutralization, Exopolysaccharides, Toxic activities

In recent years, numerous countries with tropical and subtropical climates, particularly Africa, Asia, Latin America, India, and Oceania have experienced a surge in morbidity and mortality rates associated with venomous snake bites. These countries have a high level of biodiversity, large areas of forest, indiscriminate deforestation, and overexploitation of natural resources, bringing the population into constant contact with snakes and other reptiles^{1–3}. More than 15% of all animals on Earth are considered venomous and contact with them can cause irreparable damage to human health^{1,4,5}. Globally, the death rate of snake bite envenomation (SBE) is approximately 0.5–1%; males are more affected than females, and more children (25%) than adults experienced SBE (12%)^{6,8}. Poorer populations are more susceptible to SBE because of inadequate bite treatment and prevention and environmental conditions, leading to pathological consequences, including local edema, tissue necrosis (myotoxicity), neuromuscular paralysis, systemic hemorrhage, coagulopathy, and death, as well as economic, social, and physiological consequences⁷. This led the World Health Organization (WHO) to add snakebites to the list of priority neglected tropical diseases (NTDs) in 2017^{7,9,10}.

Snake bites affect approximately 10 million people worldwide each year^{7,9}. Asian countries have the highest rate of incidents and complications from SBE, accounting for more than 70% of all cases worldwide. India ranks first among Asian countries in terms of deaths, with over 45,000 deaths in 2020, and Africa and the Middle East rank second, with more than 35,000 deaths¹⁰. Brazil has a high rate of death, with approximately 27,000 fatalities reported annually, and pit viper species account for the highest frequency of snakebite accidents (99%)^{11,12}. Among vipers, the genus *Bothrops* accounts for 87% of the registered incidents and is the largest genus with 47 species, of which *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* are among the most important due to their association with high numbers of SBE events¹².

In general, the high mortality and morbidity rates of SBE caused by pit vipers and true vipers result from the presence of three dominant protein families in the venom: including: 26.4% phospholipases A₂ (PLA₂s), 24% snake venom metalloproteinases (SVMPs), and 12% snake venom serine proteinases (SVSPs)^{13,14}. The main

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components of snake venom (up to 95%) are proteins and peptides with a wide range of toxic activities that may cause pain, inflammation, edema, intense hemorrhage, renal and heart failure, myonecrosis, and death¹⁵. *Bothrops* sp. has a high content of active proteins in their venom, including SVMs, representing as high as 74% in some species¹⁶. Other biomolecules including SVSPs, PLA₂s, LAAOs, and hyaluronidases; other proteins lack catalytic activity, e.g., bradykinin-potentiating peptides (BPPs), but are associated with devastating toxic effects, including vasodilatation, decreased blood pressure, and circulation shock^{14–17}. However, the complications, symptoms, and protein compositions may vary according to the snake species, age, diet, sex, and environmental conditions^{12,18,19}.

Currently, SBE is effectively treated by intravenously administering immunoglobulin-based antivenoms to the victim. To date, antivenom is the only SBE treatment that has been recognized by the WHO and Ministries of Health worldwide for SBE^{7,9}. Antivenoms efficiently prevent death in most cases and thus reduce mortality rates. However, the use of antivenoms has disadvantages, such as high production costs, side effects (rash, severe malaise, fever, and allergic reactions), the inability to block tissue necrosis, and the potential for untimely administration or incorrect dosages²⁰. Victims must be quickly transported to a health center and administered antivenom early to maximize its effects. Despite these adverse reactions, antivenoms are still considered the only available and effective treatment for SBE. In recent years, the WHO has become concerned about the use of antivenoms, including their safety, efficacy, adverse effects, distribution, and production. A roadmap was developed to improve the quality of antivenoms and mechanisms to develop complementary solutions for the treatment and prevention of SBE^{21,22}.

Natural resources are a source of active compounds of pharmacological interest to industry and communities because of their availability, cost-effectiveness, cultural acceptability, and fewer side effects versus antivenoms²³. Plants have long been used to treat symptoms of SBE, especially in remote areas^{24–27}. In addition to medicinal plants, seaweeds and microalgae are rich sources of biologically active substances^{28,29}. Although abundant in the marine environment, microalgae remain poorly understood as sources of compounds with pharmacological and biological relevance, particularly for use as antivenoms. No antivenom in SBE of exopolysaccharides from microalgae has been reported in the literature; however, they are a potential source of valuable chemicals with biological activities^{30,31}. Several eukaryotic and prokaryotic microalgae species produce and release large volumes of polysaccharides into the environment^{32,33}. The biological properties of exopolysaccharides produced by microalgae include anti-inflammatory, immunomodulatory, antitumor, antiviral, antiparasitic, antioxidant, hypoglycemic, and cholesterol-lowering activities^{29,30}. Microalgae are good candidates for inhibiting the main active enzymes of snake venoms, including SVMs, SVSPs, and PLA₂s, and should be prioritized in future pharmacological investigations^{15–17}.

In this study, the abilities of four exopolysaccharides extracted from the microalgae *Chlorella sorokiniana* (*C. sorokiniana*), *Scenedesmus obliquus* (*S. obliquus*), *Nannochloris* sp. Naumann, and *Scenedesmus acuminatus* (*S. acuminatus*) to inhibit the PLA₂, proteolytic, and coagulant activities of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms, as well as their physicochemical properties and in vitro toxicity, were assessed.

Results

Molecular weight and viscosity of exopolysaccharides from microalgae

Table 1 shows that among the C1, S302, 1509, and S329, samples S329 and S302 from *S. acuminatus* (8.04 mg/mL) and *S. obliquus* (7.15 mg/mL) had the highest viscosities, whereas exopolysaccharide 1509 from *Nannochloris* sp. Naumann had the lowest value (1.35), and the lowest molecular weight (13.25 × 10⁴ Da). C1 from *C. sorokiniana* had the highest molecular weight (17.71 × 10⁴ Da).

Determination of the sugar composition of exopolysaccharides from microalgae via HPLC

The chromatograms of the exopolysaccharides from the microalgae are shown in Fig. 1, including C1 from *C. sorokiniana* (Panel A), 1509 from *Nannochloris* sp. Naumann (Panel B), S302 from *S. obliquus* (Panel C), and S329 from *S. acuminatus* (Panel D). Table 2 provides the carbohydrate composition on the basis of the peak areas presented in the chromatograms. The error in calculating the concentration of the exopolysaccharides did not exceed 10%.

Chromatographic separation revealed that the exopolysaccharide S302 from *S. obliquus* is composed of arabinose, glucose, maltose, and trehalose (Table 2). The sugar composition of S329 from *S. acuminatus* was mainly fucose, xylose, arabinose, glucose, sucrose, and maltose. The exopolysaccharide C1 from *C. sorokiniana* did not contain xylose or sucrose but contained arabinose, maltose, and the highest amount of glucose and trehalose content among the samples (Table 2). The sugars galactose and raffinose were not present in the polysaccharides C1, S302, 1509, or S329 (Table 2).

Sample	Viscosity (cP)	MW (×10 ⁴ Da)
C1	2.05 ± 0.06	17.71 ± 0.66
1509	1.35 ± 0.41	13.25 ± 0.10
S302	7.15 ± 0.54	14.40 ± 0.47
S329	8.04 ± 0.47	13.54 ± 0.35

Table 1. Viscosity and molecular weights of microalgae exopolysaccharides.

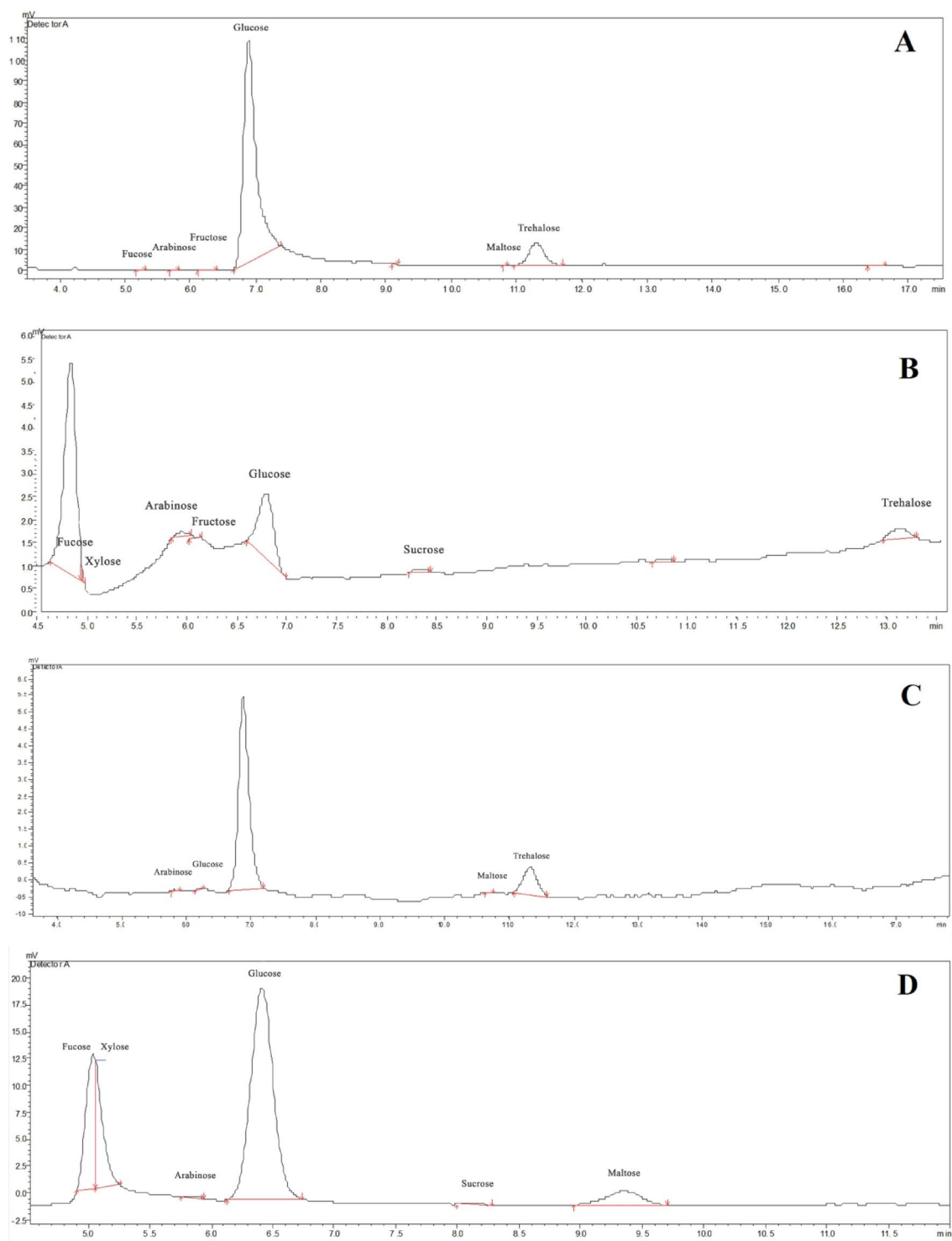


Fig. 1. Chromatograms of exopolysaccharides C1 from *C. sorokiniana* (A), 1509 from *Nannochloris* sp. Naumann (B), S302 from *S. obliquus* (C), and S329 from *S. acuminatus* (D).

Effect of exopolysaccharides on the proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom

One arbitrary unit designed effective concentration (EC) of *B. jararaca* and *B. neuwiedi* venom (30 µg/mL) or *B. jararacussu* (50 µg/mL) venom was incubated for 5 min at 37 °C with saline solution (positive control group) or with the exopolysaccharides C1, S302, 1509, and S329 at a 1:10 (wt/wt) venom-to-polysaccharide ratio. C1 and S329 did not inhibit the proteolytic activity of *B. neuwiedi* venom (Fig. 2C) but inhibited approximately 20% of the proteolysis of *B. jararaca* (Fig. 2A) and *B. jararacussu* venom (Fig. 2B). The exopolysaccharides S302 and

Carbohydrates	Polysaccharides			
	C1	1509	S302	S329
Rhamnose	ND	BDL	ND	ND
Fucose	0.111	0.097	ND	0.4234
Xylose	ND	0.142	ND	0.5459
Arabinose	0.073	0.056	0.066	0.0345
Fructose	0.083	0.034	ND	ND
Glucose	15.705	6.012	0.662	1.3742
Galactose	ND	ND	ND	ND
Sucrose	ND	0.053	ND	0.0167
Maltose	0.062	ND	0.061	0.1693
Trehalose	2.323	0.541	0.211	ND
Raffinose	ND	ND	ND	ND

Table 2. Carbohydrate composition of exopolysaccharides C1, 1509, S302, and S329. The carbohydrate compositions of the polysaccharides are expressed in µg/mL. BDL = below the detection limit; ND = not detected.

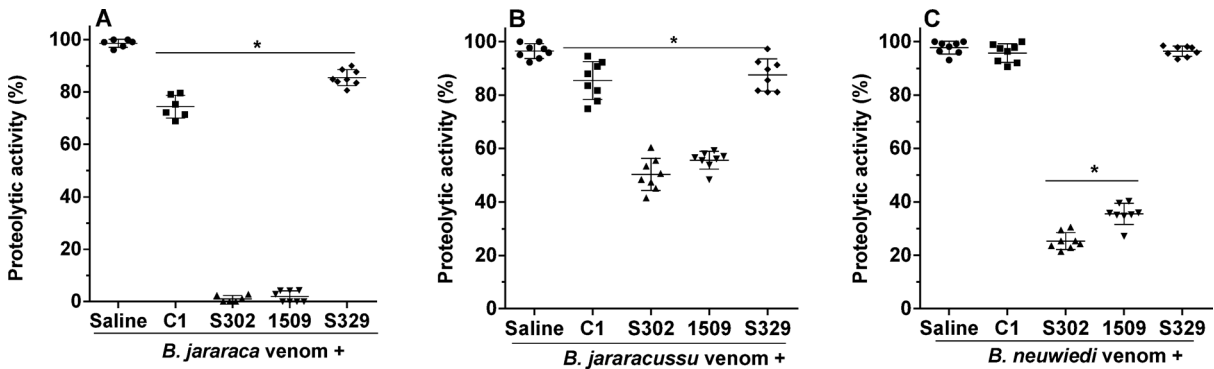


Fig. 2. Effect of exopolysaccharides on the proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *B. jararaca* (A), *B. jararacussu* (B), and *B. neuwiedi* (C) venoms were incubated with saline solution or C1, S302, 1509, or S329 for 5 min at 37 °C. Venom plus saline yielded 100% of proteolytic activity. The results are expressed as the means ± SD (n = 6). * p < 0.05 compared with venom plus saline.

1509 completely inhibited (100%) the *B. jararaca* venom-induced proteolytic activity, 50% of the *B. jararacussu* venom activity (Fig. 2B), and 72% of the *B. neuwiedi* venom activity (Fig. 2C). In the absence of venom, none of the four exopolysaccharides hydrolyzed azocasein and, thus, were devoid of proteolytic activity (data not shown).

Effect of exopolysaccharides on the coagulant activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom

B. jararaca, *B. jararacussu*, and *B. neuwiedi* venom clotted plasma in a concentration-dependent manner, and the concentration of venom able to clot plasma at approximately 60 s was considered the minimum coagulant concentration (MCC). The MCCs of *B. jararaca* (20 µg/mL), *B. jararacussu* (30 µg/mL), and *B. neuwiedi* (15 µg/mL) were incubated with the exopolysaccharides C1, S302, 1509, or S329 at the ratio of 1:10 venom/polysaccharide for 5 min at 37 °C. The plasma coagulation assay was performed as described in the methods. The exopolysaccharide C1 did not inhibit coagulation caused by *B. jararaca* (Fig. 3A), *B. jararacussu* (Fig. 3B), or *B. neuwiedi* (Fig. 3C) venom, and S329 was ineffective against *B. jararaca* (Fig. 3A) and *B. jararacussu* venom (Fig. 3B). S302 and 1509 inhibited plasma coagulation caused by all the venoms (Fig. 3) and more efficiently inhibited *B. jararacussu* venom-induced coagulation (Fig. 3B). The exopolysaccharides C1, S302, 1509, and S329 did not clot the plasma in the absence of venom (data not shown). Notably, for the venom of *B. jararaca*, the time of 82 s was chosen because higher concentrations of this venom would have been needed to clot the plasma at 60 s.

Effect of exopolysaccharides on the PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom

Venoms of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom incubated at a concentration of 50 µg/mL with saline produced readouts of approximately 0.8 at an absorbance of 850 nm, which was considered as 100% PLA₂ activity. Then, the same concentration of venom was incubated with 500 µg/mL of the exopolysaccharides C1,

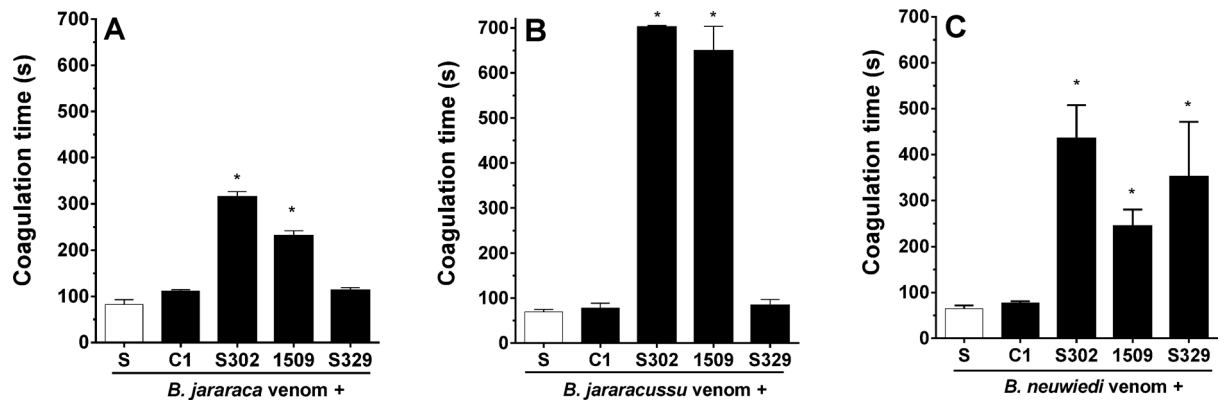


Fig. 3. Effect of exopolysaccharides on plasma coagulation caused by *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. 20 $\mu\text{g/mL}$ *B. jararaca* (A), 30 $\mu\text{g/mL}$ *B. jararacussu* (B), and 15 $\mu\text{g/mL}$ *B. neuwiedi* (C) venom samples were incubated with saline (S) or C1, S302, 1509, or S329 at a 1:10 ratio for 5 min at 37 °C, and then added to the medium. The coagulation of the plasma was then monitored. The results are expressed as the means $\pm$ SD ($n=6$). * $p < 0.05$ compared with venom plus saline.

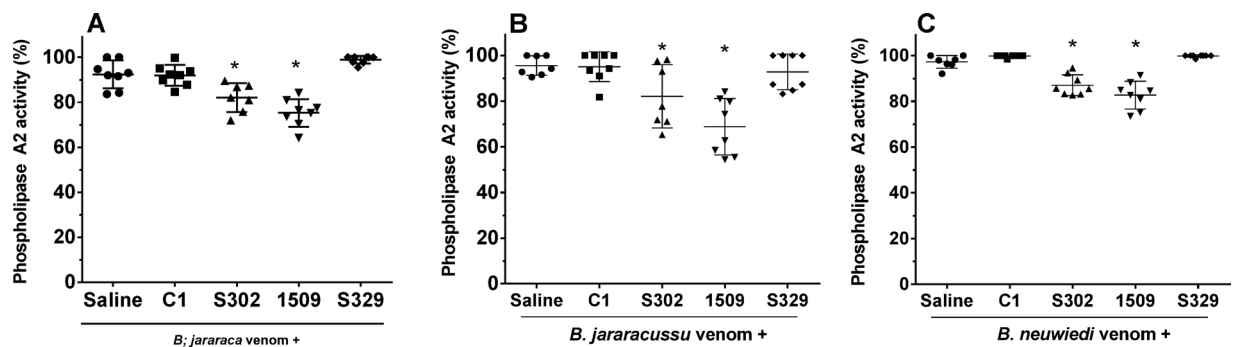


Fig. 4. Effect of exopolysaccharides on the PLA_2 activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *B. jararaca* (A), *B. jararacussu* (B), and *B. neuwiedi* (C) venom (50 $\mu\text{g/mL}$) was incubated with saline or with 500 $\mu\text{g/mL}$ C1, S302, 1509, or S329 for 5 min at 37 °C. PLA_2 activity was measured as described. The results are expressed as the means $\pm$ SD ($n=6$). * $p < 0.05$ compared with venom plus saline.

S302, 1509, or S329 (1:10 venom/polysaccharide, wt/wt) for 5 min at 37 °C. As shown in Fig. 4, C1 and S329 did not inhibit the PLA_2 activity of *B. jararaca*, *B. jararacussu*, or *B. neuwiedi* venom. S302 and 1509 inhibited PLA_2 activity by 17% and 30%, respectively, regardless of the venom tested (Fig. 4). In the absence of venom, C1, S302, 1509, and S329 did not exhibit PLA_2 activity (data not shown).

Discussion

C. sorokiniana, *S. obliquus*, *Nannochloris* sp. Naumann, and *S. acuminatus* exopolysaccharides inhibited the main toxic activities of some *Bothrops* sp. venom, including that from *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*. The exopolysaccharides neutralized the coagulant, proteolytic, and PLA_2 activities of venom from these Brazilian pit vipers. These activities are considered the most important toxic effects following SBE, resulting in severe hemorrhaging, inflammation, blood clotting disturbances, and tissue necrosis. We also demonstrated that the exopolysaccharides lysed only 5%–10% of human red blood cells; thus, they can be considered non-hemotoxic compounds. According to literature³⁴ hemolysis below 20% is acceptable in molecules being considered as candidates as medicine. Therefore, plant products may inhibit the toxic activity of envenomation, prevent death and physical sequelae caused by snake bites, and may complement current antivenom therapy with safety. On the other hand, other toxicity assessments must be done to postulate the non-toxicity of exopolysaccharides from *C. sorokiniana*, *S. obliquus*, *Nannochloris* sp. Naumann, and *S. acuminatus*, including: acute, sub-acute, and chronic toxic tests. However, microalgae have been used safely as food to humans and animals.

The exopolysaccharides demonstrated carbohydrate composition differences. The four microalgae contain neutral carbohydrates, including fucose, arabinose, and glucose, but not galactose, rhamnose, or raffinose. The carbohydrate composition may influence the inhibitory potential of polysaccharides against the toxic activities of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. The trehalose concentration was low in 1509 and S302 but high in C1. C1 failed to inhibit the proteolytic, coagulant, and PLA_2 activities of all the venoms; however, 1509 and S302 neutralized these activities. Thus, the presence of trehalose in the exopolysaccharides

of these microalgae may interfere with their binding with the dominant active enzymes (as SVMs, SVSPs, and PLA₂) that are responsible for proteolysis, coagulant activity, and PLA₂ activity or cofactors of such enzymes (such as Ca²⁺ and Zn²⁺). The complex structure and steric inhibition process (conformational changes) of exopolysaccharides may prevent their binding to the predominant active enzymes. Different glucose contents were observed among the exopolysaccharides of the microalgae. C1 and 1509 had the highest contents, whereas S302 and S329 had the lowest. However, despite having the highest concentration of glucose, C1 and 1509 had opposite inhibitory effects against proteolytic, coagulant, and PLA₂ activities. Notably, C1 has a twofold lower glucose content than 1509, S302 has twofold lower glucose than S329, and S302 has the lowest content of glucose of all exopolysaccharides from the microalgae. S302 was the most efficient antagonist of the toxic activities assessed in this study. Nonetheless, the contents of all carbohydrates from C1, 1509, S302, and S329 should be analyzed to determine the relationship between their composition and biological activity. Similar content has been reported in the literature where microalgae have been shown to synthesize monosaccharide units of hexoses and pentoses linked to form complex sugars²⁹.

Polysaccharides are found in all living organisms and a wide range of biological and pharmacological effects have been attributed to them²⁹ beyond their use in cosmetics and foods and as medicines for the treatment for several ailments^{28,34}. Seaweeds are the most prominent producers of polysaccharides, and annually approximately 26,500 tons of carrageenan, 50,000 of alginate, and 9,600 of agar are produced on the market, yielding a profit of billions of dollars^{35,36}. Alginate and agar demonstrate no or minimal^{29,37} are biocompatible at high concentrations in microalgae, are ecologically sustainable, and have stable structures³². Carrageenans are the main thickening, emulsifying, or stabilizing additives used in milkshakes, ice creams, beers, hams, and typically have molecular weights of 400–600 kDa³⁵. However, the Food and Drug Administration (FDA) has established a minimum value for the molecular weight of carrageenan (100 kDa) for human consumption due to reports of colon ulceration resulting from highly degraded carrageenans³⁵. The World Health Organization and Agriculture Organization of the United Nations Expert Committee on Food Additives revised the studies on carrageenans and established that a concentration of up to 1,000 mg/L is “not of concern” in infants. Thus, carrageenans are not toxic or associated with adverse health effects and are generally recognized as safe (GRAS) food additives. Some microalgae species have been included in the FDA list as GRAS for human consumption^{37,38} including *Chlorella vulgaris*, *Arthrospira platensis*, and *Euglena gracilis*²⁹. The composition of polysaccharides differs in structure, molecular weight, and sugar composition. Glucose is generally the most abundant component, and fructose is found in exopolysaccharides from cyanobacteria^{30,39}. Exopolysaccharides from Charophyta have mainly fucose and uronic acids, whereas exopolysaccharides from Rhodophyta are predominantly composed of xylose and galactose³⁰.

Biological and pharmacological effects, including antioxidant and antibacterial activities are attributed to the protein and lipid contents of microalgae⁴⁰. Nevertheless, polysaccharides from seaweed are of great interest because of their unique structures and pharmacological properties. In addition to biologically active proteins and lipids, sulfate groups are essential for many polysaccharides to display pharmacological activities in many cases.

The viscosity of C1, 1509, S302, and S320 does not appear to be a crucial parameter for blocking the proteolytic, coagulant, and PLA₂ activities of venoms; C1 and 1509 have the lowest viscosity, but the latter is a more effective as inhibitor than the former. Furthermore, S302 and S329 have the highest viscosity values, but significant differences on the inhibitory of the activities. Overall, S302 showed greatest percentage of venom-induced toxic activities, whereas S329 was not an efficient antagonist.

The molecular weights of the exopolysaccharides ranged from 13 to 18 × 10⁴ Da. C1 had the highest molecular weight (17.71 × 10⁴ Da) and was the least efficient inhibitor of the toxic effects of snake venom. Exopolysaccharides 1509, S302, and S329 had molecular weights of approximately 14 × 10⁴ Da. Therefore, one may speculate that the molecular weight is not a crucial feature for distinguishing the inhibitory efficacy of exopolysaccharides derived from microalgae. High-molecular-weight fucoidan polysaccharides from the brown macroalgae *Fucus vesiculosus* and *Undaria pinnatifida* have been reported to inhibit some toxic activities of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom⁴¹ and sulfate groups are required for their inhibition because of the high content of negatively charged groups within their chemical structure⁴¹. The major active enzymes found in most viper venoms, e.g. the *Bothrops* genus, are SVMs, SVSPs, and PLA₂s, which are key contributors to the toxicity of these venoms. SVMs are zinc-dependent endopeptidases containing Zn²⁺ at the catalytic center and Ca²⁺ in the C-terminal region, which are involved in structural stabilization of the protein, and basic residues, e.g., lysine at the molecule's surface. Thus, molecules with a net negative charge can bind to these venom enzymes, leading to inhibition^{41,42}. Varespladib^{42,43} CP471474⁴³, and suramin⁴⁴ are inhibitors of PLA₂s, SVSPs, and SVMs, respectively, and their net charge is vital for their interaction with specific amino acid residues or cofactors of such enzymes^{45,46}. However, further studies should be performed to elucidate the mechanism of action of exopolysaccharides using purified enzymes from venoms. Docking studies should also be performed. Inhibiting the different enzymes of snake venoms that have multiple toxic effects via one molecule or extract is challenging. Combining molecules or extracts may be a strategy to efficiently block venom toxicity.

Conclusions

Samples of exopolysaccharides isolated by precipitation from the culture fluid of the microalgae *C. sorokiniana*, *S. obliquus*, *Nannochloris* sp. Naumann, and *S. acuminatus* inhibited the PLA₂, proteolytic, and plasma coagulant activities of venom from *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*. Exopolysaccharide samples from *S. acuminatus* and *Nannochloris* sp. Naumann completely suppressed the assessed toxic actions of *B. jararaca* venom. The chemical components of these exopolysaccharides are mainly carbohydrates, such as fucose, arabinose, glucose, sucrose, maltose, and trehalose, which may be highly important for neutralizing venom toxicity. These findings may inspire scientists to explore the enormous potential of exopolysaccharides from

microalgae as a new source of valuable chemicals that inhibit snake venom proteases and PLA₂ enzymes, which are responsible for the most toxic effects of SBE.

Materials and methods

Reagents and snake venoms

Lyophilized venom of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* was provided by the serpentarium of the Ezequiel Dias Foundation (FUNED), Belo Horizonte, Minas Gerais, Brazil. Venom was diluted in physiological saline and stored at −20 °C before analysis. Snake venom was collected with authorization from the Brazilian National System for the Management of Genetic Heritage and Related Traditional Knowledge (SISGEN), process number A39CTRI 04E. The venom was dissolved in physiological saline at a concentration of 1 mg/mL. Azocasein was purchased from Sigma Chemical Co. (St. Louis, Missouri, USA). All other reagents and solvents were of research grade.

Extraction of the exopolysaccharides from microalgae

Exopolysaccharides were extracted from *C. sorokiniana* (coded as C1) with an isopropyl alcohol to dry biomass ratio of 1:3 at room temperature for 700 min; from *S. acuminatus* (coded as S329) with an ethyl alcohol to dry biomass ratio of 1:3 at −25 °C for 720 min; from *Nannochloris* sp. Naumann (coded as 1509) with an ethyl alcohol to dry biomass ratio of 1:2 at −25 °C for 720 min; and from *S. abundans* (coded as S302) with an isopropyl alcohol to dry biomass ratio of 1:3 at 25 °C for 540 min. The resulting mixtures were centrifuged (20 min, 5,000 rpm), after which the alcohol was removed by decantation, and the exopolysaccharides were dissolved in sodium phosphate buffer, pH 7.0.

The exopolysaccharides were purified via deproteinization using the Sevag method, depigmented using hexane, and subjected to dialysis to remove salts, from the culture medium and DNA, RNA, and other molecules trapped in the sample extracts during precipitation with isopropyl alcohol. After preparation, the whole exopolysaccharide extract was lyophilized, and solution samples were prepared by dissolving in physiological saline solution at 25 mg/mL.

Determination of the molecular weight and viscosity of the exopolysaccharides from microalgae

The polysaccharides' molecular weight (MW) was determined by gel permeation chromatography (GPC) using Sephadex G 150 gel as the filler⁴⁷. Dextrans of different molecular weights (T-10, T-40, T-70, and T-500) were used as molecular standards. GPC was performed via elution with phosphate-salt buffer (PBS, 0.2 M) at a flow rate of 6 mL/h. The total carbohydrate content of each sample was estimated via the phenol-sulfuric acid method⁴⁸. The molecular weight of each fraction was determined via regression of a calibration plot of the molecular masses of standard dextran samples. The average molecular weight of the exopolysaccharides was calculated via the following equation:

$$Mw = \frac{\sum M_i C_i}{\sum C_i},$$

where M_i is the molecular weight of each fraction, Da;
 C_i is the total carbohydrate concentration in each fraction, mg/mL.

For viscosity studies, the polysaccharides were evaporated in a Hei-VAP Precision HL/G3 rotary evaporator (Heidolph, Germany) and lyophilized in a Triad lyophilic drying unit (Labconco, USA). A Ubbelohde ASTM glass capillary viscometer (Fungilab, Spain) was used to evaluate viscosity. The lyophilized exopolysaccharide fractions were dissolved in 0.2 M phosphate buffer at 25.0 ± 0.5 °C, centrifuged at 3,500 rpm, and filtered through a membrane filter with a pore diameter of 0.45 µm⁴⁸. The relative viscosity (η_r) was calculated via the following equation:

$$\eta_r = \frac{\eta}{\eta_0},$$

where η_0 is the solvent viscosity, cP, and η is the solution viscosity at the same temperature, cP.

The specific viscosity (η_{sp}) was determined via the following formula:

$$\eta_{sp} = \frac{\eta - \eta_0}{\eta_0} = \eta_r - 1.$$

Determination of the monosaccharide composition of exopolysaccharides from Microalgae via HPLC

The exopolysaccharide were prepared for HPLC by acid hydrolysis per the modified method described by Dische and Shettles⁴⁷. Three milliliters of 20% trifluoroacetic acid (TFA) solution was added to a 10 mg sample of exopolysaccharide. The hydrolysis process was conducted for 1 h at 120 °C. TFA was distilled via a rotary evaporator and re-extracted by adding 500 µL of isopropyl alcohol to the reaction mixture. The purified hydrolysate was dissolved in 3 mL of distilled water, centrifuged for 5 min at 3,500 rpm, and filtered through a syringe filter with a pore diameter of 0.22 µm. The separation was performed at 30 °C in isocratic elution mode at a wavelength of 192 nm, with a flow rate of 1 mL/min. Acetonitrile and double-distilled water were used as

the mobile phase. The components were identified using a refractometer detector on the basis of their retention times and the spectra of the individual standard substances, with which the instrument was precalibrated.

Effects of exopolysaccharides on the PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom

The PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was measured as described by Marinetti et al.⁴⁹. One fresh hen egg yolk obtained from a local supermarket was filtered (yielding, on average, 15 mL) and dissolved in physiological saline to a final volume of 100 mL. The solution was centrifuged at 15,000 × g for 60 min at 10 °C to clarify the solution and remove any insoluble material. The supernatant was transferred to tubes and stored at 4 °C prior to use. One volume of the supernatant was mixed with nine parts of saline solution to produce the egg yolk substrate mixture. Five hundred micrograms of exopolysaccharides C1, S302, 1509, and S329 or saline were incubated with 50 µg of each venom mixture for 5 min at 37 °C in a final volume of 220 µL. Then, 50 µL of this mixture was added to medium containing 132 µL of saline, 20 µL of 0.4% sodium taurocholate, 5 µL of Tris-HCl (200 mM), pH 7.5, and 2 µL of CaCl₂ (0.5 M). The enzymatic reaction was triggered by adding 50 µL of the egg yolk substrate solution. After 30 min at 37 °C, the absorbance at 850 nm was read via a microplate reader (Versamax, Molecular Devices, California, USA). 100% of the PLA₂ activity of venoms was achieved by incubating venom with saline in the absence of exopolysaccharides, whereas 0% PLA₂ activity was obtained with saline alone. The negative control comprised exopolysaccharides incubated in saline solution without venom.

Effect of exopolysaccharides on the proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom

The proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was evaluated as described by Garcia et al.⁵⁰. Different concentrations of each venom mixture (10–50 µg/mL) were incubated for 90 min at 37 °C with 400 µL of azocasein (0.2% wt/vol. dissolved in 20 mM Tris-HCl, and 8 mM CaCl₂, pH 8.8). The enzymatic reaction was stopped by adding 400 µL of trichloroacetic acid (10%), and the tubes were centrifuged for 3 min at 12,000 rpm. Then, 1.0 mL of the supernatant was removed, transferred to tubes containing 500 µL NaOH 2 N, and incubated for 10 min at room temperature. The absorbance at 420 nm was read on a spectrophotometer. The concentration of venom (µg/mL) capable of producing a readout of 0.2 (i.e., 70–80% of the maximum activity) was defined as the arbitrary unit of effective concentration (EC). One EC of *B. jararaca* and *B. neuwiedi* venom (30 µg/mL) and *B. jararacussu* (50 µg/mL) venom was incubated with saline (positive control) or the exopolysaccharides C1, S302, 1509, or S329 for 5 min at 37 °C at a venom-to-exopolysaccharide ratio of 1:10 (wt/wt), resulting in a final volume of 220 µL. After incubation, an aliquot of 50 µL was removed and added to the mixture, and the proteolytic activity was measured. The negative controls comprised medium with saline or exopolysaccharides to the medium without venom.

Effect of exopolysaccharides on the coagulation caused by *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom

The coagulant activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was assessed using a pool of human plasma donated from the blood bank of the Hospital Antônio Pedro of Federal Fluminense University (HUAP) under the authorization of the Committee for Ethical in Experimentation (CEP-UFF, CAAE: 28941314.0.0000.5243). Two hundred microliters of diluted plasma (1:1 in saline solution) was incubated at 37 °C for 1 min, and then different concentrations of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom (10–100 µg/mL) were added to the medium. The coagulation time was monitored via a digital Amelung coagulometer, model KC4A (Labcon, Heppenheim, Germany). The concentration of venom (µg/mL) able to clot plasma at ~60 s was determined as the arbitrary unit minimum coagulant concentration (MCC). One MCC of each venom mixture was incubated for 5 min at 37 °C with saline (positive control) or with the exopolysaccharides C1, S302, 1509, and S329 at a ratio of 1:10 venom: polysaccharide (wt/wt), in a final volume of 220 µL. An aliquot of 50 µL was added to the plasma, and coagulation was monitored as described above. As negative controls, exopolysaccharides or saline were added to the plasma without venom. A similar procedure was described by Castro-Pinheiro et al.³⁹.

Toxicity of exopolysaccharides

The *in vitro* toxicity of exopolysaccharides was evaluated via a hemocompatibility test, as previously described³⁴. Exopolysaccharides (2 mg/mL, final concentration) and saline solution were incubated with a 13% (v/v) washed human red blood cell suspension for 3 h at 37 °C. The tubes were centrifuged for 3 min at 1,800 × g, and the volume of hemoglobin released from the washed red blood cells was measured at an absorbance of 578 nm. Hemolysis (100% and 0%) were obtained by adding water and saline, respectively, to the washed red cells.

Statistical analysis

The results are expressed as the mean ± standard deviation (SD) obtained with the number of experiments indicated in each result via the GraphPad Prism[®] program. The significance of differences between experimental groups was assessed via one-way ANOVA with a *post hoc* Dunnett's test; *p* values of <0.05 were considered to indicate significance.

Data availability

All the data generated or analyzed during this study are included in this published article.

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References

1. Fry, B. G. & Snakebite When the human touch becomes a bad touch. *Toxins* **10**, 1–24 (2018).
2. Harrison, R. A., Hargreaves, A., Wagstaff, S. C., Faragher, B. & Lalloo, D. G. Snake envenoming: A disease of poverty. *PLoS Negl. Trop. Dis.* **3**, e569–e574 (2009).
3. Kasturiratne, A. et al. The global burden of snakebite: A literature analysis and modelling based on regional estimates of envenoming and deaths. *PLoS Med.* **5**, e218 (2008).
4. Herzig, V. Animal Venoms—Curse or cure?? *Biomedicines* **9**, 413 (2021).
5. Oliveira, A. L. et al. The chemistry of snake venom and its medicinal potential. *Nat. Rev. Chem.* **6**, 451–469 (2022).
6. Robert, N. L. S. et al. Global mortality of snakebite envenoming between 1990 and 2019. *Nat. Comm.* **13**, 6160 (2022).
7. Gutiérrez, J. M. et al. *Nat. Rev. Dis. Primers* **14**, 17063 (2017).
8. Chippaux, J. P. Estimating the global burden of snakebite can help to improve management. *PLOS Med.* **5**, e221 (2008).
9. Williams, D. J. et al. Strategy for a globally coordinated response to a priority neglected tropical disease: Snake-bite envenoming. *PLoS Negl. Trop. Dis.* **13**, e0007059 (2019).
10. WHO. Snakebite envenoming. Available from: <https://www.who.int/news-room/fact-sheets/detail/snakebite-envenoming> (Accessed on August 2024).
11. Chippaux, J. P. Incidence and mortality due to snakebite in the Americas. *PLoS Negl. Trop. Dis.* **11**, e0005662 (2017).
12. Queiroz, G. P., Pessoa, L. A., Portaro, F. C. V., Furtado, M. F. D. & Tambourgi, D. V. Interspecific variation in venom composition and toxicity of Brazilian snakes from *Bothrops* genus. *Toxicon* **52**, 842–851 (2008).
13. Gutiérrez, J. M. et al. Unresolved issues in the Understanding of the pathogenesis of local tissue damage induced by snake venoms. *Toxicon* **148**, 123–131 (2018).
14. Weekers, D. J. C., Alonso, L. L., Verstegen, A. X., Slagboom, J. & Kool, J. Qualitative profiling of venom toxins in the venoms of several *Bothrops* species using High-Throughput venomomics and coagulation bioassaying. *Toxins (Basel)*. **16**, 300 (2024).
15. Cavecci-Mendonça, B. et al. Santos, L.D.D. Preliminary insights of Brazilian snake venom metalloproteomics. *Toxins (Basel)*. **15**, 648–654 (2023).
16. de Oliveira, F. N., Sachetto, A. T. A. & Santoro, M. L. Two-dimensional blue native/sds polyacrylamide gel electrophoresis for analysis of Brazilian *Bothrops* snake venoms. *Toxins (Basel)*. **14**, 661–680 (2022).
17. Casewell, N. R., Jackson, T. N. W., Laustsen, A. H. & Sunagar, K. Causes and consequences of snake venom variation. *Trends Pharmacol. Sci.* **41**, 570–581 (2020).
18. Alangode, A., Rajan, K. & Nair, B. G. Snake antivenom: challenges and alternate approaches. *Biochem. Pharmacol.* **181**, 114135 (2020).
19. Espin-Angulo, J. & Vela, D. Exploring the venom gland transcriptome of *Bothrops asper* and *Bothrops jararaca*: de Novo assembly and analysis of novel toxic proteins. *Toxins (Basel)*. **16**, 511 (2024).
20. Warrell, D. A. Snake bite. *Lancet* **375**, 77–88 (2010).
21. Rathore, A. S., Kumar, R. & Tiwari, O. S. Recent advancements in snake antivenom production. *Int. J. Biol. Macromol.* **240**, 124478 (2023).
22. Williams, D. et al. The global snake bite initiative: an antidote for snake bite. *Lancet* **375**, 89–91 (2010).
23. Singh, P., Yasir, M., Hazarika, R., Sugunan, S. & Shrivastava, R. A review on venom enzymes neutralizing ability of secondary metabolites from medicinal plants. *J. Pharmacopunct.* **20**, 173–178 (2017).
24. Gómez-Betancur, I., Gogineni, V., Salazar-Ospina, A. & León, F. Perspective on the therapeutics of anti-snake venom. *Molecules* **24**, 3276 (2019).
25. Mors, W. B., Nascimento, M. C., Pereira, B. M. & Pereira, N. A. Plant natural products active against snake bite - the molecular approach. *Phytochemistry* **5**, 627–642 (2000).
26. Félix-Silva, J., Silva-Junior, A. A., Zucolotto, S. M. & Fernandes-Pedrosa, M. F. Medicinal plants for the treatment of local tissue damage induced by snake venoms: an overview from traditional use to pharmacological evidence. *Evid. Based Complement. Alternat. Med.* 5748256 (2017). (2017).
27. Puzari, U., Fernandes, P. A. & Mukherjee, A. K. Pharmacological re-assessment of traditional medicinal plants-derived inhibitors as antidotes against snakebite envenoming: A critical review. *J. Ethnopharmacol.* **28**, 115208 (2022).
28. Hamidi, M. et al. Marine bacteria versus microalgae: who is the best for biotechnological production of bioactive compounds with antioxidant properties and other biological applications? *Mar. Drugs*. **29**, 28 (2019).
29. Babich, O. et al. Dolganyuk, V. Algae: study of edible and biologically active fractions, their properties and applications. *Plants (Basel)*. **11**, 780 (2022).
30. Moreira, J. B. et al. Microalgae polysaccharides: an alternative source for food production and sustainable agriculture. *Polysaccharides* **2**, 441–457 (2022).
31. Colusse, G. A., Carneiro, J., Duarte, M. E. R., Carvalho, J. C. & Nosedá, M. D. Advances in microalgal cell wall polysaccharides: a review focused on structure, production, and biological application. *Crit. Rev. Biotechnol.* **42**, 562–577 (2022).
32. Barsanti, L. & Gualtieri, P. Is exploitation of microalgae economically and energetically sustainable? *Algal Res.* **31**, 107–111 (2018).
33. Chanda, M. J., Merghoub, N. & Arroussi, E. Microalgae polysaccharides: the new sustainable bioactive products for the development of plant bio-stimulants? *World J. Microbiol. Biotechnol.* **35**, 177–190 (2019).
34. Bauer, M. et al. Poly(2-ethyl-2-oxazoline) as alternative for the stealth polymer poly(ethylene glycol): comparison of in vitro cytotoxicity and hemocompatibility. *Macromol. Biosci.* **12**, 986–998 (2012).
35. Leandro, A., Pereira, L. & Gonçalves, A. M. M. Diverse applications of marine macroalgae. *Mar. Drugs*. **24**, 17 (2019).
36. Chudasama, N. A., Sequeira, R. A., Moradiya, K. & Prasad, K. Seaweed polysaccharide based products and materials: an assessment on their production from a sustainability point of view. *Molecules* **26**, 260 (2021).
37. Benalaya, I., Alves, G., Lopes, J. & Silva, L. R. A review of natural polysaccharides: sources, characteristics, properties, food, and pharmaceutical applications. *Int. J. Mol. Sci.* **25**, 1322 (2024).
38. Ravindran, R. & Rajauria, G. Carbohydrates derived from microalgae in the food industry. *Cult. Microalgae Food Ind.* **2**, 127–146 (2021).
39. Costa, J. A. V., Lucas, B. F., Alvarenga, A. G. P., Moreira, J. B. & Morais, M. G. Microalgae polysaccharides: an overview of production, characterization, and potential applications. *Polysaccharides* **2**, 759–777 (2021).
40. Shevelyuhina, A. et al. Antioxidant and antimicrobial activity of microalgae of the Filinskaya Bay (Baltic Sea). *Plants (Basel)*. **31**, 2264 (2022).
41. Castro-Pinheiro, C. et al. Effect of Seaweed-derived fucoidans from *Undaria pinnatifida* and *Fucus vesiculosus* on coagulant, proteolytic, and phospholipase A2 activities of snake *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *Toxins (Basel)*. **16**, 188 (2024).
42. Salvador, G. H. M., Cardoso, F. F., Lomonte, B. & Fontes, M. R. M. Inhibitors and activators for myotoxic phospholipase A(2)-like toxins from snake venoms - A structural overview. *Biochimie* **30**, S0300–S084 (2024).
43. Lewin, M., Samuel, S., Merkel, J. & Bickler, P. Varespladib (LY315920) appears to be a potent, broad-spectrum, inhibitor of snake venom phospholipase A₂ and a possible pre-referral treatment for envenomation. *Toxins* **8**, 248 (2016).

44. Preciado, L. M., Pereañez, J. A. & Comer, J. Potential of matrix metalloproteinase inhibitors for the treatment of local tissue damage induced by a type P-I snake venom metalloproteinase. *Toxins* **12**, 8 (2019).
45. Ullah, A. et al. Thrombin-like enzymes from snake venom: structural characterization and mechanism of action. *Int. J. Biol. Macromol.* **15**, 788–811 (2018).
46. Quiroz, S. et al. Inhibitory effects of varespladib, CP471474, and their potential synergistic activity on *Bothrops asper* and *Crotalus durissus cumanensis* venoms. *Molecules* **6**, 8588 (2022).
47. Xu, Y. et al. Chemically modified polysaccharides: synthesis, characterization, structure activity relationships of action. *Int. J. Biol. Macromol.* **132**, 970–977 (2019).
48. Dische, Z. & Shettles, L. B. A specific color reaction of methylpentoses and a spectrophotometric micromethod for their determination. *J. Biol. Chem.* **175**, 595–603 (1948).
49. Marinetti, G. V. The action of phospholipase A₂ on lipoproteins. *Biochim. Biophys. Acta Lipid Metab.* **98**, 554–565 (1965).
50. Garcia, E. S., Guimarães, J. A. & Prado, J. L. Purification and characterization of a sulfhydryl-dependent protease from *Rhodnius prolixus* midgut. *Arch. Biochem. Biophys.* **2**, 315–322 (1978).

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Author contributions

C.C.P. and A.L.F. designed the study, discussed the data, performed the experiments and drafted the manuscript. S.S., O.B., and S.I. analyzed the data, drafted and revised the manuscript, and acquired the data. E.F.S. received funding, discussed the data, drafted and revised the manuscript. A.L.F. conceptualized the research, analyzed and interpreted the data, received funding, wrote and drafted the manuscript, and supervised the work. All the authors have read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

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4 DISCUSSÃO

4.1 Efeito dos fucoidans derivados das algas marinhas *Undaria pinnatifida* e *Fucus vesiculosus* nas atividades coagulante, proteolítica e fosfolipase A2 do veneno das serpentes *Bothrops jararaca*, *B. jararacussu* e *B. neuwiedi*

Este estudo fornece evidências que os dois extratos de fucoidanas derivados de *F. vesiculosus* e *U. pinnatifida*, nomeadamente FVF e UPF, respectivamente, inibiram as atividades proteolítica, coagulante sobre o plasma e fibrinogênio e a atividade PLA₂ do veneno das espécies *B. jararaca*, *B. jararacussu* e *B. neuwiedi*, mas com percentuais inibitórios diferentes. Essas atividades estão diretamente envolvidas no dano tecidual local e na hemotoxicidade sistêmica, bem como estas espécies são as que causam mais acidentes ofídicos no Brasil, cerca de 70%.

As atividades proteolítica, hemorrágica e coagulante são causadas pelas famílias de enzimas SVMPs e SVSPs (Cavecci-Mendonça *et al.*, 2023; Olaoba *et al.*, 2020; Serrano, 2013), enquanto os efeitos de edema, inflamação e dano muscular devem-se principalmente às enzimas PLAs₂ (Aloulou *et al.*, 2018; de Paula *et al.* 2009). No entanto, as enzimas SVMPs, SVSPs e PLAs₂ são consideradas o grupo majoritário e o biologicamente ativo presentes nos venenos viperídeos e, portanto, são responsáveis por produzir os sintomas mais graves no envenenamento (Paula *et al.* 2009; Serrano, 2013). Portanto, a inibição dos efeitos tóxicos desta família de enzimas por estes polissacarídeos de algas é de grande importância para um tratamento mais efetivo.

Na literatura, existem diferenças na toxicidade dos venenos entre espécies de *Bothrops*. O veneno de *B. jararaca* é altamente hemorrágico devido à presença de SVMPs, por exemplo, jararhagin (Laing e Moura-da-Silva, 2005) e bothropasin (Molina *et al.*, 2018). Por outro lado, o veneno de *B. jararacussu* é altamente miotóxico devido à atividade de duas enzimas com estrutura PLA₂: a bothropstoxina I (Bthtx-I) cataliticamente inativa e a bothropstoxina II, cataliticamente ativa (Borges *et al.*, 2021; Cardoso *et al.*, 2024); enquanto que o veneno de *B. neuwiedi* tem alta atividade coagulante, hemorrágica e miotóxica (Belo *et al.*, 2022).

Os antivenenos não neutralizam essas atividades tóxicas causadas pelos venenos botrópicos (Belo *et al.*, 2022) ou outras víboras (Thumtecho *et al.*, 2023) e podem até induzir reações hipersensíveis em pacientes (Solano *et al.*, 2024). Assim, moléculas capazes de neutralizar as atividades tóxicas dos venenos de serpentes são necessárias para melhorar a recuperação das vítimas de envenenamento ou para prevenir sequelas.

As famílias SVMPS e SVSPs incluem muitas enzimas com características quimicamente únicas e mecanismos de ação intrigantes (Olaoba *et al.*, 2020; Serrano, 2013). Portanto, bloquear essas enzimas do veneno, que têm ações múltiplas e complexas, usando uma única molécula é um desafio. As fucoidanas, polissacarídeos sulfatados derivados de algas marinhas, podem ser uma boa escolha devido à sua segurança e eficácia em humanos e propriedades físicas e químicas atrativas (Haggag *et al.*, 2023). Além disso, a facilidade de desenvolvimento de produtos farmacêuticos usando fucoidanas significa que elas são boas candidatas para desenvolver medicamentos (Haggag *et al.*, 2023; Trujillo *et al.*, 2024). Como os antivenenos convencionais têm limitações (Silva *et al.*, 2020; Solano *et al.*, 2024) desenvolveram um gel contendo um polissacarídeo galactano da alga vermelha *Chondrus crispus* que protegeu camundongos da hemorragia causada pelo veneno de *B. jararaca* e *B. jararacussu* quando o gel foi aplicado topicamente antes ou depois da injeção do veneno.

A literatura também descreve o uso de uma pomada ou creme contendo fucoidana das algas *F. vesiculosus* e *U. pinnatifida* para tratar doenças inflamatórias da pele, trombose, feridas e queimaduras (Pozharitskaya *et al.*, 2019). Esses autores também estudaram a farmacocinética da fucoidana após a aplicação tópica de tais formulações e mostraram que a fucoidana penetrou na pele de forma eficiente e teve uma meia-vida longa no plasma, músculo estriado e pele, e sem acúmulo no plasma mesmo após a aplicação repetida da pomada contendo a fucoidana por 5 dias consecutivos (Pozharitskaya *et al.*, 2019).

O envenenamento por serpentes ocasiona um distúrbio inflamatório, produzindo sérios efeitos tóxicos locais em humanos, e assim uma formulação transdérmica usando fucoidana é uma boa estratégia para auxiliar antivenenos

comerciais. Nosso grupo desenvolveu um gel usando carragenina de *C. crispus*, (Silva *et al.*, 2020) e o desenvolvimento desse tipo de formulação usando FVF ou UPF é um caminho promissor para tratar os efeitos tóxicos locais de envenenamento.

Fucoidanas de diferentes pesos moleculares foram investigadas para descobrir como são metabolizadas nos processos de absorção, distribuição, metabolismo e excreção de compostos derivados de esponja marinha, fúngicos e algas (Shikov *et al.*, 2020). Além disso, as fucoidanas têm alta estabilidade química e física, biocompatibilidade e biodegradabilidade e não apresentam efeitos colaterais em humanos (George e Shrivastav, 2023). Por outro lado, uma desvantagem das fucoidanas é a falta de capacidade de gelificação, mas essa limitação pode ser superada combinando fucoidanas com polímeros, como quitosana, o que pode lhes fornecer uma carga positiva.

O efeito do FVF e do UPF contra as atividades proteolítica, coagulante e PLAs₂ do veneno de *B. jararaca*, *B. jararacussu* e *B. neuwiedi* pode ser devido à presença e quantidade de grupos sulfato em sua estrutura. Eles possuem uma estrutura carregada negativamente, permitindo que se liguem às regiões ativas carregadas positivamente das enzimas SVMPS, SVSPs e PLAs₂ do veneno de *Bothrops*.

As atividades tóxicas das enzimas PLA₂ dos venenos de serpente dependem do cofator Ca²⁺, bem como de alguns resíduos de aminoácidos básicos específicos de seu sítio ativo, como a histidina na posição 48 e outros localizados na região C-terminal das enzimas PLAs₂ (Aloulou *et al.*, 2018; Paula *et al.*). Por outro lado, as SVMPS são os componentes dominantes desses venenos e contêm Zn²⁺ no sítio ativo. As SVSPs afetam a cascata de coagulação devido à ação de enzimas semelhantes à trombina que possuem atividade fibrinogenolítica, levando à formação de fibrina através da clivagem do fibrinogênio (Serrano, 2013; Ullah *et al.*, 2018). No entanto, outras abordagens para inibir as SVSPs têm sido investigadas, como anticorpos monoclonais murinos (mAbs) e peptídeos sintéticos baseados na sequência de seus substratos ou inibidores naturais (Belo *et al.*, 2022). No entanto, os mAbs não conseguiram neutralizar as toxinas dos venenos de *B. alternatus* e *B. neuwiedi* (Belo *et al.*, 2022).

Da Silva *et al.* (Silva *et al.*, 2021) projetaram dois peptídeos sintéticos que inibiram as SVSPs do veneno bruto de *B. jararaca* e a serinoprotease purificada batroxobina, mas não a tripsina humana, que é uma serinoprotease. Sem dúvida, tais peptídeos são uma ferramenta terapêutica promissora para melhorar o tratamento do envenenamento causado por espécies de *Bothrops*.

Os grupos sulfato ligados à estrutura do FVF e do UPF são cruciais para que as fucoidanas inibam a coagulação do plasma causada pelos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*. A estrutura carregada negativamente das fucoidanas pode permitir que elas se liguem aos metais Ca^{2+} e Zn^{2+} dentro do sítio ativo das enzimas. Alguns compostos comerciais capazes de quelar metais divalentes, como EDTA e O-fenantrolina, inibem os efeitos tóxicos das SVMPs. Outros reagentes, como o brometo de p-bromofenacila e o EDTA, reagem irreversivelmente com as enzimas PLAs₂, inibindo assim os efeitos tóxicos dependentes da atividade catalítica da PLAs₂, ou seja, a miotoxicidade, a hemólise e a agregação plaquetária induzida por ADP ou colágeno (Fuly *et al.*, 2000; Fuly *et al.*, 2002). No entanto, tais inibidores comerciais são considerados tóxicos para humanos ou, em alguns casos, para ambientes aquáticos.

Neste manuscrito, a atividade PLAs₂ dos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi* foi inibida pelo EDTA. Portanto, o FVF e o UPF podem compartilhar o mecanismo inibitório com tais reagentes comerciais, sem produzir toxicidade para humanos ou animais. O peso molecular do FVF e do UPF pode ser uma característica importante no bloqueio das enzimas do veneno de serpente. É razoável postular que cadeias maiores de polissacarídeos teriam uma quantidade maior de grupos sulfato e, portanto, a capacidade do FVF e do UPF de se ligar às enzimas do veneno seria maior. No geral, o mecanismo de inibição dos compostos FVF e UPF contra SVMPs, SVSPs e enzimas PLAs₂ parece ocorrer por meio da ligação a metais divalentes e, portanto, essa informação deve ser considerada no desenvolvimento de compostos com efeitos antiveneno.

Um inibidor sintético das enzimas PLAs₂, o “Varespladib” (LY315920) foi considerado em 2019 pela Administração de Alimentos e Medicamentos dos Estados Unidos (FDA) como um medicamento para envenenamento por serpentes (Lewin *et*

al., 2016). O “Varespladib” é capaz de se ligar ao canal hidrofóbico das enzimas PLA₂ enzimaticamente ou não enzimaticamente ativas, levando à inibição da miotoxicidade, hemorragia e edema causados por essas enzimas (Lewin *et al.*, 2016; Salvador *et al.*, 2023). Assim, uma combinação de alguns compostos com FVF e UPF pode ser uma boa estratégia para aumentar a eficácia da neutralização dos sintomas do envenenamento por serpentes.

Assim, a ação quelante das fucoidanas FVF e UPF e o tamanho de suas cadeias são relevantes para sua atividade antiveneno. No entanto, o mecanismo exato de inibição por FVF e UPF não foi abordado neste trabalho porque os experimentos foram realizados usando veneno bruto em vez de enzimas purificadas. Experimentos futuros explorarão o mecanismo de ação de FVF e UPF usando enzimas SVMs, SVSPs e PLAs₂ purificadas. Neste trabalho, 1500 µg/mL de FVF ou UPF lisaram cerca de 2% das hemácias, o que é um valor aceitável para uma molécula candidata (Bauer *et al.*, 2012). A empresa de biotecnologia australiana Marinova fornece fucoidanas para mercados como pesquisa alimentar, médica e farmacêutica e, portanto, um protocolo meticuloso de garantia de qualidade é realizado para garantir a pureza e segurança para consumo humano.

FVF e UPF são produzidos no laboratório analítico acreditado pela Organização Internacional de Normalização 9001 (ISO 9001) da Marinova e caracterizados quimicamente usando os métodos de análise de conteúdo de carboidratos e ligação descritos pela FDA. A FVF e a UPF foram aprovadas pela FDA como pertencentes à categoria de ingredientes alimentares “Geralmente Reconhecido como Seguro” (GRAS). Além disso, em alguns países europeus, como Áustria, Bélgica, França, Polônia, Espanha e Reino Unido, preparações contendo *F. vesiculosus* também são permitidas (Citkowska *et al.*, 2019). Uma fucoidana da alga *Cladosiphon okamuranus* foi administrada por via oral a humanos e não mostrou toxicidade em doses de 4 g por dia durante duas semanas. Além disso, outras vias de administração de fucoidana, como injeção subcutânea, também foram testadas e não tiveram nenhum impacto negativo na homeostase (Citkowska *et al.*, 2019).

4.2 Explorando exopolissacarídeos isolados de microalgas para bloquear os efeitos tóxicos in vitro do veneno de *Bothrops* sp.

Os exopolissacarídeos isolados das algas *C. sorokiniana* (C1), *S. obliquus* (S302), *Nannochloris* sp. Naumann (1509) e *S. acuminatus* (S329) inibiram as principais atividades tóxicas de alguns venenos de *Bothrops* sp., incluindo os de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*. Os exopolissacarídeos neutralizaram as atividades coagulante, proteolítica e de PLA₂ do veneno dessas serpentes brasileiras. Essas atividades são consideradas os efeitos tóxicos mais importantes após envenenamento, resultando em hemorragia grave, inflamação, distúrbios da coagulação sanguínea e necrose tecidual.

Também pudemos observar que os exopolissacarídeos isolados lisaram apenas 5%-10% das hemácias humanas; portanto, podem ser considerados compostos não hemotóxicos. De acordo com a literatura (Bauer *et al.*, 2012), hemólise abaixo de 20% é aceitável em moléculas consideradas como candidatas a medicamento. Portanto, produtos naturais podem inibir a atividade tóxica do envenenamento, prevenir a morte e sequelas físicas causadas por picadas de serpente e podem complementar com segurança a terapia antiveneno atual.

Por outro lado, outras avaliações de toxicidade devem ser feitas para postular a não toxicidade dos exopolissacarídeos isolados de *C. sorokiniana*, *S. obliquus*, *Nannochloris* sp. Naumann e *S. acuminatus*, incluindo testes de toxicidade aguda, subaguda e crônica. No entanto, as microalgas têm sido usadas com segurança como alimento para humanos e animais. Os exopolissacarídeos demonstraram diferenças na composição de carboidratos. As quatro microalgas contêm carboidratos neutros, incluindo fucose, arabinose e glicose, mas não galactose, ramnose ou rafinose. A composição de carboidratos pode influenciar o potencial inibitório dos polissacarídeos contra as atividades tóxicas do veneno de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*. A concentração de trealose foi baixa em 1509 e S302, mas alta em C1. C1 não conseguiu inibir as atividades proteolítica, coagulante e PLAs₂ de todos os venenos; no entanto, 1509 e S302 neutralizaram essas atividades.

Assim, a presença de trealose nos exopolissacarídeos isolados dessas microalgas pode interferir na sua ligação com as enzimas ativas dominantes (como SVMs, SVSPs e PLAs₂) que são responsáveis pela proteólise, atividade coagulante e atividade PLAs₂ ou cofatores dessas enzimas (como Ca²⁺ e Zn²⁺). A estrutura complexa e o processo de inibição estérica (mudanças conformacionais) dos exopolissacarídeos podem impedir sua ligação às enzimas ativas predominantes (Babich, 2022).

Diferentes teores de glicose foram observados entre os exopolissacarídeos isolados das microalgas. C1 e 1509 tiveram os maiores teores, enquanto S302 e S329 tiveram os menores. No entanto, apesar de terem a maior concentração de glicose, C1 e 1509 tiveram efeitos inibitórios opostos contra as atividades proteolítica, coagulante e PLAs₂. Notavelmente, C1 tem um teor de glicose duas vezes menor que 1509, S302 tem glicose duas vezes menor que S329, e S302 tem o menor teor de glicose de todos os exopolissacarídeos das microalgas. S302 foi o antagonista mais eficiente das atividades tóxicas avaliadas neste estudo (Babich, 2022).

No entanto, os teores de todos os carboidratos de C1, 1509, S302 e S329 devem ser analisados para determinar a relação entre sua composição e atividade biológica. Conteúdo semelhante foi relatado na literatura, onde as microalgas demonstraram sintetizar unidades monossacarídicas de hexoses e pentoses ligadas para formar açúcares complexos (Babich, 2022).

Os polissacarídeos são encontrados em todos os organismos vivos, e uma ampla gama de efeitos biológicos e farmacológicos tem sido atribuída a eles (Babich, 2022), além de seu uso em cosméticos, alimentos e como medicamentos para o tratamento de várias doenças (Bauer *et al.*, 2012; Hamidi, 2019). As algas marinhas são os produtores mais proeminentes de polissacarídeos, e anualmente aproximadamente 26.500 toneladas de carragenina, 50.000 de alginato e 9.600 de ágar são produzidas no mercado, gerando um lucro de bilhões de dólares (Chudasama *et al.*, 2021; Leandro *et al.*, 2019).

O alginato e o ágar apresentam nenhum ou mínimo efeito tóxico (Babich, 2022; Benalaya *et al.*, 2024), são biocompatíveis em altas concentrações, ecologicamente sustentáveis e possuem estruturas estáveis (Barsanti e Gualtieri, 2018). As

carrageninas são os principais aditivos espessantes, emulsionantes ou estabilizantes usados em milkshakes, sorvetes, cervejas e presuntos e normalmente têm pesos moleculares de 400–600 kDa. No entanto, a Administração de Alimentos e Medicamentos (FDA) estabeleceu um valor mínimo para o peso molecular da carragenina (100 kDa) para consumo humano devido a relatos de ulceração do cólon resultante de carrageninas altamente degradadas (Leandro *et al.*, 2019).

A Organização Mundial da Saúde e o Comitê de Especialistas em Aditivos Alimentares da Organização das Nações Unidas para Agricultura revisaram os estudos sobre carrageninas e estabeleceram que uma concentração de até 1.000 mg/L é “não preocupante” em bebês. Assim, as carrageninas não são tóxicas ou associadas a efeitos adversos à saúde e são geralmente reconhecidas como seguras (GRAS) como aditivos alimentares. Algumas espécies de microalgas foram incluídas na lista da FDA como GRAS para consumo humano (Benalaya *et al.*, 2024; Ravindran e Rajauria, 2021), incluindo *Chlorella vulgaris*, *Arthrospira platensis* e *Euglena gracilis* (Babich, 2022).

A composição dos polissacarídeos difere em estrutura, peso molecular e composição de açúcares. A glicose é geralmente o componente mais abundante, e a frutose é encontrada em exopolissacarídeos de cianobactérias (J.A.V. Costa *et al.*, 2021; Moreira *et al.*, 2021). Os exopolissacarídeos de Charophyta têm principalmente fucose e ácidos urônicos, enquanto os exopolissacarídeos de Rhodophyta são predominantemente compostos de xilose e galactose (Moreira *et al.*, 2021).

Efeitos biológicos e farmacológicos, incluindo atividades antioxidante e antibacteriana, são atribuídos aos conteúdos de proteína e lipídio das microalgas (Sheveluyhina, 2022). No entanto, os polissacarídeos das algas marinhas são de grande interesse devido às suas estruturas únicas e propriedades farmacológicas. Além das proteínas e lipídios biologicamente ativos, os grupos sulfato são essenciais para que muitos polissacarídeos exibam atividades farmacológicas em muitos casos. A viscosidade de C1, 1509, S302 e S320 não parece ser um parâmetro essencial para bloquear as atividades proteolítica, coagulante e PLAs₂ dos venenos; C1 e 1509 têm a menor viscosidade, mas o último é um inibidor mais eficaz do que o primeiro. Além disso, S302 e S329 têm os maiores valores de viscosidade, mas há diferenças

significativas na inibição das atividades. No geral, S302 mostrou a maior porcentagem de inibição das atividades tóxicas induzidas pelo veneno, enquanto S329 não foi um antagonista eficiente (Castro-Pinheiro *et al.*, 2024).

Os pesos moleculares dos exopolissacarídeos isolados variaram de 13 a 18×10^4 Da. C1 tinha o maior peso molecular ($17,71 \times 10^4$ Da) e foi o inibidor menos eficiente dos efeitos tóxicos do veneno de serpente. Os exopolissacarídeos 1509, S302 e S329 tinham pesos moleculares de aproximadamente 14×10^4 Da. Portanto, pode-se especular que o peso molecular não é uma característica chave para distinguir a eficácia inibitória dos exopolissacarídeos derivados de microalgas. Polissacarídeos de fucoídano de alto peso molecular das macroalgas marrons *Fucus vesiculosus* e *Undaria pinnatifida* foram relatados como inibidores de algumas atividades tóxicas do veneno de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*, e os grupos sulfato são necessários para sua inibição devido ao alto teor de grupos carregados negativamente em sua estrutura química (Castro-Pinheiro *et al.*, 2024).

As principais enzimas ativas encontradas na maioria dos venenos de serpentes, por exemplo, do gênero *Bothrops*, são SVMPs, SVSPs e PLAs₂, que são contribuintes-chave para a toxicidade desses venenos. As SVMPs são endopeptidases dependentes de zinco contendo Zn²⁺ no centro catalítico e Ca²⁺ na região C-terminal, que estão envolvidos na estabilização estrutural da proteína, e resíduos básicos, por exemplo, lisina na superfície da molécula. Assim, moléculas com uma carga líquida negativa podem se ligar a essas enzimas do veneno, levando à inibição (Castro-Pinheiro *et al.*, 2024; Salvador *et al.*, 2024).

Varespladib CP471474 (Lewin *et al.*, 2016; Salvador *et al.*, 2024) e suramina (Preciado *et al.*, 2018) são inibidores de PLAs₂, SVSPs e SVMPs, respectivamente, e sua carga líquida é vital para sua interação com resíduos de aminoácidos específicos ou cofatores de tais enzimas (Quiroz, 2022; Ullah *et al.*, 2018). No entanto, mais estudos devem ser realizados para elucidar o mecanismo de ação dos exopolissacarídeos isolados usando enzimas purificadas de venenos. Estudos de *docking* também devem ser realizados. Inibir as diferentes enzimas dos venenos de serpente, que têm múltiplos efeitos tóxicos, por meio de uma molécula ou extrato é

desafiador. Combinar moléculas ou extratos pode ser uma estratégia para bloquear eficientemente a toxicidade do veneno.

5 CONCLUSÃO

Os resultados observados neste trabalho mostraram que polissacarídeos derivados de algas marinhas e microalgas apresentam relevante potencial antiveneno *in vitro* frente às principais atividades tóxicas dos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*. As fucoidanas provenientes de *F. vesiculosus* e *U. pinnatifida* inibiram as atividades proteolítica, coagulante e fosfolipásica, destacando-se como candidatas promissoras para o desenvolvimento de estratégias terapêuticas complementares aos antivenenos convencionais.

De maneira similar, os exopolissacarídeos isolados de microalgas demonstraram capacidade de neutralizar essas atividades tóxicas, com variações de eficácia associadas à composição de carboidratos, à presença de grupos sulfato e às características estruturais dessas macromoléculas, mais do que ao peso molecular ou à viscosidade. A baixa hemotoxicidade observada reforça o potencial desses compostos como candidatos seguros para aplicação farmacológica.

Os dados que observamos neste trabalho indicam que a carga negativa conferida por grupos sulfato e a interação com cofatores metálicos essenciais às enzimas do veneno desempenham papel central no mecanismo de inibição. Apesar dos resultados promissores, estudos posteriores com enzimas purificadas, abordagens *in vivo* e análises mecanísticas detalhadas são necessários para elucidar plenamente os mecanismos de ação e viabilizar a utilização dos polissacarídeos estudados como agentes antiveneno ou adjuvantes terapêuticos no tratamento do envenenamento botrópico.

CAPÍTULO II- EFEITO DE PRODUTOS NATURAIS CONTRA AÇÕES TÓXICO-FARMACOLÓGICAS DE VENENOS DE SERPENTES DO GÊNERO *Bothrops*

1 Introdução

1.1 Plantas com potencial biotecnológico: contextualização e influência do clima na composição metabólica

As plantas constituem uma das mais antigas e mais ricas fontes de metabólitos bioativos utilizados pela humanidade. Desde civilizações pré-históricas até a medicina moderna, compostos derivados de folhas, raízes, cascas, resinas e sementes foram empregados para tratar infecções, reduzir inflamação, modular dor, controlar hemorragias e neutralizar toxinas biológicas. Essa capacidade terapêutica deriva da imensa diversidade metabólica vegetal, resultado de milhões de anos de evolução sob pressões ambientais, ecológicas e bióticas. A síntese de metabólitos secundários, como alcaloides, flavonoides, terpenos, saponinas e cumarinas, desempenha papel essencial na proteção das plantas contra herbívoros, patógenos, radiação ultravioleta e estresses abióticos. Exatamente por isso, a flora terrestre é hoje reconhecida como um dos mais importantes reservatórios naturais de moléculas com potencial farmacológico (Riaz *et al.*, 2023; Yeshi *et al.*, 2022).

A diversidade metabólica das plantas está profundamente ligada às características climáticas de seu habitat. Plantas de clima temperado, por exemplo, enfrentam ciclos bem definidos de luz e temperatura, com invernos marcantes e verões moderados. Esses fatores favorecem a síntese de compostos associados à resistência ao frio, à tolerância ao estresse hídrico e à proteção contra radiação UV, como flavonoides, catequinas, taninos condensados e lignanas. Muitas espécies temperadas apresentam metabolismo rico em polifenóis e glicosídeos, que desempenham funções antioxidantes e anti-inflamatórias intensas, características fundamentais em aplicações farmacológicas modernas (Elshafie *et al.*, 2023; Salam *et al.*, 2023).

Plantas de clima tropical, por sua vez, convivem com altos índices de radiação solar, elevada umidade e temperaturas relativamente estáveis. Esses fatores selecionam metabólitos voltados à defesa contra herbivoria, fungos e bactérias, além do controle de estresse oxidativo causado pela exposição solar contínua. Assim, plantas tropicais são especialmente ricas em terpenos, alcaloides complexos, óleos essenciais, cumarinas, saponinas e flavonoides altamente oxigenados. A extraordinária riqueza fitoquímica das florestas tropicais, especialmente a Amazônia, é amplamente reconhecida como uma das principais fontes mundiais para a descoberta de novos fármacos, incluindo moléculas investigadas como agentes antiveneno (Almada-Vilhena *et al.*, 2024; Avila-Sosa *et al.*, 2019; Xiao *et al.*, 2025).

Já plantas de clima subtropical, que ocupam regiões de transição entre ambientes tropicais e temperados, apresentam perfis metabólicos intermediários, combinando características químicas típicas dos dois extremos. Essas espécies frequentemente acumulam polifenóis e terpenos em proporções equilibradas, além de apresentarem biocompostos altamente adaptativos, capazes de responder a variações térmicas mais amplas e a regimes de umidade variáveis. Isso as torna particularmente interessantes para estudos de bioprospecção, pois o balanço entre estabilidade metabólica e diversidade química tende a gerar moléculas com propriedades terapêuticas relevantes (Berini *et al.*, 2018; Saini *et al.*, 2019; Sampaio *et al.*, 2016; Weinhold *et al.*, 2022).

A relação entre clima e composição metabólica é especialmente importante no contexto toxinológico. As classes químicas mais encontradas em plantas tropicais e subtropicais, como terpenoides, alcaloides e flavonoides, já demonstraram capacidade de modular ou inibir atividades enzimáticas associadas a toxinas de venenos de serpentes. SVMPs, PLAs₂ e SVSPs são alvos frequentes de compostos fenólicos e terpenos naturais, que podem agir por diferentes mecanismos: ligação direta ao sítio catalítico, quelação do íon metálico ou estabilização de membranas celulares para prevenir lise induzida por PLAs₂ (Qaderi *et al.*, 2023; Romanelli *et al.*, 2025).

Portanto, compreender a influência do clima sobre o metabolismo especial vegetal é fundamental para orientar a escolha das espécies estudadas nesta tese.

Assim como as algas marinhas foram abordadas no capítulo anterior pelo seu potencial antiveneno, agora se estabelece o universo das plantas como uma fonte ainda mais vasta e diversa de moléculas capazes de interferir em vias fisiopatológicas associadas ao envenenamento por serpentes do gênero *Bothrops*. Esse conjunto de fatores justifica a exploração profunda das famílias botânicas e espécies selecionadas para esta investigação.

1.2 Medicina tradicional, famílias vegetais com potencial antiveneno e metabólitos bioativos de interesse toxinológico

O uso de plantas na medicina tradicional é uma prática universal, registrada em praticamente todas as culturas humanas e profundamente enraizada em sistemas médicos antigos, como a Ayurveda, a Medicina Tradicional Chinesa e o fitoterapismo indígena nas Américas. A OMS estima que mais de 80% da população mundial utiliza plantas como primeira linha de cuidado em saúde, especialmente em regiões tropicais e subtropicais, onde o acesso aos serviços médicos é limitado. O conhecimento tradicional acumulado ao longo de séculos permitiu a seleção empírica de espécies capazes de combater inflamações, modular dor, promover cicatrização e neutralizar toxinas. Em grande parte das regiões com maior incidência de acidentes ofídicos, como África, sudeste asiático e América Latina, a dependência de plantas medicinais é ainda mais acentuada, tanto por fatores culturais quanto pela distância física dos centros de atenção hospitalar (Akiro, 2025; Rahiman *et al.*, 2025; Ralte *et al.*, 2024).

Diversas famílias botânicas são amplamente empregadas na medicina tradicional pela presença de compostos de alta reatividade biológica. Entre as mais utilizadas destacam-se *Fabaceae*, *Asteraceae*, *Rutaceae*, *Lamiaceae*, *Apocynaceae*, *Meliaceae* e *Piperaceae*, todas reconhecidas pela abundância de metabólitos como alcaloides, flavonoides, terpenoides, taninos e cumarinas. No contexto toxinológico, algumas dessas famílias despertam especial interesse devido ao uso histórico empírico contra picadas de serpentes, como *Apocynaceae*, *Euphorbiaceae*, *Melastomataceae*, *Fabaceae*, *Annonaceae*, *Aristolochiaceae* e *Siparunaceae*. Muitas dessas espécies são empregadas sob a forma de raízes maceradas, extratos em água quente (“chás”), aplicações tópicas ou preparados resinosos utilizados por populações

ribeirinhas e comunidades tradicionais da Amazônia, Cerrado e Mata Atlântica (Kumar *et al.*, 2025; Mogha *et al.*, 2022).

O potencial antiveneno de plantas está diretamente relacionado às classes químicas que elas produzem. Os taninos condensados e hidrolisáveis, por exemplo, têm longa história de uso como agentes “adstringentes”, mas do ponto de vista bioquímico, sua capacidade de precipitar proteínas e formar complexos estáveis pode neutralizar enzimas proteolíticas presentes em venenos, como SVMs e SVSPs. Da mesma forma, flavonoides como quercetina, canferol e derivados metoxilados possuem propriedades de inibição de PLAs₂ por mecanismos que incluem quelação de íons essenciais e bloqueio do sítio catalítico da enzima. Cumarinas e derivados, frequentemente encontrados em espécies como *Melilotus officinalis*, apresentam atividade antitrombótica, antioxidante e anti-inflamatória, modulando vias que também estão envolvidas na hemostasia alterada por venenos de *Bothrops* (Kumar *et al.*, 2025; Lättig *et al.*, 2007; Sachetto *et al.*, 2022)

Além disso, muitos terpenoides presentes em plantas aromáticas da família Lamiaceae já demonstraram, em modelos experimentais, capacidade de reduzir edema, modular respostas inflamatórias e proteger contra lesão muscular, sintomas comuns em envenenamentos botrópicos. Alcaloides presentes em famílias como *Apocynaceae* e *Annonaceae*, por sua vez, mostram alta afinidade por proteínas e receptores celulares, podendo interferir em processos de hemólise, coagulação e necrose tecidual induzidos pelos venenos. Embora esses mecanismos ainda não sejam completamente elucidados, evidências crescentes mostram que o potencial antiofídico das plantas se baseia em múltiplos níveis de ação, envolvendo desde a neutralização direta de toxinas até a modulação de vias inflamatórias sistêmicas (Adrião *et al.*, 2022; Félix-Silva *et al.*, 2017).

1.3 *Aesculus hippocastanum*

Aesculus hippocastanum, conhecida como castanheira-da-Índia, é uma espécie arbórea pertencente à família *Sapindaceae*, amplamente distribuída na Europa Central e nos Bálcãs, embora hoje seja cultivada globalmente devido ao seu interesse ornamental e farmacêutico. Originária das regiões montanhosas da Grécia, Albânia e Macedônia, é uma árvore de grande porte, facilmente reconhecida pelo tronco robusto, copa arredondada e folhas compostas palmadas. Desde o século XVII, extratos de suas sementes, cascas e folhas passaram a ser utilizados na medicina tradicional europeia para tratar distúrbios circulatórios, inflamação, varizes, edema e dores reumáticas (Dudek-Makuch e Studzińska-Sroka, 2015; Jasim, 2021).



Figura 12. *Aesculus hippocastanum* popularmente conhecido como castanha da Índia (<https://www.treesandshrubsonline.org/articles/aesculus/aesculus-hippocastanum/> acessado em 05/02/26)

O perfil fitoquímico de *A. hippocastanum* é amplamente estudado e revela uma complexa combinação de metabólitos biologicamente ativos, entre eles triterpenos, saponinas, flavonoides, taninos, cumarinas, ácido gálico e derivados terpênicos menores. A escina é o principal composto farmacológico da espécie e apresenta propriedades anti-inflamatórias, anti-exsudativas, venotônicas, antioxidantes e vasoprotetoras comprovadas. Esses efeitos são reforçados pela presença de

flavonoides, que atuam como antioxidantes potentes e moduladores da permeabilidade vascular (Idris *et al.*, 2020; Yuman Wang *et al.*, 2023).

A relevância toxinológica da castanheira-da-Índia emerge principalmente de sua ação moduladora sobre processos inflamatórios, permeabilidade capilar e danos teciduais, exatamente os mecanismos fisiopatológicos exacerbados durante o envenenamento por serpentes dos gêneros *Bothrops*. Estudos observam que a escina reduz extravasamento plasmático, estabiliza membranas endoteliais e inibe a liberação de mediadores inflamatórios como prostaglandinas, histamina e citocinas pró-inflamatórias de vias NF- κ B dependentes. Esses efeitos são altamente relevantes no contexto dos acidentes botrópicos, caracterizados pela presença de SVMP e PLA₂, que aumentam drasticamente a permeabilidade microvascular e induzem edema, necrose e hemorragia local (Domanski *et al.*, 2016; Mamede *et al.*, 2020; Yan *et al.*, 2025).

Outro mecanismo de interesse é a atividade anti-hialuronidase reportada para extratos da castanheira-da-Índia. A hialuronidase é uma enzima presente nos venenos viperídeos que aumenta a difusão das toxinas no tecido; portanto, compostos capazes de inibir essa atividade podem reduzir a propagação das toxinas pelo interstício. Assim, mesmo que *A. hippocastanum* não possua histórico de uso etnomédico relacionado a acidentes ofídicos, seus mecanismos farmacológicos justificam plenamente sua investigação em modelos toxinológicos (Bala *et al.*, 2018; Melo *et al.*, 2025).

1.4 *Melilotus officinalis*

Melilotus officinalis, conhecida como meliloto-amarelo, é uma planta herbácea bienal pertencente à família Fabaceae, amplamente distribuída na Europa, Ásia Ocidental e Oriente Médio, e naturalizada em praticamente todos os continentes devido ao uso medicinal e agrícola. Tradicionalmente, é empregada na fitoterapia europeia para tratar inflamações, congestão linfática, edemas, flebites e distúrbios associados à vasodilatação inadequada. Suas flores e folhas são ricas em compostos aromáticos e apresentam fragrância suave típica de cumarina, que historicamente favoreceu seu uso como erva medicinal e aromatizante (Liu *et al.*, 2018).



Figura 13. *Melilotus officinalis* popularmente conhecido como meliloto amarelo ou trevo de cheiro (https://jb.utad.pt/especie/Melilotus_officinalis acessado em 05/02/26)

A composição química da espécie é marcada pela presença abundante de cumarinas, flavonoides, taninos, triterpenos, derivados fenólicos e saponinas. A cumarina é seu metabólito mais característico e está associada a propriedades anti-inflamatórias, vasoprotetoras, antiedematogênicas e antioxidantes amplamente descritas na farmacologia vegetal. Embora o dicumarol seja conhecido como antagonista da vitamina K, esse composto só é produzido em quantidades significativas após processos fermentativos; portanto, extratos padronizados usados em estudos farmacológicos apresentam baixo risco de anticoagulação clinicamente relevante (Borhani *et al.*; Garg *et al.*, 2020; Liu *et al.*, 2018).

Do ponto de vista toxinológico, o *Melilotus officinalis* é particularmente interessante devido ao conjunto de propriedades que interferem diretamente em vias

fisiopatológicas ativadas pelos venenos de serpentes. A cumarina possui efeito anti-inflamatório robusto, modulando a liberação de citocinas como TNF- α , IL-1 β e IL-6, que são intensamente produzidas após acidentes botrópicos e laquéticos. Além disso, há evidências de que derivados cumarínicos apresentam atividade antioxidante e estabilizadora de membranas, limitando o dano induzido por PLAs₂ miotóxicas e por radicais livres formados durante necrose muscular (Pleşca-Manea *et al.*, 2002).

Flavonoides presentes em *Melilotus officinalis* exibem relevante capacidade quelante de metais, incluindo Fe³⁺ e Zn²⁺, cofatores essenciais para a atividade hemorrágica das metaloproteinases. Desta maneira, extratos ricos em flavonoides podem, teoricamente, reduzir a atividade proteolítica dessas enzimas, modulando hemorragia local e degradação de matriz extracelular. A propriedade venotônica da planta, somada ao efeito redutor de edema, torna-a ainda mais relevante para tratar manifestações locais, que frequentemente são pouco responsivas à soroterapia convencional (Preciado *et al.*, 2018).

Apesar desse conjunto promissor de atividades, não há estudos específicos avaliando *Melilotus officinalis* como agente antiveneno de serpentes. Considerando a sinergia metabólica entre cumarinas, flavonoides e taninos, o *Melilotus officinalis* representa uma candidata extremamente promissora para investigação em modelos de envenenamento botrópico.

2. OBJETIVOS

2.1 Objetivo geral

Avaliar o efeito neutralizante de extratos de plantas contra atividades tóxicas *in vitro* (fosfolipásica, coagulante e proteolítica) causadas pelo veneno das serpentes *B. jararaca*, *B. jararacussu* ou *B. neuwiedi*.

2.2 Objetivos específicos:

- A. Avaliar o efeito dos extratos de *Aesculus hippocastanum* e *Melilotus officinalis* contra as atividades coagulante, proteolítica e fosfolipásica dos venenos de *B. jararaca*, *B. jararacussu* ou *B. neuwiedi*.

3. ARTIGO REFERENTE AO CAPÍTULO II

Os resultados referentes ao capítulo II estão apresentados na forma de artigo publicado em revista científica indexada.

O artigo publicado no periódico *Rodriguésia* (Bioactive extracts from *Aesculus hippocastanum* and *Melilotus officinalis* as inhibitors of the toxic effects of *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* snake venom. Gonçalves R da C, Castro-Pinheiro C, Sanchez EF, et al.. *Rodriguésia*. 2025;76:e00392025. doi:10.1590/2175-7860202576056) descreve o efeito dos extratos brutos das plantas *Aesculus hippocastanum* (AH) e *Melilotus officinalis* (MO) na inibição das atividades tóxicas dos venenos de *Bothrops*.



Bioactive extracts from *Aesculus hippocastanum* and *Melilotus officinalis* as inhibitors of the toxic effects of *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* snake venom

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Key words:

- antivenom
- *Bothrops*
- plants
- snakebite envenomation

Abstract

Snakebite envenomation (SBE) is a neglected disease causing annually 138,000 deaths worldwide. In Brazil, the genus *Bothrops* is responsible for 87% of these SBE. In this work, we assessed the extracts from the plants *Aesculus hippocastanum* (AH) and *Melilotus officinalis* (MO) to inhibit the plasma coagulant, proteolytic and phospholipase A₂ (PLA₂) activities of *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms, as well as chemical characterization. The extracts from AH and MO were incubated with each venom for 30 min at 37 °C and then activities were assessed. AH inhibited 38% and 55% of the proteolytic activity of *B. jararaca* and *B. neuwiedi* venoms, respectively and MO inhibited 70% of this activity; but they failed to inhibit *B. jararacussu* venom. AH and MO inhibited more efficiently the *B. jararaca*- and *B. neuwiedi* venom-induced coagulation. AH and MO inhibited 60% of PLA₂ of *B. jararaca*, 30% of *B. jararacussu* and 40% of *B. neuwiedi* activities. Overall, AH and MO inhibited with more efficacy the activities of *B. jararaca* and *B. neuwiedi* venoms. The extracts contain flavonoids, saponins, rosmarinic acid, quercetin, and phenolic acids. Thus, the work provided potential of these plants to complement commercial antivenoms against envenomation by *Bothrops* species.

Palavras-chave:

- antiveneno
- *Bothrops*
- plantas
- envenenamento ofídico

Resumo

O envenenamento ofídico é uma doença negligenciada com 138.000 mortes no mundo. No Brasil, o gênero *Bothrops* causa 87% dos incidentes, levando à necrose tecidual, hemorragia, inflamação e óbito. Neste trabalho, avaliamos o efeito dos extratos das plantas *Aesculus hippocastanum* (AH) e *Melilotus officinalis* (MO) em inibir atividade coagulante, proteolítica e fosfolipásica A₂ do veneno de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*. AH e MO foram incubadas com cada veneno por 30 min a 37 °C e as atividades realizadas. AH e MO inibiram 38% e 55% a atividade proteolítica do veneno de *B. jararaca* e *B. neuwiedi*, respectivamente e MO inibiu 70% para o veneno de *B. jararaca* e *B. neuwiedi*; mas AH e MO não inibiram o veneno de *B. jararacussu*. AH e MO inibiram de forma mais eficiente a

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coagulação causada pelo veneno de *B. jararaca* e *B. neuwiedi*. AH e MO inibiram 60% a atividade fosfolipásica A_2 de *B. jararaca* e 30% e 40% para o veneno de *B. jararacussu* e *B. neuwiedi*, respectivamente. Os extratos contêm flavonóides, saponinas, ácido rosmarínico, quercetin e ácidos fenólicos. Portanto, estas plantas podem ser úteis para tratar o envenenamento por estas espécies.

Introduction

Snakebite envenomation (SBE) since 2017 is considered a neglected tropical disease (NTD) by the World Health Organization (WHO) (World Health Organization 2019). Globally, SBE incidents affect 2.7 million people, causing 138,000 deaths and 400,000 amputations each year, with a death rate of approximately 0.5%–1% (World Health Organization 2019; Chippaux 2008). However, it is likely that these numbers may be higher due to underreporting (Chippaux 2008). Tropical and subtropical impoverished regions have by far the highest number of cases, particularly Africa, Asia, India, Oceania, and Latin America (Gutiérrez 2017; Chippaux 2017). Moreover, biodiversity and extensive forested areas may contribute to the higher rates of SBE (Fry 2018).

Brazil has the highest rate of SBE in South America, on average 30,000 cases are reported annually with a lethality index of 0.4%, causing serious economic, social, and physiological consequences for society (Duque 2023; Schneider 2021). Vipers are responsible for the highest SBE (approximately 99% of the registered cases) and lethality (Chippaux 2017) of incidents with *Bothrops* the major genus with forty-seven species and 87% of the registered SBE (Silva-Neto *et al.* 2020). Among such species, *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* are the main species of medical importance in Brazil, Paraguay, Argentina, and Bolivia (Chippaux 2017) due to devastating toxic effects on victims, including pain, inflammation, edema, bleeding, tissue damage, disturbances in the blood coagulation system, heart and kidney failure, hypotension, and eventually death (Cavecci-Mendonça 2023).

Snake venoms are mainly composed of active proteins and enzymes (95%) functioning in defense, immobilization, and digestion of the prey (El-Aziz *et al.* 2019). In addition, there is a smaller amount of a non-protein constituents (approximately 5% of the dry weight), such as metals ions (acting as cofactors to the enzymes), lipids, carbohydrates, riboflavin, nucleotides, nucleosides, small peptides, amines, aminoacids, which are less biologically active (El-Aziz *et al.* 2019; Tasoulis & Isbister 2023). In general, snake venoms are classified into forty-two protein families, according to their abundance and occurrence: four are dominant, six are secondary,

fourteen are minor, and eighteen are rare (Gutiérrez *et al.* 2017; Tasoulis & Isbister 2023). The dominant, principal, secondary, and minor families account for 96% of snake venom composition. The most prominent medically relevant toxic effects of SBE are mainly caused by PLA₂, snake venom metalloprotease (SVMP), three-finger toxins (3FTx), and snake venom serine protease (SVSP). However, other compounds of the secondary family should also be considered, such as the L-amino acid oxidases (LAOs), C-type lectins (CTLs), natriuretic peptides (NPs), and disintegrins (DIS) (Tasoulis & Isbister 2023). Overall, SBE symptoms are a result of toxic actions of these principal components of venoms (Gutiérrez *et al.* 2017; Tasoulis & Isbister 2023) and neutralizing most of such effects is needed to treat with efficacy SBE.

The WHO and Ministries of Health worldwide recommend commercial antivenoms as the only available treatment for SBE, and these are produced through equine hyperimmunization. In Brazil, antivenoms have been used for more than a century and are produced by four public research centers, including the Institute Vital Brazil (IVB, Niterói, Rio de Janeiro), Institute Butantan (IB, São Paulo, São Paulo), Ezequiel Dias Foundation (FUNED, Belo Horizonte, Minas Gerais), and the Immunological Production and Research Center (Piraquara, Paraná) (Fan & Monteiro 2018). Polyvalent antivenom is produced using a mixture of venoms from *B. jararaca* (50%), *B. jararacussu* (12.5%), *B. alternatus* (12.5%), *B. moojeni* (12.5%), and *B. neuwiedi* (12.5%) and is preferable to monovalent antivenom in most of SBE as it can treat several venoms from a region without the need of species identification (Grego *et al.* 2021; Squaiella-Baptistão 2018). Antivenoms are effective in preventing death of the victims as indicated by the low global mortality index of 0.5–1.0% (Saladini *et al.* 2024). However, antivenoms have disadvantages, including high production costs, side effects (such as fever or allergic reactions), and the need for storage at + 2 °C to +10 °C. Furthermore, the inefficacy of antivenoms to neutralize some venom constituents is the most negative issue that in turn may lead to amputation and deformity of the affected limbs (Estevão-Costa *et al.* 2016). As a consequence of physical morbidity, depression and post traumatic stress frequently occur in the victims of SBE (Gutiérrez *et al.*, 1998).

Thus, complementary treatments to antivenoms have been sought and the WHO has founded a roadmap to improve the quality of antivenoms and encourage research for complement treatments SBE treatments (Warrel & Gutiérrez 2024). In this regard, plants are a rich source of biologically active molecules derived of primary and secondary metabolism that have been used for human consumption (Simsek *et al.* 2024) to treat and alleviate a wide range of diseases (Kamal *et al.* 2020; Elshafie *et al.* 2023), such as SBE (Rodrigues *et al.* 2025; Simsek *et al.* 2024; Simas Pereira *et al.* 2020). The metabolites of primary metabolism include amino acids, proteins, lipids, and carbohydrates that are involved in development, reproduction, and growth of plants. The secondary metabolites, including alkaloids, flavonoids, carotenoids, xanthones, phenolic acids, and terpenoids are involved in defense against predators, plant signaling, environmental adaptation and resistance, and ecological functions (Elshafie *et al.*, 2023). In addition, secondary metabolites have been used in different fields, including nutrition, pharmaceuticals, agriculture, medicine, and cosmetics (Elshafie *et al.* 2023; Idris *et al.* 2020), and have biotechnological potential in many areas of science.

Bioactive compounds of plants are used to prepare herbal extracts or medicines for many illnesses (Woo *et al.* 2012) and, according to the WHO, 80% of the population worldwide uses plants to ameliorate and promote health conditions (Idris *et al.* 2020). However, traditional medicine (TM) or the use of complementary medicines (CAM) should be used with caution and scientific validation of such practices is of great importance to society. The inclusion of such procedures into formal health systems would benefit citizens worldwide and the WHO has defined a global traditional medicine strategy regarding the role of TM or CAM (World Health Organization 2019).

The plant *A. hippocastanum* (AH) belongs to the family Sapindaceae and sub-family Hippocastanaceae and is commonly referred to as horse chestnut or buckeye, and widely cultivated as ornamental tree. It is renowned for its medicinal applications in the prevention and treatment of various ailments and exhibits remarkable tolerance to urban pollution (Idris *et al.* 2020; Bencsik *et al.* 2024). The main phytoconstituents of *A. hippocastanum* are escin and prosapogenin, but other compounds are also present in the leaves and flowers, including carbohydrate, anthocyanin, coumarins (esculetin, scopoletin, sculin, and frutexin), flavonoids, saponins, amino acids, amines, phytosterols, fatty acids, and kaempferol derivatives (Makuch *et al.* 2011; Savarino *et al.* 2023;

Bielarska *et al.* 2022). Based on its constituents, *A. hippocastanum* has pharmacological actions, such as antiinflammatory, antidiabetic, gastro protective, antibacterial, radical scavenging, antithrombotic, antiviral, antiedematogenic, antioxidant, and to treat hemorrhoids, capillary fragility, and burnt skin inflammation (Srijayanta *et al.* 1999; Bielarska *et al.* 2022). Due to the healing properties of *A. hippocastanum*, it was included in the European Medicines Agency (EMA) catalogue and the Community Herbal monograph (Calapai *et al.* 2014).

Melilotus officinalis (yellow sweet clover) is a widely distributed annual herb belonging to the family Fabaceae and sub-family Faboideae (Sowa-Borowiec *et al.* 2025). *M. officinalis* (MO) has been used in TM as infusion to treat some illnesses, including digestive issues, bruises, skin and joint inflammation, neuralgias, and circulatory imbalance. In addition, *M. officinalis* has diuretic, sedative, and emollient properties (Paun *et al.* 2020). The chemical composition of the extract of *M. officinalis* suggests the presence of rosmarinic acid, chlorogenic acid, coumarins, flavonoids, vitamin C, tannins, allatoin, and phenolic acids (Sowa-Borowiec *et al.* 2025). Although the chemical composition of the extracts from *A. hippocastanum* and *M. officinalis* contain constituents with antivenom effects, such as saponins, flavonoids, and phenolic acids (Simas Pereira *et al.* 2022), no reports of such plants as inhibitors of toxic activities of snake venoms have been described. On the other hand, therapeutic applications of both plants have been described (Owczarek *et al.* 2021; Idris *et al.* 2020; Liu *et al.* 2018).

Thus, the aim of the present study was to investigate the inhibition potential of the extracts from *A. hippocastanum* (AH) and *M. officinalis* (MO) on the proteolytic, plasma coagulant, and PLA₂ activities of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom, as well as a chemical characterization of the plants extracts.

Material and Methods

Preparation of venom from Bothrops jararaca, B. jararacussu, and B. neuwiedi

Bothrops jararaca, *B. jararacussu*, and *B. neuwiedi* venoms were supplied by the Ezequiel Dias Foundation (FUNED) and collection was conducted under the authorization of the Brazilian System for Genetic Management of Heritage and Associated Traditional Knowledge (SISGEN), process number A39CD4E. The lyophilized venom was diluted in 0.9 % physiological saline solution (0.9 g of NaCl (Sigma Chemical Co.) in 100 mL of Milli-Q® water) to 1 mg/mL and stored at −20 °C.

Collecting and preparation of the extracts from *A. hippocastanum* and *M. officinalis*

The flowers of AH and stems with leaves of MO were collected during July and August 2022 from the Kaliningrad region of Russia, and chosen as the objects of study. The taxonomic affiliation of the plant material was confirmed according to protocol No. 6/2022 by the head of the herbarium of the Higher School of Living Systems of the Immanuel Kant Baltic Federal University, Dr. Pungin. Methanol (chemically pure, LenReaktiv, Russia) was used to prepare plant extracts using the Soxhlet method. Extraction was performed for 8 hr (15 cycles), then, the material was concentrated in a Rotavapor R-300 vacuum rotary evaporator (Buchi Labortechnik AG, Switzerland) and dried in a Labconco Triad freeze dryer (Labconco, Kansas City, Missouri, USA). The flowers of AH and stems with leaves of MO were simultaneously dissolved in dimethylsulfoxide (DMSO, from Sigma Chemical, USA) at the concentration of 50 mg/mL and stored at -20 °C prior to use.

Chemical parameters and quantitative yield of AH and MO extracts

The content of total phenols in the extract was determined by the Folin-Ciocalteu method with some modifications (Adzhiahmetova *et al.* 2021). A series of calibration solutions were prepared from a standard solution of gallic acid (CAS 149-91-7, Sigma-Aldrich (USA)) at the concentration of 1 mg/mL. For this, a 20 µL of extract sample, calibration solution or blank sample solvent and 100 µL of Folin-Ciocalteu reagent were added to the plate. The mixture was incubated for 4 min, after which 75 µL of 7.5% Na₂CO₃ was added. The mixture was further incubated for 30 min and readouts were taken on a CLARIOstar microplate reader with monochromator and optical filters (BMG LABTECH, Germany) at an absorbance of 765 nm. The concentration of the sum of polyphenols was calculated using a graduation graph (straight line equation: $y = 5.8667 \cdot x - 0.0069$, $R^2 = 0.997$), the results were expressed in mg of gallic acid equivalent/g dry extract. The total (quantitative) yield of polyphenols in the AH and MO extract samples was determined gravimetrically after solvent removal on a Rotavapor R-300 rotary vacuum evaporator (Buchi Labortechnik AG, Switzerland). The extract yield was calculated using the following formula: Extract yield (%) = (dry extract weight (g) / dry plant weight (g)) * 100 %. The pH of extract samples was measured using a Starter ST5000-F stationary pH meter (OHAUS, Switzerland). The content of solids was not evaluated because the calculations were based on the dry mass of extracts.

Determination of the content of polyphenols compounds from AH and MO extracts

High-performance liquid chromatography (HPLC) of the extracts from AH and MO was performed on an LC-20AB Shimadzu Prominence Chromatograph (Shimadzu, Kyoto, Japan) with a binary pump; an SPD-M20A diode array detector (Shimadzu, Kyoto, Japan); and a 4.6 mm × 250 mm 5 µm Zorbax 300SB-C18 column (Agilent, Santa Clara, CA, USA). The separation was performed at a column temperature of 40 °C in gradient elution mode. Mobile phase composition was: eluent A–0.1% TFA in distilled water, B–acetonitrile. A gradient mode was used during separation: 0 min–95% A, 5% B; 3 min–88% A, 12% B; 46 min–75% A, 25% B; 49.5 min–10% A, 90% B; 52 min–10% A, 90% B; 52.7 min–95% A, 5% B; 59 min–95% A, 5% B. Sample volume was 5 µL, flow rate 1 mL/min, and the analytical wavelengths were 254, 280, and 325 nm (Babich *et al.* 2023). The total run time of the assay was 60 min. Quantitative analysis of flavonoids was carried out using calibration graphs plotted in the concentration range of 1–100 µg/mL. The components of both plant species were identified by retention times and the spectra of individual standard substances. The error of concentration determination is 3%–5%. The data for method validation (Tab. 1) confirmed the suitability of the method for studying the phytochemical composition of plant extracts.

The analytical standards for polyphenols compounds purchased from Sigma-Aldrich (USA) were: 3,4-dihydroxybenzoic acid (CAS 99-50-3); caffeic acid (CAS 331-39-5); chlorogenic acid (CAS 327-97-9); *n*-coumaric acid (CAS 501-98-4); ferulic acid (CAS 1135-24-6); hyperoside (CAS 482-36-0); rutin (CAS 250249-75-3); quercetin-3D-glucoside (CAS 482-35-9); luteolin-7-glucoside (CAS 5373-11-5); astragalin (CAS 480-10-4); rosmarinic acid (CAS 20283-92-5); acacetin, 97.0% (CAS 480-44-4); apigenin-7-*o*-glucoside (CAS 578-74-5); caftaric acid (CAS 67879-58-7); chicoric acid (CAS 6537-80-0); catechin (CAS 225937-10-0); ellagic acid (CAS 476-66-4); gallic acid (CAS 149-91-7).

Inhibition of the proteolytic activity of *Bothrops venom* by AH and MO extracts

The proteolytic activity *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was determined as in Garcia *et al.* (1978), using azocasein (Sigma Chemical Co., USA) as the substrate. Different concentrations of venoms (10–70 µg/mL) were incubated for 90 min at 37 °C with 0.2 % v/v of azocasein (Sigma Chemical Co., USA) diluted in 20 mM Tris-HCl, 8 mM CaCl₂, pH 8.8. The reaction was stopped by adding 10 % of trichloroacetic acid (Merck KGaA, Germany) followed by

Table 1 – Method validation parameters.

Standard	Precision (%)	Repeatability (% R.S.D)	Intermediate accuracy (% R.S.D.)	Detection limit (mg/mL)	Quantification limit (mg/mL)	Linear range (µg/mL)	Determination coefficient (R ²)
3,4-dihydroxybenzoic acid	97.9	3.1	3.6	0.3	1.0	1–100	0.9997
<i>n</i> -coumaric acid	97.3	2.8	3.4	0.3	1.0	1–100	0.9993
hyperoside	97.6	3.4	3.8	0.3	1.0	1–100	0.9995
quercetin-3D-glucoside	98.1	2.8	3.3	0.3	1.0	1–100	0.9970
astragalin	97.5	3.3	3.7	0.3	1.0	1–100	1.0000
luteolin-7-glucoside	98.4	2.9	3.4	0.3	1.0	1–100	0.9983
apigenin-7-O- glucoside	97.1	3.5	3.8	0.3	1.0	1–100	1.0000
acacetin	98.2	2.8	3.6	0.3	1.0	1–100	0.9993

centrifugation at 16,740 RFC for 3 min at 10 °C. Then, 1 mL of supernatant was transferred to tubes containing 500 µL NaOH 2M and kept for 10 min at room temperature. The tubes were read at an absorbance of 410 nm via a spectrophotometer (Kasvi, model K37-VIS, bandwidth 4 nm, wavelength 320~1020 nm, SN YA5020241023, Weissópolis, Pinhais, Brazil) using plastic cuvettes. The amount of venom (µg/mL) able to produce readouts of ~0.2 that represented approximately 100% of maximal activity was determined as the arbitrary unit of effective concentration (EC). Effective concentrations of *B. jararaca* (10 µg/mL), *B. jararacussu* (20 µg/mL), and *B. neuwiedi* venoms (15 µg/mL) were incubated for 30 min at 37 °C with extracts of AH and MO at a ratio weight-weight (wt/wt) of venom-to-plants of 1:10, in a final volume of 220 µL. After incubation, an aliquot of 50 µL of each mixture was added to medium and the proteolytic activity was performed. A total of 100% of proteolytic activity was achieved with venom in saline, and the negative controls contained the plants extracts AH and MO incubated with saline solution or 0.9% v/v (final concentration) of dimethyl sulfoxide (DMSO, from Sigma Chemical Co.) without venom. Each experimental group was performed with three replicates of three individual experiments, yielding n = 9.

Inhibition of the plasma coagulant activity of *Bothrops* venom by AH and MO extracts

The coagulant activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom on plasma was assessed via a four-channel digital coagulation analyzer (Model URIT-610, SN 610ET01173, Information Industry District, High-Tech Zone, Guilin, Guangxi, China), using a pool of donated

citratated plasma from the blood bank of the Antônio Pedro Hospital of the Federal Fluminense University (HUAP) under the authorization of the Committee for Ethical in Experimentation (CEP-UFF, CAAE: 28941314.0.0000.5243). Two hundred microliters of diluted plasma (1:1 v/v in saline solution) was incubated at 37 °C for 1 min in the coagulometer, and the plasma coagulation was initiated by adding different concentrations of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms (10–100 µg/mL) to a final volume of 250 µL, and then, plasma coagulation was monitored in seconds. The arbitrary unit, minimum coagulant concentration (MCC) of venom was determined as the concentration of venom (µg/mL) able to clot the plasma approximately at ~60 s. The MCCs of the venoms of *B. jararaca* (25 µg/mL), *B. jararacussu* (50 µg/mL), and *B. neuwiedi* (25 µg/mL) were incubated for 30 min at 37 °C with saline solution, DMSO (0.9% final concentration, v/v), or extracts of AH and MO at the ratio 1:10 of venom:plant (wt/wt). Then, an aliquot of 50 µL of each mixture was added to 200 µL of the plasma, and coagulation time was monitored, as previously described. As negative control, the extracts of AH, MO, saline, and DMSO (0.9% v/v, final concentration) were added to plasma in the absence of venoms and plasma coagulation was monitored during 800 s. Each experimental group was performed with three replicates of two individual experiments, yielding n = 6.

Inhibition of the phospholipase A₂ activity of *Bothrops* venom by AH and MO extracts

The PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was determined as described by Marinetti *et al.* (1965). One fresh

hen egg yolk obtained from a local supermarket was filtered and diluted (yielding, on average, 15 mL) in physiological saline to a final volume of 100 mL. This material was centrifuged at 16,740 RFC for 60 min at 10 °C to remove insoluble compounds and the supernatant, the clarified egg solution was transferred to plastic tubes and stored at 4 °C before the use. The positive controls contained the egg yolk substrate mixture incubated with *B. jararaca* and *B. neuwiedi* venom (50 µg/mL) or *B. jararacussu* venom (30 µg/mL) and solvents (saline solution or DMSO, 0.9% v/v, final concentration) for 30 min at 37 °C. In parallel, venoms were incubated with AH and MO extracts (at the ratio of 1:10 venom:plant, wt/wt) through the same experimental protocol. Then, aliquots of 50 µL of each mixture were added to a reaction medium containing 580 µL saline, 25 µL sodium taurocholate (Sigma Chemical Co., USA) 0.4%, 45 µL Tris-HCl (200 mM), 20 µL CaCl₂ (0.5 M), pH 7.5. The enzymatic reaction was initiated by adding 300 µL of the clarified egg yolk substrate solution, and after 30 min of reaction the tubes were read at an absorbance of 740 nm via a spectrophotometer (Kasvi, model K37-VIS, bandwidth 4 nm, wavelength 320~1020 nm, SN YA5020241023, Weissópolis, Pinhais, Brazil) using plastic cuvettes. The readouts of approximately 0.8 on this absorbance achieved by the positive control were considered as 100% of PLA₂ activity, whereas 0% of PLA₂ activity was achieved with saline solution alone. Negative controls comprised the extracts from AH and MO mixed with saline solution or DMSO without venom. Each experimental group was performed with three replicates of three individual experiments, yielding n = 9.

Statistical analysis

The results are expressed as the means ± standard deviation (SD) and were analyzed via ANOVA using GraphPad Prism® program (version 9.0, Software Inc., San Diego, CA, USA), followed by the Dunnet's *post hoc* test. *p* values < 0.05 were considered statistically significant.

Results

Physicochemical parameters and quantitative yields of AH and MO extracts

The results of standardization and quantitative yield of methanolic extracts of *M. officinalis* and *A. hippocastanum* are presented in Table 2. The calculations were based on the weight of the dry extract; the content of solids was not estimated.

Chemical analyses of AH and MO extracts

The results of the qualitative analysis of the biologically active polyphenols extracted from AH and MO are presented in the Figures 1 and 2 and Tables 1 and 2.

AH and MO extracts inhibited the proteolytic activity of venoms

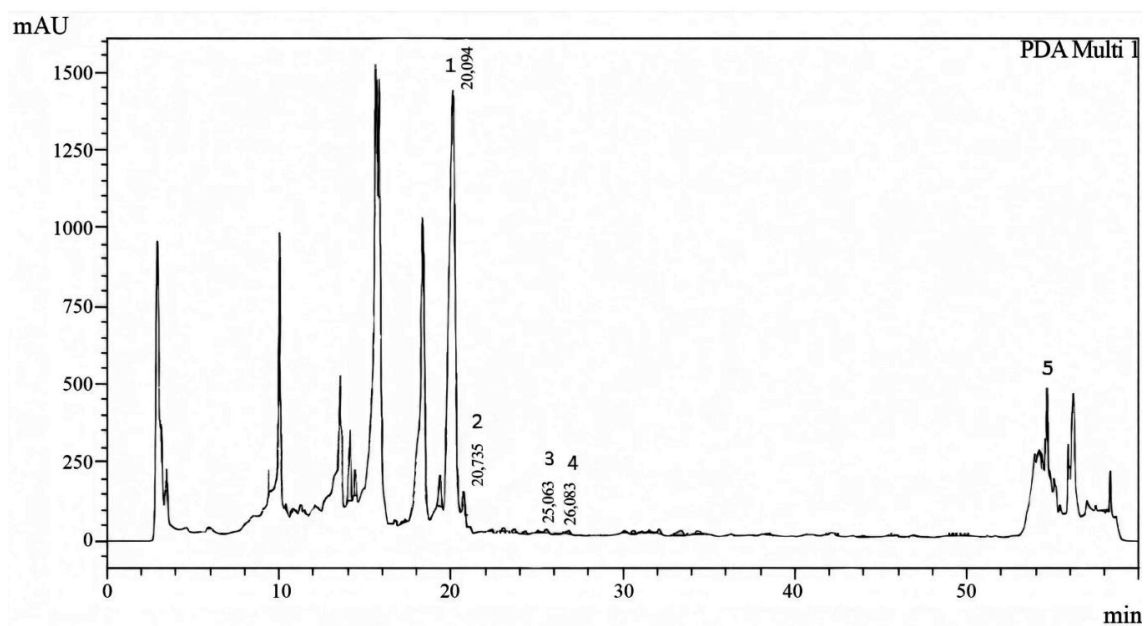
As seen in Figure 3, 100% of proteolytic activity was achieved using one EC of *B. jararaca*, *B. jararacussu*, or *B. neuwiedi* venom mixed with saline. The incubation of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom with DMSO did not interfere with this activity (Fig. 3). The AH extract inhibited 38% and 55% of the proteolytic activity of *B. jararaca* (Fig. 3a) and *B. neuwiedi* (Fig. 3c) venoms, respectively. A similar inhibitory percentage of approximately 70% was observed for the extract MO extract against *B. jararaca* (Fig. 3a) and *B. neuwiedi* venoms (Fig. 3c). The AH and MO extracts did not inhibit the proteolytic activity of *B. jararacussu* venom (Fig. 3b). In the controls, the AH and MO extracts did not hydrolyze the azocasein substrate in the absence of venom, under the experimental conditions (data not shown).

AH and MO extracts delayed plasma coagulation of venoms

One MCC of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom mixed with saline clotted plasma approximately at 60 s (Fig. 4) and was considered a value of 1.0. Treatment of *B. jararaca* (Fig. 4a) and *B. neuwiedi* (Fig. 4c) venom with the AH and MO extracts delayed plasma coagulation *ca* two fold. It should be noted that DMSO interfered a little with the plasma coagulation caused by *B. jararaca* and

Table 2 – Physicochemical parameters and quantitative yield of AH and MO extracts.

Indicator	Value	
	MO	AH
Extract yield (%)	14.49 ± 0.43	27.13 ± 0.81
Total phenolic content, mg gallic acid equivalent/g dry extract	24.09 ± 0.04	14.84 ± 2.42
pH	6.23 ± 0.18	6.95 ± 0.20

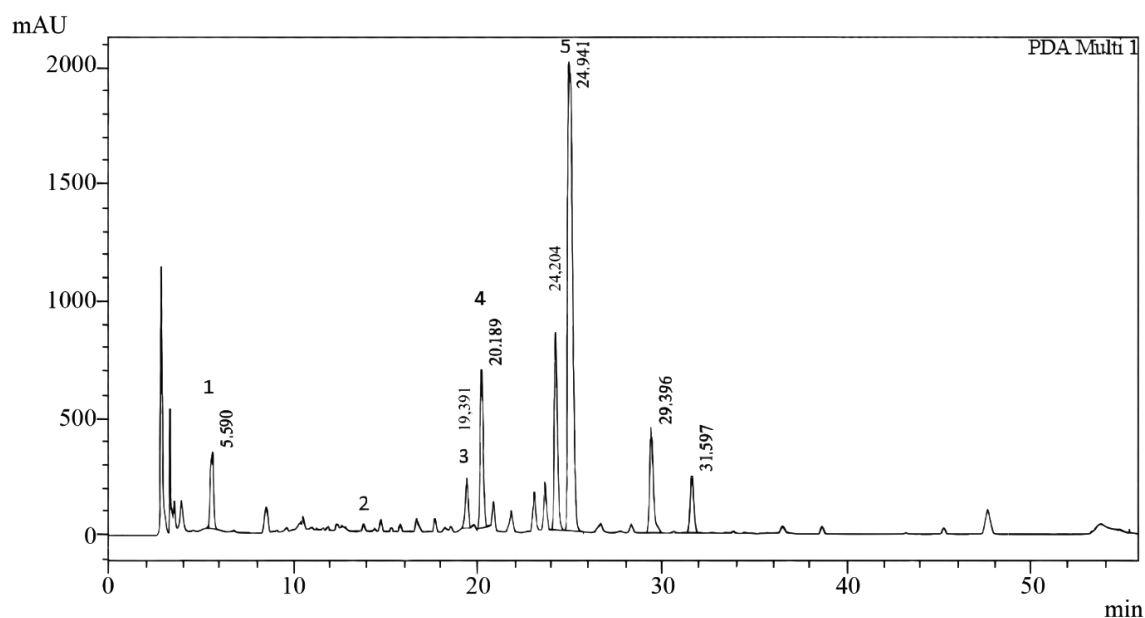


1 PDA Multi 1 / 254nm 4nm

Figure 1 – HPLC chromatogram of the extract from *M. officinalis*. 1 = quercetin-3D-glucoside; 2 = luteolin-7-glucoside; 3 = astragalin; 4 = apigenin-7-*o*-glucoside; 5 = acacetin.

B. neuwiedi venoms (approximately 20%), but the values of the AH and MO extracts were significantly higher (approximately 90% and 110%, respectively) than venoms mixed with DMSO (Fig. 4a,c). The inhibition of the AH and MO extracts on the plasma

coagulation caused by *B. jararacussu* venom was less efficient (approximately 25%) (Fig. 4b). In the absence of venoms, the AH and MO extracts did not clot plasma and, thus, they are devoid of procoagulant effect (data not shown).



1 PDA Multi 1 / 254nm 4nm

Figure 2 – HPLC chromatogram of the extract from *A. Hippocastanum*. 1 = 3,4-dihydroxybenzoic acid; 2 = *p*-coumaric acid; 3 = hyperoside; 4 = quercetin-3D-glucoside; 5 = astragalin.

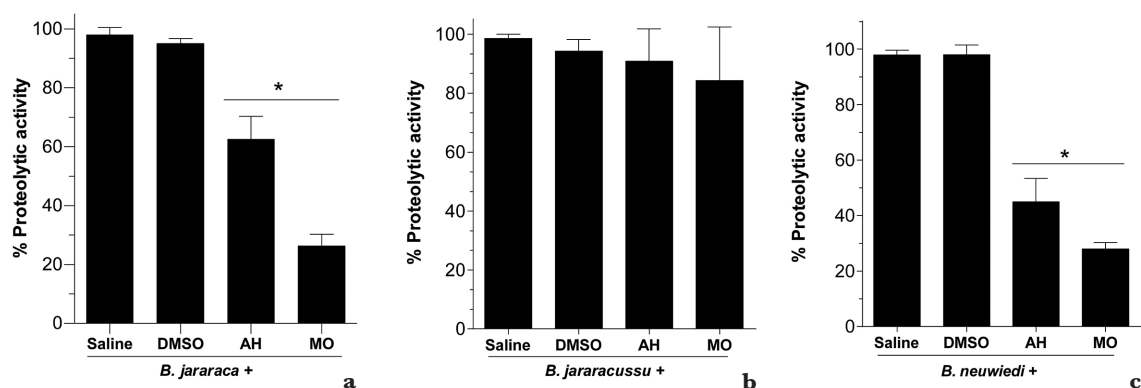


Figure 3 – a-c. Inhibition of AH and MO extracts on the proteolytic activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms were incubated with saline, DMSO, or AH and MO, as described in method. The results are expressed as % of proteolytic activity with the means $\pm$ SD (n = 8) of two individual experiments. *, $p < 0.05$ compared with venom plus saline and DMSO.

AH and MO extracts inhibited the PLA_2 activity of venoms

The clarified egg yolk solution enriched in phospholipids has been used as substrate to assess the activity of the PLA_2 enzymes in the venoms. *B. jararaca* (Panel A), *B. jararacussu* (Panel B), and *B. neuwiedi* (Panel C) venoms mixed with saline were considered as 100 % of PLA_2 activity, and the DMSO solvent did not interfere with the PLA_2 activity of the venoms (Fig. 5). The AH and MO extracts inhibited 60% of the PLA_2 activity of the *B. jararaca* venom (Fig. 5a), and 30% and 40% for *B. jararacussu* (Fig. 5b) and *B. neuwiedi* (Fig. 5c) venoms, respectively. In the absence of venoms, no intrinsic PLA_2 activity was detected in the AH and MO extracts under the experimental conditions (data not shown).

Discussion

This manuscript showed for the first time that extracts of the flowers from AH and stems with leaves from MO inhibited the proteolytic, plasma coagulant, and PLA_2 activities of *B. jararaca* and *B. neuwiedi* venom, but failed to inhibit such activities of *B. jararacussu* venom. The purified SVMs, bothropasin and jararahagin, from *B. jararaca* venom have hemorrhagic activity and alter the blood clotting system; myotoxicity of *B. jararacussu* venom is due to the presence of the PLA_2 s bothropstoxin I (Bthtx-I) and bothropstoxin II (Bthtx-II) (Ayres *et al.* 2015). Additional experiments using such purified enzymes could be performed to address the mechanism of action of AH and MO extracts. Although the species assessed in the present work belong to the same genus, literature

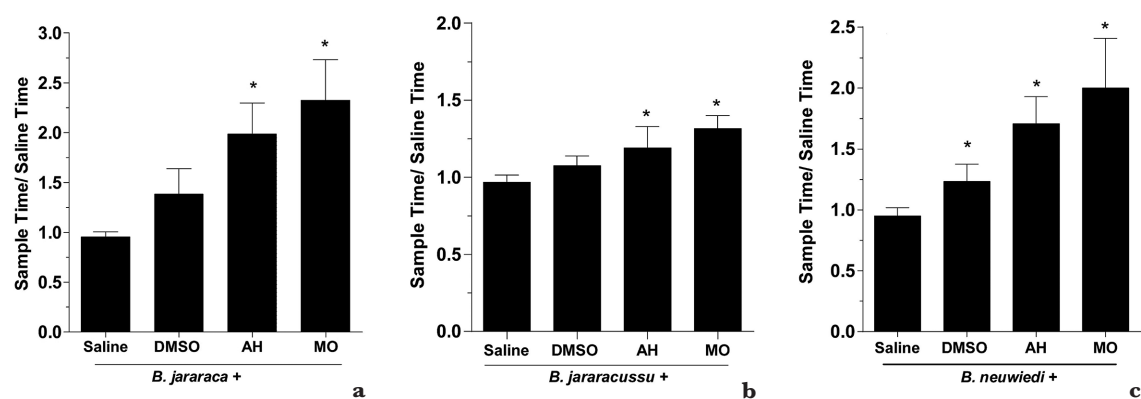


Figure 4 – a-c. Inhibition of AH and MO extracts on the plasma coagulation of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom was mixed with saline, DMSO, or the extracts AH and MO. Then, aliquot of each mixture was added to the plasma and coagulation time was monitored. The results are expressed as sample time/ sample saline with the means $\pm$ SD (n = 12) of three individual experiments. *, $p < 0.05$ compared with venom plus saline and DMSO.

has shown intra- and interspecific variations of venoms, including according to diet, sex, age, and habitat among the biologically active components of venoms (Queiroz *et al.* 2008). Thus, the development of a treatment for SBE using molecules or crude extracts from natural sources able to neutralize all of these active proteins and peptides is a difficult issue.

The antivenom effects of the AH and MO extracts against *in vitro* venom activity was quite different, probably due to slight variations between the chemical composition of both plant extracts. For example, the compounds hyperoside and 3,4-dihydroxybenzoic acid that are absent in MO, but present in AH in considerable quantities (15.45 mg/kg and 29.54 mg/kg, respectively). Hyperoside is a molecule present in the extract from the plant *Hypericum brasiliensis* that has antivenom effect (Assafim *et al.* 2011). The compound quercetin is also a constituent of *H. brasiliensis* that has a wide range of pharmacological activities, including anti-inflammatory, reduction of heart failure, and anticancer (Perazzo *et al.* 2008). However, in the work conducted by Assafim *et al.* (2011), quercetin did not inhibit hemorrhagic, lethal, and edematogenic activities of *B. jararaca* venom as did the crude extract of *H. brasiliensis* (Assafim *et al.* 2011). Thus, inhibition of AH and MO extracts of the proteolytic and coagulant activities of *B. jararaca* and *B. neuwiedi* venom may be not due only to the content of quercetin that is known to bind divalent metals, such as Ca^{+2} and Zn^{+2} , which are crucial for the enzymatic activity of PLA_2 s and SVMPs, respectively. Moreover, such compounds of plants are able to bind amino acids of the active site of such enzymes. However, the content of quercetin in the extracts of *H. brasiliensis* in the work of Assafim *et al.* (2011) was not described, and the period of collecting plants may alter the content

of compounds and alter the inhibitory action against toxins of venoms. Thus, this lack of data may limit comparison among results describing antivenom of plants. In addition, the compound apigenin-7-O-glucoside is present in low quantity in MO (28.83 mg/kg) and undetectable in AH and thus, antivenom activity is not related to this compound; quercetin-3D-glucoside is present in both extracts but higher quantities in MO. The AH extract has 55 fold higher of the flavonoid glycoside, astragalin than MO; and this compound has been used as template to develop drugs against some diseases, including obesity, cancer, osteoarthritis, osteoporosis, and diabetes mellitus (Ruan *et al.* 2024; Chen *et al.* 2023).

It was found that the Soxhlet method allows preparation of extract samples with practically neutral pH values (pH of MO extract - 6.95 ± 0.20 , pH of AH extract - 6.23 ± 0.18) and is characterized by high concentration of polyphenols (Tab. 2). It has been found that the extracts from AH and MO contained quercetin-3D-glucoside in the amounts of 99.35 mg/kg and 62.65 mg/kg, respectively, and astragalin in amounts of 8.76 mg/kg and 469.33 mg/kg, respectively. However, higher amounts of some compounds in the extracts from AH and MO are not directly involved in the inhibitory efficacy of them against toxic effects of venoms. One may speculate that higher amounts of compound(s) may have opposite effects, instead of inhibiting toxic effect(s) of venoms; a stimulant action may occur or even produce toxicity and undesirable effects in victims. Furthermore, the chemical composition of plants varies according to the seasons, in which undoubtedly influence on the biological and pharmacological properties of plants (Pott *et al.* 2000). The empirical data is supported by the study of Yildiz *et al.* (2024) using the LC-MS/MS method

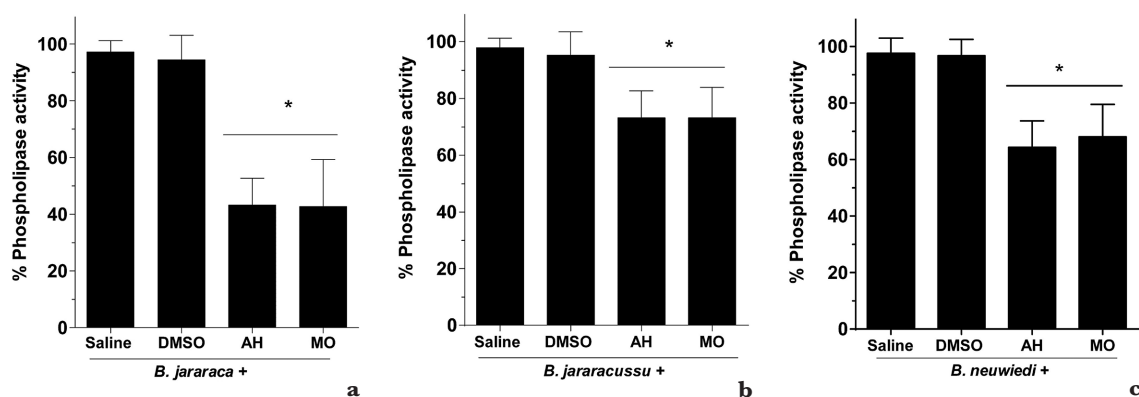


Figure 5 – a-c. Effect of AH and MO extracts on the PLA_2 activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* venoms were incubated with saline, DMSO, or AH and MO. Then, PLA_2 activity was performed as described in method. The results are expressed as the means $\pm$ SD ($n = 10$) of two individual experiments. * $p < 0.05$ when compared with venom plus saline and DMSO.

that revealed the presence of numerous polyphenols in the MO extracts, including isoquercitrin that acts as inhibitor of PLA₂ and hemorrhagic activities of *B. jararacussu* and *B. neuwiedi* venoms (Vale *et al.* 2011); while for astragalin, there is no scientific evidence as antivenom. However, as mentioned before, astragalin is a bioactive compound from plants with various beneficial effects to humans (Kangra *et al.* 2025). Nevertheless, it worth notice that the works describing antivenom effects of plants did not analyze the content of polyphenols, as well as snake venoms may suffer variation on chemical composition. Thus, compare data of literature with the present work is a challenge, including the period of collecting plants and snake venoms, as well as the biotic and abiotic factors. Moreover, the use of solvents to prepare extracts from plants may alter their chemical composition, and in turn, may alter the inhibitory efficacy of plants.

In addition to the aforementioned components, the extract from MO has significant amounts of luteolin-7-glucoside (28.83 mg/kg), apigenin-7-O-glucoside (3.33 mg/kg), and acacetin (2.28 mg/kg), as well as traces of hyperoside, rutin, rosmarinic, and chicoric acids. Our research further confirmed that MO extract is rich in flavonoids and phenolic acids that have therapeutic potential (Liu *et al.* 2018). A similar conclusion was described by Sheikh *et al.* (2017), who found that the methanolic and aqueous extracts of MO contained high amounts of polyphenols. The literature has shown antivenom effect of tannic and gallic acids (Costa *et al.* 2021; Mahadeswaraswamy *et al.* 2011) and docking studies showed that inhibitory action of these compounds may occur through interaction with amino acid in the active site of PLA₂ and others of the N-terminal region (Toyama *et al.* 2002). In addition, phenolic acids have attractive features in contrast to commercial antivenom, including low production costs, ease of synthesis, do not require refrigeration, stable chemical, and stable chemical composition. Sowa-Borowiec *et al.* (2020) have determined that MO extracts contains umbelliferone, melilot acid, coumarin, *p*-coumaric and *o*-coumaric acids, quercetin, and kaempferol derivatives (quercetin 3-rhamnosyl-(1-6)-galactoside 7-*o*-rhamnoside and kaempferol 3-galactosyl-(1-6)-glucoside 7-*o*-rhamnoside), so that this plant has a intriguing chemical composition and novel antivenom compounds worth investigating further. The richer polyphenols composition of the extract obtained by other authors can be explained by the different extraction methods and climatic conditions in which the plant grew (Sowa-Borowiec *et al.* 2025).

The methanolic extract from AH was found to contain 29.59 mg/kg of 3,4-dihydroxybenzoic acid and 15.45 mg/kg of hyperoside. This extract was

also enriched with coumaric acid and apigenin-7-O-glucoside. Due to these vital phytocomponents, horse chestnut has been used as phytomedicine to prevent and treat various diseases (Celik & Aslantürk 2023). The results obtained by analyzing the polyphenolic profile are consistent with the findings reported by other authors who identified some classes of components, including flavonoids, gum, phenols, and saponins (Liaquat *et al.* 2022; Felipe *et al.* 2013). The results of qualitative phytochemical screening showed that the methanolic extract from AH contains phenols, tannins, flavonoids, and saponins but not alkaloids and anthraquinones (Çelik & Aslantürk 2023), and as mentioned such compounds have antivenom effects (Simas Pereira *et al.* 2020).

The difference in inhibition profile among *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* activities may be due to variability in the venom composition of such species. As seen in the literature, the extracts from AH and MO have been used worldwide to treat venous insufficiency, vascular problems, wounds, and hemorrhoids (Küçükkurt *et al.* 2010) and this is a favorable issue to develop medicines from both plants. One of the most promising components of AH extract is the triterpenoid saponin β-escin that has multiple pharmacological actions, such as antitumoral, antiviral, anti-inflammatory and with less adverse effects and higher safety than other compounds (Wang *et al.* 2023). These favorable issues led the approval of seeds of AH by the American Food Drug Administration (FDA) for clinical usage (Wang *et al.* 2023). Moreover, it is important to note that the plants AH and MO are cultivated worldwide in a wide range of temperature and climate conditions and have been used in the traditional medicine to venous insufficiency and both plants are included in Europe Pharmacopoeia (Shikov *et al.* 2021).

The extracts from AH and MO inhibited the PLA₂ activity of *B. jararaca*, *B. jararacussu* and *B. neuwiedi* venoms, but that *B. jararaca* activity was the most inhibited at approximately 60%. The inhibitory percentage against *B. jararacussu* and *B. neuwiedi* venom was approximately 30%, and, thus despite such species having different venom compositions, a similar inhibition was achieved. The PLA₂ enzymes cause toxic effects, such as myonecrosis, inflammation, edema, disturbances on blood coagulation, and hemolysis that undoubtedly contribute to local and systemic toxic symptoms of SBE. Thus, blocking these enzymes of venoms by the extracts from AH and MO should be considered. The anti-inflammatory activity of the extract from MO was previously observed in rats injected with oil of turpentine that is a volatile terpene isolated from pines (Pleşca-Manea *et al.* 2002). The authors observed that the anti-inflammatory effect of *M.*

officinalis occurred by reducing both the activation of circulating phagocytes and the production of citrulline. This is in accordance to the inhibitory effect of AH and MO against the PLA₂ activity of *Bothrops* spp venoms and thus, the plants might have complementary action on the inflammatory process that occurs in SBE (Pleşca-Manea *et al.* 2002). The knowledge about the chemical composition of AH and MO extracts is extremely important to develop novel antivenom compounds from these plants. As demonstrated in this work, AH and MO have compounds that act as inhibitors of toxins of *Bothrops* spp venoms, and thus, could be used alongside commercial antivenom where this therapeutic is unavailable or insufficient.

Thus, to our knowledge, the present paper describes for the first time inhibitory effects of the extracts from AH and MO on the *in vitro* toxic activities caused by the venoms of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi*, although the mechanism of action was not addressed. The isolation of active compounds of plants and additional toxic *in vivo* tests of venoms (such as hemorrhage, edema, lethality, and myotoxicity) are needed to understand the effectiveness and consider these plants as a complementary treatment for SBE by these Brazilian endemic snakes. However, these goals have limitations, since *Bothrops* spp venoms are a mixture of active enzymes with multiple isoforms and thus, assessing the neutralizing effects of the extracts from AH and MO against such targets will be long road ahead.

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Data availability statement

In accordance with Open Science communication practices, the authors inform that all data are available within the manuscript.

References

- Adzhiahmetova SL, Chervonnaja NM, Pozdnjakov DI & Oganessian ET (2021) The content of phenols (including flavonoids) and antioxidants in the leaves of *Viscum album* L. and *Pyrus communis* L. Problems of Biological Medical and Pharmaceutical Chemistry 24: 15-22.
- Assafim M, Coriolano EC, Benedito SE, Fernandes CP, Lobo JE, Sanchez EF, Rocha LM & Fuly AL (2011) *Hypericum brasiliense* plant extract neutralizes some biological effects of *Bothrops jararaca* snake venom. Journal of Venom Research 25: 11-16.
- Ayres R, Feijó PRO, Adelia CO, Cintra, Tomaz MA, Melo PA, Cunha VM & Quintas LEM (2015) Different effects of myotoxins bothropstoxin-I and II from *Bothrops* snake venom on cation transport ATPases from murine fast twitch skeletal muscle. Toxicon 103: 80-88. DOI: 10.1016/j.toxicon.2015.06.009
- Babich O, Larina V, Krol O, Ulrikh E, Sukhikh S, Gureev MA & Ivanova S (2023) *In vitro* study of biological activity of *Tanacetum vulgare* extracts. Pharmaceutics 15: 616. DOI: 10.3390/pharmaceutics15020616
- Bencsik T, Balázs VL, Farkas A, Csikós E, Horváth A, Ács K, Kocsis M, Doseděl M, Fialová B, Czeglé S, Nagy M, Tóth J, Protti M, Mercolini L, Mladěnká P, Szentpéteri J & Horváth G (2024) Herbal drugs in chronic venous disease treatment: an update. Fitoterapia 179: 106256. DOI: 10.1016/j.fitote.2024.106256
- Bielarska AM, Jasek JW, Kazimierzczak R & Hallmann E (2022) Red horse chestnut and horse chestnut flowers and leaves: a potential and powerful source of polyphenols with high antioxidant capacity. Molecules 27: 2279. DOI: 10.3390/molecules27072279
- Calapai G, Miroddi M, Minciullo PL, Caputi AP, Gangemi S & Schmidt RJ (2014) Contact dermatitis as an adverse reaction to some topically used European herbal medicinal products - part 1: *Achillea millefolium*-*Curcuma longa*. Contact Dermatitis 71: 1-12. DOI: 10.1111/cod.12222
- Cavecci-Mendonça B, Luciano KM, Vaccas T, Oliveria LA, Clemente EF, Rossini BC, Vieira JCS, Barros LC, Biondi I, Magalhães PP & Santos LD (2023) Preliminary insights of Brazilian snake venom metalloproteomics. Toxins 15: 648-657. DOI: 10.3390/toxins15110648
- Çelik TA & Aslantürk ÖS (2023) Phytochemical screening and *in vitro* assessments of antioxidant and cytotoxic potentials of extracts from *Aesculus hippocastanum* L. green fruit mesocarps. International Journal of Secondary Metabolite 10: 245-256.
- Costa TR, Francisco AF, Cardoso FF, Moreira-Dill LS, Fernandes CAH, Gomes AAS, Guimarães CLS, Marcussi S, Pereira PS, Oliveira HC, Fontes MRM, Silva SL, Zuliani JP & Soares AM (2021) Gallic acid anti- myotoxic activity and mechanism of action, a snake venom phospholipase A₂ toxin inhibitor, isolated from the medicinal plant *Anacardium humile*. International Journal of Biological Macromolecular. 31: 494-512. DOI: 10.1016/j.ijbiomac.2021.06.163
- Chen J, Zhong K, Qin S, Jing Y, Liu S, Li D & Peng C (2023) Astragalin: a food-origin flavonoid with therapeutic effect for multiple diseases. Frontiers in Pharmacology 14: 1265960. DOI: 10.3389/fphar.2023.1265960

- Chippaux JP (2008) Estimating the global burden of snakebite can help to improve management. *PLOS Medicine* 5: e221. DOI: 10.1371/journal.pmed.0050221
- Chippaux JP (2017) Incidence and mortality due to snakebite in the Americas. *PLOS Neglected Tropical Diseases* 11: e0005662. DOI: 10.1371/journal.pntd.0005662
- Duque BR, Bruno SF, Ferreira V, Guedes TB, Machado C & Hamdan B (2023) Venomous snakes of medical importance in the Brazilian state of Rio de Janeiro: habitat and taxonomy against ophidism Brazilian *Journal of Biology* 83: e272811. DOI: 10.1590/1519-6984.272811
- El-Aziz TMA, Soares AG & Stockand JD (2019) Snake venoms in drug discovery: valuable therapeutic tools for life saving. *Toxins (Basel)* 11: 564. DOI: 10.3390/toxins1100564
- Elshafie HS, Camele I & Mohamed AA (2023) A comprehensive review on the biological, agricultural and pharmaceutical properties of secondary metabolites based-plant origin. *International Journal of Molecular Sciences* 24: 3266. DOI: 10.3390/ijms24043266
- Estevão-Costa MI, Gontijo SS, Correia BL, Yarleque A, Vivas-Ruiz D, Rodrigues E, Chavez-Olorteguic C, Oliveira LC & Sanchez EF (2016) Neutralization of toxicological activities of medically-relevant *Bothrops* snake venoms and relevant toxins by two polyvalent bothropic antivenoms produced in Peru and Brazil. *Toxicon* 122: 67-77. DOI: 10.1016/j.toxicon.2016.09.010
- Fan HW & Monteiro WM (2018) History and perspectives on how to ensure antivenom accessibility in the most remote areas in Brazil. *Toxicon* 151: 5-23. DOI: 10.1016/j.toxicon.2018.06.070
- Felipe MBM, Carvalho FM, Félix-Silva J, Fernandes-Pedrosa ME, Scortecchi KC, Agnez-Lima LF & Medeiros SRB (2013) Evaluation of genotoxic and antioxidant activity of an *Aesculus hippocastanum* L. (Sapindaceae) phytotherapeutic agent. *Biomedicine & Preventive Nutrition* 3: 261-266. DOI: 10.1016/j.bionut.2012.10.014
- Fry BG (2018) Snakebite: when the human touch becomes a bad touch. *Toxins* 10: 1-24. DOI: 10.3390/toxins10040170
- Garcia ES, Guimarães JA & Prado JL (1978) Purification and characterization of a sulfhydryl-dependent protease from *Rhodnius prolixus* midgut. *Archives of Biochemistry and Biophysics* 188: 315-322. DOI: 10.1016/S0003-9861(78)80015-0
- Grego KF, Vieira SEM, Vidueiros JP, Serapicos EO, Barbarini CC, Silveira GPM, Rodrigues FS, Alves LCF, Stuginski DR, Rameh-de-Albuquerque LC, Furtado MFD, Tanaka-Azevedo AM, Morais-Zani K, Rocha MMT, Fernandes W & Sant'Anna SS (2021) Maintenance of venomous snakes in captivity for venom production at Butantan Institute from 1908 to the present: a scoping history. *Journal of Venomous Animals Toxins Including Tropical Diseases* 22: e20200068. DOI: 10.1590/1678-9199-JVATITD-2020-0068
- Gutiérrez JM, Calvete JJ, Habib AG, Harrison RA, Williams DJ & Warrell DA (2017) Snakebite envenoming. *Nature Reviews Disease Primers* 3: 17063-17084. DOI: 10.1038/nrdp.2017.63
- Gutiérrez JM, León G, Rojas G, Lomonte B, Rucavado A & Chaves A (1998) Neutralization of local tissue damage induced by *Bothrops asper* (terciopelo) snake venom. *Toxicon* 36: 1549-1560. DOI: 10.1016/S0041-0101(98)00145-7
- Idris S, Mishra A & Khushtar M (2020) Phytochemical, ethnomedicinal and pharmacological applications of escin from *Aesculus hippocastanum* L. towards future medicine. *Journal of Basic and Clinical Physiology and Pharmacology* 31: 19-32. DOI: 10.1515/jbcpp-2019-0115
- Kamal RK, Sahu N, Rahul J & Singh SP (2015) Snake bite, venom, antivenom production and antivenom activity of medicinal plants: a review. *International Journal of Pharmaceutical Sciences and Research* 30: 227-234.
- Kangra K, Kakkar S, Mittal V, Kumar V, Aggarwal N, Chopra H, Malik T & Garg V (2025) Incredible use of plant-derived bioactives as anticancer agents. *RSC Advances* 15: 1721-1746. DOI: 10.1039/d4ra05089d
- Liaqat A, Mallhi TH, Khan YH, Khokhar A, Chaman S & Ali M (2022) Anti-snake venom properties of medicinal plants: a comprehensive systematic review of literature. *Brazilian Journal of Pharmaceutical Sciences* 58: e191124. DOI: /10.1590/s2175-97902022e191124
- Kedzierski B, Koch WK, Widelski J & Głowniak K (2016) Impact of harvest time of *Aesculus hippocastanum* seeds on the composition, antioxidant capacity and total phenolic content. *Industrial Crops and Products* 86: 68-72. DOI: 10.1016/j.indcrop.2016.03.034
- Kim SE, Kim TH, Park SA, Kim WT, Park YW, Ahn JS, Jeong M, Kim MY & Seo K (2017) Efficacy of horse chestnut leaf extract ALH-L1005 as a matrix metalloproteinase inhibitor in ligature-induced periodontitis in canine model. *Journal of Veterinary Science* 18: 245-251. DOI: 10.4142/jvs.2017.18.2.245
- Küçükkurt I, Ince S, Keleş H, Akkol EK, Avci G, Yeşilada E & Bacak E (2010) Beneficial effects of *Aesculus hippocastanum* L. seed extract on the body's own antioxidant defense system on subacute administration. *Journal of Ethnopharmacology* 129: 18-22. DOI: 10.1016/j.jep.2010.02.017
- Liu YT, Gong PH, Xiao FQ, Shao S, Zhao DQ, Yan MM & Yang XW (2018) Chemical constituents and antioxidant, anti-inflammatory and anti-tumor activities of *Melilotus officinalis* (Linn.) Pall. *Molecules* 23: 271. DOI: 10.3390/molecules23020271
- Mahadeswaraswamy YH, Kumar MS, Gowtham YJ, Nagaraju S, Girish KS & Kemparaju K (2011) The polyphenol 3, 4, 5 - tri-hydroxy benzoic acid inhibits indian daboia russelli venom and its hemorrhagic complex induced local toxicity. *Current Topics Medicinal Chemistry* 11: 2520-2530. DOI: 10.2174/156802611797633447
- Liaqat A, Mallhi TH, Khan YH, Khokhar A, Chaman S & Li M (2022) Anti-snake venom properties of medicinal plants: a comprehensive systematic review of literature. *Brazilian Journal of Pharmaceutical Sciences* 58: e191124. DOI: 10.1590/s2175-97902022e191124
- Makuch MD & Matlawska I (2011) Flavonoids from the flowers of *Aesculus hippocastanum*. *Acta Poloniae Pharmaceutica - Drug Research* 68: 403-408.
- Marinetti GV (1965) The action of phospholipase A on lipoproteins. *Biochimica et Biophysica Acta*

- Lipids and Lipid Metabolism 98: 554-565. DOI: 10.1016/0005-2760(65)90152-9
- Owczarek A, Kołodziejczyk-Czepas J, Marczuk P, Siwek J, Wąsowicz K & Olszewska MA (2021) Bioactivity potential of *Aesculus hippocastanum* L. flower: Phytochemical profile, antiradical capacity and protective effects on human plasma components under oxidative/nitrative stress *in vitro*. *Pharmaceuticals* 14: 1301. DOI: 10.3390/ph14121301
- Paun G, Neagu E, Albu C, Savin S & Radu GL (2020) *In vitro* evaluation of antidiabetic and anti-inflammatory activities of polyphenolic-rich extracts from *Anchusa officinalis* and *Melilotus officinalis*. *ACS Omega* 5: 13014-13022. DOI: 10.1021/acsomega.0c00929
- Perazzo FF, Lima LM, Padilha MM, Rocha LM, Pergentino JC & Carvalho JCT (2008) Anti-inflammatory and analgesic activities of *Hypericum brasiliense* (Willd) standardized extract. *Revista Brasileira de Farmacognosia* 18: 320-325. DOI: 10.1590/S0102-695X2008000300002
- Pleşca-Manea L, Pârnu AE, Pârnu M, Tămaş M, Buia R & Puia M (2002) Effects of *Melilotus officinalis* on acute inflammation. *Phytotherapy Research* 16: 316-319. DOI: 10.1002/ptr.875
- Pott DM, Vallarino JG & Osorio S (2020) Metabolite changes during postharvest storage: effects on fruit quality traits. *Metabolites* 10: 187. DOI: 10.3390/metabo10050187
- Queiroz GP, Pessoa LA, Portaro FCV, Furtado MFD & Tambourgi DV (2008) Interspecific variation in venom composition and toxicity of Brazilian snakes from *Bothrops* genus. *Toxicon* 52: 842-851. DOI: 10.1016/j.toxicon.2008.10.002
- Rodrigues SA, Pietroluongo M, Rodrigues da Silva AC, Sanchez EF, Amaral BS, Anholeti MC, Lobão AQ, Valverde AL, Paiva SR & Fuly AL (2025) Assessment of the effects of harvesting period of *Clusia fluminensis* Planch. and Triana fruits on neutralization of the toxic activities of *Bothrops jararacussu* snake venom. *Chemistry and Biodiversity* 3: e202402941. DOI: 10.1002/cbdv.202402941
- Ruan J, Shi Z, Cao X, Dang Z, Zhang Q, Zhang W, Wu L, Zhang Y & Wang T (2024) Research progress on anti-inflammatory effects and related mechanisms of astragalins. *International Journal of Molecular Sciences* 25: 4476. DOI: 10.3390/ijms25084476
- Saladini LY, Magalhães Júnior MJ, Silva CCF, Oliveira PGC, Kodama RT, Gomes L, Nishiyama Júnior MY, Spencer PJ, Silva WD & Portaro FCV (2024) Evaluation of the inhibitory potential of synthetic peptides homologous to CDR3 regions of a monoclonal antibody against bothropic venom serine proteases. *International Journal of Molecular Sciences* 25: 5181-5199. DOI: 10.3390/ijms25105181
- Savarino P, Colson E, André J & Gerbaux P (2023) Horse chestnut saponins - escins, isoescins, transescins, and desacylescins. *Molecules* 28: 2087. DOI: 10.3390/molecules28052087
- Schneider MC, Min KD, Hamrick PN, Montebello LR, Ranieri TM, Mardini L, Camara VM, Raggio Luiz R, Liese B, Vuckovic M, Moraes MO & Lima NT (2021) Overview of snakebite in Brazil: possible drivers and a tool for risk mapping. *PLoS Neglected Tropical Disease* 15: e0009044. DOI: 10.1371/journal.pntd.0009044
- Silva-Neto AV, Mouta GS, Balieiro AAS, Cordeiro JS, Balieiro PCS, Ramos TCA, Baia-da-Silva DC, Rocha ES, Endo PT, Lynn T, Monteiro WM & Sampaio VS (2024) Data descriptor of snakebites in Brazil from 2007 to 2020. *Data* 9: 91. DOI: 10.3390/data9080091
- Sheikh NA (2017) Phyto-physicochemical and high performance thin layer chromatography investigation of *Melilotus officinalis* Linn. *International Journal of Green Pharmacy* 11: 108-112. DOI: 10.22377/ijgp.v11i02.1036
- Shikov AN, Narkevich IA, Flisyuk EV, Luzhanin VG & Pozharitskaya ON (2021) Medicinal plants from the 14th edition of the Russian Pharmacopoeia, recent updates. *Journal of Ethnopharmacology* 25: 113685. DOI: 10.1016/j.jep.2020.113685
- Simas Pereira LCJ, Oliveira EC, Rorig TDV, Araújo PIP, Sanchez EF, Garrett R, Palazzo de Mello JC & Fuly AL (2020) The plant *Stryphnodendron adstringens* (Mart.) Coville as a neutralizing source against some toxic activities of *Bothrops jararacussu* snake venom. *Toxicon* 186: 182-190. DOI: 10.1016/j.toxicon.2020.08.011
- Simsek M & Whitney K (2024) Examination of primary and secondary metabolites associated with a plant-based diet and their impact on human health. *Foods* 13: 1020. DOI: 10.3390/foods13071020
- Sowa-Borowiec P, Czernicka M, Jarecki W & Dżugan M (2025) Sweet clover (*Melilotus spp.*) as a source of biologically active compounds. *Molecules* 30: 526. DOI: 10.3390/molecules30030526
- Squaiella-Baptistão CC, Sant'Anna OA, Marcelino JR & Tambourgi DV (2018) The history of antivenoms development: beyond Calmette and Vital Brazil. *Toxicon* 150: 86-95. DOI: 10.1016/j.toxicon.2018.05.008
- Srijayanta S, Raman A & Goodwin BLA (1999) Comparative study of the constituents of *Aesculus hippocastanum* and *Aesculus indica*. *Journal of Medicinal Food* 2: 45-50. DOI: 10.1089/jmf.1999.2.45
- Tasoulis T & Isbister GK (2023) A current perspective on snake venom composition and constituent protein families. *Archives of Toxicology* 97: 133-153. DOI: 10.1007/s00204-022-03420-0
- Toyama MH, Rogero A, Moraes LLF, Fernandes GA, Cruz Costa CR, Belchor MN, Carli AM & Oliveira MA (2022) Gallic acid as a non-selective inhibitor of -hydrolase fold enzymes involved in the inflammatory process: the two sides of the same coin. *Pharmaceutics* 14: 368-383. DOI: 10.3390/pharmaceutics14020368
- Vale LH, Mendes MM, Fernandes RS, Costa TR, S Hage-Melim LJ, A Sousa M, Hamaguchi A, Homs-Brandeburgo MI, Franca SC, Silva CH, Pereira PS, Soares AM & Rodrigues VM (2011) Protective effect of *Schizolobium parahyba* flavonoids against snake venoms and isolated toxins. *Current Topics in Medicinal Chemistry* 11: 2566-2577. DOI: 10.2174/156802611797633438
- Wang Y, Han X, Wan X, Niu F & Zhou C (2023) β -Escin: an updated review of its analysis, pharmacology, pharmacokinetics, and toxicity. *The American Journal of Chinese Medicine* 51: 2095-2120. DOI: 10.1142/S0192415X23500908

- Warrell DA & Gutiérrez JM (2024) Reply to 'Antivenom treatment for snakebite envenoming'. *Natural Review Disease Primers* 10: 60. DOI: 10.1038/s41572-024-00544-8
- World Health Organization (2019) Snakebite envenoming. Available at <<https://www.who.int/en/news-room/fact-sheets/detail/snakebite-envenoming>>. Access on 2 June 2020.
- Woo CSJ, Lau JSH & El-Nezami H (2012) Herbal medicine: toxicity and recent trends in assessing their potential toxic effects. *Advances in Botanical Research* 62: 365-384. DOI: 10.1016/B978-0-12-394591-4.00009-X
- Yildiz İ, Başar Y, Erenler R, Alma MH & Calimli MH (2024) A phytochemical content analysis and antioxidant activity evaluation using a novel method on *Melilotus officinalis* flower. *South African Journal of Botany* 174: 686-693. DOI: 10.1016/j.sajb.2024.09.060
- Zhang Z, Li S & Lian XY (2010) An overview of genus *Aesculus* L.: ethnobotany, phytochemistry, and pharmacological activities. *Pharmaceutical Crops* 1: 24-51. DOI: 10.2174/2210290601001010024

4. DISCUSSÃO

Este manuscrito mostrou pela primeira vez que os extratos das flores de AH e dos caules com folhas de MO inibiram as atividades proteolítica, coagulante plasmática e PLA₂ do veneno de *B. jararaca* e *B. neuwiedi*, mas não conseguiram inibir tais atividades do veneno de *B. jararacussu*. As SVMPs purificadas, bothropasina e jararagina, do veneno de *B. jararaca* possuem atividade hemorrágica e alteram o sistema de coagulação sanguínea; a miotoxicidade do veneno de *B. jararacussu* deve-se à presença das PLAs₂ bothropstoxina I e bothropstoxina II (Ayres *et al.*, 2015). Experimentos adicionais utilizando tais enzimas purificadas poderiam ser realizados para abordar o mecanismo de ação dos extratos de AH e MO. Embora as espécies avaliadas no presente trabalho pertençam ao mesmo gênero, a literatura tem mostrado variações intra e interespecíficas dos venenos, inclusive de acordo com dieta, sexo, idade e habitat entre os componentes biologicamente ativos dos venenos (Queiroz *et al.*, 2008). Assim, o desenvolvimento de um tratamento para envenenamento usando moléculas ou extratos brutos de fontes naturais capazes de neutralizar todas essas proteínas e peptídeos ativos é uma questão difícil.

Os efeitos antiofídicos dos extratos AH e MO contra as atividades dos venenos *in vitro* foram bastante diferentes, provavelmente devido a pequenas variações entre a composição química de ambos os extratos vegetais. Por exemplo, os compostos hiperosídeo e ácido 3,4-di-hidroxibenzoico estão ausentes no MO, mas presentes no AH em quantidades consideráveis (15,45 mg/kg e 29,54 mg/kg, respectivamente). O hiperosídeo é uma molécula presente no extrato da planta *Hypericum brasiliensis* que possui efeito antiofídico (Assafim *et al.*, 2011). O composto quercetina também é um constituinte do *H. brasiliensis* que possui uma ampla faixa de atividades farmacológicas, incluindo anti-inflamatória, redução da insuficiência cardíaca e anticâncer (Perazzo *et al.*, 2008). No entanto, no trabalho conduzido por Assafim *et al.*, a quercetina não inibiu as atividades hemorrágica, letal e edematogênica do veneno de *B. jararaca* como fez o extrato bruto do *H. brasiliensis*.

Assim, a inibição dos extratos AH e MO das atividades proteolítica e coagulante dos venenos da *B. jararaca* e da *B. neuwiedi* pode não se dever apenas ao conteúdo

de quercetina, que é conhecida por se ligar a metais divalentes, como Ca^{2+} e Zn^{2+} , que são cruciais para a atividade enzimática das PLAs₂ e SVMPs, respectivamente. Além disso, tais compostos das plantas são capazes de se ligar a aminoácidos do sítio ativo de tais enzimas. No entanto, o conteúdo de quercetina nos extratos do *H. brasiliensis* no trabalho de Assafim *et al.* (2011) não foi descrito, e o período de coleta das plantas pode alterar o conteúdo de compostos e alterar a ação inibitória contra toxinas dos venenos. Assim, essa falta de dados pode limitar a comparação entre os resultados que descrevem o antiveneno de plantas.

Além disso, o composto apigenina-7-O-glicosídeo está presente em baixa quantidade no MO (28,83 mg/kg) e indetectável no AH e, portanto, a atividade antiofídica não está relacionada a este composto; a quercetina-3D-glicosídeo está presente em ambos os extratos, mas em quantidades maiores no MO. O extrato AH possui 55 vezes mais do glicosídeo flavonoide astragalina do que o MO; e este composto tem sido usado como modelo para desenvolver medicamentos contra algumas doenças, incluindo obesidade, câncer, osteoartrite, osteoporose e diabetes *mellitus* (Chen *et al.*, 2023; Ruan *et al.*, 2024).

Verificou-se que o método Soxhlet permite a preparação de amostras de extrato com valores de pH praticamente neutros (pH do extrato MO – $6,95 \pm 0,20$; pH do extrato AH – $6,23 \pm 0,18$) e é caracterizado por alta concentração de polifenóis. Constatou-se que os extratos de AH e MO continham quercetina-3D-glicosídeo nas quantidades de 99,35 mg/kg e 62,65 mg/kg, respectivamente, e astragalina nas quantidades de 8,76 mg/kg e 469,33 mg/kg, respectivamente. No entanto, quantidades maiores de alguns compostos nos extratos de AH e MO não estão diretamente envolvidas na eficácia inibitória deles contra os efeitos tóxicos dos venenos. Pode-se especular que quantidades maiores de compostos podem ter efeitos opostos, em vez de inibir os efeitos tóxicos dos venenos; uma ação estimulante pode ocorrer ou até mesmo produzir toxicidade e efeitos indesejáveis nas vítimas.

Além disso, a composição química das plantas varia de acordo com as estações, o que sem dúvida influencia as propriedades biológicas e farmacológicas das plantas (Pott *et al.*, 2020). Os dados empíricos são corroborados pelo estudo de (Yildiz *et al.*, 2024) usando o método LC-MS/MS, que revelou a presença de

numerosos polifenóis nos extratos de MO, incluindo isoquercitrina, que atua como inibidora das atividades PLAs₂ e hemorrágicas dos venenos de *B. jararacussu* e *B. neuwiedi* (Vale *et al.*, 2011.); enquanto para a astragalina não há evidência científica como antiveneno. No entanto, como mencionado anteriormente, a astragalina é um composto bioativo de plantas com vários efeitos benéficos para os humanos (Kangra *et al.*, 2025.).

Vale notar que os trabalhos que descrevem os efeitos antiofídicos das plantas não analisaram o conteúdo de polifenóis, assim como os venenos de serpentes podem sofrer variação na composição química. Assim, comparar dados da literatura com o presente trabalho é um desafio, incluindo o período de coleta das plantas e dos venenos de serpente, bem como os fatores bióticos e abióticos. Além disso, o uso de solventes para preparar extratos de plantas pode alterar sua composição química e, por sua vez, pode alterar a eficácia inibitória das plantas.

Além dos componentes mencionados anteriormente, o extrato de MO tem quantidades significativas de luteolina-7-glicosídeo (28,83 mg/kg), apigenina-7-O-glicosídeo (3,33 mg/kg) e acacetina (2,28 mg/kg), bem como traços de hiperosídeo, rutina e ácidos rosmarínico e chicórico. Nossa pesquisa confirmou ainda que o extrato MO é rico em flavonoides e ácidos fenólicos que têm potencial terapêutico (Liu *et al.*, 2018). Uma conclusão semelhante foi descrita por (Sheikh, 2017), que descobriram que os extratos metanólicos e aquosos de MO continham altas quantidades de polifenóis. A literatura mostrou efeito antiveneno dos ácidos tânico e gálico (T. R. Costa *et al.* 2021; Mahadeswaraswamy *et al.*, 2011.), e estudos de *docking* mostraram que a ação inibitória desses compostos pode ocorrer através da interação com aminoácidos no sítio ativo da PLA₂ e outros da região N-terminal (Toyama *et al.*, 2022). Além disso, os ácidos fenólicos têm características atraentes em contraste com o antiofídico comercial, incluindo baixos custos de produção, facilidade de síntese, não requerem refrigeração, são quimicamente estáveis e têm composição química estável.

Sowa-Borowiec *et al.* (2020) determinaram que os extratos de MO contêm umbeliferona, ácido melilótico, cumarina, ácidos p-cumárico e o-cumárico, quercetina e derivados de kaempferol (quercetina 3-ramnosil-(1-6)-galactosídeo 7-O-ramnosídeo e kaempferol 3-galactosil-(1-6)-glicosídeo 7-O-ramnosídeo), de modo

que esta planta tem uma composição química intrigante e novos compostos antiofídicos que valem a pena investigar mais. A composição mais rica em polifenóis do extrato obtido por outros autores pode ser explicada pelos diferentes métodos de extração e condições climáticas em que a planta cresceu (Sowa-Borowiec *et al.*, 2025).

O extrato metanólico de AH foi encontrado contendo 29,59 mg/kg de ácido 3,4-di-hidroxibenzoico e 15,45 mg/kg de hiperosídeo. Este extrato foi também enriquecido com ácido cumárico e apigenina-7-O-glucósido. Devido a esses fitocomponentes vitais, a castanha-da-índia tem sido usada como fitomedicina para prevenir e tratar várias doenças (Çelik e Aslantürk, 2023). Os resultados obtidos pela análise do perfil polifenólico são consistentes com os achados relatados por outros autores que identificaram algumas classes de componentes, incluindo flavonoides, goma, fenóis e saponinas (Felipe *et al.*, 2013; Liaqat *et al.*, 2022). Os resultados da triagem fitoquímica qualitativa mostraram que o extrato metanólico de AH contém fenóis, taninos, flavonoides e saponinas, mas não alcaloides e antraquinonas (Çelik e Aslantürk, 2023), e, como mencionado, tais compostos têm efeitos antiveneno (Simas Pereira Junior *et al.*, 2020).

A diferença no perfil de inibição entre as atividades de *B. jararaca*, *B. jararacussu* e *B. neuwiedi* pode ser devida à variabilidade na composição do veneno dessas espécies. Como visto na literatura, os extratos de AH e MO têm sido usados mundialmente para tratar insuficiência venosa, problemas vasculares, feridas e hemorroidas (Küçükkurt *et al.*, 2010), e essa é uma questão favorável para desenvolver medicamentos a partir de ambas as plantas. Um dos componentes mais promissores do extrato AH é a saponina triterpenoide β -escina, que tem múltiplas ações farmacológicas, como antitumoral, antiviral, anti-inflamatória e com menos efeitos adversos e maior segurança do que outros compostos (Yunyu Wang *et al.*, 2023). Essas questões favoráveis levaram à aprovação das sementes de AH pela FDA para uso clínico (Yunyu Wang *et al.*, 2023). Além disso, é importante observar que as plantas AH e MO são cultivadas mundialmente em uma ampla faixa de temperatura e condições climáticas e têm sido usadas na medicina tradicional para insuficiência venosa, e ambas as plantas estão incluídas na Farmacopeia Europeia (Shikov *et al.*, 2021).

Os extratos de AH e MO inibiram a atividade PLA₂ dos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*, mas a atividade de *B. jararaca* foi a mais inibida, em aproximadamente 60%. O percentual de inibição contra os venenos de *B. jararacussu* e *B. neuwiedi* foi de aproximadamente 30% e, portanto, apesar de tais espécies terem composições do veneno diferentes, uma inibição semelhante foi alcançada. As enzimas PLAs₂ causam efeitos tóxicos, como mionecrose, inflamação, edema, distúrbios na coagulação sanguínea e hemólise, que sem dúvida contribuem para os sintomas tóxicos locais e sistêmicos do envenenamento. Assim, o bloqueio dessas enzimas dos venenos pelos extratos de AH e MO deve ser considerado.

A atividade anti-inflamatória do extrato de MO foi observada anteriormente em ratos injetados com óleo de terebintina, que é um terpeno volátil isolado de pinheiros (Pleșca-Manea *et al.*, 2002). Os autores observaram que o efeito anti-inflamatório de *M. officinalis* ocorreu pela redução tanto da ativação de fagócitos circulantes quanto da produção de citrulina. Isso está de acordo com o efeito inibitório de AH e MO contra a atividade PLA₂ dos venenos de *Bothrops* spp. e, assim, as plantas podem ter ação complementar no processo inflamatório que ocorre no envenenamento (Pleșca-Manea *et al.*, 2002).

O conhecimento sobre a composição química dos extratos de AH e MO é extremamente importante para desenvolver novos compostos antiveneno a partir dessas plantas. Como demonstrado neste trabalho, AH e MO possuem compostos que atuam como inibidores de toxinas dos venenos de *Bothrops* spp. e, assim, poderiam ser usados juntamente com o antiveneno comercial quando este tratamento não estiver disponível ou for insuficiente. Assim, até onde sabemos, o presente trabalho descreve pela primeira vez os efeitos inibitórios dos extratos de AH e MO sobre as atividades tóxicas *in vitro* causadas pelos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*, embora o mecanismo de ação não tenha sido abordado. O isolamento dos compostos ativos das plantas e testes tóxicos adicionais *in vivo* dos venenos (como hemorragia, edema, letalidade e miotoxidade) são necessários para compreender a eficácia e considerar essas plantas como tratamento complementar para o envenenamento causado por essas serpentes endêmicas brasileiras. No entanto, esses objetivos têm limitações, uma vez que os venenos de *Bothrops* spp. são uma mistura de enzimas ativas com múltiplas isoformas e, portanto, avaliar os

efeitos neutralizantes dos extratos de AH e MO contra tais alvos será um longo caminho a percorrer.

5. CONCLUSÃO

No presente estudo observou-se que os extratos de *Aesculus hippocastanum* e *Melilotus officinalis* demonstraram atividade inibitória significativa *in vitro* contra os venenos de *B. jararaca* e *B. neuwiedi*, embora tenham apresentado menor eficácia frente ao veneno de *B. jararacussu*, evidenciando a influência da variabilidade interespecies na composição dos venenos. A diversidade química dos extratos, especialmente o conteúdo de flavonoides, ácidos fenólicos e saponinas, parece desempenhar papel central na neutralização das toxinas, ainda que a eficácia não esteja diretamente relacionada à maior concentração isolada de determinados compostos.

Em conjunto, os dados obtidos fornecem respaldo científico para o uso tradicional dessas plantas e indicam seu potencial como fontes promissoras de compostos bioativos com aplicação no desenvolvimento de terapias complementares ao antiveneno. No entanto, estudos adicionais envolvendo o isolamento de princípios ativos, a elucidação dos mecanismos de ação e a avaliação *in vivo* são necessários para melhor compreender sua eficácia e viabilidade terapêutica frente ao envenenamento botrópico.

CONCLUSÃO GERAL

Os resultados observados nesta tese demonstram que produtos naturais constituem fontes relevantes de moléculas com potencial antiveneno frente aos efeitos tóxico-farmacológicos induzidos por venenos de serpentes do gênero *Bothrops*. Os extratos de plantas e polissacarídeos sulfatados testados inibiram a proteólise, coagulação e fosfolipase A₂ dos venenos de *B. jararaca*, *B. jararacussu* e *B. neuwiedi*.

As diferentes classes de metabólitos atuam sobre alvos comuns dos venenos por mecanismos distintos e complementares. Compostos fenólicos, flavonoides e taninos presentes nas plantas interagem com proteínas e podem quelar íons metálicos, contribuindo para a inibição de metaloproteinases e serinoproteases, enquanto polissacarídeos sulfatados atuam por interações eletrostáticas com as toxinas, modulando sua atividade enzimática. A variação na eficácia entre os compostos avaliados reforça a influência da origem e da estrutura química na capacidade de neutralização das toxinas, destacando a diversidade metabólica como fator determinante para a atividade biológica observada.

Em conjunto, os resultados confirmam que metabólitos de plantas e polissacarídeos de algas e microalgas apresentam atividade antiveneno significativa, atuando em diferentes etapas do envenenamento e se destacando como candidatos promissores para uso como agentes auxiliares à soroterapia, especialmente no controle dos efeitos locais. Esses achados reforçam o potencial biotecnológico dos produtos naturais e sustentam sua relevância no desenvolvimento de novas abordagens terapêuticas capazes de ampliar a eficácia do tratamento dos envenenamentos por serpentes do gênero *Bothrops*.

REFERÊNCIAS BIBLIOGRÁFICAS

- Adrião AAX, dos Santos AO, de Lima EJSP, *et al.* Plant-Derived Toxin Inhibitors as Potential Candidates to Complement Antivenom Treatment in Snakebite Envenomations. *Front Immunol.* 2022;13.
- Alangode A, Rajan K, Nair BG. Snake antivenom: Challenges and alternate approaches. *Biochem Pharmacol.* 2020;181:114-135.
- Alcântara JA, Bernarde PS, Sachett J, *et al.* Stepping into a dangerous quagmire: Macroecological determinants of Bothrops envenomings, Brazilian Amazon. *PLOS ONE.* 2018;13(12):e0208532.
- Alencar POC, Lima GC, Barros FCN, *et al.* A novel antioxidant sulfated polysaccharide from the algae *Gracilaria caudata*: In vitro and in vivo activities. *Food Hydrocoll.* 2019;90:28-34.
- Almada-Vilhena AO de, Santos OVM dos, Machado M de A, Nagamachi CY, Pieczarka JC. Prospecting Pharmacologically Active Biocompounds from the Amazon Rainforest: In Vitro Approaches, Mechanisms of Action Based on Chemical Structure, and Perspectives on Human Therapeutic Use. *Pharmaceuticals.* 2024;17(11).
- Aloulou A, Rahier R, Arhab Y, Noiriél A, Abousalham A. Phospholipases: An Overview. *Methods Mol Biol.* Published online 2018:1835,69-105.
- Angulo Y, Lomonte B. Inhibitory effect of fucoidan on the activities of crotaline snake venom myotoxic phospholipases A2. *Biochem Pharmacol.* 2003;66(10):1993-2000.
- Apostolova E, Lukova P, Baldzhieva A, *et al.* Immunomodulatory and Anti-Inflammatory Effects of Fucoidan: A Review. *Polymers.* 2020;12(10).
- Arora N, Laurens LML, Sweeney N, Pruthi V, Poluri KM, Pienkos PT. Elucidating the unique physiological responses of halotolerant *Scenedesmus* sp. cultivated in sea water for biofuel production. *Algal Res.* 2019;37:260-268.
- Assafim M, de Coriolano EC, Benedito SE, *et al.* Hypericum brasiliense plant extract neutralizes some biological effects of Bothrops jararaca snake venom. *J Venom Res.* 2011;2:11-16.
- Avila-Sosa R, Montero-Rodríguez AF, Aguilar-Alonso P, *et al.* Antioxidant Properties of Amazonian Fruits: A Mini Review of In Vivo and In Vitro Studies. *Oxid Med Cell Longev.* 2019;2019(1):8204129.
- Ayres R, Feijó PRO, Cintra ACO, *et al.* Different effects of myotoxins bothropstoxin-I and II from Bothrops snake venom on cation transport ATPases from murine fast twitch skeletal muscle. *Toxicon.* 2015;103:80-84.

Babich O. Algae: study of edible and biologically active fractions, their properties and applications. *Plants Basel*. 2022;11:780.

Babich O, Ivanova S, Tupitsyn A, *et al*. Study of the polysaccharide production by the microalgae C-1509 *Nannochloris* sp. *Naumann. Biotechnol Rep*. 2023;40:e00818.

Bala E, Hazarika R, Singh P, Yasir M, Shrivastava R. A biological overview of Hyaluronidase: A venom enzyme and its inhibition with plants materials. *Mater Today Proc*. 2018;5(2, Part 1):6406-6412.

Banskota AH, Stefanova R, Hui JPM, *et al*. Comprehensive Analysis of Biomass from *Chlorella sorokiniana* Cultivated with Industrial Flue Gas as the Carbon Source. *Molecules*. 2024;29(14).

Barbosa FBA, Raad R de S, Ibiapina HNS, *et al*. Dermatopathological findings of *Bothrops atrox* snakebites: A case series in the Brazilian Amazon. *PLoS Negl Trop Dis*. 2024;18(12):e0012704.

Barsanti L, Gualtieri P. Is exploitation of microalgae economically and energetically sustainable? *Algal Res*. 2018;31:107-111.

Bauer M, Lautenschlaeger C, Kempe K, Tauhardt L, Schubert US, Fischer D. Poly(2-ethyl-2-oxazoline) as alternative for the stealth polymer poly(ethylene glycol): Comparison of in vitro cytotoxicity and hemocompatibility. *Macromol Biosci*. 2012;12:986-998.

Begum R, Howlader S, Mamun-Or-Rashid ANM, *et al*. Antioxidant and Signal-Modulating Effects of Brown Seaweed-Derived Compounds against Oxidative Stress-Associated Pathology. *Oxid Med Cell Longev*. 2021;2021(1):9974890.

Belo AA, Souza DLN, Melo-Braga MN, *et al*. Production of a murine mAb against *Bothrops alternatus* and *B. neuwiedi* snake venoms and its use to isolate a thrombin-like serine protease fraction. *Int J Biol Macromol*. 2022;214:530-541.

Benalaya I, Alves G, Lopes J, Silva LR. A review of natural polysaccharides: sources, characteristics, properties, food, and pharmaceutical applications. *Int J Mol Sci*. 2024;25:1322.

Berini JL, Brockman SA, Hegeman AD, *et al*. Combinations of Abiotic Factors Differentially Alter Production of Plant Secondary Metabolites in Five Woody Plant Species in the Boreal-Temperate Transition Zone. *Front Plant Sci*. 2018;9.

Borges RJ, Salvador GHM, Campanelli HB, *et al*. BthTX-II from *Bothrops jararacussu* venom has variants with different oligomeric assemblies: An example of snake venom phospholipases A2 versatility. *Int J Biol Macromol*. 2021;191:255-266.

Borhani G, Mazandarani M, Abbaspour H. Antioxidant, Antibacterial Activity, Ethnopharmacology, Phytochemical in Different Extracts of *Melilotus officinalis* L. as

an Anti-infection and Anti-diabetic in Traditional Uses of Two Northern Provinces From Iran. *Crescent J Med Biol Sci.* 11(2).

Brovko O, Palamarchuk I, Gorshkova N, Chukhchin D, Eliseeva I. Effect of Polyelectrolyte Properties of Fucoidans on Antimicrobial Activity. *J Polym Environ.* 2025;33(5):2138-2148.

Cao S, He X, Qin L, *et al.* Anticoagulant and Antithrombotic Properties in Vitro and in Vivo of a Novel Sulfated Polysaccharide from Marine Green Alga *Monostroma nitidum*. *Mar Drugs.* 2019;17(4).

Cardoso FF, Salvador GHM, Cavalcante WLG, Dal-Pai M, Fontes MRM. BthTX-I, a phospholipase A2-like toxin, is inhibited by the plant cinnamic acid derivative: Chlorogenic acid. *Biochim Biophys Acta Proteins Proteom.* Published online 2024:1872,140988-141001.

Castro-Pinheiro C, Junior LCSP, Sanchez EF, *et al.* Effect of Seaweed-Derived Fucoidans from *Undaria pinnatifida* and *Fucus vesiculosus* on Coagulant, Proteolytic, and Phospholipase A2 Activities of Snake *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* Venom. *Toxins.* 2024;16(4):188.

Cavalcante JS, Almeida DEG de, Santos-Filho NA, *et al.* Crosstalk of Inflammation and Coagulation in *Bothrops* Snakebite Envenoming: Endogenous Signaling Pathways and Pathophysiology. *Int J Mol Sci.* 2023;24(14).

Cavecci-Mendonça B, Luciano KM, Vaccas T, *et al.* Preliminary Insights of Brazilian Snake Venom Metalloproteomics. *Toxins.* 2023;15(11).

Çelik TA, Aslantürk ÖS. Phytochemical screening and in vitro assessments of antioxidant and cytotoxic potentials of extracts from *Aesculus hippocastanum* L. green fruit mesocarps. *Int J Second Metab.* 2023;10(2):245-256.

Chauton MS, Forbord S, Mäkinen S, *et al.* Sustainable resource production for manufacturing bioactives from micro- and macroalgae: Examples from harvesting and cultivation in the Nordic region. *Physiol Plant.* 2021;173(2):495-506.

Chen J, Zhong K, Qin S, *et al.* Astragalin: a food-origin flavonoid with therapeutic effect for multiple diseases. *Front Pharmacol.* 2023;14.

Cheong KL, Chen W, Wang M, Zhong S, Veeraperumal S. Therapeutic Prospects of *Undaria pinnatifida* Polysaccharides: Extraction, Purification, and Functional Activity. *Mar Drugs.* 2025;23(4).

Chippaux JP. Incidence and mortality due to snakebite in the Americas. *PLoS Negl Trop Dis.* 2017;11(6):e0005662.

Chudasama NA, Sequeira RA, Moradiya K, Prasad K. Seaweed polysaccharide based products and materials: an assessment on their production from a sustainability point of view. *Molecules.* 2021;26:260.

Cichoński J, Chrzanowski G. Microalgae as a Source of Valuable Phenolic Compounds and Carotenoids. *Molecules*. 2022;27(24).

Citkowska A, Szekalska M, Winnicka K. Possibilities of Fucoidan Utilization in the Development of Pharmaceutical Dosage Forms. *Mar Drugs*. 2019;17:458.

Costa JAV, Lucas BF, Alvarenga AGP, Moreira JB, Morais MG. Microalgae polysaccharides: an overview of production, characterization, and potential applications. *Polysaccharides*. 2021;2:759-777.

Costa TR, Francisco AF, Cardoso FF, *et al.* Gallic acid anti-myotoxic activity and mechanism of action, a snake venom phospholipase A2 toxin inhibitor, isolated from the medicinal plant *Anacardium humile*. *Int J Biol Macromol*. 2021;185:494-512.

Coulombier N, Nicolau E, Déan LL, Antheaume C, Jauffrais T, Lebouvier N. Impact of Light Intensity on Antioxidant Activity of Tropical Microalgae. *Mar Drugs*. 2020;18(2).

Cremonez CM, Leite FP, Bordon K de CF, *et al.* Experimental *Lachesis muta rhombeata* envenomation and effects of soursop (*Annona muricata*) as natural antivenom. *J Venom Anim Toxins Trop Dis*. 2016;22.

Cumashi A, Ushakova NA, Preobrazhenskaya ME, *et al.* A comparative study of the anti-inflammatory, anticoagulant, antiangiogenic, and antiadhesive activities of nine different fucoidans from brown seaweeds. *Glycobiology*. 2007;17(5):541-552.

Dahiya S, Chowdhury R, Tao W, Kumar P. Biomass and Lipid Productivity by Two Algal Strains of *Chlorella sorokiniana* Grown in Hydrolysate of Water Hyacinth. *Energies*. 2021;14(5).

Damm M, Hempel BF, Süssmuth RD. Old World Vipers—A Review about Snake Venom Proteomics of Viperinae and Their Variations. *Toxins*. 2021;13(6). doi:10.3390/toxins13060427

Dean PR, Hurd CL. Seasonal growth, erosion rates, and nitrogen and photosynthetic ecophysiology of *Undaria pinnatifida* (Heterokontophyta) in southern New Zealand. *J Phycol*. 2007;43(6):1138-1148.

Diniz-Sousa R, Moraes J do N, Rodrigues-da-Silva TM, Oliveira CS, Caldeira CA da S. A brief review on the natural history, venomics and the medical importance of bushmaster (*Lachesis*) pit viper snakes. *Toxicon X*. 2020;7:100053.

Domanski D, Zegrocka-Stendel O, Perzanowska A, *et al.* Molecular Mechanism for Cellular Response to β -Escin and Its Therapeutic Implications. *PLOS ONE*. 2016;11(10):e0164365.

Domínguez-Gámez M, Romo-Sáenz CI, Gomez-Flores R, *et al.* In Vitro Antitumor, Antioxidant, and Hemolytic Activities of *Chlorella sorokiniana* Methanol Extracts and Collective Fractions. *Appl Sci*. 2024;14(20).

Dudek-Makuch M, Studzińska-Sroka E. Horse chestnut – efficacy and safety in chronic venous insufficiency: an overview. *Rev Bras Farmacogn.* 2015;25(5):533-541.

Duque BR, Bruno SF, Ferreira V, Guedes TB, Machado C, Hamdan B. Venomous snakes of medical importance in the Brazilian state of Rio de Janeiro: habitat and taxonomy against ophidism. *Braz J Biol.* 2023;83:e272811.

El Asri S, Ben Mrid R, Zouaoui Z, *et al.* Advances in structural modification of fucoidans, ulvans, and carrageenans to improve their biological functions for potential therapeutic application. *Carbohydr Res.* 2025;549:109358.

El-Baz FK, Salama A, Ali SI, Elgohary R. Lutein isolated from *Scenedesmus obliquus* microalga boosts immunity against cyclophosphamide-induced brain injury in rats. *Sci Rep.* 2022;12(1):22601.

El-Naggar NEA, Hussein MH, Shaaban-Dessuuki SA, Dalal SR. Production, extraction and characterization of *Chlorella vulgaris* soluble polysaccharides and their applications in AgNPs biosynthesis and biostimulation of plant growth. *Sci Rep.* 2020;10(1):3011.

Elshafie HS, Camele I, Mohamed AA. A Comprehensive Review on the Biological, Agricultural and Pharmaceutical Properties of Secondary Metabolites Based-Plant Origin. *Int J Mol Sci.* 2023;24(4).

Faculty of Science and Technology Kampala International University Uganda, Akiro F. T. Assessing the Feasibility of Cultivating Medicinal Plants for Local Health Needs. *NEWPORT Int J Res Med Sci.* 2025;6(2):154-162.

Felipe MBMC, de Carvalho FM, Félix-Silva J, *et al.* Evaluation of genotoxic and antioxidant activity of an *Aesculus hippocastanum* L. (Sapindaceae) phytotherapeutic agent. *Biomed Prev Nutr.* 2013;3(3):261-266.

Félix-Silva J, Silva-Junior AA, Zucolotto SM, Fernandes-Pedrosa M de F. Medicinal Plants for the Treatment of Local Tissue Damage Induced by Snake Venoms: An Overview from Traditional Use to Pharmacological Evidence. *Evid Based Complement Alternat Med.* 2017;2017(1):5748256.

Figuerola FA, Abdala-Díaz R, Cabrera-Barjas G, *et al.* Latitudinal variation in chemical composition and biological activities of fucoidans from *Macrocystis pyrifera* (brown algae): first insights along the Chilean coast. *Int J Biol Macromol.* 2025;318:144978.

Fuly AL, Calil-Elias S, Zingali RB, Guimarães JA, Melo PA. Myotoxic activity of an acidic phospholipase A2 isolated from *Lachesis muta* (Bushmaster) snake venom. *Toxicon.* 2000;38:961-972.

Fuly AL, Miranda AL, Zingali RB, Guimarães JA. Purification and characterization of a phospholipase A2 isoenzyme isolated from *Lachesis muta* snake venom. *Biochem Pharmacol.* 2002;63:1589-1597.

Garg SS, Gupta J, Sharma S, Sahu D. An insight into the therapeutic applications of coumarin compounds and their mechanisms of action. *Eur J Pharm Sci.* 2020;152:105424.

George A, Shrivastav PS. Fucoidan, a brown seaweed polysaccharide in nanodrug delivery. *Drug Deliv Transl Res.* Published online 2023:13,2427-2446.

Ghaliaoui N, Hazzit M, Mokrane H. Seaweeds as a Potential Source of Bioactive Compounds. *Res Biotechnol Environ Sci.* 2024;3(1):1-8.

Ghallab DS, Ibrahim RS, Mohyeldin MM, Shawky E. Marine algae: A treasure trove of bioactive anti-inflammatory compounds. *Mar Pollut Bull.* 2024;199:116023.

Ghany LMAA, Beshay BY, Youssef Moustafa AM, *et al.* Design, synthesis, anti-inflammatory evaluation, and molecular modelling of new coumarin-based analogs combined curcumin and other heterocycles as potential TNF- α production inhibitors via upregulating Nrf2/HO-1, downregulating AKT/mTOR signalling pathways and downregulating NF- κ B in LPS induced macrophages. *J Enzyme Inhib Med Chem.* 2023;38(1):2243551.

Ghosh S, Sarkar T, Pati S, Kari ZA, Edinur HA, Chakraborty R. Novel Bioactive Compounds From Marine Sources as a Tool for Functional Food Development. *Front Mar Sci.* 2022;9.

Goya L, Mateos R. Antioxidant and Anti-inflammatory Effects of Marine Phlorotannins and Bromophenols Supportive of Their Anticancer Potential. *Nutr Rev.* 2025;83(3):e1225-e1242.

GRAS. Re: GRAS Notice, No. GRN 000661. Available online: <https://www.fda.gov/downloads/Food/IngredientsPackagingLabeling/GRAS/NoticeInventory/ucm549588.pdf> (accessed on 10 February 2024).

Gupta SK, Ansari FA, Shrivastav A, Sahoo NK, Rawat I, Bux F. Dual role of *Chlorella sorokiniana* and *Scenedesmus obliquus* for comprehensive wastewater treatment and biomass production for bio-fuels. *J Clean Prod.* 2016;115:255-264.

Gutiérrez JM, Albulescu LO, Clare RH, *et al.* The Search for Natural and Synthetic Inhibitors That Would Complement Antivenoms as Therapeutics for Snakebite Envenoming. *Toxins.* 2021;13(7).

Gutiérrez JM, Calvete JJ, Habib AG, Harrison RA, Williams DJ, Warrell DA. Snakebite envenoming. *Nat Rev Dis Primer.* 2017;3(1):17063.

Gutiérrez JM, Escalante T, Rucavado A. Experimental pathophysiology of systemic alterations induced by *Bothrops asper* snake venom. *Toxicon.* 2009;54(7):976-987.

Gutiérrez JM, Rucavado A, Chaves F, Díaz C, Escalante T. Experimental pathology of local tissue damage induced by *Bothrops asper* snake venom. *Toxicon.* 2009;54(7):958-975.

Haggag YA, Elrahman AAA, Ulber R, Zayed A. Fucoidan in Pharmaceutical Formulations: A Comprehensive Review for Smart Drug Delivery Systems. *Mar Drugs*. 2023;21:112.

Hamidi M. Marine bacteria versus microalgae: who is the best for biotechnological production of bioactive compounds with antioxidant properties and other biological applications? *Mar Drugs*. 2019;29:28.

Hamza M, Knudsen C, Gnanathanan CA, *et al.* Clinical management of snakebite envenoming: Future perspectives. *Toxicon X*. 2021;11:100079.

Haque N, Parveen S, Tang T, Wei J, Huang Z. Marine Natural Products in Clinical Use. *Mar Drugs*. 2022; 20(8):528

Hosny S, Elshobary ME, El-Sheekh MM. Unleashing the power of microalgae: a pioneering path to sustainability and achieving the sustainable development goals. *Environ Sci Pollut Res*. 2025;32(29):17312-17342.

Hulatt CJ, Wijffels RH, Bolla S, Kiron V. Production of Fatty Acids and Protein by *Nannochloropsis* in Flat-Plate Photobioreactors. *PLOS ONE*. 2017;12(1):e0170440.

Idris S, Mishra A, Khushtar M. Phytochemical, ethanomedicinal and pharmacological applications of escin from *Aesculus hippocastanum* L. towards future medicine. *J Basic Clin Physiol Pharmacol*. 2020;31(5).

Inwongwan S, Sriwari S, Pumas C. Metabolomic Insights into the Adaptations and Biotechnological Potential of *Euglena gracilis* Under Different Trophic Conditions. *Plants*. 2025;14(11).

Jasim S. An Overview of Genus *Aesculus* L.: Ethnobotany, Phytochemistry, and Pharmacological Activities. *AACE Clin Case Rep*. 2021;7(1):1.

Kangra K, Kakkar S, Mittal V, *et al.* Incredible use of plant-derived bioactives as anticancer agents. *RSC Adv*. 15(3):1721-1746.

Kautsky L, Qvarfordt S, Schagerström E. *Fucus vesiculosus* adapted to a life in the Baltic Sea: impacts on recruitment, growth, re-establishment and restoration. *Bot Mar*. 2019;62(1):17-30.

Koh HSA, Lu J, Zhou W. Structure characterization and antioxidant activity of fucoidan isolated from *Undaria pinnatifida* grown in New Zealand. *Carbohydr Polym*. 2019;212:178-185.

Küçükkurt I, Ince S, Keleş H, *et al.* Beneficial effects of *Aesculus hippocastanum* L. seed extract on the body's own antioxidant defense system on subacute administration. *J Ethnopharmacol*. 2010;129(1):18-22.

Kumar A, Banjara RA, Aneshwari RK, Khan J, Bernarde PS. A comprehensive review on recent advances in the use of ethnomedicinal plants and their metabolites in snake bite treatment. *Front Pharmacol*. 2025;16.

Laing GD, Moura-da-Silva AM. Jararhagin and its multiple effects on hemostasis. *Toxicon*. 2005;45:987-996.

Larréché S, Chippaux JP, Chevillard L, *et al*. Bleeding and Thrombosis: Insights into Pathophysiology of Bothrops Venom-Related Hemostasis Disorders. *Int J Mol Sci*. 2021;22(17).

Lättig J, Böhl M, Fischer P, *et al*. Mechanism of inhibition of human secretory phospholipase A2 by flavonoids: rationale for lead design. *J Comput Aided Mol Des*. 2007;21(8):473-483.

Leandro A, Pereira L, Gonçalves AMM. Diverse applications of marine macroalgae. *Mar Drugs*. 2019;24:17.

Lewin M, Samuel S, Merkel J, Bickler P. Varespladib (LY315920) Appears to Be a Potent, Broad-Spectrum, Inhibitor of Snake Venom Phospholipase A2 and a Possible Pre-Referral Treatment for Envenomation. *Toxins (Basel)*. 2016;8(9):248. Published 2016 Aug 25.

Li C, Li P, Fu H, *et al*. A comparative study of the accumulation and detoxification of copper and zinc in *Chlamydomonas reinhardtii*: The role of extracellular polymeric substances. *Sci Total Environ*. 2023;871:161995.

Li S, Guo W, Zhang M, Zeng M, Wu H. Microalgae polysaccharides exert antioxidant and anti-inflammatory protective effects on human intestinal epithelial cells in vitro and dextran sodium sulfate-induced mouse colitis in vivo. *Int J Biol Macromol*. 2024;254:127811.

Liaqat A, Mallhi TH, Khan YH, Khokhar A, Chaman S, Ali M. Anti-Snake Venom Properties of Medicinal Plants: A Comprehensive Systematic Review of Literature. *Braz J Pharm Sci*. 2022;58:e191124.

Liu YT, Gong PH, Xiao FQ, *et al*. Chemical Constituents and Antioxidant, Anti-Inflammatory and Anti-Tumor Activities of *Melilotus officinalis* (Linn.) Pall. *Molecules*. 2018;23(2).

Liu YT, Gong PH, Xiao FQ, *et al*. Chemical Constituents and Antioxidant, Anti-Inflammatory and Anti-Tumor Activities of *Melilotus officinalis* (Linn.) Pall. *Molecules*. 2018;23(2).

Lizzul AM, Lekuona-Amundarain A, Purton S, Campos LC. Characterization of *Chlorella sorokiniana*, UTEX 1230. *Biology*. 2018;7(2).

Lomonte B, Díaz C, Chaves F, *et al.* Comparative characterization of Viperidae snake venoms from Perú reveals two compositional patterns of phospholipase A2 expression. *Toxicon* X. 2020;7:100044.

Lourenço-Lopes C, Fraga-Corral M, Soria-Lopez A, *et al.* Fucoxanthin's Optimization from *Undaria pinnatifida* Using Conventional Heat Extraction, Bioactivity Assays and In Silico Studies. *Antioxidants*. 2022;11(7).

Machado H, Machado JP, Alves C, *et al.* Exploring the Pharmacological Landscape of *Undaria pinnatifida*: Insights into Neuroprotective Actions and Bioactive Constituents. *Nutraceuticals*. 2025;5(3).

Mahadeswaraswamy YH, Kumar MS, Gowtham YJ, Nagaraju S, Girish KS, Kemparaju K. The Polyphenol 3, 4, 5 - Tri-hydroxy Benzoic Acid Inhibits Indian *Daboia russelli* Venom and Its Hemorrhagic Complex Induced Local Toxicity. *Curr Top Med Chem*. 11(20):2520-2530.

Mamede CCN, de Sousa Simamoto BB, da Cunha Pereira DF, de Oliveira Costa J, Ribeiro MSM, de Oliveira F. Edema, hyperalgesia and myonecrosis induced by Brazilian bothropic venoms: overview of the last decade. *Toxicon*. 2020;187:10-18.

Manikandan R, Parimalanandhini D, Mahalakshmi K, *et al.* Studies on isolation, characterization of fucoidan from brown algae *Turbinaria decurrens* and evaluation of it's in vivo and in vitro anti-inflammatory activities. *Int J Biol Macromol*. 2020;160:1263-1276.

Markou G, Kougia E, Arapoglou D. *Nannochloris* sp. JB17 as a Potential Microalga for Carbon Capture and Utilization Bio-Systems: Growth and Biochemical Composition Under High Bicarbonate Concentrations in Fresh and Sea Water. *Bioengineering*. 2024;11(12).

Matin M, Koszarska M, Atanasov AG, *et al.* Bioactive Potential of Algae and Algae-Derived Compounds: Focus on Anti-Inflammatory, Antimicrobial, and Antioxidant Effects. *Molecules*. 2024;29(19).

Matos RR, Ignotti E. Incidência de acidentes ofídicos por gêneros de serpentes nos biomas brasileiros. *Ciênc Saúde Coletiva*. 2020;25:2837-2846.

Maurya AK, Ahmed HA, DeWitt A, Shami AA, Misra SK, Pomin VH. Structure and Binding Properties to Blood Co-Factors of the Least Sulfated Galactan Found in the Cell Wall of the Red Alga *Botryocladia occidentalis*. *Mar Drugs*. 2024;22(2).

Melo PA, Nogueira-Souza PD, Romanelli MA, *et al.* Plant-Derived Lapachol Analogs as Selective Metalloprotease Inhibitors Against Bothrops Venom: A Review. *Int J Mol Sci*. 2025;26(9).

Mogha NG, Kalokora OJ, Amir HM, Kacholi DS. Ethnomedicinal plants used for treatment of snakebites in Tanzania – a systematic review. *Pharm Biol*. 2022;60(1):1925-1934.

Molina DA, Guerra-Duarte C, Souza DL, *et al.* Identification of a linear B-cell epitope in the catalytic domain of bothropasin, a metalloproteinase from *Bothrops jararaca* snake venom. *Mol Immunol.* 2018;104:20-26.

Monteiro-Machado M, Tomaz MA, Fonseca RJC, *et al.* Occurrence of sulfated fucose branches in fucosylated chondroitin sulfate are essential for the polysaccharide effect preventing muscle damage induced by toxins and crude venom from *Bothrops jararacussu* snake. *Toxicon.* 2015;98:20-33.

Moreira V, Leiguez E, Janovits PM, Maia-Marques R, Fernandes CM, Teixeira C. Inflammatory Effects of *Bothrops* Phospholipases A2: Mechanisms Involved in Biosynthesis of Lipid Mediators and Lipid Accumulation. *Toxins.* 2021;13(12).

Obluchinskaya ED, Pozharitskaya ON, Shikov AN. In Vitro Anti-Inflammatory Activities of Fucoidans from Five Species of Brown Seaweeds. *Mar Drugs.* 2022;20(10).

Offor BC, Muller B, Piater LA. A Review of the Proteomic Profiling of African Viperidae and Elapidae Snake Venoms and Their Antivenom Neutralisation. *Toxins.* 2022;14(11).

Ojuka P, Nyamato GS, Santos CBR, Kimani NM. A review and in silico screening of plant-derived snake venom/toxin inhibitors: ADMET, drug-likeness, and medicinal chemistry profiling. *PLoS Negl Trop Dis.* 2025;19(10):e0013579.

Olaoba OT, Karina dos Santos P, Selistre-de-Araujo HS, Ferreira de Souza DH. Snake Venom Metalloproteinases (SVMPs): A structure-function update. *Toxicon X.* 2020;7:100052.

Páez E, Galvis-Pérez Y, Pereañez JA, Preciado LM, Henao-Castañeda IC. *Coffea arabica* Extracts and Metabolites with Potential Inhibitory Activity of the Major Enzymes in *Bothrops asper* Venom. *Pharmaceuticals.* 2025;18(8).

Pandeirada CO, Maricato É, Ferreira SS, *et al.* Structural analysis and potential immunostimulatory activity of *Nannochloropsis oculata* polysaccharides. *Carbohydr Polym.* 2019;222:114962.

Panicucci G, Chiellini C, Sbrana C, Echeverría C, Guglielminetti L, Reyes TH. Phylogenetic Proximity vs. Environmental Adaptation: Exploring Photosynthetic Performances in Mediterranean and Andean Isolated Microalgae Under Different Light Intensities. *Phycology.* 2025;5(2).

Park JS, Han JM, Park SW, *et al.* Subcritical Water Extraction of *Undaria pinnatifida*: Comparative Study of the Chemical Properties and Biological Activities across Different Parts. *Mar Drugs.* 2024;22(8).

Patil L, Kaliwal BB. Microalga *Scenedesmus bajacalifornicus* BBKLP-07, a new source of bioactive compounds with in vitro pharmacological applications. *Bioprocess Biosyst Eng.* 2019;42(6):979-994.

Patwal T, Baranwal M. Scenedesmus acutus extracellular polysaccharides produced under increased concentration of sulphur and phosphorus exhibited enhanced proliferation of peripheral blood mononuclear cells. 3 Biotech. 2021;11(4):171.

Paula RC, Castro HC, Rodrigues CR, Melo PA, Fuly AL. Structural and pharmacological features of phospholipases A2 from snake venoms. Protein Pept Lett. 2009;16(8):899-907.

Peipei L, Qinghong Z, Yin C, Pengfei H, Junjie Z. Structure and anticoagulant activity of a galactoarabinan sulfate polysaccharide and its oligosaccharide from the green alga, Codium fragile. Int J Biol Macromol. 2024;279:135255.

Perazzo FF, Lima LM, Padilha M de M, Rocha LM, Sousa PJC, Carvalho JCT. Anti-inflammatory and analgesic activities of Hypericum brasiliense (Willd) standardized extract. Rev Bras Farmacogn. 2008;18:320-325.

Phinyo K, Pekkoh J, Peerapornpisal Y. Distribution and ecological habitat of Scenedesmus and related genera in some freshwater resources of Northern and North-Eastern Thailand. Biodiversitas J Biol Divers. 2017;18(3):1092-1099.

Piasecka A, Baier A. Metabolic and Proteomic Analysis of Chlorella sorokiniana, Chloroidium saccharofilum, and Chlorella vulgaris Cells Cultured in Autotrophic, Photoheterotrophic, and Mixotrophic Cultivation Modes. Molecules. 2022;27(15).

Pleşca-Manea L, Pârvu AE, Pârvu M, Taaş M, Buia R, Puia M. Effects of Melilotus officinalis on acute inflammation. Phytother Res. 2002;16(4):316-319.

Pleşca-Manea L, Pârvu AE, Pârvu M, Taaş M, Buia R, Puia M. Effects of Melilotus officinalis on acute inflammation. Phytother Res. 2002;16(4):316-319. doi:10.1002/ptr.875

Pott DM, Vallarino JG, Osorio S. Metabolite Changes during Postharvest Storage: Effects on Fruit Quality Traits. Metabolites. 2020;10(5).

Pozharitskaya ON, Shikov AN, Obluchinskaya ED, Vuorela H. The pharmacokinetics of fucoidan after topical application to rats. Mar Drugs. 2019;17:687.

Pradhan B, Nayak R, Patra S, Jit BP, Ragusa A, Jena M. Bioactive Metabolites from Marine Algae as Potent Pharmacophores against Oxidative Stress-Associated Human Diseases: A Comprehensive Review. Molecules. 2020;26(1).

Preciado LM, Comer J, Núñez V, Rey-Suárez P, Pereañez JA. Inhibition of a Snake Venom Metalloproteinase by the Flavonoid Myricetin. Molecules. 2018;23(10).

Qaderi MM, Martel AB, Strugnell CA. Environmental Factors Regulate Plant Secondary Metabolites. Plants. 2023;12(3).

Qin L, He M, Yang Y, *et al.* Anticoagulant-active sulfated arabinogalactan from *Chaetomorpha linum*: Structural characterization and action on coagulation factors. *Carbohydr Polym.* 2020;242:116394.

Queiroz GP, Pessoa LA, Portaro FCV, Furtado M de FD, Tambourgi DV. Interspecific variation in venom composition and toxicity of Brazilian snakes from *Bothrops* genus. *Toxicon.* 2008;52(8):842-851.

Quiroz S. Inhibitory effects of varespladib, CP471474, and their potential synergistic activity on *Bothrops asper* and *Crotalus durissus cumanensis* venoms. *Molecules.* 2022;6:8588.

Rahiman) FF (née, Africa C, Klaasen J, Fisher R. South African Medicinal Plants Traditionally Used for Wound Treatment: An Ethnobotanical Systematic Review. *Plants.* 2025;14(5).

Ralte L, Sailo H, Singh YT. Ethnobotanical study of medicinal plants used by the indigenous community of the western region of Mizoram, India. *J Ethnobiol Ethnomedicine.* 2024;20(1):2.

Rao S, Reghu N, Nair BG, Vanuopadath M. The Role of Snake Venom Proteins in Inducing Inflammation Post-Envenomation: An Overview on Mechanistic Insights and Treatment Strategies. *Toxins.* 2024;16(12).

Ravindran R, Rajauria G. Carbohydrates derived from microalgae in the food industry. *Cult Microalgae Food Ind.* 2021;2:127-146.

Resiere D, Kallel H, Florentin J, *et al.* *Bothrops* (Fer-de-lance) snakebites in the French departments of the Americas (Martinique and Guyana): Clinical and experimental studies and treatment by immunotherapy. *PLoS Negl Trop Dis.* 2023;17(2):e0011083.

Riaz M, Khalid R, Afzal M, *et al.* Phytobioactive compounds as therapeutic agents for human diseases: A review. *Food Sci Nutr.* 2023;11(6):2500-2529.

Rocha JEC, Gomides SC. The Amazonian snakebite burden: Unveiling seasonal dynamics in a region with tenfold higher incidence compared to the Brazilian average. *Trop Med Int Health.* 2024;29(12):1041-1050.

Romanazzi M, Filardi ETM, Pires GMM, *et al.* The Versatility of Serine Proteases from Brazilian *Bothrops* Venom: Their Roles in Snakebites and Drug Discovery. *Biomolecules.* 2025;15(2).

Romanelli MA, Guerrero TN, Brito E, *et al.* Plant-derived secondary metabolites against *Bothrops* envenomation: A review. *Toxicon.* 2025;258:108340.

Ruan J, Shi Z, Cao X, *et al.* Research Progress on Anti-Inflammatory Effects and Related Mechanisms of Astragalin. *Int J Mol Sci.* 2024;25(8).

Sachett J de AG, Marinho APS, Santos MM de O, Fan HW, Bernarde PS, Monteiro WM. When to think about a *Lachesis muta* envenomation in the Western Brazilian Amazon: Lessons from a case report. *Rev Soc Bras Med Trop*. 2022;55:e0027.

Sachetto ATA, Miyamoto JG, Tashima AK, de Souza AO, Santoro ML. The Bioflavonoids Rutin and Rutin Succinate Neutralize the Toxins of *B. jararaca* Venom and Inhibit its Lethality. *Front Pharmacol*. 2022;13. doi:10.3389/fphar.2022.828269

Saini MK, Capalash N, Kaur C, Singh SP. Targeted metabolic profiling indicates differences in primary and secondary metabolites in Kinnow mandarin (*C. nobilis* × *C. deliciosa*) from different climatic conditions. *J Food Compos Anal*. 2019;83:103278.

Salam U, Ullah S, Tang ZH, *et al*. Plant Metabolomics: An Overview of the Role of Primary and Secondary Metabolites against Different Environmental Stress Factors. *Life*. 2023;13(3).

Salvador GHM, Cardoso FF, Lomonte B, Fontes MRM. Inhibitors and activators for myotoxic phospholipase A(2)-like toxins from snake venoms - A structural overview. *Biochimie*. 2024;30:0300-9084.

Salvador GHM, Pinto ÊKR, Ortolani PL, *et al*. Structural basis of the myotoxic inhibition of the Bothrops pirajai PrTX-I by the synthetic Varespladib. *Biochimie*. 2023;207:1-10.

Sampaio BL, Edrada-Ebel R, Da Costa FB. Effect of the environment on the secondary metabolic profile of *Tithonia diversifolia*: a model for environmental metabolomics of plants. *Sci Rep*. 2016;6(1):29265.

Sanders CK, Hanschen ER, Biondi TC, *et al*. Phylogenetic analyses and reclassification of the oleaginous marine species *Nannochloris* sp. "desiccata" (Trebouxiophyceae, Chlorophyta), formerly *Chlorella desiccata*, supported by a high-quality genome assembly. *J Phycol*. 2022;58(3):436-448.

Serrano SMT. The long road of research on snake venom serine proteinases. *Toxicon*. 2013;62:19-26.

Sheikh NA. Phyto-physicochemical and high performance thin layer chromatography investigation of *Melilotus officinalis* Linn. *Int J Green Pharm IJGP*. 2017;11(02).

Shevel'yuhina A. Antioxidant and antimicrobial activity of microalgae of the Filinskaya Bay (Baltic Sea). *Plants Basel*. 2022;31:2264.

Shikov AN, Flisyuk EV, Obluchinskaya ED, Pozharitskaya ON. Pharmacokinetics of marine-derived drugs. *Mar Drugs*. 2020;18:557.

Shikov AN, Narkevich IA, Flisyuk EV, Luzhanin VG, Pozharitskaya ON. Medicinal plants from the 14th edition of the Russian Pharmacopoeia, recent updates. *J Ethnopharmacol*. 2021;268:113685.

Silva ACR, Pereira KKG, Critchley AT, Sanchez EF, Fuly AL. Potential utilization of a lambda carrageenan polysaccharide, derived from a cultivated, clonal strain of the red seaweed *Chondrus crispus* (Irish moss) against toxic actions of venom of *Bothrops jararaca* and *B. jararacussu* snakes. *J Appl Phycol*. 2020;32:4309-4320.

Silva ACRD, Duarte MER, Nosedá MD, *et al*. Potential Utilization of a Polysaccharide from the Marine Algae *Gayralia oxysperma*, as an Antivenom for Viperidae Snakebites. *Mar Drugs*. 2018;16(11).

Silva GM, Souza DHB, Waitman KB, *et al*. Design, synthesis, and evaluation of *Bothrops* venom serine protease peptidic inhibitors. *J Venom Anim Toxins Incl Trop Dis*. 2021;27:20200066.

de Silva HA, Ryan NM, de Silva HJ. Adverse reactions to snake antivenom, and their prevention and treatment. *Br J Clin Pharmacol*. 2016;81(3):446-452.

Simas Pereira Junior LC, Coriolano de Oliveira E, Valle Rorig TD, *et al*. The plant *Stryphnodendron adstringens* (Mart.) Coville as a neutralizing source against some toxic activities of *Bothrops jararacussu* snake venom. *Toxicon*. 2020;186:182-190.

Skriptsova AV, Shevchenko NM, Zvyagintseva TN, Imbs TI. Monthly changes in the content and monosaccharide composition of fucoidan from *Undaria pinnatifida* (Laminariales, Phaeophyta). *J Appl Phycol*. 2010;22(1):79-86.

Solano G, Ainsworth S, Sánchez A, *et al*. Analysis of commercially available snake antivenoms reveals high contents of endotoxins in some products. *Toxicon X*. 2024;211:100187-100189.

Solovchenko A, Lukyanov A, Solovchenko O, Didi-Cohen S, Boussiba S, Khozin-Goldberg I. Interactive effects of salinity, high light, and nitrogen starvation on fatty acid and carotenoid profiles in *Nannochloropsis oceanica* CCLA 804. *Eur J Lipid Sci Technol*. 2014;116(5):635-644.

South PM, Floerl O, Forrest BM, Thomsen MS. A review of three decades of research on the invasive kelp *Undaria pinnatifida* in Australasia: An assessment of its success, impacts and status as one of the world's worst invaders. *Mar Environ Res*. 2017;131:243-257.

Sowa-Borowiec P, Czernicka M, Jarecki W, Dżugan M. Sweet Clover (*Melilotus* spp.) as a Source of Biologically Active Compounds. *Molecules*. 2025;30(3).

Stengel DB, Connan S, Popper ZA. Algal chemodiversity and bioactivity: Sources of natural variability and implications for commercial application. *Biotechnol Adv*. 2011;29(5):483-501.

Su J, Guo K, Huang M, *et al*. Fucoxanthin, a Marine Xanthophyll Isolated From *Conticribra weissflogii* ND-8: Preventive Anti-Inflammatory Effect in a Mouse Model of Sepsis. *Front Pharmacol*. 2019;10.

Takolander A. Seasonal ecophysiology of *Fucus vesiculosus* (Phaeophyceae) in the Northern Baltic Sea. *Eur J Phycol.* 2023;58(3):300-314. doi:10.1080/09670262.2022.2110288

Tasoulis T, Isbister GK. A current perspective on snake venom composition and constituent protein families. *Arch Toxicol.* 2023;97(1):133-153.

Thumtecho S, Burlet NJ, Ljungars A, Laustsen AH. Towards better antivenoms: Navigating the road to new types of snakebite envenoming therapies. *J Venom Anim Toxins Incl Trop Dis.* 2023;29:20230057.

Toyama MH, Rogero A, Moraes LLF de, *et al.* Gallic Acid as a Non-Selective Inhibitor of α/β -Hydrolase Fold Enzymes Involved in the Inflammatory Process: The Two Sides of the Same Coin. *Pharmaceutics.* 2022;14(2).

Trujillo E, Monreal-Escalante E, Ramos-Vega A, Ângulo C. Macroalgae: Marine players in vaccinology. *Algal Res.* 2024;78:103392-103403.

Ullah A, Masood R, Ali I, *et al.* Thrombin-like enzymes from snake venom: Structural characterization and mechanism of action. *Int J Biol Macromol.* 2018;15:788-811.

Usov AI, Bilan MI, Ustyuzhanina NE, Nifantiev NE. Fucoidans of Brown Algae: Comparison of Sulfated Polysaccharides from *Fucus vesiculosus* and *Ascophyllum nodosum*. *Mar Drugs.* 2022;20(10).

Vale LHF, Mendes MM, Fernandes RS, *et al.* Protective effect of schizolobium parahyba flavonoids against snake venoms and isolated toxins. *Curr Top Med Chem.* 2011;11(20):2566-2577.

Vasilakis G, Marka S, Ntzouvaras A, *et al.* Wound Healing, Antioxidant, and Antiviral Properties of Bioactive Polysaccharides of Microalgae Strains Isolated from Greek Coastal Lagoons. *Mar Drugs.* 2025;23(2).

Wang L, Oliveira C, Li Q, *et al.* Fucoidan from *Fucus vesiculosus* Inhibits Inflammatory Response, Both In Vitro and In Vivo. *Mar Drugs.* 2023;21(5).

Wang Y, Chu T, Pan X, Bian Y, Li J. Escin ameliorates inflammation via inhibiting mechanical stretch and chemically induced Piezo1 activation in vascular endothelial cells. *Eur J Pharmacol.* 2023;956:175951.

Wang Y, Han X, Wan X, Niu F, Zhou C. β -Escin: An Updated Review of Its Analysis, Pharmacology, Pharmacokinetics, and Toxicity. *Am J Chin Med.* 2023;51(08):2095-2120.

Wang Y, Xing M, Cao Q, Ji A, Liang H, Song S. Biological Activities of Fucoidan and the Factors Mediating Its Therapeutic Effects: A Review of Recent Studies. *Mar Drugs.* 2019;17(3).

Weinhold A, Döll S, Liu M, *et al.* Tree species richness differentially affects the chemical composition of leaves, roots and root exudates in four subtropical tree species. *J Ecol.* 2022;110(1):97-116.

Xiao R, Zheng Y. Overview of microalgal extracellular polymeric substances (EPS) and their applications. *Biotechnol Adv.* 2016;34(7):1225-1244.

Xiao Y, Xia W, Yang Z, Zhou J, Luo J. Advances in the analysis and application of metabolites from tropical plants. *Curr Opin Plant Biol.* 2025;85:102728.

Yan L, Zhang Y, Dai D, Li Y, Xiao W. Pharmacological Activities of *Aesculus hippocastanum* and its Bioactive Constituents in Hemostasis: A Mechanistic Review. *Pharmacogn Mag.* 2025;21(3):727-737.

Yan Y, Lv Q, Zhou F, *et al.* Discovery of an effective anti-inflammatory agent for inhibiting the activation of NF- κ B. *J Enzyme Inhib Med Chem.* 2023;38(1):2225135.

Ye S, Gao L, Zhao J, An M, Wu H, Li M. Simultaneous wastewater treatment and lipid production by *Scenedesmus* sp. HXY2. *Bioresour Technol.* 2020;302:122903.

Yesli K, Crayn D, Ritmejerlytė E, Wangchuk P. Plant Secondary Metabolites Produced in Response to Abiotic Stresses Has Potential Application in Pharmaceutical Product Development. *Molecules.* 2022;27(1).

Yildiz İ, Başar Y, Erenler R, Alma MH, Calimli MH. A phytochemical content analysis, and antioxidant activity evaluation using a novel method on *Melilotus officinalis* flower. *South Afr J Bot.* 2024;174:686-693.

Zeng J, Luan F, Hu J, *et al.* Recent research advances in polysaccharides from *Undaria pinnatifida*: Isolation, structures, bioactivities, and applications. *Int J Biol Macromol.* 2022;206:325-354.

Zhao J, Liu G, Shi X, Huang C. Combination of Rhamnetin and RXP03 Mitigates Venom-Induced Toxicity in Murine Models: Preclinical Insights into Dual-Target Antivenom Therapy. *Toxins.* 2025;17(6).

Zhao Y, Han C, Wu Y, *et al.* Extraction, structural characterization, and antioxidant activity of polysaccharides from three microalgae. *Sci Total Environ* 2024;931:172567.

APÊNDICE

Artigos publicados durante o desenvolvimento desta tese, não incluído nos capítulos principais.

1. Alkaloids from Siparuna (Siparunaceae) are predicted as the inhibitors of proteolysis and plasma coagulation caused by snake venom and potentially counteract phospholipase A2 activity of *Bothrops jararaca*. Fernandes DA, Gomes BA, Mendonça SC, Pinheiro CC, Sanchez EOF, Leitão SG, Fuly AL, Leitão GG. J Ethnopharmacol. 2024 Oct 5;332:118349. doi: 10.1016/j.jep.2024.118349\



Alkaloids from *Siparuna* (Siparunaceae) are predicted as the inhibitors of proteolysis and plasma coagulation caused by snake venom and potentially counteract phospholipase A₂ activity of *Bothrops jararaca*

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ABSTRACT

Ethnopharmacological relevance: Snakebite envenomation (SBE) is the world's most lethal neglected tropical disease. *Bothrops jararaca* is the species that causes the greatest number of SBEs in the South and Southeastern of Brazil. The main symptoms are local (inflammation, edema, hemorrhage, and myonecrosis) and systemic (hemorrhage, hemostatic alterations with consumptive coagulopathy, and death) effects. Species of the genus *Siparuna*, Siparunaceae, are used in folk and traditional medicine to treat SBE. However, limited information is available concerning Brazilian *Siparuna* species against SBE.

Aim of the study: To investigate the correlation between the compounds present in the extracts of five *Siparuna* species as potential agents against proteolytic activity, plasma coagulation, and phospholipase A₂ (PLA₂) activity caused by *B. jararaca* venom, using data obtained by UHPLC-MS/MS, biological activity, and multivariate statistics.

Materials and methods: The ethanol extracts from leaves of *S. ficoides*, *S. decipiens*, *S. glycyarpa*, *S. reginae*, and *S. cymosa* were fractionated by liquid-liquid extraction using different solvents of increasing polarity (hexane, dichloromethane, ethyl acetate, and *n*-butanol), affording their respective extracts, totaling 25 samples that were assayed through *in vitro* plasma coagulation and proteolytic activity assays. Moreover, the extracts were analyzed by UHPLC-MS/MS, using electrospray ionization (ESI) and atmospheric-pressure chemical ionization (APCI) in negative and positive ionization modes. The data was processed in MZmine v. 2.53 and evaluated by multivariate statistical tests (PLS) using the software UnscramblerX v. 10.4. These data were also used to build molecular networks (GNPS), and some ions of interest could be annotated using the library of molecules on the GNPS platform.

Abbreviations: APCI, Atmospheric pressure ionization; BuOH, butanol; DCM, dichloromethane; ESI, Electrospray ionization; EtOAc, ethyl acetate; GNPS, Global Natural Product Social Molecular Networking; Hex, hexane; MeOH, methanol; MS/MS, Tandem mass spectrometry; PLA₂, phospholipase A₂; PLS, Partial least squares; SC, *Siparuna cymosa*; SD, *Siparuna decipiens*; SF, *Siparuna ficoides*; SG, *Siparuna glycyarpa*; SR, *Siparuna reginae*; SBE, Snakebite envenomation; SEM, Standard Error Mean; SVMPs, Snake venom metalloproteinases; SVSPs, Snake venom serine proteinases; UHPLC, Ultra-high performance liquid chromatography; WHO, World Health Organization.

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Results: A total of 19 extracts inhibited *B. jararaca*-induced plasma coagulation, with emphasis on *S. cymosa* and *S. reginae* (800 s). The inhibition of the proteolytic activity was also promising, ranging from 16% (*S. glycyarpa*) to 99% (*S. cymosa*, *S. decipiens*, and *S. reginae*). In addition, most extracts from *S. cymosa* and *S. reginae* inhibited 70–90% of PLA₂ activity. Based on data from positive mode APCI analyses, it was possible to obtain a statistic model with reliable predictive capacity which exhibited an average R² of 0.95 and a Q² of 0.88, indicating a robust fit. This process revealed five ions, identified as the alkaloids: coclaurine (1), stepholidine (2) *O*-methylyllopinine (3), nornantenine (4) and lauroilsine (5). This is the first study to evidence the potential antivenom of alkaloids from *Siparuna* species.

Conclusions: Altogether, our results give support to the popular use of *Siparuna* extracts in SBE accidents, suggesting their potential as an alternative or complementary strategy against envenoming by *B. jararaca* venom. The predicted ions in the chemometric analysis for the assayed activities can also be correlated with the blocking activity and encourage the continuation of this study for possible isolation and testing of individual compounds on the used models.

1. Introduction

Snakebite envenomation (SBE) is an important public health challenge worldwide, being included in the list of neglected tropical diseases by the World Health Organization (WHO). It is estimated to affect close to 2.7 million people globally every year, especially in rural areas of tropical and subtropical countries (Binorkar and Jani, 2012; Feitosa et al., 2015; Adrião et al., 2022). According to an estimate of WHO, around 100,000 people annually die and 400,000 suffer disability due to SBE (Schneider et al., 2021a; Liaqat et al., 2022).

Most snakebite cases occur in Africa, Asia, and Latin America. Brazil, with its large forest areas and great variety of venomous snakes, is among the countries with the highest risk of SBE in South America, with about 26,000–29,000 cases per year, followed by Venezuela (7,000), Colombia (3,000), Ecuador (1,500), Peru (1,500), and Bolivia (1,000) (Roriz et al., 2018; Schneider et al., 2021a, 2021b).

Species of the genus *Bothrops* are responsible for 90% of the SBE in Latin America. Most of the Brazilian snakebite accident notifications are related to specimens of the Viperidae family, which comprise the genera *Bothrops* (known as ‘jararaca’) (83.3%), followed by *Crotalus* (‘cascavel’) (8.5%), and *Lachesis* (‘surucucu-pico-de-jaca’) (3.4%). In the Elapidae family, the genus *Micrurus* (true ‘coral’) is responsible for 0.8% of these accidents (Feitosa et al., 2015; Farias et al., 2018; Schneider et al., 2021a; Konrath et al., 2022). Clinical manifestations of bothropic accidents are characterized by prominent local and systemic effects in the victims, such as inflammation, edema, hemorrhage, renal or cardiac failure, muscle necrosis, and eventually death (Otero et al., 2000; Jorge et al., 2017).

Bothrops jararaca (Wied-Neuwied, 1824) is the species that causes the greatest number of snakebites in South and Southeastern Brazil, around 60% in areas at sea level as well as over 1000 m altitude, or even in large cities (Moura et al., 2015; Machado et al., 2016; Jorge et al., 2017; Farias et al., 2018). The venom of *B. jararaca* is a mixture of active protein and peptides, including snake venom metalloproteinases (SVMPs), snake venom serine proteinases (SVSPs), phospholipases A₂ (PLA₂), L-amino acid oxidases (LAAOs), hyaluronidases, and bradykinin potentiators among others, and some of them are responsible for the most toxic activities (Kisaki et al., 2021).

The treatment recommended by WHO and the Ministry of Health of Brazil is performed by the intravenous injection of commercial antivenoms that are produced by hyperimmunization of equines. Antivenom has been available for 120 years and is highly effective in preventing mortality, especially if administered early in an adequate dose (Estevo-Costa et al., 2016). However, the utilization of antivenoms to treat morbidity and disability as consequences of snakebites has been limited by multiple factors, including adverse effects profile (fever or anaphylactic shock), stringent storage conditions, specificity issues, risk of immunological reactions, and high cost of production (Félix-Silva et al., 2017; Vásquez et al., 2013; Liaqat et al., 2022).

The inefficacy of antivenoms to neutralize the local effect of SBE may have serious consequences for recovery of victims, affecting the quality of life. Thus, seeking alternative treatments for SBE deserves deeper investigation. Over the years, many attempts have been made to find snake venom’s antagonists from plants, since these organisms produce many compounds with unique and intriguing chemical structure. Some of these compounds may be useful as supplement or alternative treatments of SBE, and with less adverse events, better tolerability, and safety profile (Binorkar and Jani, 2012; Félix-Silva et al., 2017). The phytochemical analysis of various plants has revealed that phenols, alkaloids, triterpenoids, and steroids possess promising effects against the toxic activities caused by snake venom (Janardhan et al., 2014; Singh et al., 2017; Liaqat et al., 2022).

Currently, modern mass spectrometry techniques are widely employed for the identification and structural analysis of compounds in complex mixtures (Popov et al., 2022). In parallel, Computer-Aided Drug Design has emerged as a pivotal method for integrating the intricate complexities of biological systems with the predictive power of computational algorithms (Khan et al., 2020; Mostofa et al., 2023; Niazi and Mariam, 2024). Utilizing this data, fingerprint analysis can forecast which metabolites are likely contributors to specific biological activities even before isolating the target compounds. This approach complements experimental analysis and aids in the rational design of novel therapeutic agents (Darwish et al., 2022; Sweilam et al., 2022).

Siparuna species, Siparunaceae family, thrive in the tropical and subtropical regions of the Southern hemisphere. These plants hold significance in folk and traditional medicine due to their therapeutic properties, as evidenced by documented use in treating various illnesses (Leitão et al., 1999, 2000; Silva et al., 2021). In remote areas where access to prompt medical care is limited, patients often experience exacerbated conditions, leading to fatalities. Such circumstances prompt at-risk populations to develop community-based solutions to address their healthcare needs. For instance, in indigenous communities in Colombia, macerated leaves of *Siparuna gesnerioides* (Kunth) A.DC. are utilized to combat SBE (Vásquez et al., 2013, 2015; Félix-Silva et al., 2017), while decoctions or macerations of leaves, branches, and stems of *Siparuna thecaphora* (Poepp. & Endl.) A.DC. are employed in the form of drinks, external baths, and poultices to prevent SBE-related hemorrhaging (Otero et al., 2000; Gomes et al., 2010; Binorkar and Jani, 2012). Moreover, marginalized communities in the Amazon region of Brazil commonly rely on *Siparuna apiosyce* (Mart.) A.DC., (synonym: *Siparuna brasiliensis*), *Siparuna guianensis* Aublet and *Siparuna ficoides* S. S. Renner and Hausner, locally known as ‘Limão bravo’, ‘Negramina’, and ‘Yekuna’ respectively, in traditional medicine to mitigate the effects of snake bites (Leitão et al., 2000; Renner and Hausner, 2005; Rieder, 2013).

Based on these reports of *Siparuna* species against snakebite, and given the lack of identification of compounds in these plants responsible for such antivenom effect, this study aimed to investigate the correlation

between the chemical compounds present in the extracts of *Siparuna ficoides* Renner & Hausner (originally identified as *S. cristata* Poepp. & Endl. A.DC.), *Siparuna decipiens* (Tul.) A.DC., *Siparuna glycyarpa* (Ducke) S.S. Renner & Hausner, *Siparuna reginae* (Tul.) A. DC., and *Siparuna cymosa* Tolm (originally identified as *S. sarmentosa* Perkins) (INPA, 2024), as potential candidates against the proteolytic and plasma coagulant by ultra-high performance liquid chromatography tandem mass spectrometry (UHPLC-MS/MS), and multivariate statistics. Furthermore, it was also aimed to evaluate the potential of the extracts in inhibiting PLA₂ activity, allowing inference of possible mechanisms of action of the extracts.

2. Materials and Methods

2.1. Plant material

Leaves of *S. ficoides* (SF), *S. decipiens* (SD), *S. glycyarpa* (SG), *S. reginae* (SR), and *S. cymosa* (SC), were collected at Reserva Adolpho Ducke/Manaus/Brazil, in August 2015. Voucher specimens are deposited at Instituto Nacional de Pesquisas da Amazônia (INPA) herbarium (Manaus, Brazil) under the registration numbers INPA: 269731, 269734, 269732, 269735, and 269733, respectively. The plants names (species) have been checked with worldfloraonline.org and reflora.jbrj.gov.br website, in February 2024. This work was authorized by the Directing Council of Heritage (Conselho de Gestão do Patrimônio Genético, CGEN) by the authorization Genetic A3C04CB.

Dried and ground leaves were extracted by percolation with ethanol 96 °GL. Part of the crude ethanol extract (CEE) was fractionated by liquid-liquid extraction between water-methanol 7:3 (v/v) and hexane (HEX), dichloromethane (DCM), ethyl acetate (EtOAc) and butanol (BuOH), affording their respective extracts. This method was performed according to Leal et al. (2021).

2.2. Venom of *B. jararaca*

Lyophilized *B. jararaca* snake venom was kindly supplied by Ezequiel Dias Foundation, Belo Horizonte, Minas Gerais, Brazil. The venom was kept at −20 °C, and then dissolved in physiological saline at 1 mg/mL concentration to perform the toxic assays. Snake venom collection was conducted under the authorization of the Brazilian National System for Genetic Heritage Management and Associated Traditional Knowledge (SISGEN), process number A39CTRI 04E.

2.3. Antiproteolytic activity of *Siparunas* extracts

Proteolytic activity of *B. jararaca* venom was determined according to the method of Garcia et al. (1978), using azocasein as substrate (0.2%, w/v, in 20 mM Tris-HCl, 8 mM CaCl₂, pH 8.8). Aliquots of *B. jararaca* venom (5–50 µg mL^{−1}) were incubated with 0.4 mL azocasein at 37 °C for 90 min in a total volume of 1.2 mL. The enzymatic reaction was stopped by the addition of trichloroacetic acid (5%, v/v, final concentration). The tubes were centrifuged at 15,000 rpm for 2 min and the supernatant was removed and mixed with 2N NaOH. After 10 min, tubes were read at the absorbance of 420 nm. An Effective Concentration (EC) was defined as the amount of venom (µg.mL^{−1}) able to produce a variation of approximately 0.2 OD units at 420 nm. For inhibition experiments, one EC of *B. jararaca* venom (24 µg mL^{−1}) was incubated for 30 min at 25 °C with the *Siparuna* extracts (240 µg mL^{−1}), saline (S) or with 0.9% DMSO (DM), final concentration. Then, an aliquot of each mixture was added to medium reaction, and proteolytic activity was performed, as described. The negative controls of such activity contained solely the *Siparuna* extracts or solvents, in the absence of venom. Then, the anti-proteolytic activity was determined.

2.4. Anticoagulant activity of *Siparunas* extracts

A pool of citrated normal human plasma (diluted with an equal volume of saline) from healthy volunteers of the hospital of Federal Fluminense University (UFF) was mixed with *B. jararaca* venom (2.5–70 µg mL^{−1}) and the coagulation time was assessed using a digital coagulometer (Amelung, model KC4A, Labcon, Germany). The concentration of venom (20 µg mL^{−1}) able to clot plasma approximately in 60 s was designated as the minimum coagulant concentration (MCC). To evaluate the inhibitory effect of the plant, *Siparuna* extracts (200 µg mL^{−1}) were incubated for 30 min at 25 °C with one MCC of venom (20 µg mL^{−1}) or with saline or with 0.9% DMSO, final concentration. Then, an aliquot of each mixture was added to plasma and coagulation time was registered in the coagulometer. The negative controls were performed in parallel by mixing *Siparuna* extracts or solvents with plasma, in the absence of *B. jararaca* venom.

2.5. Antiphospholipase A₂ activity of *Siparunas* extracts

PLA₂ activity of *B. jararaca* venom was measured according to Marinetti (1965) and Castro-Pinheiro et al. (2024). *B. jararaca* venom (40 µg mL^{−1}) was incubated with solvents (saline or 0.9% DMSO) or with *Siparuna* extracts (400 µg mL^{−1}) for 30 min at 25 °C, and then an aliquot of this mixture was added to medium containing 580 µL saline, 25 µL sodium taurocholate 0.4%, 25 µL Tris-HCl (200 mM), pH 7.5, 20 µL CaCl₂ (0.5 M). The enzymatic reaction was initiated by the addition of the egg yolk substrate, and after 30 min of reaction at 37 °C, the tubes were read at the absorbance of 740 nm. PLA₂ activity (100%) of the venom was determined using venom incubated with solvents in the absence of *Siparuna* extracts, and negative controls of PLA₂ activity contained only solvents (saline and DMSO).

2.6. Statistical analysis

Results are presented as means ± S.E.M. The statistical significance of variance using ANOVA and Dunnett post-hoc were performed using GraphPad Prism® software. *P* values of <0.05 were considered statistically significant.

2.7. UHPLC-MS/MS analysis

The extracts (2 µg mL^{−1}) were analyzed in a UHPLC Dionex Ultimate 3000 (ThermoFischer Scientific, Waltham, MA, USA) coupled to a LCQfleet (ThermoFischer Scientific, Waltham, MA, USA). The analyzes were performed using a Acquity BEH C18 column (1.7 µm, 2.1 × 100 mm, 100 Å) with a flow rate of 0.45 mL min^{−1} according to the elution gradient: (0 min, 5%B), (20 min, 95%B), (25 min, 95%B), (31 min, 100%B), (37 min, 95%B) and (5 min, 0%B). The mass spectrometer, equipped with atmospheric pressure chemical ionization (APCI) and electrospray (ESI) was operated in positive and negative ionization modes. MS spectra were acquired with a range of *m/z* 100–1000 and the normalized collision energy of the collision-induced dissociation (CID) cell was set at 35 eV.

2.8. Chemometric and molecular networking analysis

The LC-MS/MS data were converted to the.mzML format using the Proteowizard toolkit (MSConvert v. 3.02) and processed utilizing MZmine v. 2.53. Subsequently, the MZmine results were uploaded to the Global Natural Products Social Molecular Networking (GNPS) online platform, together with the openly accessible converted files (.mzML) and metadata table. Default standards for unity resolution data were applied for both data mining and molecular network construction processes. The molecular network data were then downloaded and analyzed in Cytoscape v. 3.9.1. The processed chemical and biological data were exported to UnscramblerX software version 10.4.

The study aimed to predict potential active compounds inhibiting both the proteolytic and coagulant activities of the venom of *B. jararaca*. Chemometric analyses were conducted, and a Partial Least Squares Regression model was utilized to correlate LC-MS/MS (ESI and APCI [+/-]) data with inhibitory activities (multitarget - antiproteolytic and anticoagulant). The PLS regression identified sample patterns, classifying them into two binary groups (active [1] and non-active [-1]) based on average activity values. The predictions of interest in chemometric analysis were annotated using a custom database, library search on the GNPS platform, and fragmentation pattern analysis.

3. Results and discussion

Plant extracts are widely used in traditional medicine as therapy for snakebite, and especially in tropical regions where plant resources are diverse and plentiful (Nishijima et al., 2009; Carvalho et al., 2013; Dellacassa et al., 2014; Oliveira et al., 2014). In this scenario, various plant species are used as local first aid for treating snakebite accidents in Brazil, and some of them can effectively inhibit lethality, neurotoxicity, hemorrhage, and venom enzyme activities (Konrath et al., 2022).

Siparuna is recognized as the major and most complex genus of the Siparunaceae family with several species with medicinal and economical interest. These plants produce a variety of bioactive compounds, being flavonoids and alkaloids the most characteristic (Costa et al., 2013; Leal et al., 2021). Previous investigations of extracts from plants of this genus have reported antispasmodic, antitumoral (Renner and Hausner, 2005; dos Santos et al., 2023), and antiviral activities (Leal et al., 2021a; Leal et al., 2021b; Leal et al., 2022).

Based on traditional uses of plants of this genus to treat snakebites (Leitão et al., 2000; Renner and Hausner, 2005; Rieder, 2013), CEE from the leaves of five *Siparuna* species (SC, SD, SF, SG, and SR) and their four extracts obtained after liquid-liquid extraction (HEX, DCM, EtOAc, and BuOH) were assayed against the proteolytic, coagulant, and PLA₂ activity of *B. jararaca* venom.

3.1. Antiproteolytic activity of *Siparuna* extracts

Fig. 1 shows that the inhibition of the *Siparuna* extracts on the proteolytic activity of *B. jararaca* venom occurred with different inhibitory percentages. The HEX extracts from SF, SD, and SG, as well as the CEE and BuOH from SF did not inhibit proteolysis (Fig. 1); while all other *Siparunas* extracts affected proteolysis from 30 to 96 %. Overall, it seemed that the extracts of SF inhibited less the proteolysis caused by *B. jararaca* venom, and SC inhibited more efficiently, near to 95%. The

DCM and EtOAc extracts from SD inhibited proteolysis with values of 90%, as well (Fig. 1).

All the lowest polarity hexane extracts of *Siparuna* species (except the one from SC – 40% and SR – 34%) did not inhibit proteolysis. The extracts CEE, EtOAc, and BuOH from SG inhibited proteolysis at 50, 70, and 80%, respectively. It is worth saying that only the five extracts from SR (CEE = 59%, HEX = 34%, DCM, EtOAc, and BuOH = 85% each) and SC (CEE = 94%, HEX = 40%, DCM = 48%, EtOAc, and BuOH = 96% each) inhibited proteolysis; On the other hand, at least one or two extracts of SF, SD, and SG, failed to inhibit proteolysis caused by *B. jararaca* venom (Fig. 1). All *Siparuna* extracts, in the absence of *B. jararaca* venom did not hydrolyze azocasein, and, thus, were devoid of proteolytic activity (data not shown).

Natural compounds like terpenes, phenolics, and alkaloids, commonly found in medium and high polarity extracts (DCM, EtOAc, and BuOH) constitute the predominant group of compounds that are responsible for the inhibition of toxic activities of snake venoms. Therefore, they may provide therapeutic molecular models against SBE to supplement the conventional serum therapy or for the development of a novel antivenom therapeutic (Samy et al., 2012; Carvalho et al., 2013; Baraldi et al., 2016; Adrião et al., 2022).

A study carried by Oliveira et al. (2016) showed that the extracts of *Erythroxylum ovalifolium* Peyr. and *Erythroxylum subsessile* (Mart.) O.E. Schulz, from where terpenes and flavonoids were isolated inhibited the proteolytic activity of *Lachesis muta* (Lineu, 1766) venom, but with different potencies. The DCM and BuOH extracts of *E. ovalifolium* and aqueous extracts of *E. subsessile* inhibited more than 80% of the proteolytic activity, while the CEE, EtOAc, and aqueous extracts of *E. ovalifolium*, and the DCM extract of *E. subsessile* inhibited proteolysis from 40 to 75%. In contrast, the HEX and EtOAc extracts of *E. subsessile* inhibited proteolysis below 10%.

The DCM, HEX, and EtOAc extracts from the marine seaweed *Placodium brasiliense* Grev. M. Howe & W.R. Taylor inhibited proteolytic activity of *B. jararaca* venom, so that these extracts may be interacting with proteases of such venom. In contrast, the extract of this seaweed prepared in hydroalcoholic solution failed to inhibit proteolysis (Silva et al., 2015). The fruit extracts (aqueous, methanol-MeOH, and acetone) from *Clusia fluminensis* Planch & Triana exhibited the most prominent antiproteolytic activity on proteolysis induced by *B. jararaca* venom, followed by extracts from the stems (MeOH) and leaves (aqueous). Moreover, the aqueous extract of the fruits of *C. fluminensis* inhibited more efficiently the proteolysis caused by the venom of *B. jararacus* (da Silva et al., 2019; Faria et al., 2023).

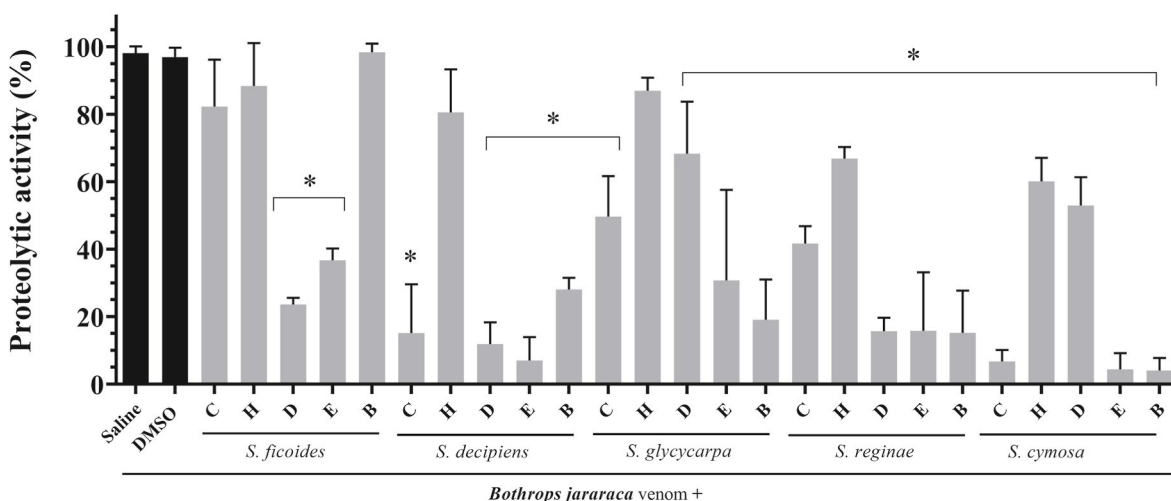


Fig. 1. Inhibition of the proteolytic activity of *B. jararaca* venom by *Siparuna* extracts. C: Crude ethanol extract; H: HEX extract; D: DCM extract; E: EtOAc extract; B: BuOH extract. Data are expressed as means $\pm$ S.E.M (n = 6). *, means with $P < 0.05$ when compared to *B. jararaca* venom + Saline or + DMSO.

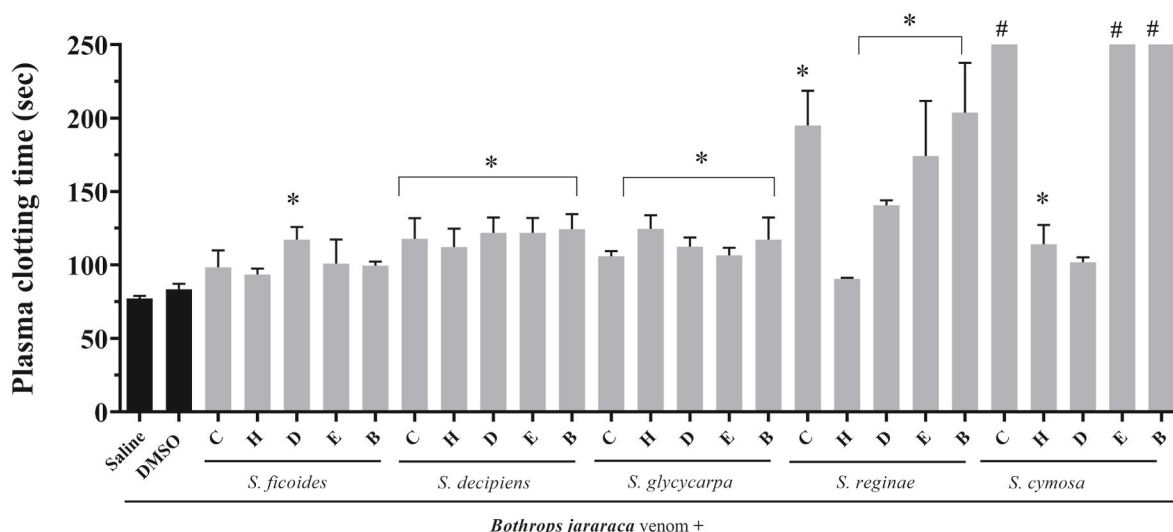


Fig. 2. Effect of *Siparuna* extracts on plasma clotting time induced by *B. jararaca* venom. C: Crude ethanol extract; H: HEX extract; D: DCM extract; E: EtOAc extract; B: BuOH extract. Data are expressed as means $\pm$ S.E.M (n = 6). #, means that plasma did not clot until 800 s of observation. *, means with $P < 0.05$ when compared to *B. jararaca* venom + Saline or + DMSO.

3.2. Anticoagulant effect of *Siparuna* extracts

As shown in Fig. 2, most of the extracts inhibited plasma coagulation induced by the venom of *B. jararaca*, except for the extracts from SF (CEE, HEX, EtOAc, and BuOH), SG (CEE), SR (HEX), and SC (DCM). However, the extracts CEE, EtOAc, and BuOH of SC fully inhibited coagulation of plasma, since no plasma coagulation occurred during 800 s which was the maximal period of observation of coagulation (Fig. 2).

Moreover, the five extracts from SD (CEE, HEX, DCM, EtOAc, and BuOH), four extracts from SG (HEX, DCM, EtOAc, and BuOH) and three from SR (DCM, EtOAc, and BuOH) inhibited plasma coagulation. On the other hand, only one extract from SF (DCM) and SC (HEX) inhibited coagulation (Fig. 2). Thus, these extracts from *Siparuna* would prevent bleeding or vascular disturbances caused by *B. jararaca* bites and, undoubtedly the polarity of extracts interferes in the effectiveness of inhibition. All the *Siparuna* extracts, in the absence of *B. jararaca* venom did not clot plasma, and, thus, were devoid of procoagulant effect (data not shown).

This profile was also observed in a previous study by da Silva et al. (2019), where the plasma coagulation caused by *B. jararaca* venom was attenuated by the crude extracts of *C. fluminensis*; while the highest anticlotting activities were recorded for acetone and MeOH extracts of the stems of *Clusia*. The fruit (acetone, MeOH, aqueous) and flower (HEX) extracts had similar anticlotting properties (da Silva et al., 2019). Regarding the *B. jararacussu* venom-induced coagulation, the aqueous extracts of the leaves, stems, and fruits of *C. fluminensis* inhibited coagulation, as well (da Silva et al., 2019; Faria et al., 2023). Terpenoids, sterols, and phenolic compounds were isolated and identified in the HEX and MeOH extracts of this species (Mazza et al., 2019; Camara et al., 2018; Guimarães et al., 2021).

The DCM, BuOH or water extracts of *E. ovalifolium*, as well as the HEX, EtOAc, or DCM extracts of *E. subsessile* did not inhibit the coagulation activity of *L. muta* venom. In contrast, the CEE or EtOAc extracts of *E. ovalifolium* and the CEE, BuOH, or aqueous extracts of *E. subsessile* significantly delayed plasma coagulation induced by *L. muta* venom. Additionally, neither the extracts nor solvents induced coagulation of plasma (Oliveira et al., 2016).

A study carried by Silva et al. (2015) showed that the DCM extract of the seaweed *P. brasiliense* inhibited coagulation induced by *B. jararaca* venom, preventing fibrin production. In contrast, the hydroalcoholic extract failed to inhibit coagulation provoked by *B. jararaca* venom. The

studies with plants or seaweed extracts obtained with solvents of different polarities, demonstrate the great chemical variety and broad profile of biological activity of such extracts against proteolytic and coagulant activities caused by snake venoms.

3.3. Antiphospholipase A₂ effect of *Siparuna* extracts

The extracts from SF (HEX, DCM and BuOH) and SC (HEX) failed to inhibit significantly PLA₂ activity of *B. jararaca* venom. All extracts from SD, as well as the extracts from SF (CEE and EtOAc), and SR (HEX) inhibited 30%–40%. On the other hand, SG (CEE, HEX and DCM) and SC (DCM) extracts inhibited around 60%, while those from SR (CEE, DCM, EtOAc and BuOH), and SC (CEE, EtOAc and BuOH) inhibited 70–90% the PLA₂ activity of *B. jararaca* venom (Fig. 3). These results corroborate what was previously observed for the anticoagulant and antiproteolytic activities of *S. cymosa* and *S. reginae*, highlighting their EtOAc and BuOH extracts.

The literature widely acknowledges that snake venoms comprise a complex blend of active proteins and peptides, particularly the families of SVMPs, SVSPs, and PLA₂, which elicit a wide range of toxic effects such as disrupting blood coagulation, hemolysis, edema and muscle necrosis (Leitão et al., 1996; Kasaki et al., 2021). Previous works with antivenom plant extracts also address inhibition of PLA₂ as target. Extracts from leaves of *Azima tetraacantha* Lam. (Petroleum ether, HEX, chloroform, EtOAc, MeOH, and water) (100 μ g mL⁻¹) rich in flavonoids, phytosterols and proteins, were tested against different toxic enzymes from the venoms of *Bungarus caeruleus* (Schneider, 1801) and *Daboia russelli* (Shaw & Nodder, 1977), where chloroform and EtOAc extracts showed *B. caeruleus* PLA₂ inhibition of 81% and 61%, respectively (Janardhan et al., 2014). Kusr et al. (2024) analyzed anti-snake venom potential (*Naja naja*) of *Vitex negundo* L. root extracts, through various *in vitro* assays targeting the PLA₂ enzyme. The results demonstrated that all extracts effectively neutralized the hydrolytic action of phospholipases, with inhibition values ranging between 59.83% and 69.33%, with IC₅₀ values ranging from 0.278 mg/mL (for MeOH) to 0.392 mg/mL (for DCM). These results corroborate our findings, which indicate better activities for the medium and high polarity fractions.

The investigation of extracts' capacity to inhibit the proteases and PLA₂ enzymes found in snake venoms offer valuable insights into the ability of these products to mitigate tissue degradation or disruptions in the coagulation cascade, thereby reducing the risk of local hemorrhages (Adrião et al., 2022; Sampat et al., 2023). Furthermore, assessing their

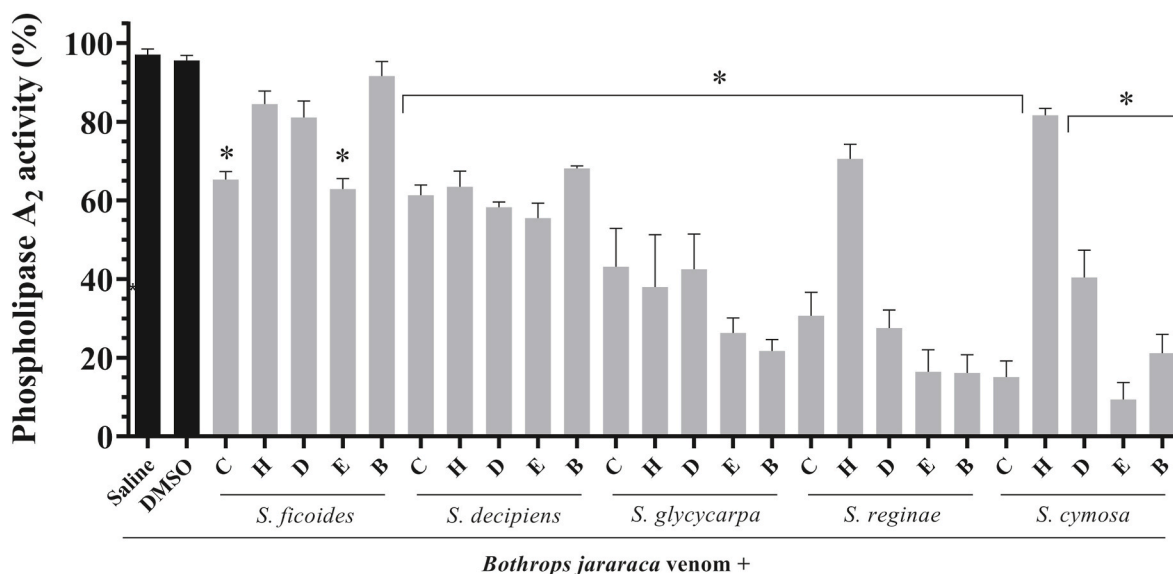


Fig. 3. Effect of *Siparuna* extracts on PLA₂ activity of *B. jararaca* venom. C: Crude ethanol extract; H: HEX extract; D: DCM extract; E: EtOAc extract; B: BuOH extract. Data are expressed as means $\pm$ S.E.M (n = 6). *, means with $P < 0.05$ when compared to *B. jararaca* venom + Saline or + DMSO.

potential to interfere with blood coagulation is crucial for determining their efficacy in preventing or treating coagulopathies induced by snakebites (Sampat et al., 2023).

Previous studies with extracts derived from the leaves, branches, and stems of *S. techaphora* demonstrated antihemorrhagic properties against *B. asper* and *B. atrox* venoms (Otero et al., 2000; Lomonte et al., 2009). However, this is the first study to investigate and demonstrate the inhibitory potential of plasma coagulation, proteolytic, and PLA₂ activity of species within the genus *Siparuna*. The inhibition of such activities is undoubtedly important to ameliorate the snake venom-induced hemorrhaging in victims since SVSPs, SVMPs, and PLA₂ enzymes contribute to this toxic effect.

Particularly noteworthy are the inhibiting plasma coagulation, PLA₂ and proteolysis percentages of SC (CEE, EtOAc, and BuOH), SD (CEE, DCM, EtOAc, and BuOH), SR (DCM, EtOAc, and BuOH), SG (EtOAc and BuOH), and SF (DCM) extracts, confirming the ethnomedicinal potential of species within the genus against snake venom (Leitão et al., 2000; Renner and Hausner, 2005; Rieder, 2013). Previous studies conducted by our research group led to the isolation and identification of flavonoids and alkaloids from the DCM and BuOH extracts of these species (Leal et al., 2021a, 2021b, 2022). Recent studies highlight that these classes of specialized metabolites have shown prominence in combating the effects resulting from the bite of *Bothrops* species (Adrião et al., 2022; Sachetto et al., 2022; Torres et al., 2023).

Medicinal plant extracts are a result of the combined action of multiple compounds with synergistic, additive, or antagonistic activity, being necessary future research to determine if either the crude extracts or the isolated compounds are more efficacious as antivenom (Fernandes et al., 2018; Vaou et al., 2022). However, as known from literature, snake venoms are a mixture of active proteins or peptides, and, to neutralize efficiently all these compounds is a difficult task and a challenge.

In this context, the assessed extracts of *Siparuna* showcased a potential to inhibit plasma coagulation, proteolysis, and PLA₂ activity, thereby mitigating the toxic effects of snake envenomation. Therefore, conducting these *in vitro* assays is essential for pinpointing promising compounds with therapeutic potential against both the localized and systemic effects of snake venoms, ultimately facilitating the advancement of more effective and safer treatments for individuals bitten by snakes.

3.4. Chemometric analyses using the PLS regression model

To predict compounds potentially associated with the inhibition of proteolytic and coagulant activities of *B. jararaca* venom, we conducted a liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) analysis of the 25 extracts originating from the five *Siparuna* species. The results of the anti-PLA₂ activity were excluded from the chemometric analyses due to its statistical relevance compared to the responses of other activities. PLA₂ may be associated with coagulant and proteolytic effects, being identified as one of the targets responsible for the toxic effects of venom (Marcussi et al., 2007). This could lead to redundancy in model responses or increase the statistical weight of one of the targets (anticoagulant and antiproteolytic) already evaluated, resulting in increased intrinsic errors and reduced robustness of the multitarget model. Therefore, this assay was conducted solely to investigate the possible mechanisms of action involved in the venom-neutralizing activity of the investigated extracts. This analysis employed two ionization sources, both in positive and negative modes (LC-ESI/APCI-(+/-)-MS/MS). Furthermore, continuous values representing the antiproteolytic and anticoagulant activities of *Siparuna* extracts were also employed for chemometric analyses.

For the development of a correlation model, we established a Partial Least Squares Regression (PLS) analysis, linking the LC-MS dataset of the 25 extracts as independent variables (X), with the dependent variables (Y) representing the activities against the targets. This algorithm integrates principal component analysis (PCA) and linear regression, with the aim of identifying latent variables that enhance the correlation between predictors (m/z ions) and biological activity.

We also modeled all four chemical datasets (data not shown), and the data obtained from positive APCI ionization exhibited the highest explained variance, R^2 and Q^2 values, as well as the lowest root mean square error (RMSE) (Fig. 4A–B).

The PLS model achieved a calibration explained variance of 98.3% using ten factors. The model exhibited an average R^2 of 0.95 and a Q^2 of 0.88, indicating a robust fit. Furthermore, it displayed low RMSE values, with RMSECV (cross-validation error) at 0.29, slightly higher than RMSEC (calibration error) at 0.19. This relationship of RMSECV > RMSEC confirmed the representativeness of the calibration data and the absence of overfitting in the model. Based on the PLS analyses, patterns in the sample distribution were identified, allowing classification into two groups based on average activity values across of the two activities:

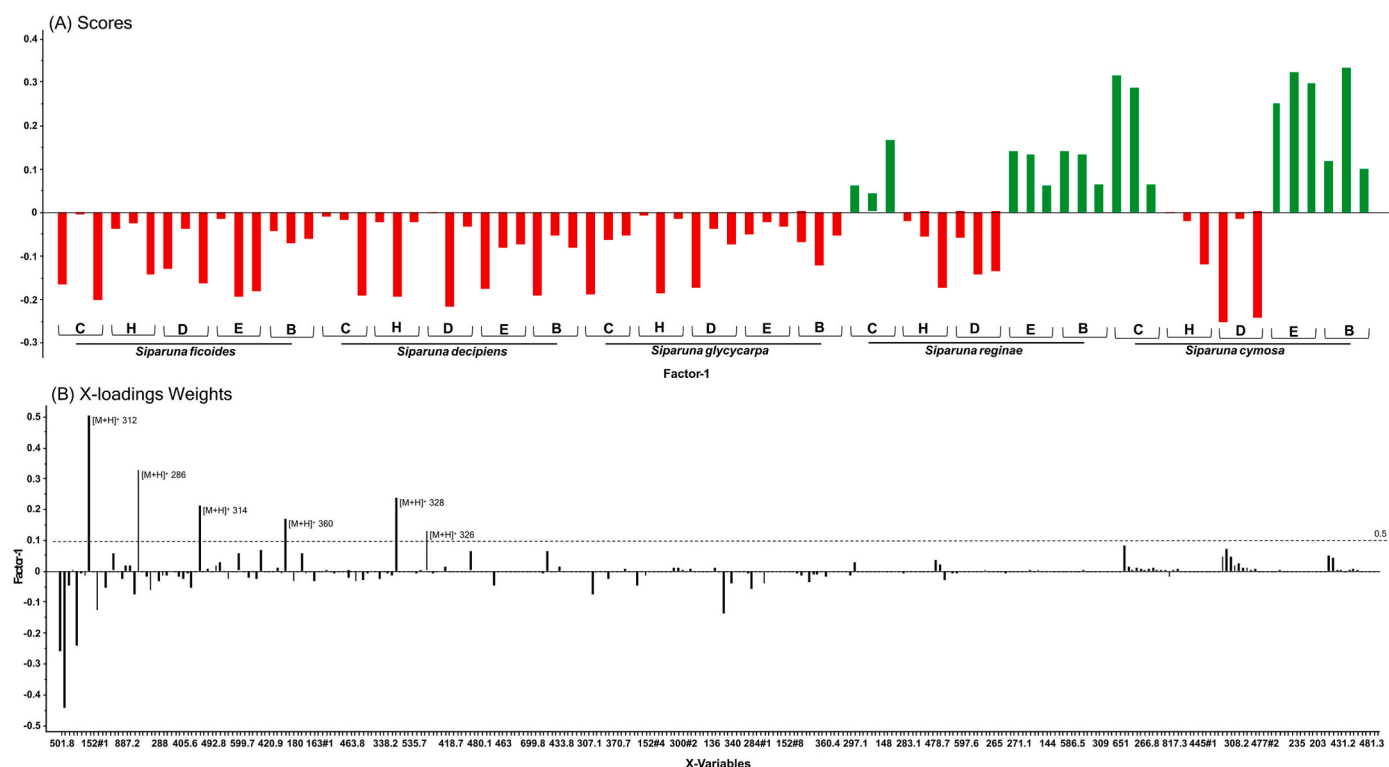


Fig. 4. Scores (A) Green: Activity multitarget; Red: Inactivity multitarget, and X-loadings (B) - detail of predict ions plots for the PLS model based on the LC-MS data obtained in the APCI ionization in positive mode of the 25 samples of five species of Siparunaceae family, and the values of the extracts antiproteolytic and anti-coagulant activities. The dashed line indicates the cutoff for ions with Y-explained contributions above 0.5. C: Crude ethanol extract; H: HEX extract; D: DCM extract; E: EtOAc extract; and B: BuOH extract. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

proteolytic inhibition ($\geq 50\%$ - Activity [1] or $< 50\%$ - Inactivity [-1]), and clotting time (≥ 175 s - Activity [1] or < 175 s - Inactivity [-1]). Multitarget activity was considered when the extracts were active in both assays (proteolytic inhibition [1] and clotting time [1]), and not multitarget when the extracts were considered inactive in at least one of the assays [-1,1 or 1, -1].

Fig. 4A (linear score plot) shows that SC and SR extracts (CEE, EtOAc, and BuOH) are with a positive trend in such activities. This result indicates the potential correlation of these extracts with the anti-proteolytic and anticoagulant effects of *B. jararaca* venom.

The X-loadings plot (Fig. 4B) illustrates the ions that contribute to the explanation (multitarget) of both antiproteolytic and anticoagulant

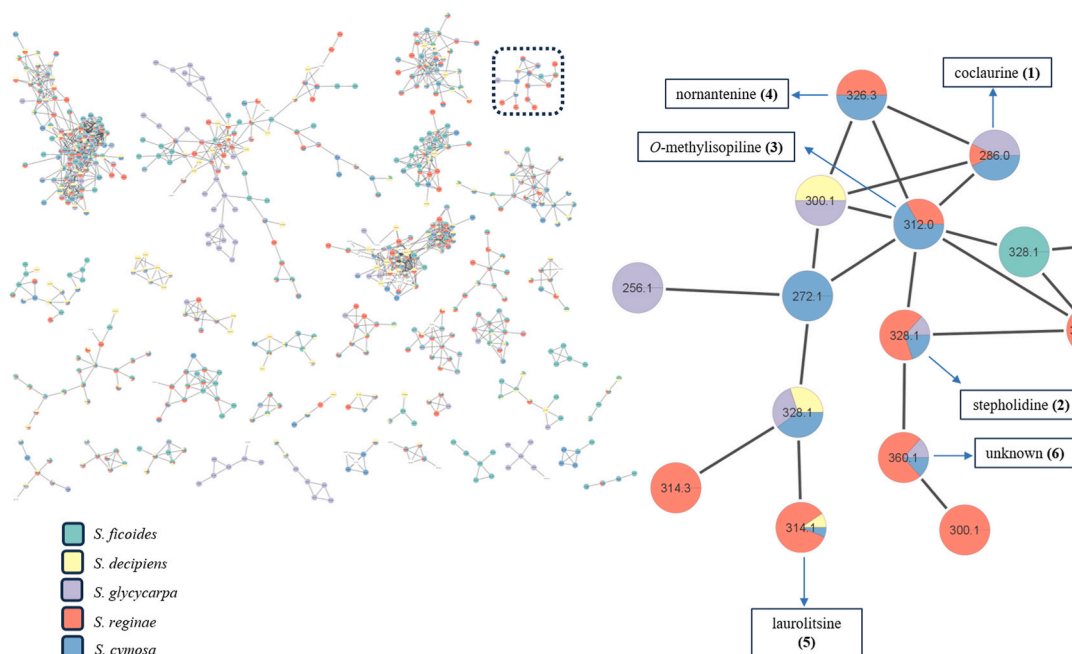


Fig. 5. Molecular network formed through UHPLC-APCI(+)-MS/MS data with emphasis on the network containing the predicted ions.

activities of these extracts. Predicted ions were examined to identify those with the highest positive contribution to the desired classification, meaning that they are ions that are exclusively present and/or in higher abundance in the extract with higher activity. This process led to the identification of five ions (contribution Y-explained above 0.5) with values of $[M + H]^+$ m/z 286, 312, 314, 326, 328 and 360.

3.5. Annotated compounds using molecular networking analyses

In natural products (NP) chemistry the use of advanced computational techniques is common to expedite the discovery of bioactive NP, organizing and extracting insights from large datasets (Yang et al., 2013; Silva et al., 2021; Gogoi et al., 2021). In this context, molecular networks are capable of successfully structuring extensive mass spectrometry data and sample metadata, presenting them in a user-friendly visual format known as spectral similarity networks (Pilon et al., 2021).

So after processing the UHPLC-APCI(+)-MS/MS data in MZmine, they were analyzed on the GNPS online platform for the creation of molecular networks. Several networks were formed, revealing the chemical complexity of the extracts. The predicted ions could be identified within the highlighted network depicted in Fig. 5.

It is interesting to note that the predicted ions shown in the chemometric analyses for both evaluated activities (antiproteolytic and anticoagulant) appeared prominently in more active species, SR and SC, red and blue colors, respectively. It is also possible to notice the presence of some of the predicted molecules in the extracts of SG and SD (purple and yellow, respectively); however, these extracts did not inhibit proteolysis and plasma coagulation caused by *B. jararaca* venom. This can be explained either by the lower abundance (area signal) of these compounds in these extracts, or by the absence of compounds with higher activity, as well as a possible synergy among some of the predicted active molecules.

UHPLC-APCI(+)-MS/MS analysis of the extracts and MS/MS fragments of predicted ions strongly suggest that they are alkaloids (Table 1), already described in the literature for *Siparuna* species (Leitão et al., 1999; Leal et al., 2021a, 2022).

Data obtained in the positive ionization mode were effective in pointing out alkaloids of the benzyloquinoline (coclaurine-1), protoberberine (stepholidine-2), and aporphine classes (*O*-methylisopiline-3, nornantenine-4, lauroilsine-5) (Fig. 6).

Compound 1 at m/z 286 $[M+H]^+$ was annotated based on diagnostic fragment ions (m/z 107 and m/z 192) and neutral losses (NH_3 or CH_3NH_2 , H_2O , CO , CH_3OH) characteristic for the benzyloquinoline skeleton (Leal et al., 2022). Coclaurine has been previously reported in dereplication studies by our group for *S. glycyarpa*, *S. decipiens*, *S. reginae* and *S. cymosa* (Leal et al., 2021a, 2022). Concerning compound 2, m/z 328 $[M+H]^+$, MS/MS data showed typical fragments of the protoberberine class of alkaloids mainly due to the presence of the diagnostic fragment ion at m/z 178, formed via a retro-Diels-Alder ring opening (Lima et al., 2020).

On the other hand, the compounds 3, 4 and 5, aporphine derivatives, were annotated for m/z 312, 326 and 314 $[M+H]^+$ based on their fragmentation pattern, i.e., the initial losses of NH_3 (−17Da) or CH_3NH_2 (31Da) followed by CO (−28Da) when an CH_2O (−30Da) followed by CO there is a methylenedioxy group (compound 4), OH and an OCH_3 group when are vicinal in the aromatic ring (compound 5) (Stévigny et al., 2004).

Some of these compounds have already been described by our group in BuOH extracts of *Siparuna* species through LC-MS/MS data like compound 1 (SG and SC-originally identified as *S. sarmentosa*), compounds 2 and 3 (SG), compounds 4 and 5 (SC-originally identified as *S. sarmentosa*) (Leal et al., 2021a, 2022). In addition, compound 3 was isolated from *S. poeppigii* (Marti et al., 2013), and compound 4 has already been isolated from *S. tonduziana* (Lopez et al., 1990) and *S. guianensis* (Simas et al., 2001).

Finally, compound 6 was not identified, but its fragmentation pattern is like that of the benzyloquinoline class. Additionally, it is in the same network as the predicted ions, strongly suggesting that it is also an alkaloid as the annotated compounds.

Alkaloids have numerous biological activities, but only a few reports describe their inhibitory activities against the enzymes of snake venoms (Adrião et al., 2022). The glycosylated alkaloid benzopyranylpiperidinone (schumanniofoside) isolated from the stem bark of *Schumanniophyton magnificum* Harms (Rubiaceae) reduced *in vivo* the lethal effect of *Naja melanoleuca* Hallowell, 1857 venom (Akunyili and Akubue, 1986).

An indole alkaloid (12-methoxy-4-methyl-voachalotine) isolated from the root bark of *Tabernaemontana catharinensis* A.DC. (Apocynaceae) inhibited 100% of the lethality of *Crotalus durissus terrificus* venom, besides showing antimyotoxic activity. Furthermore, this compound inhibited PLA₂-like proteins from *B. jararacussu* (Batina et al., 2000; Borges et al., 2023). Seven steroidal alkaloids isolated from leaves of *Solanum campaniforme* Roem. & Schult. (Solanaceae) showed anti-myotoxic, antihemorrhagic, and antinecrotizing activity after *in vitro* incubation of the venom of *B. pauloensis* Amaral, 1925; Torres et al. (2011); Torres et al. (2013).

The nitrophenanthrene carboxylic acid alkaloid (aristolochic acid), found in roots of *Aristolochia indica* L. and stems of *A. sprucei* L. (Aristolochiaceae) has been reported as a promising antivenom compound against the venom of *D. russelii*, *B. jararacussu*, *Bothrops asper* (Garman, 1883), *Bothrops pirajai* Amaral, 1923 and *Naja naja* Laurenti, 1768 through the inhibition of edematogenic, myotoxic, and hemolytic activities (Vishwanath and Gowda, 1987; Chandra et al., 2002; Fernandes et al., 2015; Bhattacharjee et al., 2017). Noteworthy is the fact that plants of the genus *Aristolochia* are also used in Brazilian Folk Medicine to treat inflammation, asthma, different kinds of cancer, and snakebites (Bhattacharjee and Bhattacharyya, 2013; Giovannini and Howes, 2017).

Berberine, an isoquinoline alkaloid obtained from the stem bark of *Cardiospermum halicacabum* L. (Sapindaceae), was discovered to be a PLA₂ inhibitor of the venom of *D. russelii* (Chandra et al., 2011).

Table 1

LC-MS/MS data of predicted ions and proposed annotated compounds.

Compound	Rt (min)	Molecular Formula	$[M+H]^+$ m/z	MS/MS	Proposed compound	Relative Area SR	RelativeArea SC	Ref.
1	9.48	C ₁₇ H ₁₉ NO ₃	286	269, 237, 209, 175, 192, 137, 107	coclaurine	2.5E3	2.0E5	Leal et al. (2022)
2	10.66	C ₁₉ H ₂₁ NO ₄	328	313, 298, 285, 265.5, 253, 238, 209, 182, 178	stepholidine	3.4E3	8.8E2	Leal et al. (2022)
3	11.61	C ₁₉ H ₂₁ NO ₃	312	295, 280, 263	<i>o</i> -methylisopiline	2.6E3	2.0E5	Leal et al. (2022)
4	12.17	C ₁₉ H ₁₉ NO ₄	326	295, 263, 237	nornantenine	9.6E3	1.8E4	Leal et al. (2021a)
5	12.92	C ₁₈ H ₁₉ NO ₄	314	297, 177, 145, 117	lauroilsine	9.2E3	1.1E3	Leal et al. (2021a)
6	27.55	–	360	343, 311, 283, 206, 149	NI*	5.7E4	2.5E3	–

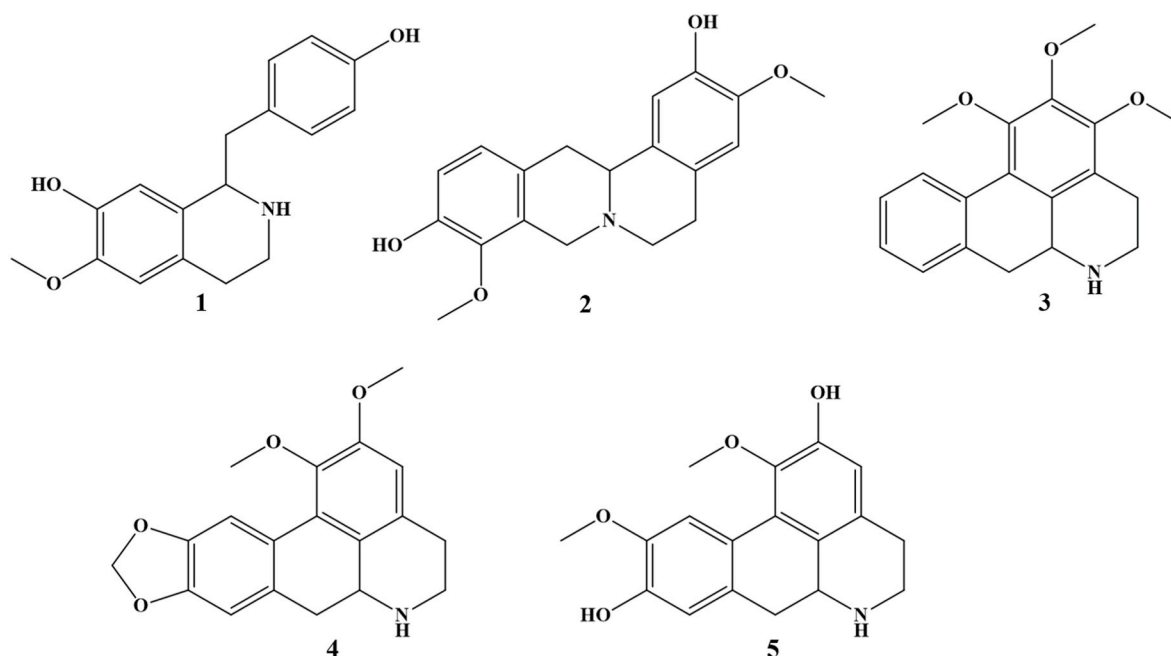


Fig. 6. Structure of alkaloids identified in *Siparuna* species with potential against *B. jararaca* venom.

Piperine, a derivative of piperidine and the primary alkaloidal component found in the flowers of *Piper longum* L. (Piperaceae), could serve as an effective treatment for Russell's viper snake bites (*D. russelii*, Viperidae). It may be applied both systemically and locally, demonstrating the inhibition of lethality, hemorrhage, necrosis, defibrinogenation, and edema (Shenoy et al., 2013).

Plant alkaloids showed activity against the toxic effects of snake venom, being potential candidates as an auxiliary in antivenom therapy, particularly when dealing with the local toxicity (Santhosh et al., 2013; Konrath et al., 2022).

This is the first study to demonstrate the potential of alkaloids from *Siparuna* species against proteolytic, coagulant and PLA₂ activities of *B. jararaca* venom, considered to be the most toxic effects of this species of snake. This research supports the traditional use of *Siparuna* as an alternative to treat snake envenoming (Leitão et al., 2000; Renner and Hausner, 2005; Rieder, 2013).

4. Conclusion

Our results showed that the EtOAc and BuOH extracts from *S. cymosa* and *S. reginae* stood out for showing the best profile of inhibiting for proteolysis, PLA₂, and plasma coagulation caused by *B. jararaca* venom, thus showing the ability of such species as potential alternative or complementary treatment strategy against SBE. The predicted ions in *Siparuna* extracts point alkaloids of the benzyloquinoline, aporphine, and protoberberine types with high antivenom potential, which may be correlated to the observed activity. Therefore, such results encourage the continuation of this study aiming the isolation of the pure compounds and assay against proteolysis and coagulation. This is the first report of the antivenom effect of these types of alkaloids.

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CRediT authorship contribution statement

Diégina Araújo Fernandes: Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Data curation, Conceptualization. **Brendo Araujo Gomes:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Simony Carvalho Mendonça:** Writing – original draft, Software, Formal analysis, Data curation. **Camila de Castro Pinheiro:** Writing – original draft, Formal analysis, Data curation. **Eladio Oswaldo Flores Sanchez:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Data curation, Conceptualization. **Suzana Guimarães Leitão:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Data curation, Conceptualization. **André Lopes Fuly:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Data curation, Conceptualization. **Gilda Guimarães Leitão:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Adrião, A.A.X., Santos, A.O., Lima, E.J.S.P., Maciel, J.B., Paz, W.H.P., Silva, F.M.A., Pucca, M.B., Silva, A.M.M., Monteiro, W.M., Sartim, M.A., Koolen, H.H.F., 2022. Plant-derived toxin inhibitors as potential candidates to complement antivenom treatment in snakebite envenomations. *Front. Immunol.* 13, 842576. <https://doi.org/10.3389/fimmu.2022.842576>.
- Akunyili, D.N., Akubue, P.I., 1986. Schumanniofoside, the Antisnake Venom principle from the stem bark of *Schumanniophyton magnificum* Harms. *J. Ethnopharmacol.* 18, 167–172. [https://doi.org/10.1016/0378-8741\(86\)90028-0](https://doi.org/10.1016/0378-8741(86)90028-0).
- Baraldi, P.T., Magro, A.J., Matioli, F.F., Marcussi, S., Lemke, N., Calderon, L.A., Stabeli, R.G., Soares, A.M., Correa, A.G., Fontes, M.R.M., 2016. A novel synthetic quinolinone inhibitor presents proteolytic and hemorrhagic inhibitory activities against snake venom metalloproteases. *Biochimie* 121, 179–188. <https://doi.org/10.1016/j.biochi.2015.11.031>.
- Bhattacharjee, P., Bhattacharyya, D., 2013. Characterization of the aqueous extract of the root of *Aristolochia indica*: evaluation of its traditional use as an antidote for snake bites. *J. Ethnopharmacol.* 14, 220–226. <https://doi.org/10.1016/j.jep.2012.10.056>.
- Bhattacharjee, P., Bera, I., Chakraborty, S., Ghoshal, N., Bhattacharyya, D., 2017. Aristolochic acid and its derivatives as inhibitors of snake venom lamino acid oxidase. *Toxicol.* 138, 1–17. <https://doi.org/10.1016/j.toxicol.2017.08.003>.
- Batina, M.F., Cintra, A.C.O., Veronese, E.L.G., Lavrador, M.A., Giglio, J.R., Pereira, P.S., Dias, D.A., França, S.C., Sampaio, S.V., 2000. Inhibition of the lethal and myotoxic activities of *Crotalus durissus terrificus* venom by *Tabernaemontana catharinensis*: identification of one of the active components. *Planta Med.* 66, 424–428. <https://doi.org/10.1055/s-2000-8577>.
- Binorkar, S.V., Jani, D.K., 2012. Profile of medicinal plants with anti-ophidian properties. *J. Pharmaceut. Sci. Innovat.* 1, 13–20. Available online: <http://jpsonline.com/admin/php/uploads/118.pdf.pdf>. (Accessed 3 October 2023).
- Borges, R.J., Cardoso, F.F., Carvalho, C., Marino, I., Pereira, P.S., Soares, A.M., Dal-Pai-Silva, M., Usón, I., Fontes, M.R.M., 2023. Structural and functional studies of a snake venom phospholipase A₂-like protein complexed to an inhibitor from *Tabernaemontana catharinensis*. *Biochimie* 206, 105–115. <https://doi.org/10.1016/j.biochi.2022.10.011>.
- Camara, C.A.G.D., Marsaioli, A.J., Bittrich, V., 2018. Chemical constituents of apolar fractions from fruit latex of twelve *Clusia* species (Clusiaceae). *An. Acad. Bras. Cienc.* 90, 1919–1927. <https://doi.org/10.1590/0001-3765201820170257>.
- Castro-Pinheiro, C., Junior, L.C.S.P., Sanchez, E.F., da Silva, A.C.R., Dwan, C.A., Karpiniec, S.S., Critchley, A.T., Fuly, A.L., 2024. Effect of seaweed-derived fucoidans from *Undaria pinnatifida* and *Fucus vesiculosus* on coagulant, proteolytic, and phospholipase A₂ activities of snake *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *Toxins* 12, 188–190. <https://doi.org/10.3390/toxins16040188>.
- Carvalho, B.M.A., Santos, J.D.L., Xavier, B.M., Almeida, J.R., Resende, L.M., Martins, W., Marcussi, S., Marangoni, S., Stabeli, R.G., Calderon, L.A., Soares, A.M., Da Silva, S.L., Marchi-Salvador, D.P., 2013. Snake venom PLA₂s inhibitors isolated from Brazilian plants: synthetic and natural molecules. *BioMed Res. Int.* 153045, 1–8. <https://doi.org/10.1155/2013/153045>.
- Chandra, V., Jasti, J., Kaur, P., Srinivasan, A., Betzel, C.H., Singh, T.P., 2002. Structural basis of phospholipase A₂ inhibition for the synthesis of prostaglandins by the plant alkaloid aristolochic acid from a 1.7 Å crystal structure. *Biochem.* 41, 10914–10919. <https://doi.org/10.1021/bi0258593>.
- Chandra, D.N., Prasanth, G.K., Singh, N., Kumar, S., Jithesh, O., Sadasivan, C., Sharma, S., Singh, T.P., Haridas, M., 2011. Identification of a novel and potent inhibitor of phospholipase A₂ in a medicinal plant: crystal structure at 1.93 Å and surface plasmon resonance analysis of phospholipase A₂ complexed with berberine. *Biochim. Biophys. Acta, Proteins Proteomics* 1814, 657–663. <https://doi.org/10.1016/j.bbapap.2011.03.002>.
- Costa, F.N., Garrard, I., Da-Silva, A.J.R., Leitão, G.G., 2013. Changes in the mobile phase composition on a stepwise counter-current chromatography elution for the isolation of flavonoids from *Siparuna glycyrcarpa*. *J. Separ. Sci.* 36, 2253–2259. <https://doi.org/10.1002/jssc.201201054>.
- da Silva, A.R., Anholeti, M.C., Pietroluongo, M., Sanchez, E.F., Valverde, A.L., de Paiva, S.R., Figueiredo, M.R., Kaplan, M.A.C., Fuly, A.L., 2019. Utilization of the plant *Clusia Fluminensis* Planch & Triana against some toxic activities of the venom of *Bothrops jararaca* and *B. jararacussu* snake venom toxic activities. *Curr. Top. Med. Chem.* 19, 1990–2002. <https://doi.org/10.2174/1568026619666190724160711>.
- Darwish, R.S., El-Banna, A.A., Ghareeb, D.A., El-Hosseny, M.F., Seadawy, M.G., Dawood, H.M., 2022. Chemical profiling and unraveling of anti-COVID-19 biomarkers of red sage (*Lantana Camara* L.) cultivars using UPLC-MS/MS coupled to chemometric analysis, *in vitro* study and molecular docking. *J. Ethnopharmacol.* 291, 115038. <https://doi.org/10.1016/j.jep.2022.115038>.
- Dellacassa, E., Torres, A.M., Ricciardi, G.A.L., Camargo, F.J., Tressens, S.G., Ricciardi, A.I.A., 2014. Anti-venom activity of medicinal plants from South America. In: *Utilisation and Management of Medicinal Plants*. Daya Publishing House, New Delhi, pp. 1–11. Available online: <http://repositorio.unne.edu.ar/handle/123456789/28837>. (Accessed 8 October 2023).
- Dos Santos, M.B.V., Gontijo, D.C., Nascimento, M.F.A., Paula, R.C., Bellei, J.C.B., Raimundo, F.O., Scopel, K.K.G., Oliveira, A.B., Mourão, R.H.V., 2023. *In vitro* and *in vivo* antimalarial activity, cytotoxicity and phytochemical HRMS2 profile of plants from the Western Pará State, Brazilian Amazonia. *Chem. Biodiversity*, e202301082. <https://doi.org/10.1002/cbdv.202301082>, 1–10.
- Estevao-Costa, M.I., Gontijo, S.S., Correia, B.L., Yarleque, A., Vivas-Ruiz, D., Rodrigues, E., Chávez-Olortegui, C., Oliveira, L.C., Sanchez, E.F., 2016. Neutralization of toxicological activities of medically-relevant *Bothrops* snake venoms and relevant toxins by two polyvalent bothropic antivenoms produced in Peru and Brazil. *Toxicon* 122, 67–77. <https://doi.org/10.1016/j.toxicon.2016.09.010>.
- Faria, G.M., Lemos, A.P.A., Anholeti, M.C., Paiva, S.R., Amorim, L.M.F., 2023. The bioprospecting potential of *Clusia fluminensis* Planch. & Triana: a scoping review. *An. Acad. Bras. Cienc.* 95, e20211605. <https://doi.org/10.1590/0001-3765202320211605>.
- Farias, I.B., Morais-Zani, K., Serino-Silva, C., Sant'Anna, S.S., Rocha, M.M.T., Grego, K.F., Andrade-Silva, D., Serano, S.M.T., Azevedo, A.M.T., 2018. Functional and proteomic comparison of *Bothrops jararaca* venom from captive specimens and the Brazilian Bothropic Reference venom. *J. Proteomics* 174, 36–46. <https://doi.org/10.1016/j.jpropt.2017.12.008>.
- Feitosa, E.S., Sampaio, V., Sachett, J., Castro, D.B., Noronha, M.D.N., Lozano, J.L.L., Munis, E., Ferreira, L.C.L., Lacerda, M.V.G., Monteiro, W.M., 2015. Snakebites as a largely neglected problem in the Brazilian Amazon: highlights of the epidemiological trends in the State of Amazonas. *Rev. Soc. Bras. Med. Trop.* 48, 34–41. <https://doi.org/10.1590/0037-8682-0105-2013>.
- Félix-Silva, J., Silva-Junior, A.A., Zucolotto, S.M., Fernandes-Pedrosa, M.F., 2017. Medicinal plants for the treatment of local tissue damage induced by snake venoms: an overview from traditional use to pharmacological evidence. *Evidence Based Complem. Alternative Med.* 1–52. <https://doi.org/10.1155/2017/5748256>.
- Fernandes, C.A., Cardoso, F.F., Cavalcante, W.G., Soares, A.M., Dal-Pai, M., Gallacci, M., Fontes, M.R.M., 2015. Structural basis for the inhibition of a phospholipase A₂-like toxin by caffeic and aristolochic acids. *PLoS One* 10, 1–19. <https://doi.org/10.1371/journal.pone.0133370>.
- Fernandes, D.A., Souza, M.S.R., Teles, Y.C.F., Oliveira, L.H.G., Lima, J.B., Conceição, A.S., Nunes, F.C., Silva, T.M., Souza, M.F.V., 2018. New sulphated flavonoids and larvicidal of *Helicoverpa zea* K. Schum (Sterculiaceae). *Molecules* 23, 2784. <https://doi.org/10.3390/molecules23112784>.
- Garcia, E.S., Guimaraes, J.A., Prado, J.L., 1978. Purification and characterization of a sulphhydryl-dependent protease from *Rhodnius prolixus* midgut. *Arch. Biochem. Biophys.* 188, 315–322. [https://doi.org/10.1016/S0003-9861\(78\)80015-0](https://doi.org/10.1016/S0003-9861(78)80015-0).
- Giovannini, P., Howes, M.J.R., 2017. Medicinal plants used to treat snakebite in Central America: review and assessment of scientific evidence. *J. Ethnopharmacol.* 199, 240–256. <https://doi.org/10.1016/j.jep.2017.02.011>.
- Gogoi, B., Gogoi, D., Gogoi, N., Mahanta, S., Buragohain, A.K., 2021. Network pharmacology based high throughput screening for identification of multi targeted anti-diabetic compound from traditionally used plants. *J. Biomol. Struct. Dyn.* 40, 8004–8017. <https://doi.org/10.1080/07391102.2021.1905554>.
- Gomes, A., Das, R., Sarkhel, S., Mishra, R., Mukherjee, S., Bhattacharya, S., Gomes, A., 2010. Herbs and herbal constituents are active against snake bites. *Indian J. Exp. Biol.* 48, 865–878. Available online: <https://pubmed.ncbi.nlm.nih.gov/21506494/>. (Accessed 7 October 2023).
- Guimaraes, A., Vieira, R., Vieira, A., 2021. Structure of leaf galls in *Clusia fluminensis* Planch. and Triana (Clusiaceae): sex-biased development in a dioecious host plant. *Plants* 10, 1–19. <https://doi.org/10.3390/plants10010020>.
- INPA - Instituto Nacional de Pesquisas da Amazônia, 2024. Herbário INPA. Available online: <https://specieslink.net/search/>. (Accessed 10 April 2024).
- Janardhan, B., Shrikanth, V.M., Mirajkar, K.K., More, S.S., 2014. *In vitro* screening and evaluation of antivenom phytochemicals from *Azima tetraantha* Lam. leaves against *Bungarus caeruleus* and *Viper russelli*. *J. Venomous Animals Toxins* including Trop. Dis. 20 (12), 1–8. <https://doi.org/10.1186/1678-9199-20-12>.
- Jorge, R.J.B., Jorge, A.R.C., Menezes, R.R.P.P.B., Mello, C.P., Lima, D.B., Oliveira, J.A.M., Alves, N.T.Q., Marinho, A.D., Ximenes, R.M., Corres-Neto, C., Machado, L.G., Zingali, B., Martins, A.M.C., Monteiro, H.S.A., 2017. Differences between renal effects of venom from two *Bothrops jararaca* populations from southeastern and southern Brazil. *Toxicon* 125, 84–90. <https://doi.org/10.1016/j.toxicon.2016.11.249>.
- Khan, M.F., Kader, F.B., Arman, M., Ahmed, S., Lyzu, C., Sakib, S.A., Tanzil, S.M., Zim, A.F.M.I., Imran, A.S., Venneri, T., Romano, B., Haque, A., Capasso, R., 2020. Pharmacological insights and prediction of lead bioactive isolates of Dita bark through experimental and computer-aided mechanism. *Biomed. Pharmacother.* 131, 110774. <https://doi.org/10.1016/j.biopha.2020.110774>.
- Kisaki, C.Y., Arcos, S.S.S., Montoni, F., Santos, W.S., Calacina, H.M., Lima, I.F., Carvalho, D.C., Ferro, E., Nishiyama-Jr, M.Y., Iai, L.K., 2021. *Bothrops Jararaca* snake venom modulates key cancer-related proteins in breast tumor cell lines. *Toxins* 13 (8), 519. <https://doi.org/10.3390/toxins13080519>.
- Konrath, E.L., Strauch, I., Boeff, D.D., Arbo, M.D., 2022. The potential of Brazilian native plant species used in the therapy for snakebites: a literature review. *Toxicon* 217, 17–40. <https://doi.org/10.1016/j.toxicon.2022.08.002>.
- Kusar, S., Saddique, Z., Asad, M.H.H.B., Ali, F., Kirmani, F., 2024. Phytochemical characterization and phospholipase A₂ inhibitory effect of *Vitex negundo* L. root extracts. *J. Ethnopharmacol.* 323, 117671. <https://doi.org/10.1016/j.jep.2023.117671>.
- Leal, C.M., Simas, R.C., Miranda, M., Campos, M.F., Gomes, B.A., Siqueira, M.M., Vale, G., Almeida, C.V.G., Leitão, S.G., Leitão, G.G., 2021a. Amazonian Siparuna extracts as potential anti-influenza agents: metabolic fingerprinting. *J. Ethnopharmacol.* 270, 113788. <https://doi.org/10.1016/j.jep.2021.113788>.
- Leal, C.M., Leitão, S.G., Sausset, R., Mendonça, S.C., Nascimento, P.H.A., Cheohen, P.F.A.R., Esteves, M.E.A., Silva, M.L., Gondim, T.S., Monteiro, M.E.S., Tucci, A.R.,

- Fintelman-Rodrigues, N., Siqueira, M.M., Miranda, M., Costa, F.N., Leitão, G.G., 2021b. Flavonoids from *Siparuna cristata* as potential inhibitors of SARS-CoV-2 replication. *Braz. J. Pharmacog.* 31, 658–666. <https://doi.org/10.1007/s43450-021-00162-5>.
- Leal, C.M., Leitão, S.G., De Mello, L.L.O., Rangel, I.C., Da Silva, C.V.A., Miranda, M.D., Tucci, A.R., De Assis, C.B., Sacramento, C.Q., Fintelman-Rodrigues, N., Koolen, H.H.F., Vaz, B.G., Simas, R.C., Leitão, G.G., 2022. Bioassay-guided fractionation of *Siparuna glycyarpa* n-butanol extract with inhibitory activity against influenza A (H1N1)pdm09 virus by Centrifugal Partition Chromatography (CPC). *Molecules* 27, 399–413. <https://doi.org/10.3390/molecules27020399>.
- Leitão, S.G., Silva, M.H., Kaplan, M.A.C., Delle Monache, F., 1996. Constituents of some *Aegiphila* species and neutralization of *Crotalus durissus terrificus* venom by *Aegiphila lhotzkyana*. *Fitoterapia* 67, 377–379.
- Leitão, G.G., Simas, N.K., Soares, S.S.V., de Brito, A.P.P., Claros, B.M.G., Brito, T.B.M., Monache, F.D., 1999. Chemistry and pharmacology of monimiaceae: a special focus on Siparuna and Mollinedia. *J. Ethnopharmacol.* 65, 87–102. [https://doi.org/10.1016/S0378-8741\(98\)00233-5](https://doi.org/10.1016/S0378-8741(98)00233-5).
- Leitão, G.G., Soares, S.S.V., Brito, T.B.M., Monache, F.D., 2000. Kaempferol glycosides from *Siparuna apiosyce*. *Phytochemistry* 55, 679–682. [https://doi.org/10.1016/S0031-9422\(00\)00222-3](https://doi.org/10.1016/S0031-9422(00)00222-3).
- Liaquat, A., Mallhi, T.H., Khan, Y.H., Khokhar, A., Chaman, S., Ali, M., 2022. Anti-snake venom properties of medicinal plants: a comprehensive systematic review of literature. *Braz. J. Pharm. Sci.* 58, 1–16. <https://doi.org/10.1590/s2175-97902022e191124>.
- Lima, B.R., Da Silva, F.M.A., Soares, E.R., De Almeida, R.A., Da Silva-Filho, F.A., Barison, A., Costa, E.V., Koolen, H.H.F., De Souza, A.D.L., Pinheiro, M.L.B., 2020. Integrative approach based on leaf spray mass spectrometry, HPLC-DAD-MS/MS, and NMR for comprehensive characterization of isoquinoline-derived alkaloids in leaves of *Onychopetalum amazonicum* R. E. Fr. *J. Braz. Chem. Soc.* 31, 79–89. <https://doi.org/10.21577/0103-5053.20190125>.
- Lomonte, B., Leon, G., Angulo, Y., Rucavado, A., Núñez, V., 2009. Neutralization of *Bothrops asper* venom by antibodies, natural products and synthetic drugs: contributions to understanding snakebite envenomings and their treatment. *Toxicon* 54, 1012–1028. <https://doi.org/10.1016/j.toxicon.2009.03.015>.
- Lopez, J.A., Laurito, J.G., Lin, F., Duah, F.K., Sharaf, M., Aly, Y., Wong, L.K., Shiff Jr, P.L., 1990. Alkaloids of *Siparuna tonduziana*. *Planta Med.* 56, 492. <https://doi.org/10.1055/s-2006-961021>.
- Machado, L.G., Pla, D., Sanz, L., Jorg, R.J.B., Araujo, M.L., Alves, M.L.M., Alvares, D.J., Miranda, J., Nowatzki, J., Zani, K.M., ernandes, W., Azevedo, A.M.T., Fernandez, J., Zingali, R.B., Gutierrez, J.M., Correa-Neto, C., Calvete, J.J., 2016. Combined venomomics, venom gland transcriptomics, bioactivities, and antivenomics of two *Bothrops jararaca* populations from geographic isolated regions within the Brazilian Atlantic rainforest. *J. Proteomics* 135, 73–89. <https://doi.org/10.1016/j.jprot.2015.04.029>.
- Marcussi, S., Sant'Ana, C.D., Oliveira, C.Z., Quintero Rueda, A., Menaldo, D.L., Bebeloni, R.O., Satabeli, R.G., Giglio, J.R.M., Fontes, M.R., Soares, A.M., 2007. Snake venom phospholipase A₂ inhibitors: medicinal chemistry and therapeutic potential. *Curr. Top. Med. Chem.* 7, 743–756. <https://doi.org/10.2174/156802607780487614>.
- Marinetti, G.V., 1965. The action of phospholipase A₂ on lipoproteins. *Biochim. Biophys. Acta Lipids Lipid. Metabol.* 98, 554–565. [https://doi.org/10.1016/0005-2760\(65\)90152-9](https://doi.org/10.1016/0005-2760(65)90152-9).
- Marti, G., Eparvier, V., Morleo, B., Le Ven, J., Apel, C., Bodo, B., Amand, S., Dumontet, V., Lozach, O., Meijer, L., Guéritte, F., Litaudon, M., 2013. Natural aristolactams and aporphine alkaloids as inhibitors of CDK1/Cyclin B and DYRK1A. *Molecules* 18, 3018–3027. <https://doi.org/10.3390/molecules18033018>.
- Mazza, K.E.L., Santiago, M.C.P.A., Pacheco, S., Nascimento, L.S.M., Braga, E.C.O., Martins, V.C., Cunha, C.P., Godoy, R.L.O., Borguini, R.G., 2019. Determination of bioactive compounds in *Clusia fluminensis* Planch. & Triana fruit's arils. *Rev Virtual Química* 11, 3–17. <https://doi.org/10.21577/1984-6835.20190002>.
- Mostofa, M.G., Reza, A.S.M.A., Khan, Z., Munira, M.S., Khatoon, M.M., Kabir, S.R., Sadik, M.G., Agagündüz, D., Capasso, R., Kazi, M., Alam, A.K., 2023. Apoptosis-inducing anti-proliferative and quantitative phytochemical profiling with *in silico* study of antioxidant-rich *Leea aquata* L. leaves. *Heliyon* 10, e23400. <https://doi.org/10.1016/j.heliyon.2023.e23400>.
- Moura, V.M., Sousa, L.A.F., Santos, M.C., Raposo, J.D.A., Lima, A.E., Oliveira, R.B., Silva, M.N., Mourão, R.H.V., 2015. Plants used to treat snake bites in Santarém, western Pará, Brazil: an assessment of their effectiveness in inhibiting hemorrhagic activity induced by *Bothrops jararaca* venom. *J. Ethnopharmacol.* 161, 224–232. <https://doi.org/10.1016/j.jep.2014.12.020>.
- Niaz, S.K., Mariam, Z., 2024. Computer-Aided Drug design and Drug discovery: a prospective analysis. *Pharmaceuticals* 17, 22. <https://doi.org/10.3390/ph17010022>.
- Nishijima, C.M., Rodrigues, C.M., Silva, M.A., Lopes-Ferreira, M., Vilegas, W., Hiruma-Lima, C.A., 2009. Anti-hemorrhagic activity of four Brazilian vegetable species against *Bothrops jararaca* venom. *Molecules* 14, 1072–1080. <https://doi.org/10.3390/molecules14031072>.
- Oliveira, E.C., Fernandes, A.P., Sanchez, E.F., Rocha, L.R., Fuly, A.L., 2014. Inhibitory effect of plant *Manilkara subsericea* against biological activities of *Lachesis muta* snake venom. *BioMed Res. Int.* 408068, 1–7. <https://doi.org/10.1155/2014/408068>.
- Oliveira, E.C., Cruz, R.A.S., Amorim, N.M., Santos, M.G., Pereira-Junior, L.C.S., Sanchez, E.O.F., Fernandes, C.P., Garrett, R., Rocha, L.M., Fuly, A.L., 2016. Protective effect of the plant extracts of *Erythroxylum* sp. against toxic effects induced by the venom of *Lachesis muta* snake. *Molecules* 21, 1–14. <https://doi.org/10.3390/molecules21101350>.
- Otero, R., Núñez, V., Barona, J., Fonnegra, R., Jiménez, S.L., Osorio, R.G., Saldarriaga, M., Díaz, A., 2000. Snakebites and ethnobotany in the northwest region of Colombia part III: neutralization of the hemorrhagic of *Bothrops atrox* venom. *J. Ethnopharmacol.* 73, 233–241. [https://doi.org/10.1016/S0378-8741\(00\)00321-4](https://doi.org/10.1016/S0378-8741(00)00321-4).
- Pilon, A.C., Vieira, N.C., Amaral, J.G., Monteiro, A.F., da Silva, R.R., Spindola, L.S., Castro-Gamboa, I., Lopes, N.P., 2021. Redes Moleculares: uma análise sobre anotações e descoberta de novos ativos. *Quim. Nova* 44, 1168–1179. <https://doi.org/10.21577/0100-4042.20170777>.
- Popov, R.S., Ivanchina, N.V., Dmitrenko, P.S., 2022. Application of MS-based metabolomic approaches in analysis of starfish and sea cucumber bioactive compounds. *Mar. Drugs* 20, 320. <https://doi.org/10.3390/md20050320>.
- Renner, S.S., Hausner, G., 2005. New species of *Siparuna* (Siparunaceae) IV. A new subcanopy tree from white-sand areas in Brazil and Venezuela. *Novon* 15, 202–206.
- Rieder, A., 2013. Medicinal plants used to treat snakebites by people in the neighborhood watersheds Platium and Amazon, Brazil. *Plant. Med.* 79 <https://doi.org/10.1055/s-0033-1352066>.
- Roriz, K.R.P., Zaqueo, K.D., Setubal, S.S., Katsuragawa, T.H., Silva, R.R., Fernandes, C.F. C., Cardoso, L.A.P.C., Rodrigues, M.M.S., Soares, A.M., Stábili, R.G., Zuliani, J.P., 2018. Epidemiological study of snakebite cases on Brazilian Western Amazonia. *Rev. Soc. Bras. Trop.* 51, 338–346. <https://doi.org/10.1590/0037-8682-0489-2017>.
- Sachetto, A.T.A., Miyamoto, J.G., Tashima, A.K., Souza, A.O., Santoro, M.L., 2022. The bioflavonoids rutin and rutin succinate neutralize the toxins of *B. jararaca* venom and inhibit its lethality. *Front. Pharmacol.* 13, 828269 <https://doi.org/10.3389/fphar.2022.828269>.
- Santhosh, M.S., Hemshekhar, M., Sunitha, K., Thushara, R.M., Jnaneshwari, S., Kemparaju, K., Girish, K.S., 2013. Snake venom induced local toxicities: plant secondary metabolites as an auxiliary therapy. *Mini-Rev. Med. Chem.* 13, 106–123. <https://doi.org/10.2174/1389557511307010106>.
- Sampat, G.H., Hiremath, K., Dodakallanavar, J., Patil, V.S., Harish, D.R., Biradar, P., Mahadevamurthy, R.K., Barvaliya, M., Roy, S., 2023. Unraveling snake venom phospholipase A₂: an overview of its structure, pharmacology, and inhibitors. *Pharmacol. Rep.* 75, 1454–1473. <https://doi.org/10.1007/s43440-023-00543-8>.
- Samy, R.P., Gopalakrishnakone, P., Chow, V.T., 2012. Therapeutic application of natural inhibitors against snake venom phospholipase A₂. *Bioinformation* 8, 48–57. <https://doi.org/10.6026/97320630008048>.
- Schneider, M.S., Min, K., Hamrick, P.N., Montebello, L.R., Ranieri, T.M., Mardini, L., Camara, V.M., Luiz, R.R., Liese, B., Vuckovic, M., Moraes, M.O., Lima, N.T., 2021a. Overview of snakebite in Brazil: possible drivers and toll for risk mapping. *PLoS Neglected Trop. Dis.* 15, 1. <https://doi.org/10.1371/journal.pntd.0009044>.
- Schneider, M.C., Vuckovic, M., Montebello, L., Sarpy, C., Huang, Q., Galan, D.I., Min, K., Camara, V., Luiz, R.R., 2021b. Snakebites in rural areas of Brazil by race: indigenous the most exposed group. *Int. J. Environ. Res. Publ. Health* 18, 9365. <https://doi.org/10.3390/ijerph18179365>.
- Shenoy, P.A., Nipate, S.S., Sonpetkar, J.M., Salvi, N.C., Waghmare, A.B., Chadhari, P.D., 2013. Anti-snake venom activities of ethanolic extract of fruits of *Piper longum* L. (Piperaceae) against Russell's viper venom: characterization of piperine as active principle. *J. Ethnopharmacol.* 147, 373–382. <https://doi.org/10.1016/j.jep.2013.03.022>.
- Silva, G.A., Domingos, T.F.S., Fonseca, R.R., Sanchez, E.F., Teixeira, V.L., Fuly, A.L., 2015. The red seaweed *Plocamium brasiliense* shows anti-snake venom toxic effects. *J. Venom. Anim. Toxins Incl. Trop. Dis.* 21, 1–9. <https://doi.org/10.1186/s40409-015-0002-2>.
- Silva, M.V.T., Garrett, R., Simas, D.L.R., Konno, T.U.P., Muzitano, M.F., Pinto, S.C., Barth, T., 2021. Chemical profile of *Stachytarpheta schottiana* by LC-HRMS/MS dereplication and molecular networking. *Rodriguesia* 72, e00772020. <https://doi.org/10.1590/2175-7860202172099>.
- Silva, I.F., Oliveira, F.F., Oliveira, R.A., 2021. *Siparuna* Aublet genus (Siparunaceae): from folk medicine to chemical composition and biological activity. *Trends Phytochem. Res* 5, 168–189. <https://doi.org/10.30495/tp.2021.1934361.1211>.
- Simas, N.K., Ferrari, S.F., Pereira, S.N., Leitão, G.G., 2001. Chemical ecological characteristics of herbivory of *Siparuna guianensis* seeds by Buffy-Headed Marmosets (*Callithrix flaviceps*) in the Atlantic forest of Southeastern Brazil. *J. Chem. Ecol.* 27, 93–107. <https://doi.org/10.1023/A:1005672017673>.
- Singh, P., Yasir, M., Hazarika, R., Sugunan, S., Shrivastava, R., 2017. A review on venom enzymes neutralizing ability of secondary metabolites from medicinal plants. *J. Pharmacopuncture* 20, 173–178. <https://doi.org/10.3831/KPL.2017.20.020>.
- Stévigny, C., Jiwan, J.L.H., Rozenberg, R., Hoffmann, E., Quetin-Leclercq, J., 2004. Key fragmentation patterns of aporphine alkaloids by electrospray ionization with multistage mass spectrometry. *Rapid Commun. Mass Spectrom.* 18, 523–528. <https://doi.org/10.1002/rcm.1343>.
- Sweilam, S.H., Alqarni, M.H., Youssef, F.S., 2022. Antimicrobial alkaloids from marine-derived fungi as Drug leads versus COVID-19 infection: a computational approach to explore their anti-COVID-19 activity and ADMET properties. *Evid Based Comp. Alternat Med.* 8, 5403757 <https://doi.org/10.1155/2022/5403757>.
- Torres, M.C., das Chagas, L., Pinto, F., Braz-Filho, R., Silveira, E.R., Pessoa, O.D., Jorge, R.J., Ximenes, R.M., Monteiro, H.S.A., Evangelista, J.S.A.M., Diz-Filho, E.B.S., Toyama, M.H., 2011. Antiophidic solanidane steroidal alkaloids from *Solanum campaniforme*. *J. Nat. Prod.* 74, 2168–2173. <https://doi.org/10.1021/np200479a>.

- Torres, M.C., Jorge, R.J., Ximenes, R.M., Alves, N.T., Santos, J.V., Marinho, A.D., Monteiro, H.S.A., Toyama, M.H., Braz-Filho, R., Silveira, E.R., Pessoa, O.D.L., 2013. Solanidane and iminosolanidane alkaloids from *Solanum campaniforme*. *Phytochemistry* 96, 457–464. <https://doi.org/10.1016/j.phytochem.2013.09.007>.
- Torres, A.M., Ojeda, G.A., Angelina, E., Bustillo, S., Peruchena, N., Tonidandel, L., Larcher, R., Nardin, T., Dellacassa, E., 2023. The anti-snake activity of *Nectandra angustifolia* flavonoids on phospholipase A₂: *In vitro* and *in silico* evaluation. *J. Ethnopharmacol.* 302, 115889 <https://doi.org/10.1016/j.jep.2022.115889>.
- Vásquez, J., Jiménez, S.L., Gómez, I.C., Rey, J., Henao, A., Marín, D.N., Romero, J., Alarcón, J., 2013. Snakebites and ethnobotany in the Eastern region of Antioquia, Colombia - the traditional use of plants. *J. Ethnopharmacol.* 146, 449–455. <https://doi.org/10.1016/j.jep.2012.12.043>.
- Vásquez, J., Alarcón, J.C., Jiménez, S.L., Jaramillo, G.I., Gómez-Betancur, I.C., Rey-Suárez, J.P., Jaramillo, K.M., Muñoz, D.C., Marín, D.M., Romero, J.O., 2015. Main plants used in traditional medicine for the treatment of snake bites in the regions of the department of Antioquia, Colombia. *J. Ethnopharmacol.* 170, 158–166. <https://doi.org/10.1016/j.jep.2015.04.059>.
- Vaou, N., Stavropoulou, E., Voidarou, C.C., Tsakris, Z., Rozos, G., Tsigalou, C., Bezirtzoglou, E., 2022. Interactions between medical plant-derived bioactive compounds: focus on antimicrobial combination effects. *Antibiotics* 11, 1014. <https://doi.org/10.3390/antibiotics11081014>.
- Vishwanath, B.S., Gowda, T.V., 1987. Interaction of aristolochic acid with *Vipera russelli* phospholipase A₂: its effect on enzymatic and pathological activities. *Toxicon* 25, 929–937. [https://doi.org/10.1016/0041-0101\(87\)90155-3](https://doi.org/10.1016/0041-0101(87)90155-3).
- Yang, J.Y., Sanchez, L.M., Rath, C.M., Liu, X., Boudreau, P.D., Bruns, N., Glukhov, E., Wodtke, A., de Felicio, R., Fenner, A., Wong, W.W., Linington, R.G., Zhang, L., Debonsi, H.M., Gerwick, W.H., Dorrestein, P.C., 2013. Molecular networking as a dereplication strategy. *J. Nat. Prod.* 76, 1686–1699. <https://doi.org/10.1021/np400413s>.

2. Fucoidans from the brown seaweeds *Undaria pinnatifida* and *Fucus vesiculosus* as inhibitors of coagulant and phospholipase A2 activities of pit viper *Lachesis muta* venom and purified enzymes. Camila Castro-Pinheiro, Eladio F. Sanchez, Corinna A. Dwan, Barbara C. Wimmer, Alan T. Critchley, André L. Fuly. Journal of Applied Phycology, 2026, <https://doi.org/10.1007/s10811-026-03838-4>

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Abstract	Snakebite envenomation is a neglected tropical disease and incidents of the pit viper <i>Lachesis muta</i> cause hemorrhage, muscle damage, blood coagulation disturbances, and even death; the proteases and phospholipase A ₂ (PLA ₂) of snake venoms contribute to such toxic effects. This work evaluated the inhibitory effect of commercial fucoidans from two species of brown seaweeds <i>Fucus vesiculosus</i> (FVF) and <i>Undaria pinnatifida</i> (UPF) against the coagulant and PLA ₂ activity of <i>Lachesis muta</i> venom and two purified enzymes, a thrombin-like enzyme and PLA ₂ , denoted TLE-1 and LM-PLA ₂ -I,
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respectively. *Fucus vesiculosus* and *U. pinnatifida* fucoidans were incubated with *L. muta* venom and enzymes for 5 min at 37 °C, followed by assays, including the plasma and fibrinogen coagulation, hydrolysis of the substrates of proteases (S-2238 and S-2288), and hydrolysis of the substrate of PLA₂ enzymes, NBD-PC. *Fucus vesiculosus* and *U. pinnatifida* fucoidans inhibited the plasma coagulation of *L. muta* venom, but not the coagulation of TLE-1. However, both fucoidans fully prevented the coagulation of fibrinogen caused by *L. muta* venom and TLE-1. *Fucus vesiculosus* and *U. pinnatifida* fucoidans inhibited 60% and 40% hydrolysis of S-2238 of *L. muta* venom and TLE-1, respectively. Using the substrate S-2288, the fucoidans did not affect the hydrolysis of *L. muta* venom and inhibited 40% of hydrolysis caused by TLE-1. *Fucus vesiculosus* and *U. pinnatifida* fucoidans inhibited 40% and 20% of the PLA₂ activity of *L. muta* venom and LM-PLA₂-I, respectively. These two fucoidans should be investigated as complementary or potential adjuvants for SBE inflicted by *L. muta* venom in more detail.

Keywords (separated by '-') Phaeophyceae - *Undaria pinnatifida* - *Fucus vesiculosus* - *Lachesis muta* - Fucoidan - Snake bite envenomation

Footnote Information



Fucoidans from the brown seaweeds *Undaria pinnatifida* and *Fucus vesiculosus* as inhibitors of coagulant and phospholipase A₂ activities of pit viper *Lachesis muta* venom and purified enzymes

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Abstract

Snakebite envenomation is a neglected tropical disease and incidents of the pit viper *Lachesis muta* cause hemorrhage, muscle damage, blood coagulation disturbances, and even death; the proteases and phospholipase A₂ (PLA₂) of snake venoms contribute to such toxic effects. This work evaluated the inhibitory effect of commercial fucoidans from two species of brown seaweeds *Fucus vesiculosus* (FVF) and *Undaria pinnatifida* (UPF) against the coagulant and PLA₂ activity of *Lachesis muta* venom and two purified enzymes, a thrombin-like enzyme and PLA₂, denoted TLE-1 and LM-PLA₂-I, respectively. *Fucus vesiculosus* and *U. pinnatifida* fucoidans were incubated with *L. muta* venom and enzymes for 5 min at 37 °C, followed by assays, including the plasma and fibrinogen coagulation, hydrolysis of the substrates of proteases (S-2238 and S-2288), and hydrolysis of the substrate of PLA₂ enzymes, NBD-PC. *Fucus vesiculosus* and *U. pinnatifida* fucoidans inhibited the plasma coagulation of *L. muta* venom, but not the coagulation of TLE-1. However, both fucoidans fully prevented the coagulation of fibrinogen caused by *L. muta* venom and TLE-1. *Fucus vesiculosus* and *U. pinnatifida* fucoidans inhibited 60% and 40% hydrolysis of S-2238 of *L. muta* venom and TLE-1, respectively. Using the substrate S-2288, the fucoidans did not affect the hydrolysis of *L. muta* venom and inhibited 40% of hydrolysis caused by TLE-1. *Fucus vesiculosus* and *U. pinnatifida* fucoidans inhibited 40% and 20% of the PLA₂ activity of *L. muta* venom and LM-PLA₂-I, respectively. These two fucoidans should be investigated as complementary or potential adjuvants for SBE inflicted by *L. muta* venom in more detail.

Keywords Phaeophyceae · *Undaria pinnatifida* · *Fucus vesiculosus* · *Lachesis muta* · Fucoidan · Snake bite envenomation

Introduction

Snake bite envenomation (SBE) is classified by the World Health Organization (WHO) as a neglected tropical disease, causing 2.7 million accidents, 100,000 deaths, and 400,000 disabilities annually worldwide; however, due of underreporting, the true numbers of these incidents are expected to be 40% higher (WHO 2019). SBE is a public health issue that mainly affects rural developing tropical, subtropical, and impoverished countries such as Asia, Africa, and Latin America (Gutiérrez et al. 2017; WHO 2019). The Viperidae (pit vipers or vipers) is considered the most venomous snake family of medical relevance in Latin America; and Brazil has approximately 360 species that cause approximately 25,000 incidents a year, accounting for 99% of the registered incidents (Roriz et al. 2018; SINAN 2018). In Brazil, four genera of vipers are found, namely *Bothrocephalus*, *Bothrops*, *Lachesis*, and *Crotalus* and among them,

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the genus *Bothrops* accounts for 87% of SBE (Roriz et al. 2018; SINAN 2018). Despite having the lowest rate of SBE (~1.5–2%), envenomation caused by the genus *Lachesis*, it has the highest mortality rates among vipers, approximately 1%, mainly due to substantial venom delivered (averaging 168–552 mg per snake) (Sanchez et al. 1992; SINAN 2018). *Lachesis*, commonly known as bushmaster, has four species distributed within the forests of Central and South America. This genus is the longest among vipers of the Americas reaching up to 2.65 m in length, has nocturnal habitats, and unique that lay eggs, approximately 5 and 19 with an incubation period of circa 70 days (Cañas et al. 2023) (Fig. 1). Brazil has one species, *Lachesis muta*, that is found in the Amazon and northern Atlantic forests from Ceará to Rio de Janeiro. Given these areas have low population density, incidents caused by *Lachesis* are rare (Magalhães et al. 2019; Sachett et al. 2022). Recent studies based on high-resolution genomics, venomomics, and morphological and ecological features recognized the Amazon populations as *L. muta*; while the Atlantic Forest as *L. rhombeata* (Hamdan et al. 2024).

Lachesis muta venom comprises a mixture of active proteins, peptides, and enzymes, which are responsible for a wide range of pathophysiological effects. The symptoms and signs of envenoming are characterized by local pain and inflammation, tissue damage, blistering, hypotension, neurotoxicity, heart and kidney failure, and systemic hemostatic disturbances e.g., hemorrhage, incoagulable blood and eventually death. (Fig. 1d) In the Amazon region of some countries of South America, *L. muta* and *B. atrox* are the major species of clinical importance, and responsible for most SBE with similar clinical features (Sanchez et al. 2006).

In general, snake venom serine proteases (SVSPs), snake venom metalloproteases (SVMPs), phospholipases (PLA₂s), hyaluronidases, L-amino acid oxidases (LAAOs) are a family of enzymes highly associated with venom toxicity and symptoms of SBE (Gutiérrez et al. 2017; Tasoulis and Isbister 2023). Three kind of SVSPs are reported in *L. muta* venom, including the fibrinogen-clotting (thrombin-like enzyme) that is involved in incoagulable blood, fibrin(ogen) proteolysis, and resembles at least in part the mechanism of action of the enzyme α -thrombin (Diniz and Oliveira 1992). The second is the plasminogen activator (LV-PA) that mainly affect several steps of the fibrinolytic system through plasminogen activation (Sanchez et al. 2000) and the last enzyme is the kallikrein-like (LV-Ka) that releases bradykinin from kininogen and contribute to hypotension, neurotoxic, and shock in human victims (Felicori et al. 2003; Weinberg et al. 2004). Thus, the action of such SVSPs may explain at least partially the pathophysiological effects of SBE by bushmasters.

Among biologically active enzymes, snake venom thrombin-like enzymes (SVTLEs) and PLA₂s deserve attention since most prominent toxic effects are attributed to them (Sampat et al. 2023; Srinivasan et al. 2024; Romanazzi et al. 2025). SVTLEs and PLA₂s are widely present in snake venoms and can activate the blood clotting cascade system, i.e., they cleave fibrinogen into fibrin and induce platelet aggregation leading to clot formation (Romanazzi et al. 2025). Some hemorrhage symptoms of SBE are due to thrombosis and consumption of plasma fibrinogen (Sampat et al. 2023; Romanazzi et al. 2025). At present, over forty SVTLEs have been purified from snake venoms, and thirty

Fig. 1 Representative images of a species of *Lachesis muta*, located at the Atlantic forest of Itacaré, State of Bahia, Brazil (A), laying eggs (C, D), and hemorrhage effect of envenomation by this species (D) provided by Dr. Rodrigo Sousa, Serra Grande Center (NSG) for captive breeding of *Lachesis muta*, located at the city of Itacaré, State of Bahia, Brazil



of which have their primary structure determined with key amino acid residues of the catalytic site, including His57, Asp102, and Ser195 (Castro et al. 2004). PLA₂ enzymes are a large family of enzymes found in mammalian tissues and in animal venoms, including snakes and are classified into several groups based on structural homology, disulphide pattern, and catalytic activity (Suranse et al. 2022). Furthermore, the presence of several isoforms of svPLA₂s reflects the diverse set of pathophysiological actions on human victims (Suranse et al. 2022). The catalytically active (Asp49 PLA₂s) and inactive (Lys49 PLA₂s) svPLA₂s have a central role in victims of SBE, and are involved in disrupting the cell membrane's integrity, tissue damage, edema, inflammation, pain, hemolysis, blood clotting disturbances, and anticoagulation and cardiorespiratory arrest; thus, they contribute to overall venom toxicity (Križaj 2023). These enzymes cleave phospholipids resulting in the formation of arachidonic acid and lysophospholipids that are responsible for the above mentioned toxic effects (Srinivasan et al. 2024). Despite being widely distributed in snake venoms, significant variability of toxic activity, as well as primary, secondary and tertiary structures, are reported in SVTLEs (Castro et al. 2004) and svPLA₂s (Srinivasan et al. 2024). One basic PLA₂ (LmTX-I) and one acidic PLA₂ (LM-PLA₂-I) were isolated from *L. muta* venom and some pharmacological and physico-chemical features were described (Damico et al. 2006; Fuly et al. 2022).

According to the WHO and Ministries of Health of many countries, heterologous antivenom is the only available treatment for SBE (WHO 2019). Antivenoms effectively prevent death but are less effective in protecting against muscle damage leading to an increase of physical sequelae, including amputations and disabilities (Camey et al. 2002). Moreover, antivenoms are costly and take time to produce and they may even cause fever or anaphylactic reactions. The effective recovery of bite victims relies on early and timely administration of such antivenoms (Isbister 2024). Therefore, seeking alternatives against the toxic effects of venom would greatly benefit the victims of SBE.

Oceans contain many living organisms able to produce a wide array of compounds with intriguing and novel chemical structures, as compared to the terrestrial environment (Abdelmohsen et al. 2025). A wide range of beneficial effects have been attributed to extracts of seaweeds, such as antiviral, antifungal, antidiabetic, and others (Mayer et al. 2024; Bouafir et al. 2025); these outstanding bioactivities have drawn attention of nutraceutical and pharmaceutical industries worldwide. In general, seaweeds contain an array of compounds derived from primary (e.g., proteins, lipids, and polysaccharides) and secondary metabolism (e.g., flavonoids, terpenoids, alkaloids, carotenoids, and polyphenols) (Bouafir et al. 2025). Brown algae are a large group of multicellular algae of the class

Phaeophyceae with 1,500–2,000 species worldwide which includes a wide array of morphologies and sizes (Mayer et al. 2024). They are the most widely researched group of macroalgae, due to the presence in high amounts of bioactive molecules; various species are located in tropical, temperate, and polar marine environments (Mayer et al. 2024; Bouafir et al. 2025).

The brown seaweed *Fucus vesiculosus* (i.e., Bladderwrack; Fucales) is geographically distributed in the North Pacific and North Atlantic oceans, shores of Europe, the Baltic Sea, Canary Islands, and Northern Russia. *Fucus vesiculosus* has been used in traditional medicine to treat symptoms of hypothyroidism, obesity, atherosclerosis, circulatory system disorders, and menopause (Catarino et al. 2018, 2017). *Undaria pinnatifida* (Laminariales) is an invasive species also known as “wakame” regularly used as food and in traditional medicine of Asian countries, including Japan, South Korea, and China (Wang et al. 2018; Cheong et al. 2025). Fucoidans or fucose-rich sulfated polysaccharides are derived from various brown seaweeds, such as *U. pinnatifida* and *F. vesiculosus*; these sulphated polysaccharides have low toxicity and many bioactivities have been ascribed, including immunomodulation, antitumoral, antiviral, anti-inflammatory, neuroprotective effects anticoagulant, and antivenom properties (Pozharitskaya et al. 2020; Xing et al. 2023; Castro-Pinheiro et al. 2024). Depending on the species from which an extraction has been made and the extraction method used, fucoidans have molecular weights (MW) from five to several hundred kDa with noted variations on the sugar composition, glycosidic bonds, branching, and degree of sulfatation and such issues contribute to the widespread biomedical application of fucoidans (Lomartire and Gonçalves 2022; Abbas et al. 2025). The diversity in structure and MW contributes to differences in bioactivities and subsequent biomedical application of fucoidan polysaccharide (Abbas et al. 2025). As such, there is no one type of fucoidan and the term fucose-containing polymer (FCP) is perhaps the most appropriate. Previous reports of our research group have evidenced that FVF and UPF are non-hemotoxic molecules and inhibitory ability of FVF and UPF fucoidans against the proteolytic, plasma and fibrinogen coagulant, and PLA₂ activities of *Bothropus jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. In order to continue assessing fucoidans as inhibitors, it is of great interest assessing their potential to inhibit other crude venoms and purified enzymes of pit vipers.

Thus, because of antivenom potential of such molecules, the present work aimed to investigate the effects of commercially available fucoidan polysaccharides from *Fucus vesiculosus* and *Undaria pinnatifida* against the plasma and fibrinogen coagulant activity and PLA₂ activity of *L. muta* crude venom and two purified enzymes, TLE-1 and LM-PLA₂-I which were derived from this venom.

Material and methods

Snake venom, purified enzymes, and materials

Lachesis muta lyophilized venom was kindly provided by the serpentarium of Ezequiel Dias Foundation (FUNED), Belo Horizonte, Minas Gerais, Brazil. The venom was dissolved in saline solution at the concentration of 1 mg mL⁻¹ and stored at -20 °C prior to use. Snake venom collection was conducted under the authorization of the Brazilian National System for Genetic Heritage Management and Associated Traditional Knowledge (SISGEN), process number A39CTRI 04E. The purified enzymes LM-PLA₂-I and TLE-1 (at concentrations of 0.5 mg mL⁻¹) were isolated from the *L. muta* venom, accordingly to Fuly et al. (1997, 2022) and Magalhães et al. (2003), respectively.

Fibrinogen was purchased from Sigma Chemical Co and dissolved in saline solution concentration of 0.9% w/v to a stock concentration of 4 mg mL⁻¹. Chromogenic substrates S-2238 and S-2288 were bought from Chromogenix (Italy) and dissolved in Milli Q water to make up a stock solution of 4 mM. The fluorescent phospholipid substrate acyl-NBD-phosphatidylcholine (18:1-06:0 NBD-PC, code (810132P -1 mg) was purchased from Avanti Lipid Polar (USA). Powder NBD-PC was reconstituted in 200 µL of 100% chloroform, dried in glass tube under a slow stream of pure nitrogen gas (N₂), and dissolved in a saline solution to a concentration of 1 mg mL⁻¹. Tubes containing NDP-PC were sonicated at 4 °C three times, at intervals of 1 min and stored at -20 °C prior to use. All other reagents were of the best analytical grades.

Fucoidan extracts

Fucus vesiculosus fucoidan (FVF) and *Undaria pinnatifida* fucoidan (UPF), batch number DPFVF2021001 and DPGFS2019537 respectively were kindly supplied by Marinova (Cambridge, Tasmania, Australia). These commercial samples were produced using a proprietary aqueous extraction method by Marinova with a purity grade of > 95%. A stock solution of FVF and UPF of 10 mg mL⁻¹ dissolved in a physiological saline solution was prepared and stored at -20 °C prior to use. The mass and chemical compositions of FVF and UPF have been elucidated as described in Castro-Pinheiro et al. (2024).

Effects of FVF and UPF on coagulant activity of *L. muta* venom and TLE-1

The coagulant activity of *L. muta* venom and TLE-1 was measured using a digital Amelung coagulometer, model KC4A

(Labcon, Germany). 200 µL fibrinogen (4 mg mL⁻¹) or human plasma were incubated in the coagulometer for 1 min at 37 °C and aliquots of 50 µL containing different concentrations of *L. muta* venom (5–50 µg mL⁻¹) were added to plasma or fibrinogen and the coagulation time was monitored in seconds (s). The concentration of *L. muta* venom (µg mL⁻¹) able to clot plasma or fibrinogen at approximately 60 s was denoted as the arbitrary unit minimum coagulant concentration (MCC). A concentration–effect curve was not performed for TLE-1 due to a limited availability of this enzyme. Thus, a single challenge concentration of TLE-1 that clotted plasma or fibrinogen was approximately 50 µg mL⁻¹. One MCC of *L. muta* venom (7 µL, 30 µg mL⁻¹) or TLE-1 (11 µL, 50 µg mL⁻¹) was incubated for 5 min at 37 °C with either saline solution (positive control), UPF or FVF at the ratio of 1:10 venom-to-fucoidan (wt/wt) ratio, in a final volume of 220 µL adjusted with physiological saline solution. Then, an aliquot (50 µL) of each mixture was added to a reaction medium containing 200 µL plasma or fibrinogen. Coagulation time was registered as described above. Negative controls contained saline solution, FVF, or UPF, in the absence of venom or TLE-1. Each treatment group was assessed in quadruplicate in two independent replicates, yielding n = 8. The pooled human plasma was obtained from the blood bank of the Hospital Universitário Antônio Pedro (HUAP) under the authorization of the Committee for Ethics in Experimentation (CEP-UFF, CAAE: 28,941,314.0.0000.5243).

Hydrolysis of chromogenic substrates S-2238 and S-2288

Hydrolysis of chromogenic substrates S-2238 and S-2288 was monitored via a microplate reader from Biotek Synergy H1 Hybrid Multimode Reader (Agilent, USA), using Gen5 software. 1.8 µL of FVF and UPF samples (100 µg mL⁻¹) were incubated with 6 µL *L. muta* venom (30 µg mL⁻¹) and 18 µL TLE-1 (50 µg mL⁻¹) for 5 min at 37 °C in a medium containing 20 mM Tris-HCl, 150 mM NaCl, pH 7.5 in a final volume of 180 µL. The enzymatic activity was triggered by adding 20 µL of S-2238 and S-2288 (0.4 mM, final concentration) and the hydrolysis of substrates was monitored at the absorbance of 405 nm for 5 min at 37 °C. Negative replicated controls were performed by adding saline solution or fucoidans in the absence of venom and TLE-1. The final volume of reaction mixture was 200 µL, and each experimental group was assessed in triplicate of two independent replicates, yielding n = 6.

Effect of FVF and UPF on the PLA₂ activity of *L. muta* venom and LM-PLA₂-I

PLA₂ activity of *L. muta* venom and the purified enzyme LM-PLA₂-I was assessed using the fluorescent phospholipid

substrate NBD-PC. The assay was performed using a Biotek Synergy H1 Hybrid Multimode Reader (Agilent, USA) using Gen5 software. 6 μL *L. muta* venom (10 $\mu\text{g mL}^{-1}$) or 12 μL LM-PLA₂-I (30 $\mu\text{g mL}^{-1}$) were incubated with saline solution (positive control), 1.8 μL FVF, and UPF (100 $\mu\text{g mL}^{-1}$) for 5 min at 37 °C in a medium containing 8 mM CaCl₂, 20 mM Tris-HCl, pH 7.5, in a final volume of 180 μL . After this, enzymatic reaction was triggered by adding 5 μL of NBD-PC (27 $\mu\text{g mL}^{-1}$, final concentration) and the kinetic of reaction was monitored for 5 min at 37 °C at the excitation and emission wavelengths of 460 and 534 nm, respectively in a final volume of 200 μL adjusted with 20 mM Tris-HCl, pH 7.5. An initial record of 1-min time scan was firstly taken with all the components, except *L. muta* venom or LM-PLA₂-I, and the readings were subtracted from this initial reading. Each experimental group was assessed as three replicates in two independent experiments, yielding $n = 6$.

Statistical analysis

Results were expressed as the means $\pm$ standard deviation (SD) and analysis of variance (ANOVA), followed by Tukey's post hoc test and graphs were constructed using Graph Prism version 9.0. $p < 0.05$ were considered to indicate statistical significance.

Results

Effects of UPF and FVF on the plasma and fibrinogen coagulation

LACHESIS muta venom (30 $\mu\text{g mL}^{-1}$) mixed in saline solution (positive control group) clotted the human plasma (Fig. 2A, black columns), and fibrinogen (Fig. 2A, white columns) at approximately 25 s and 10 s, respectively.

Incubation of *L. muta* venom with fucoidan UPF or FVF (100 $\mu\text{g mL}^{-1}$) fully prevented the venom-induced coagulation on fibrinogen over the entire period of monitoring coagulation, which was 900 s (Fig. 2A, white columns). On the other hand, in the plasma, the venom-induced coagulation occurred at approximately 50 s in the presence of FVF and UPF (Fig. 2A, black column); thus coagulation time was delayed twice. As seen in Fig. 2B (black columns), FVF and UPF (100 $\mu\text{g mL}^{-1}$) did not prevent plasma coagulation of TLE-1 (50 $\mu\text{g mL}^{-1}$) but inhibited the coagulation of fibrinogen caused by TLE-1 (Fig. 2B, white columns). The clot formed by the action of TLE-1 on plasma was visually softer than that formed by *L. muta* venom (data not shown) and this difference may be related to inefficacy of TLE-1 to activate the coagulation XIII factor which is responsible for stabilizing the fibrin clot. As negative control, UPF or FVF neither clotted plasma nor fibrinogen in the absence of *L. muta* venom and TLE-1 under the experimental conditions (data not shown).

Effects of FVF and UPF on hydrolysis of chromogenic substrates

The chromogenic substrate S-2238 is specific for thrombin and S-2288 is sensitive to a wide range of serine proteases. These substrates have been used to detect the presence of such enzymes in a medium, as well as to assess the effects of inhibitors of thrombin-like enzymes and serine proteases. As seen in Fig. 3, a total of 100% of activity on S-2238 (A,B) and S-2288 (C,D) was achieved with *L. muta* venom (30 $\mu\text{g/mL}$) and TLE-1 (50 $\mu\text{g/mL}$). FVF and UPF inhibited hydrolysis of S-2238 caused by *L. muta* venom (Fig. 3A) and TLE-1 (Fig. 3B) with percentages of 38% and 54%, respectively. On the other hand, FVF and UPF did not inhibit hydrolysis of S-2288 caused by *L. muta* venom (Fig. 3C) but inhibited 60% of such activity caused by TLE-1 (Fig. 3D).

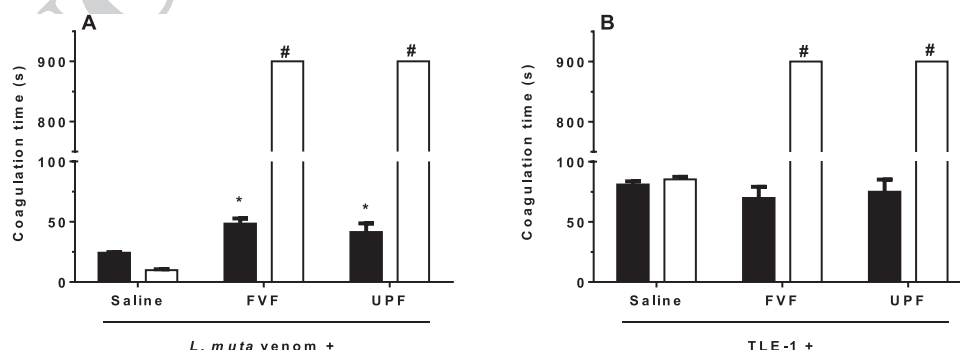
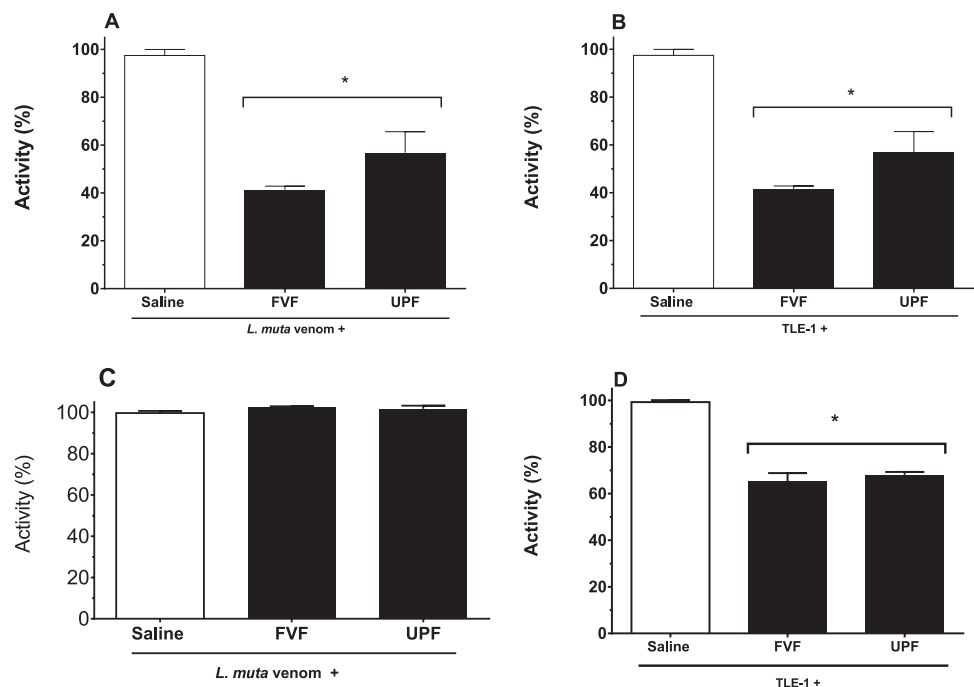


Fig. 2 Effect of UPF or FVF on coagulant activity of *Lachesis muta* venom and TLE-1. *Lachesis muta* venom (A) and TLE-1 (B) were incubated with saline, FVF, or UPF for 5 min at 37 °C. An aliquot of the mixture was added to plasma (black columns) and fibrinogen

(white columns), and the coagulation time was monitored in seconds. Results are expressed as means $\pm$ SEM ($n = 8$). # means that fibrinogen did not clot until 900 s of monitoring. * $p < 0.05$ as compared with *L. muta* venom or TLE-1 plus saline

Fig. 3 Effect of FVF and UPF on the hydrolysis of S-2238 and S-2288 of *Lachesis muta* venom and TLE-1. *Lachesis muta* venom and TLE-1 were incubated with saline (control groups, white columns), FVF, or UPF for 5 min at 37 °C. Then, an aliquot was added to the medium and the hydrolysis of S-2238 (A, B) and S-2288 (C, D) was monitored for 5 min at 37 °C as described in the method section. A total of 100% activity was obtained with the readouts at 50 s by the control group and 0% was obtained with fucoidans or saline solution without *L. muta* venom and TLE-1. Results are expressed as means $\pm$ SEM (n=6) of two individual experiments. * $p < 0.05$ as compared with *L. muta* venom or TLE-1 plus saline



In the absence of *L. muta* venom or TLE-1, FVF and UPF (100 $\mu\text{g mL}^{-1}$) did not hydrolyze S-2238 and S-2288 under the experimental conditions (data not shown).

Effects of FVF and UPF on hydrolysis of fluorescent substrate

Lachesis muta venom (10 $\mu\text{g mL}^{-1}$) and LM-PLA₂-I (30 $\mu\text{g mL}^{-1}$) incubated with saline solutions, were found to hydrolyze the phospholipid substrate NBD-PC and this was considered as 100% of activity (Fig. 4). This lipid substrate is employed to test enzymatic activity of PLA₂ enzymes of different sources. In the presence of FVF or UPF (100 $\mu\text{g mL}^{-1}$), an inhibition of 40% and 10–20% was

achieved for *L. muta* venom (Fig. 4A) and LM-PLA₂-I (Fig. 4B), respectively. Both FVF and UPF did not hydrolyze the substrate NBD-PC in the absence of *L. muta* venom or LM-PLA₂-I under the experimental conditions (data not shown).

Discussion

The ability of fucoidans extracted from the brown seaweeds *F. vesiculosus* (FVF) and *U. pinnatifida* (UPF) to inhibit coagulant, proteolytic, and PLA₂ activity of *B. jararaca*, *B. jararacussu*, and *B. neuwiedi* crude venom was previously demonstrated by Castro-Pinheiro et al. (2024). This

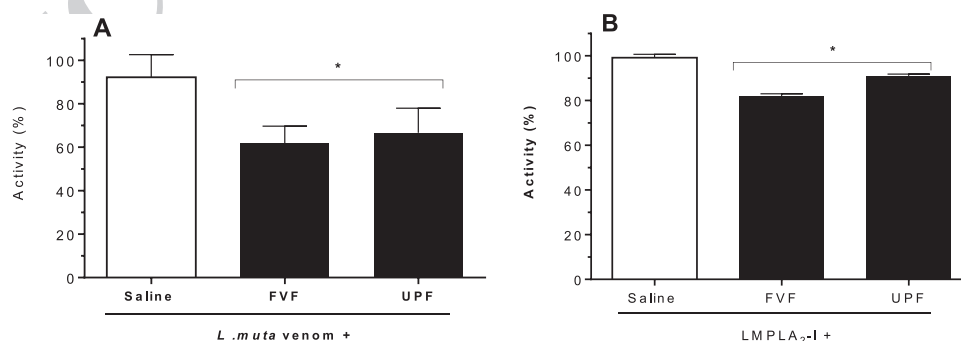


Fig. 4 Effect of FVF and UPF on the hydrolysis of NBD-PC caused by *Lachesis muta* venom and LM-PLA₂-I. *Lachesis muta* venom (A) and LM-PLA₂-I (B) were incubated with saline (positive control, white columns), FVF, or UPF for 5 min at 37 °C. NBD-PC solution was added to the medium and the reaction was monitored for a

further 10 min, as described in the method section. A total of 100% activity was obtained with readouts achieved at the 50 s time point of the reaction, in the presence of venom, or TLE-1. Results are expressed as means $\pm$ SD (n=6). * $p < 0.05$ as compared with *L. muta* venom or LMPLA₂-I plus saline

research was specifically on the *Bothrops* genus of pit viper snakes which are endemic to Brazil and other South American countries. A study conducted by Azofeifa et al. (2008) and Angulo and Lomonte (2003) described antimyotoxic effect of fucoidan extracted from *F. vesiculosus* against *B. asper* venom and purified myotoxins. However, in the present work, the antivenom effect of species-specific fucoidans against the most lethal specie of Brazil, *L. muta*, as well as two purified enzymes from this venom, the thrombin-like enzyme TLE-1 and PLA₂ (LM-PLA₂-I) are described for the first time. Undoubtedly, it is urgent to inhibit the toxic effects caused by the major groups of venomous snakes that are responsible for the highest death rates after envenomation. The genera *Bothrops* and *Lachesis* are responsible for majority of SBE reported in Brazil (SINAN 2018). This new approach of using specific fucoidans against *L. muta* venom and purified enzymes opens up possibilities to develop antivenoms against SBE by vipers and can also be a step towards assessing the mechanisms of action against the main toxic classes of enzymes, including SVTLEs and PLA₂s. TLEs and PLA₂s have a pivotal role in the symptoms of SBE since they are myotoxic, hemolytic, and activate and/or inhibited platelet aggregation induced by collagen, arachidonic acid, and adenosine diphosphate (Fuly et al. 1997; 2022; Magalhães et al. 2003). As mentioned in the introduction, SBE caused by *L. muta* venom produces a wide range of toxic symptoms in victims and the purified enzymes, TLE-1 and LM-PLA₂-I significantly contribute to the toxicity of *L. muta* venom. Thus, inhibition of the activity of such enzymes should be of great interest to relieve local venom toxicity and aid commercial antivenoms to treat victims of SBE.

In general, SVTLEs are glycoproteins and the carbohydrate moiety of such a family plays an important role on the catalytic activity, structural stabilization, and interaction with substrates (Ullah et al. 2018). As thrombin, SVTLEs have unique structures, causing cleave of the fibrinogen, they have a primary structure homology, and contain key amino acid residues of catalytic site (as Ser195, His57, and Asp102) (Ullah et al. 2018). Thrombin has two positively charged regions *i.e.*, the anion-binding exosites I and II that are located at a certain distance from the catalytic site of the enzyme (Fredenburgh and Weitz 2025). In general, exosites I is referred to interact with fibrinogen, heparin cofactor II, and the blood proteins of the coagulate cascade factors V and Va; while exosite II is responsible for enzymatic activity of thrombin and interaction with heparin (Fredenburgh and Weitz 2025). Moreover, some SVTLEs have been commercially used as medicines, including Ancrod (Arvin™, Knoll, Ludwigshafen, Germany) which is isolated from *Calloselasma rhodostoma*, Defibrase™ from *B. moojeni*, Reptilase™ from *B. atrox*, Crotalase from *Crotalus adamanteus*, and a TLE

purified from *Agkistrodon contortrix* (Castro et al. 2004); though the use of them as medicines has been discontinued due to immunological reactions and high production costs (Castro et al. 2004). Literature has described inhibition of thrombin and SVTLEs activity through chemicals, such as phenylmethylsulfonyl fluoride (PMSF) that is able to bind serine residue at position 182 of the active site of such proteases (Vilca-Quispe et al. 2023). Some commercial anticoagulants, including heparin and hirudin (a leech-derived thrombin inhibitor) inhibit the procoagulant effect of thrombin and SVTLEs either through binding to exosites II or I, which do not block the enzymatic activity (Troisi et al. 2021). The negatively charged structure of heparin as well as the sulfate groups of it allows a very strong interaction of this polysaccharide with thrombin (Troisi et al. 2021; Chan et al. 2024). FVF and UPF have negatively charged chemical groups within their structure, but different from heparin, they have high molecular weight (Castro-Pinheiro et al. 2024). Despite having high molecular weights, FVF and UPF (up to concentrations of 1,500 µg mL⁻¹) are considered non-toxic molecules through the hemocompatibility assays and thus, having potential to the development of medicines (Castro-Pinheiro et al. 2024). Bothrojaracin, a C-type lectin isolated from *B. jararaca* venom is another promising molecule to be evaluated as an inhibitor since it interacts with exosite I and allosterically inhibits the catalytic activity of thrombin measured by the cleavage of S-2238 (Zingali et al. 2005). As seen in the present work, FVF and UPF inhibited the hydrolysis of S-2238 and S-2288 caused by the purified TLE-1 with similar inhibitory percentages (60%). Curiously, fucoidans were unable to inhibit hydrolysis of S-2288 by *L. muta* venom and inhibited less hydrolysis of S-2238 (38%). As seen in literature, SVTLEs are able to hydrolyze such substrates and are inhibited by classical protease inhibitors, including PMSF and benzamidine (Zaqueo et al. 2014). The negatively charged regions of such compounds from different sources are crucial to interact with thrombin and SVTLEs leading to an inhibitory effect of the pharmacological effects and enzymatic activity of these enzymes (Fuglestad et al. 2012). Therefore, a similar mechanism of action may be attributed to FVF and UPF against TLE-1, due to the negative chemical groups within their structures as evidenced in the work of Castro-Pinheiro et al. (2024). Synthetic peptides based on the chemical structure of their substrates and natural inhibitors have been assessed to inhibit SVSPs from *B. jararaca* and *B. atrox* venoms (Saladini et al. 2024), but without significant results. Therefore, the polysaccharides FVF and UPF have a negatively charged structure that would allow them bind to cationic aminoacids (Castro-Pinheiro et al. 2024), but a docking approach may be a valuable strategy for future research to further investigate the structural-activity

relationship, as well as an important tool to develop new molecules as classes of inhibitors of pathogenic toxins, such as SVTLEs and PLA₂s (Gutiérrez et al. 2025).

As seen in this work, FVF and UPF inhibited the PLA₂ activity of *L. muta* crude venom and purified LM-PLA₂-I as assessed by using the commercial fluorescent lipid substrate, NBD-PC. PLA₂s enzymes have key amino acid residues, including His48, Asp49, Tyr52, Tyr73, and Asp99 that comprise the catalytic domain (Chioato and Ward 2003; Salvador et al. 2024) as well as others that facilitate binding to the phospholipids of cellular membranes. Divalent Ca⁺² is an essential cofactor to PLA₂s exert their enzymatic activity (Castro-Amorim et al. 2023). One may speculate that the inhibitory effects of fucoidan, against PLA₂ occurred due to the negatively charged structure, especially the sulfate groups, thereby allowing fucoidans to react with such positively charged residues, including Ca⁺². Varespladib, a synthetic PLA₂ inhibitor was referred by the U.S. Food and Drug Administration (FDA) as a drug for SBE, and its mode of action occurs through binding to hydrophobic groove and Ca⁺²-binding active site, thus inhibiting the toxic effects of PLA₂s of different snake species, including bleeding, hemorrhage, neurotoxicity, myonecrosis, edema, and death (Lewin et al. 2022; Salvador et al. 2023; Sampat et al. 2023). On the other hand, Varespladib at lower concentrations inhibited PLA₂ activity of *L. muta rhombeata* but did not inhibit metalloproteases of this venom (Sampat et al. 2023). As a strategy to increase effectiveness of antivenom, Varespladib was mixed with some venom neutralizing antibodies as a cocktail to rescue mice against lethality of Elapidae venoms (Glanville et al. 2025). In general, molecules derived from primary and secondary metabolites from specific plants and seaweeds act against the toxic activities of snake venoms through the chelation of metals, protein precipitation, or interaction of amino acid residues of the active site, including His48 and Asp49 (Sampat et al. 2023). Data from the literature showed that FVF fucoidans are approved by the FDA in the category of Food Ingredients as “Generally Recognized As Safe” (GRAS); while UPF is sometimes employed for thyroid disorders related to iodine deficiency. However, due to its high iodine content, the use of this seaweed as supplement or herbal preparation should be taken into account because of the possibility of adverse effects and interactions with other medicines, including lithium carbonate or antihypertensive drugs (Gras 2018; Citkowska et al. 2019). Although the mechanism of action of PLA₂s is not fully understood, FVF and UPF are promising inhibitors of their enzymatic activities. Some classical inhibitors have been reported, as p-bromophenacyl bromide (p-BPB) that reacts irreversibly with His48 of enzymatically active PLA₂s (Fuly et al. 1997, 2022; Soares and Giglio 2003); however, it is considered toxic to humans and thus patently not feasible for medical use to treat SBE.

In many locations, SBE represents an occupational and environmental disease and causes a significant ongoing negative economic impact to those communities which reside mainly in impoverished and rural areas (Roriz et al. 2018; Junkes et al. 2024). Moreover, physiological damages associated with SBE are increasing and strongly affecting health-care system and workers (Junkes et al. 2024). Therefore, specific fucoidan polysaccharides, as FVF and UPF may open up a possibility for drug design or their use in combination with existing commercial antivenom as inhibitors of SVSPs and PLA₂s that are considered the most toxic of snake venoms; thus, may contribute to the relief of toxic symptoms of envenomation by *L. muta* venom thereby saving lives.

Author contribution CCP: Investigation, Methodology, Writing – original draft preparation, Writing – review & editing EFS: Investigation, Resources, Writing – review & editing CAD and BCW: Investigation, Resources, Writing – review & editing ATC: Investigation, Writing – review & editing ALF: Data curation, Conceptualization, Funding acquisition, Project administration, Supervision, Investigation, Methodology, Writing – original draft preparation, Writing – review & editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflicts of interest The authors declare no competing interests.

References

- Abbas MF, Karim DK, Kareem HR, Kamil MM, Al-Musawi MH, Asker MH, Ghanami M, Shahriari-Khalaji M, Sattar M, Mirhaj M, Sharifianjazi F, Tavamaishvili K, Mohabbatkah M, Soheily A, Noory P, Tavakoli M (2025) Fucoidan and its derivatives: from extraction to cutting-edge biomedical applications. *Carbohydr Polym* 357:123468
- Abdelmohsen UR, Balasubramanian S, Oelschlaeger TA, Grkovic T, Pham NB, Quinn RJ, Hentschel U (2025) Potential of marine natural products against drug-resistant fungal, viral, and parasitic infections. *Lancet Infect Dis* 17:e30–e41
- Angulo Y, Lomonte B (2003) Inhibitory effect of fucoidan on the activities of crotaline snake venom myotoxic phospholipases A(2). *Biochem Pharmacol* 66:1993–2000
- Azofeifa K, Angulo Y, Lomonte B (2008) Ability of fucoidan to prevent muscle necrosis induced by snake venom myotoxins:

- comparison of high- and low-molecular weight fractions. *Toxicon* 51:373–380
- Bouafir Y, Bouhenna MM, Nebbak A, Belfarhi L, Aouzal B, Boufahja F, Bendif H, Bruno M (2025) Algal bioactive compounds: a review on their characteristics and medicinal properties. *Fito-terapia* 183:106591
- Camey KU, Velarde DT, Sanchez EF (2002) Pharmacological characterization and neutralization of the venoms used in the production of Bothropic antivenom in Brazil. *Toxicon* 40:501–509
- Cañas CA, Castaño-Valencia S, Castro-Herrera F (2023) The Colombian bushmasters *Lachesis acrochorda* (Garcia, 1896) and *Lachesis muta* (Linnaeus, 1766): snake species, venoms, envenomation, and its management. *Toxicon* 230:107152
- Castro HC, Zingali RB, Albuquerque MG, Pujol-Luz M, Rodrigues CR (2004) Snake venom thrombin-like enzymes: from Reptilase to now. *Cell Mol Life Sci* 61:843–856
- Castro-Amorim J, de Novo Oliveira A, Da Silva SL, Soares AM, Mukherjee AK, Ramos MJ, Fernandes PA (2023) Catalytically active snake venom PLA₂ enzymes: an overview of its elusive mechanisms of reaction. *J Med Chem* 66:5364–5376
- Castro-Pinheiro C, Junior LCSP, Sanchez EF, da Silva ACR, Dwan CA, Karpiniec SS, Critchley AT, Fuly AL (2024) Effect of seaweed-derived fucoidans from *Undaria pinnatifida* and *Fucus vesiculosus* on coagulant, proteolytic, and phospholipase A₂ activities of snake *Bothrops jararaca*, *B. jararacussu*, and *B. neuwiedi* venom. *Toxins* 16:188
- Catarino MD, Silva AMS, Cardoso SM (2018) Phytochemical constituents and biological activities of *Fucus* spp. *Mar Drugs* 16:249
- Cavecci-Mendonça B, Luciano KM, Vaccas T, de Oliveira LA, Clemente EF, Rossini BC, Vieira JCS, de Barros LC, Biondi I, de Magalhães Padilha P, Santos LDD (2023) Preliminary insights of Brazilian snake venom metalloproteomics. *Toxins* 15:648
- Chan ED, King PT, Bai X, Schoffstall AM, Sandhaus RA, Buckle AM (2024) The inhibition of serine proteases by serpins is augmented by negatively charged heparin: a concise review of some clinically relevant interactions. *Int J Mol Sci* 25:1804
- Cheong KL, Chen W, Wang M, Zhong S, Veeraperumal S (2025) Therapeutic prospects of *Undaria pinnatifida* polysaccharides: extraction, purification, and functional activity. *Mar Drugs* 23:163
- Chioato L, Ward RJ (2003) Mapping structural determinants of biological activities in snake venom phospholipases A₂ by sequence analysis and site directed mutagenesis. *Toxicon* 42:869–883
- Citkowska A, Szekalska M, Winnicka K (2019) Possibilities of fucoidan utilization in the development of pharmaceutical dosage forms. *Mar Drugs* 17:458
- Damico DC, Bueno LG, Rodrigues-Simioni L, Marangoni S, da Cruz-Höfling MA, Novello JC (2006) Functional characterization of a basic D49 phospholipase A₂ (LmTX-I) from the venom of the snake *Lachesis muta muta* (bushmaster). *Toxicon* 47:759–765
- Diniz MRV, Oliveira EB (1992) Purification and properties of a kininogenin from the venom of *Lachesis muta* (bushmaster). *Toxicon* 30:247–258
- Felicori LF, Souza CT, Velarde DT, Magalhaes A, Almeida AP, Figueiredo S, Richardson M, Diniz CR, Sanchez EF (2003) Kallikrein-like proteinase from bushmaster snake venom. *Protein Expr Purif* 30:32–42
- Fredenburgh JC, Weitz JI (2025) Exosite crosstalk in thrombin. *J Thromb Haemost* 23:1160–1168
- Fuglestad B, Gasper PM, Tonelli M, McCammon JA, Markwick PR, Komives EA (2012) The dynamic structure of thrombin in solution. *Biophys J* 103:79–88
- Fuly AL, de Miranda AL, Zingali RB, Guimarães JA (2022) Purification and characterization of a phospholipase A₂ isoenzyme isolated from *Lachesis muta* snake venom. *Biochem Pharmacol* 63:1589–1597
- Fuly AL, Machado OL, Alves EW, Carlini CR (1997) Mechanism of inhibitory action on platelet activation of a phospholipase A₂ isolated from *Lachesis muta* (Bushmaster) snake venom. *Thromb Haemost* 78:1372–1378
- Glanville J, Bellin M, Pletnev S, Zhang B, Andrade JC, Kim S, Tsao D, Verardi R, Bedi R, Liao S, Newland R, Bayless NL, Youssef S, Tully ES, Bylund T, Kim S, Hirou H, Liu T, Kwong PD (2025) Snake venom protection by a cocktail of varespladib and broadly neutralizing human antibodies. *Cell* 188:3117–3134
- GRAS Notice (2018) No. GRN 000661. Available online: <https://www.fda.gov/food/generally-recognized-safe-gras/gras-notice-inventory>
- Gutiérrez JM, Calvete JJ, Habib AG, Harrison RA, Williams DJ, Warrell DA (2017) Snakebite envenoming. *Nat Rev Dis Primers* 14:17063
- Gutiérrez JM, Casewell N, Laustsen AH (2025) Progress and challenges in the field of snakebite envenoming therapeutics. *Annu Rev Pharmacol Toxicol* 65:465–485
- Hamdan B, Bonatto SL, Rödder D, Seixas VC, Santos RMF, Santana DJ, Machado LG, Kleiz-Ferreira JM, de Freitas MA, Gonzalez RC, Cavalcante T, de Souza MB, Régis CB, Fernandes DS, Fernandes-Ferreira H, Zingali RB (2024) When a name changes everything: taxonomy and conservation of the Atlantic bushmaster (*Lachesis Daudin, 1803*) (Serpentes: Viperidae: Crotalinae). *Syst Biodivers* 22:2366215
- Isbister GK (2024) The critical time period for administering antivenom: golden hours and missed opportunities. *Clin Toxicol* 62:277–279
- Junkes VH, Matos C, Matias GS, Lermen FH, Patriarca R, Hugo Siqueira V, Giane Lenzia G (2024) Occupational risks of work in the agricultural sector: a systematic literature review. *Syst Rev Prod* 34:2024
- Lewin MR, Carter RW, Matteo IA, Samuel SP, Rao S, Fry BG, Bickler PE (2022) Varespladib in the treatment of snakebite envenoming: development history and preclinical evidence supporting advancement to clinical trials in patients bitten by venomous snakes. *Toxins* 14:783
- Lomartire S, Gonçalves AMM (2022) An overview of potential seaweed-derived bioactive compounds for pharmaceutical applications. *Mar Drugs* 20:141
- Magalhães A, Ferreira R, Richardson M, Gontijo S, Yarleque A, Magalhaes H, Bloch C, Sanchez E (2003) Coagulant thrombin-like enzymes from the venoms of Brazilian and Peruvian bushmaster (*Lachesis muta muta*) snakes. *Comp Biochem Physiol B* 136:255–266
- Magalhães SFV, Peixoto HM, Moura N, Monteiro WM, de Oliveira MRF (2019) Snakebite envenomation in the Brazilian Amazon: a descriptive study. *Trans R Soc Trop Med Hyg* 113:143–151
- Mayer AMS, Mayer VA, Swanson-Mungerson M, Pierce ML, Rodríguez AD, Nakamura F, Taglialatela-Scafati O (2024) Marine pharmacology in 2019–2021: marine compounds with antibacterial, antidiabetic, antifungal, anti-inflammatory, antiprotazoal, antituberculosis and antiviral activities; affecting the immune and nervous systems, and other miscellaneous mechanisms of action. *Mar Drugs* 22:309
- Pozharitskaya ON, Obluchinskaya ED, Shikov AN (2020) Mechanisms of bioactivities of fucoidan from the brown seaweed *Fucus vesiculosus* L. of the Barents Sea. *Mar Drugs* 18:275
- Križaj I (2023) Toxinology and pharmacology of snake venoms. *Toxins* (Basel) 15:212
- Romanazzi M, Filardi ETM, Pires GMM, Cerveja MF, Melo-Dos-Santos G, Oliveira IS, Ferreira IG, Cerni FA, Santos-Filho NA, Monteiro WM, Almeida JR, Vaiyapuri S, Pucca MB (2025) The versatility of serine proteases from Brazilian *Bothrops* venom: their roles in snakebites and drug discovery. *Biomolecules* 15:154

- Roriz KRP, Zaqueo KD, Setubal SS, Katsuragawa TH, Silva RRD, Fernandes CF, Cardoso LAP, Rodrigues MMS, Soares AM, Stábeli RG, Zuliani JP (2018) Epidemiological study of snakebite cases in Brazilian western Amazonia. *Rev Soc Bras Med Trop* 51:338–346
- Sachett JAG, Marinho APS, Santos MMO, Fan HW, Bernarde PS, Monteiro WM (2022) When to think about a *Lachesis muta* envenomation in the Western Brazilian Amazon: lessons from a case report. *Rev Soc Bras Med Trop* 55:e0027
- Saladini LY, Magalhães-Junior MJ, da Silva CCF, Oliveira PGC, Kodama RT, Gomes L, Nishiyama-Jr MY, Spencer PJ, da Silva WD, Portaro FCV (2024) Evaluation of the inhibitory potential of synthetic peptides homologous to CDR3 regions of a monoclonal antibody against Bothropic venom serine proteases. *Int J Mol Sci* 25:5181
- Salvador GHM, Cardoso FF, Lomonte B, Fontes MRM (2024) Inhibitors and activators for myotoxic phospholipase A₂-like toxins from snake venoms - a structural overview. *Biochimie* 227:231–247
- Salvador GHM, Pinto ÊKR, Ortolani PL, Fortes-Dias CL, Cavalcante WLG, Soares AM, Lomonte B, Lewin MR, Fontes MRM (2023) Structural basis of the myotoxic inhibition of the *Bothrops pirajai* PrTX-I by the synthetic varespladib. *Biochimie* 207:1–10
- Sampat GH, Hiremath K, Dodakallanavar J, Patil VS, Harish DR, Bira-dar P, Mahadevamurthy RK, Barvaliya M, Roy S (2023) Unraveling snake venom phospholipase A₂: an overview of its structure, pharmacology, and inhibitors. *Pharmacol Rep* 75:1454–1473
- Sanchez EF, Santos CI, Magalhaes A, Diniz CR, Figueiredo S, Gilroy J, Richardson M (2000) Isolation of a proteinase with plasminogen-activating activity from *Lachesis muta muta* (bushmaster) snake venom. *Arch Biochem Biophys* 378:131–141
- Sanchez EF, Freitas TV, Ferreira-Alves DL, Velarde DT, Diniz MR, Cordeiro MN, Agostini-Cotta G, Diniz CR (1992) Biological activities of venoms from South American snakes. *Toxicon* 30:95–103
- Sanchez EF, Felicori LF, Chavez-Olortegui C, Magalhaes HB, Hermogenes AL, Diniz MV, Junqueira-de-Azevedo IL, Magalhaes A, Richardson M (2006) Biochemical characterization and molecular cloning of a plasminogen activator proteinase (LV-PA) from bushmaster snake venom. *Biochim Biophys Acta* 1760:1762–1771
- SINAN. Sistema de Informação de Agravos de Notificação, 2018. Ministério da Saúde. <http://tabnet.datasus.gov.br/cgi/tabcgi.exe?sinanet/cnv/animaisbr.def>. (Accessed 2 June 2024).
- Soares AM, Giglio JR (2003) Chemical modifications of phospholipases A₂ from snake venoms: effects on catalytic and pharmacological properties. *Toxicon* 42:855–868
- Srinivasan K, Nampoothiri M, Khandibharad S, Singh S, Nayak AG, Hariharapura RC (2024) Proteomic diversity of Russell's viper venom: exploring PLA₂ isoforms, pharmacological effects, and inhibitory approaches. *Arch Toxicol* 98:3569–3584
- Suranase V, Jackson TNW, Sunagar K (2022) Contextual constraints: dynamic evolution of snake venom phospholipase A₂. *Toxins* 14:420
- Tasoulis T, Isbister GK (2023) A current perspective on snake venom composition and constituent protein families. *Arch Toxicol* 97:133–153
- Troisi R, Balasco N, Autiero I, Vitagliano L, Sica F (2021) Exosite binding in thrombin: a global structural/dynamic overview of complexes with aptamers and other ligands. *Int J Mol Sci* 22:10803
- Ullah A, Masood R, Ali I, Ullah K, Ali H, Akbar H, Betzel C (2018) Thrombin-like enzymes from snake venom: structural characterization and mechanism of action. *Int J Biol Macromol* 114:788–811
- Vilca-Quispe A, Alvarez-Risco A, Gomes Heleno MA, Ponce-Fuentes EA, Vera-Gonzales C, Zegarar-Aragon HFE, Aquino-Puma JL, Talavera-Núñez ME, Del-Aguila-Arcenales S, Yáñez JA, Ponce-Soto LA (2023) Biochemical and hemostatic description of a thrombin-like enzyme TLBro from *Bothrops roedingeri* snake venom. *Front Chem* 11:1217329
- Wang L, Park Y-J, Jeon YJ, Ryu B (2018) Bioactivities of the edible brown seaweed, *Undaria pinnatifida*: a review. *Aquaculture* 495:873–880
- Weinberg ML, Felicori LF, Bello CA, Magalhães HP, Almeida AP, Magalhães A, Sanchez EF (2004) Biochemical properties of a bushmaster snake venom serine proteinase (LV-Ka), and its kinin releasing activity evaluated in rat mesenteric arterial rings. *J Pharmacol Sci* 96:333–342
- WHO (2019) World Health Organization. <https://www.who.int/en/news-room/fact-sheets/detail/snakebite-envenoming>. (Accessed 2 June 2025)
- Xing M, Li G, Liu Y, Yang L, Zhang Y, Zhang Y, Ding J, Lu M, Yu G, Hu G (2023) Fucoidan from *Fucus vesiculosus* prevents the loss of dopaminergic neurons by alleviating mitochondrial dysfunction through targeting ATP5F1a. *Carbohydr Polym* 303:120470
- Zaqueo KD, Kayano AM, Simões-Silva R, Moreira-Dill LS, Fernandes CF, Fuly AL, Maltarollo VG, Honório KM, da Silva SL, Acosta G, Caballol MA, de Oliveira E, Albericio F, Calderon LA, Soares AM, Stábeli RG (2014) Isolation and biochemical characterization of a new thrombin-like serine protease from *Bothrops pirajai* snake venom. *Biomed Res Int* 2014:595186
- Zingali RB, Ferreira MS, Assafim M, Frattani FS, Monteiro RQ (2005) Bothrojaracin, a *Bothrops jararaca* snake venom-derived (pro) thrombin inhibitor, as an anti-thrombotic molecule. *Pathophysiol Haemost Thromb* 34:160–163

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ANEXO- Declaração de uso de inteligência artificial
Universidade Federal Fluminense
Instituto Biomédico
Programa de pós-graduação em Ciências Biomédicas
Fisiologia e Farmacologia



DECLARAÇÃO DE USO DE INTELIGÊNCIA ARTIFICIAL (IA) NA ESCRITA DE TRABALHOS DE
CONCLUSÃO (TCC)

Título do TCC: Potencial antiveneno de moléculas naturais na neutralização de atividades
tóxico-farmacológicas causadas por venenos de serpentes.

Estudante: Camila de Castro Pinheiro

Orientador(es): André Lopes Fuly

Eu, autor do TCC acima mencionado, declaro que ferramenta(s) de Inteligência Artificial (IA) foi/foram utilizada(s) no processo de elaboração de Tese Camila de Castro Pinheiro. Em conformidade com as políticas de transparência e integridade acadêmica, forneço os seguintes detalhes:

1. Ferramenta(s) de IA utilizada(s):

- o Nome da ferramenta: GPT-5.2
- o Versão: versão de janeiro de 2026
- o Desenvolvedor: OpenAI

2. Finalidade do uso: A IA foi utilizada como auxílio na revisão gramatical na sessão de introdução.

3. Extensão do uso:

- o Porcentagem estimada do conteúdo gerado ou modificado pela IA: 10%
- o Seções específicas onde a IA foi utilizada: Introdução

4. Processo de verificação: Confirmando que a contribuição intelectual humana é predominante, de que toda alteração sugerida por IA foi conferida por um humano, e que a integridade científica e originalidade do trabalho foi mantida.

Data: 01/05/2026

Assinatura do estudante

Ciência do(s) orientador(es)

